ACC FOR THE RATIO OF LOCAL VOLUME TO MINIMAL LOG DISCREPANCY

JIHAO LIU

Abstract. We prove that for klt germs of a fixed dimension whose boundary coefficients belong to a fixed DCC set, the ratios of the local volume to the minimal log discrepancy satisfy the ACC, which answers a question of Han.

Contents

1. Introduction 1
2. Preliminaries 5
3. Monotonicity of the ratio 11
4. Kollár components with bounded log discrepancy 14
5. The invariant Φ and its ACC 31
6. Degeneration to the cone of a Kollár component 50
7. Finite approximation of Φ 54
8. Finitely many families of stacks with faithful line bundles 57
9. Uniform generators of filtered algebras in families 65
10. Lifting sections and an upper bound 73
11. Valuations and rounding of coefficients on index one covers 82
12. Bounding $\widehat{\mathrm{vol}}(X \ni x, B)/a(E, X, B)$ by $\Phi(E, B_E)$ 93
13. Proof of the main theorem 100
References 102

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

[Li18] introduced the normalized volume of a valuation. For a klt germ $(X \ni x, B)$, the *local volume* $\widehat{\mathrm{vol}}(X \ni x, B)$ is the infimum of the normalized volumes of valuations on X with center x (see Definition 2.7).

[LLX20, Theorem 6.13] compares the local volume with the minimal log discrepancy: if $(X \ni x, B)$ is a klt germ of dimension $n \geq 2$ such that B is a $\mathbb{Q}$-divisor, then

$$\widehat{\mathrm{vol}}(X \ni x, B) < n^n \mathrm{mld}(X \ni x, B),$$

and there exists an open neighborhood U of x such that $(U, B|_U)$ is $\frac{\widehat{\mathrm{vol}}(X \ni x, B)}{n^n}$-lc. [LLX20, Question 6.16] asks whether n^n can be replaced by n^{n-1} when $B = 0$, that is, whether

$$\mathrm{mld}(X \ni x) \geq \frac{\widehat{\mathrm{vol}}(X \ni x)}{n^{n-1}}$$

for any klt germ $X \ni x$ of dimension n.

Date: November 10, 2026.
2020 *Mathematics Subject Classification.* 14B05, 14E30, 14J45.
Key words and phrases. local volume, minimal log discrepancy, Kollár component, ACC, normalized volume.

For a positive integer n and a finite set $\Gamma \subset [0,1]$, [LLX20, Question 6.12] asks whether 0 is the only accumulation point of the set

$$\{\widehat{\mathrm{vol}}(X \ni x, B) \mid (X \ni x, B) \text{ is a klt germ of dimension } n \text{ and } B \in \Gamma\}.$$

For $n = 3$, the affirmative answer is [Zhu24b, Corollary 1.4] (for rational coefficients, also [LMS26, Theorem 2]), and [XZ24, Theorem 1.2] answers [LLX20, Question 6.12] affirmatively in any dimension. When Γ is only assumed to be a DCC set, the set of local volumes satisfies the ACC:

Theorem 1.1 ([HLQ24, Theorem 1.2])**.** *Let n be a positive integer and $\Gamma \subset [0,1]$ a DCC set. Then the set*

$$\{\widehat{\mathrm{vol}}(X \ni x, B) \mid (X \ni x, B) \text{ is a klt germ of dimension } n \text{ and } B \in \Gamma\}$$

satisfies the ACC.

When $n = 3$, [Liu26, Theorem 1.4] proves $\widehat{\mathrm{vol}}(X \ni x) \leq 9\mathrm{mld}(X \ni x)$ for any klt germ $X \ni x$ of dimension 3, which answers [LLX20, Question 6.16] affirmatively in dimension 3. [Han26, Theorem 1.1] answers [LLX20, Question 6.16] affirmatively in any dimension $n \geq 2$, also for germs with a boundary:

Theorem 1.2 ([Han26, Theorem 1.1])**.** *Let $(X \ni x, B)$ be a klt germ of dimension $n \geq 2$. Then*

$$\widehat{\mathrm{vol}}(X \ni x, B) \leq n^{n-1}\mathrm{mld}(X \ni x, B),$$

and the equality holds if and only if $B = 0$ near x and the germ $X \ni x$ is analytically isomorphic to the cyclic quotient singularity $\frac{1}{r}(1, \ldots, 1)$ for some positive integer r.

Following [Han26, Section 1], for an integer $n \geq 2$ and a set $\Gamma \subset [0,1]$, we let

$$\mathcal{R}_{n,\Gamma} := \left\{ \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{\mathrm{mld}(X \ni x, B)} \,\middle|\, (X \ni x, B) \text{ is a klt germ of dimension } n \text{ and } B \in \Gamma \right\}. \tag{1.1}$$

By Theorem 1.2 and Lemma 2.8(1), we have

$$\mathcal{R}_{n,\Gamma} \subset (0, n^{n-1}].$$

For finite coefficient sets, [Han26] proves the following finiteness.

Theorem 1.3 ([Han26, Theorem 1.3])**.** *Let $n \geq 2$ be an integer and $\Gamma \subset [0,1]$ a finite set. Then $\mathcal{R}_{n,\Gamma} \cap [y, +\infty)$ is a finite set for any real number $y > 0$.*

In particular, $\mathcal{R}_{n,\Gamma}$ satisfies the ACC when Γ is finite. As [Han26, Section 1] points out, Theorem 1.3 is not a formal consequence of [XZ24, Theorem 1.2]: since $\mathrm{mld}(X \ni x, B)$ may tend to 0 along a sequence of germs, the fact that 0 is the only accumulation point of the set of local volumes does not by itself control the quotients in (1.1).

[Han26, Question 5.3] asks whether $\mathcal{R}_{n,\Gamma}$ satisfies the ACC for any integer $n \geq 2$ and any DCC set $\Gamma \subset [0,1]$. The goal of this paper is to answer [Han26, Question 5.3] affirmatively.

Theorem 1.4. *Let $n \geq 2$ be an integer and $\Gamma \subset [0,1]$ a DCC set. Then the set*

$$\mathcal{R}_{n,\Gamma} = \left\{ \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{\mathrm{mld}(X \ni x, B)} \,\middle|\, (X \ni x, B) \text{ is a klt germ of dimension } n \text{ and } B \in \Gamma \right\}$$

satisfies the ACC.

Remark 1.5. In Theorem 1.4, as in the definition of $\mathcal{R}_{n,\Gamma}$ in [Han26, Section 1 and Definition 2.1], X is a normal quasi-projective variety, $B \geq 0$ is an $\mathbb{R}$-divisor, and $K_X + B$ is $\mathbb{R}$-Cartier near x (Definitions 2.1 and 2.2).

Sketch of the proof. Suppose that $\mathcal{R}_{n,\Gamma}$ does not satisfy the ACC. Then there exist klt germs $(X_i \ni x_i, B_i)$ of dimension n with $B_i \in \Gamma$ such that

$$R_i := \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{\mathrm{mld}(X_i \ni x_i, B_i)}$$

is strictly increasing, and $R_\infty := \lim_{i\to+\infty} R_i \le n^{n-1}$ (Theorem 1.2). Let $\epsilon := R_1$. The proof in Section 13 has three steps, each possibly after passing to a subsequence.

(1) $\lim_{i\to+\infty} \widehat{\mathrm{vol}}(X_i \ni x_i, B_i) = 0$.

(2) There exist real numbers $C \ge 1$ and $\kappa > 0$ and Kollár components E_i of $(X_i \ni x_i, B_i)$ (Definition 2.5) such that

$$a_i := a(E_i, X_i, B_i) \le C\mathrm{mld}(X_i \ni x_i, B_i) \le \frac{C}{\epsilon}\widehat{\mathrm{vol}}(X_i \ni x_i, B_i) \quad \text{and} \quad \delta(E_i, B_{E_i}) \ge \kappa$$

(Theorem 4.7), where the second inequality holds as $R_i \ge R_1 = \epsilon$, with (E_i, B_{E_i}) the klt log Fano pair given by adjunction and δ the delta invariant (Definition 2.12). Thus $\lim_{i\to+\infty} a_i = 0$ by (1).

(3) For any $i \gg 0$, $a_i \le \theta/2$, and $\mathrm{mld}(X_i \ni x_i, B_i) = a_i$: since (E_i, B_{E_i}) is θ-lc for a real number $\theta > 0$ independent of i, any prime divisor $F \ne E_i$ over $X_i \ni x_i$ has $a(F, X_i, B_i) > \theta/2 \ge a_i$ (see Proposition 4.9(2)(5)). Thus $R_i = \widehat{\mathrm{vol}}(X_i \ni x_i, B_i)/a_i$, and for any real number $\tau > 0$ and any $i \gg 0$,

$$\Phi(E_i, B_{E_i}) \le R_i < \Phi(E_i, B_{E_i}) + \tau$$

(Theorems 6.1 and 12.1), where Φ is the invariant introduced in Definition 5.6. The set of numbers $\Phi(E, B_E)$, where E is a Kollár component of a klt germ $(X \ni x, B)$ of dimension n with $B \in \Gamma$, satisfies the ACC (Theorem 6.1(2)).

Thus $\lim_{i\to+\infty} \Phi(E_i, B_{E_i}) = R_\infty$. Since the $\Phi(E_i, B_{E_i})$ satisfy the ACC, possibly passing to a subsequence, $\Phi(E_i, B_{E_i})$ is decreasing, so

$$R_\infty \le \Phi(E_i, B_{E_i}) \le R_i < R_\infty,$$

a contradiction.

Step (1) follows [HLQ24, Proof of Theorem 5.2], with Theorem 1.3 in place of [XZ24, Theorem 1.2]: if $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)$ is bounded from below by a positive number, then raising the coefficients of B_i into a finite set (see Theorems 2.20, 2.18, and 2.19) yields klt germs whose ratios belong to $[(1-s_i)^n R_i, R_i]$ with $\lim_{i\to+\infty} s_i = 0$, so they tend to R_∞ from below, which contradicts Theorem 1.3. The upper bound R_i holds since the ratio of a fixed germ does not increase when the boundary increases (see Proposition 3.1).

In Step (2), when $\mathrm{mld}(X_i \ni x_i, B_i)$ is small (the other case follows from [HLQ24, Theorem 7.5], [HLS26, Proposition 4.3(1)], and [Bir21, Theorem 1.6]), a prime divisor E_i over $X_i \ni x_i$ with $a_i < 2\mathrm{mld}(X_i \ni x_i, B_i)$ is the exceptional divisor of an lc blow-up (see Definition 2.5 and Lemma 4.2), which is a plt blow-up, since otherwise

$$\frac{\epsilon}{2} a_i < \widehat{\mathrm{vol}}(X_i \ni x_i, B_i) \le \frac{n^n}{t_*^2} a_i^2$$

for a real number $t_* > 0$ depending only on n and Γ (see Proposition 4.4), which fails for a_i small. The bound $\delta(E_i, B_{E_i}) \ge \kappa$ follows from a lower bound of the local volumes of (E_i, B_{E_i}) at closed points (see Proposition 4.5) and [Bir21, Theorem 1.6]. Proposition 4.5 is proved on cyclic covers of degree $\lfloor 1/a_i \rfloor$ branched along E_i, taken after an index 1 cover (see Definition-Lemma 2.16), whose local volumes are at least

$$\lfloor 1/a_i \rfloor \widehat{\mathrm{vol}}(X_i \ni x_i, B_i) > \frac{\epsilon}{4} :$$

after the coefficients are moved into a finite set, their K-semistable degenerations belong to a bounded family [XZ24, Theorem 1.1] and split off a factor $\mathbb{A}^1$ for $i \gg 0$, as in [XZ24, Proof of Theorem 4.3].

In Step (3), $(X_i \ni x_i, B_i)$ degenerates to a cyclic cover of the cone $C(E_i, -qE_i|_{E_i})$ for q divisible enough (see Construction-Theorem 6.2), and $\Phi(E_i, B_{E_i})$ is the ratio $\widehat{\mathrm{vol}}/\mathrm{mld}$ at the vertex of $C(E_i, -qE_i|_{E_i})$ with the boundary induced by B_{E_i} (see Definition 5.1 and Theorem 5.9), which implies the lower bound (see Theorem 6.1(1)). For the upper bound, divisorial valuations w' on charts of the index 1 covering stack of E_i (see Definition 10.1, Theorem 11.1, and Proposition 10.5) imply

$$\frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} \leq (1-\sigma_{i,j})^{-n}\Psi_{E_i,B_{E_i}}(F_j, u_{i,j})$$

for the finitely many pairs (F_j, u_j) of Theorem 7.1, whose values $\Psi_{E_i,B_{E_i}}(F_j, u_j)$ and the number $(-(K_{E_i}+B_{E_i}))^{n-1} \geq \widehat{\mathrm{vol}}(X_i \ni x_i, B_i)/a_i$ (see Theorem 6.1(1)) have minimum less than $\Phi(E_i, B_{E_i}) + \tau/3$, where Ψ is the function of Definition 5.3, $\lim_{i\to+\infty}\sigma_{i,j} = 0$, and $\lim_{i\to+\infty} u_{i,j} = u_j$. The coefficients are moved into a finite set only on the charts, since moving them needs a positive lower bound of local volumes (see Theorem 2.19), which holds on the charts (see Lemma 11.2) but not on X_i, as $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i) \to 0$. On the charts, the error $\sigma_{i,j}$ is multiplicative in the plt log discrepancy (see Lemma 11.3).

Structure of the paper. In Section 2, we recall Kollár components and known properties of local volumes, approximate real boundaries by rational ones, recall log Fano pairs and index 1 covers of $\mathbb{Q}$-Cartier prime divisors, and recall results on klt germs whose local volumes are bounded from below. In Section 3, we show that for a fixed germ $X \ni x$, the ratio $\widehat{\mathrm{vol}}(X \ni x, B)/\mathrm{mld}(X \ni x, B)$ does not increase when B increases. In Section 4, we construct Kollár components E of klt germs $(X \ni x, B)$ such that $a(E, X, B)$ is bounded by a constant multiple of $\mathrm{mld}(X \ni x, B)$ and $\delta(E, B_E)$ is bounded from below by a positive constant, when the coefficients of B belong to a DCC set and $\widehat{\mathrm{vol}}(X \ni x, B)/\mathrm{mld}(X \ni x, B)$ is bounded from below by a positive number. In Section 5, we study cones over log Fano pairs and prove the ACC for the ratios at the vertices of such cones when the coefficients of the boundaries belong to a DCC set. In Section 6, we degenerate a klt germ $(X \ni x, B)$ to the cone of a Kollár component E, bound $\widehat{\mathrm{vol}}(X \ni x, B)/a(E, X, B)$ from below by the ratio of the cone over (E, B_E), and prove the ACC for the ratios of the cones over the pairs (E, B_E) when the coefficients of B belong to a DCC set. In Section 7, we show that, after passing to a subsequence, the pairs (E, B_E) given by Section 4 for a sequence of germs belong to one bounded family with converging coefficients, and we approximate the ratios of the cones over the pairs in a bounded family by the minimum of finitely many explicit functions of the coefficients, uniformly for coefficients in a compact set. In Section 8, we show that normal tame stacks with faithful line bundles (Definition 8.1), whose coarse spaces belong to a bounded family of varieties of Fano type, whose stabilizer orders divide a fixed integer, and whose coarse maps are isomorphisms outside a fixed closed subset of the family containing no fiber, belong to finitely many families up to twists by line bundles on the coarse spaces. In Section 9, we find uniform generators of the filtered algebras of sections in such families. In Section 10, we lift sections to the original germ and obtain an upper bound for the local volume. In Section 11, on index 1 covers, we construct divisorial valuations whose value on a given Cartier divisor is any sufficiently large integer, bound local volumes from below, and round the coefficients of boundaries up to a finite set. In Section 12, we show that for a Kollár component E as in Section 4 and any real number $\tau > 0$, $\widehat{\mathrm{vol}}(X \ni x, B)/a(E, X, B)$ is less than the ratio of the cone over (E, B_E) plus τ once $a(E, X, B)$ is small enough. In Section 13, we complete the proof of the main theorem.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and Kollár components.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. A *germ* $(X \ni x, B)$ is a pair $(X \ni x, B)$ such that x is a closed point. A germ $(X \ni x, B)$ is *klt* if (X, B) is klt near x. A prime divisor E over X is *over* $X \ni x$ if $\operatorname{center}_X E = x$. The *minimal log discrepancy* of a germ $(X \ni x, B)$ is

$$\operatorname{mld}(X \ni x, B) := \inf\{a(E, X, B) \mid E \text{ is a prime divisor over } X \ni x\}.$$

Definition 2.3. Let $\epsilon \geq 0$ be a real number. We say that a pair (X, B) is *ϵ-lc* if $a(E, X, B) \geq \epsilon$ for any prime divisor E over X. We say that (X, B) is *ϵ-plt* if $a(E, X, B) > \epsilon$ for any exceptional prime divisor E over X.

Definition 2.4. For a pair $(X, S + B)$, where S is a normal prime divisor and $B \geq 0$ has no component S, the *different* $\operatorname{Diff}_S(B)$ is defined by $K_S + \operatorname{Diff}_S(B) := (K_X + S + B)|_S$.

Definition 2.5. Let $(X \ni x, B)$ be a klt germ. A projective birational morphism $f : Y \to X$ from a normal variety Y is an *lc blow-up* (resp. a *plt blow-up*) of $(X \ni x, B)$ if f is an isomorphism over $X \backslash \{x\}$, the exceptional locus of f is a prime divisor E with $f(E) = x$, E is $\mathbb{Q}$-Cartier, $-E$ is ample over X, and $(Y, E + f_*^{-1}B)$ is lc (resp. plt) near E. We let $B_Y := f_*^{-1}B$. Then

$$K_Y + E + B_Y = f^*(K_X + B) + a(E, X, B)E.$$

If f is a plt blow-up, then E is called a *Kollár component* of $(X \ni x, B)$, and we let $B_E := \operatorname{Diff}_E(B_Y)$. Then, by adjunction, (E, B_E) is a klt pair, and $-(K_E + B_E) \sim_{\mathbb{R}} -a(E, X, B)E|_E$ is ample.

Definition 2.6. For a set $\Gamma \subset [0, 1]$ and an $\mathbb{R}$-divisor $B = \sum_{i=1}^m b_i B_i$, where $B_1, \ldots, B_m$ are the irreducible components of B, we write $B \in \Gamma$ if $b_i \in \Gamma$ for any i. We let $\|B\| := \max_i |b_i|$ if $B \neq 0$, and $\|0\| := 0$. A set $\Gamma \subset \mathbb{R}$ satisfies the *DCC* (resp. the *ACC*) if any decreasing (resp. increasing) sequence in Γ is eventually constant.

2.2. Valuations and local volumes.

Definition 2.7. Let X be a normal variety. A *valuation* on X is a map v from the set of non-zero rational functions on X to $\mathbb{R}$ such that $v(\phi\psi) = v(\phi) + v(\psi)$, $v(\phi + \psi) \geq \min\{v(\phi), v(\psi)\}$, and $v|_{\mathbb{C}^\times} = 0$, and such that v has a center $\operatorname{center}_X v$ on X. For a pair (X, B) and a valuation v on X, let $a(v, X, B) \in \mathbb{R} \cup \{+\infty\}$ be the log discrepancy of v in the sense of [Han26, Definition 2.2], which extends the log discrepancy of divisorial valuations in Definition 2.1. For an $\mathbb{R}$-divisor D

on X which is $\mathbb{R}$-Cartier near $\mathrm{center}_X v$, we let $v(D) := \sum_j r_j v(\psi_j)$, where $D = \sum_j r_j D_j$ near $\mathrm{center}_X v$ with $r_j \in \mathbb{R}$ and Cartier divisors D_j with local equations ψ_j at $\mathrm{center}_X v$. If (X, B) and $(X, B + D)$ are pairs, then, by [JM12, Corollary 4.8 and Lemma 5.3],

$$a(v, X, B + D) = a(v, X, B) - v(D). \tag{2.1}$$

Let $\eta \in X$ be a (not necessarily closed) point with $\mathrm{codim}_X \overline{\{\eta\}} = r$ and v a valuation on X with $\mathrm{center}_X v = \eta$. For a real number $\lambda \geq 0$ let

$$\mathfrak{a}_\lambda(v) := \{\phi \in \mathcal{O}_{X,\eta} \mid v(\phi) \geq \lambda\},$$

and let

$$\mathrm{vol}(v) := \lim_{\lambda \to +\infty} \frac{\ell(\mathcal{O}_{X,\eta}/\mathfrak{a}_\lambda(v))}{\lambda^r/r!};$$

the limit exists by [Cut14, Theorem 1.1]. The *normalized volume* of v, introduced in [Li18], is $\widehat{\mathrm{vol}}(v, X, B) := a(v, X, B)^r \mathrm{vol}(v)$ if $a(v, X, B) < +\infty$, and $\widehat{\mathrm{vol}}(v, X, B) := +\infty$ otherwise. For a prime divisor E over X with center η, we write $\widehat{\mathrm{vol}}(E, X, B)$ for the normalized volume of the divisorial valuation of E. If (X, B) is klt near η, then the *local volume* of $(X \ni \eta, B)$ is

$$\widehat{\mathrm{vol}}(X \ni \eta, B) := \inf\{\widehat{\mathrm{vol}}(v, X, B) \mid v \text{ is a valuation on } X \text{ with } \mathrm{center}_X v = \eta\}.$$

Lemma 2.8. *Let $(X \ni x, B)$ be a klt germ of dimension $n \geq 2$. Then:*

(1) $0 < \widehat{\mathrm{vol}}(X \ni x, B) \leq n^n$.
(2) $\widehat{\mathrm{vol}}(X \ni x, B) = \inf_E \widehat{\mathrm{vol}}(E, X, B)$, *the infimum taken over Kollár components E of $(X \ni x, B)$.*
(3) For any Kollár component E of $(X \ni x, B)$, $\widehat{\mathrm{vol}}(E, X, B) = a(E, X, B) \cdot (-(K_E + B_E))^{n-1}$.

Proof. (1) By [LLX20, Theorem 2.5], $\widehat{\mathrm{vol}}(X \ni x, B) > 0$. The inequality $\widehat{\mathrm{vol}}(X \ni x, B) \leq n^n$ is [LX19, Lemma A.1] (for real coefficients, see [Han26, Lemma 2.11]).
(2) By (3), this is [HLQ24, Section 2 and Remark 2.1].
(3) This is [Han26, Lemma 2.8]. □

2.3. **Rational approximation and local volumes.**

Lemma 2.9. *Let X be a normal variety, $x \in X$ a closed point, and $P_1, \ldots, P_m$ distinct prime divisors on X. For $b \in \mathbb{R}^m$ let $B(b) := \sum_{j=1}^m b_j P_j$, and let W be the set of $b \in \mathbb{R}^m$ such that $K_X + B(b)$ is $\mathbb{R}$-Cartier near x.*

(1) The set W is either empty or an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$.
(2) Let $b \in W$ such that $B(b) \geq 0$ and $(X \ni x, B(b))$ is klt, and let $0 < \epsilon < 1$ be a real number. Then there exist an open neighborhood U of x and a real number $c > 0$ such that $(U, B(b)|_U)$ is klt and the following hold for any $b' \in W$ with $B(b') \geq 0$ and $\|B(b') - B(b)\| < c$.
  (a) $K_U + B(b')|_U$ is $\mathbb{R}$-Cartier and $(U, B(b')|_U)$ is klt.
  (b) For any valuation v on X with $\mathrm{center}_X v \in U$,

$$(1 - \epsilon) a(v, X, B(b)) \leq a(v, X, B(b')) \leq (1 + \epsilon) a(v, X, B(b)).$$

  (c) For any (not necessarily closed) point $\eta \in U$,

$$(1 - \epsilon)^r \widehat{\mathrm{vol}}(X \ni \eta, B(b)) \leq \widehat{\mathrm{vol}}(X \ni \eta, B(b')) \leq (1 + \epsilon)^r \widehat{\mathrm{vol}}(X \ni \eta, B(b)),$$

  where $r := \mathrm{codim}_X \overline{\{\eta\}}$.
  (d) We have

$$(1 - \epsilon)\mathrm{mld}(X \ni x, B(b)) \leq \mathrm{mld}(X \ni x, B(b')) \leq (1 + \epsilon)\mathrm{mld}(X \ni x, B(b)).$$

(3) *Let $f : Y \to X$ be a plt blow-up of $(X \ni x, B(b_0))$ for some $b_0 \in W$ with $B(b_0) \geq 0$, and let W_f be the set of $b \in W$ such that $B(b) \geq 0$ and f is a plt blow-up of $(X \ni x, B(b))$. Then W_f is relatively open in $\{b \in W \mid B(b) \geq 0\}$, and b_0 is a convex combination of finitely many points $b \in W_f \cap \mathbb{Q}^m$ with* $\operatorname{Supp} B(b) \subset \operatorname{Supp} B(b_0)$.

Proof. (1) Let K_X be a canonical divisor of X and $\mathrm{Cl}_x := \mathrm{Cl}(\mathcal{O}_{X,x})$. Since an $\mathbb{R}$-divisor on X is $\mathbb{R}$-Cartier near x if and only if its class in $\mathrm{Cl}_x \otimes_{\mathbb{Z}} \mathbb{R}$ is zero, $\mathbb{R}$ is flat over $\mathbb{Q}$, and the classes of $K_X, P_1, \dots, P_m$ in $\mathrm{Cl}_x \otimes_{\mathbb{Z}} \mathbb{Q}$ span a finite-dimensional $\mathbb{Q}$-vector space, W is the solution set of a system of linear equations with rational coefficients.

(2) Let $\{w_k\}_k$ be a basis of the linear subspace $W - b$ of $\mathbb{R}^m$, and for $b' \in W$ write

$$b' - b = \sum_k t_k(b') w_k,$$

where each t_k is a linear function of $b' - b$. Since $b, b + w_k \in W$, the $\mathbb{R}$-divisor $B(w_k) = (K_X + B(b + w_k)) - (K_X + B(b))$ is $\mathbb{R}$-Cartier near x for any k. Let U be an open neighborhood of x such that $K_X + B(b)$ and $B(w_k)$ for any k are $\mathbb{R}$-Cartier on U and $(U, B(b)|_U)$ is klt. Then for any $b' \in W$,

$$K_U + B(b')|_U = K_U + B(b)|_U + \sum_k t_k(b') B(w_k)|_U$$

is $\mathbb{R}$-Cartier.

Let $h : X' \to U$ be a log resolution of $(U, \sum_j P_j|_U)$, and let D_i be the irreducible components of $\operatorname{Supp} h_*^{-1}(\sum_j P_j|_U) \cup \operatorname{Ex}(h)$, so that $\sum_i D_i$ is simple normal crossing. For $b' \in W$ define $B_{X'}(b')$ by $K_{X'} + B_{X'}(b') := h^*(K_U + B(b')|_U)$ and let $\beta_i(b') := \operatorname{mult}_{D_i} B_{X'}(b')$. Then $\operatorname{Supp} B_{X'}(b') \subset \bigcup_i D_i$ and

$$\beta_i(b') = \beta_i(b) + \sum_k t_k(b') \operatorname{mult}_{D_i} h^* B(w_k),$$

so $\beta_i(b') - \beta_i(b)$ is a linear function of $b' - b$. Since $(U, B(b)|_U)$ is klt,

$$\epsilon_0 := \min_i (1 - \beta_i(b)) > 0.$$

Thus there exists a real number $c > 0$ such that $|\beta_i(b') - \beta_i(b)| \leq \epsilon\epsilon_0$ for any i and any $b' \in W$ with $\|B(b') - B(b)\| < c$. We fix such b' with $B(b') \geq 0$.

Let v be a valuation on X with $\operatorname{center}_X v \in U$. Since h is proper, v has a center on X', and $v(D_i) \geq 0$ for any i. Since $K_{X'} + B_{X'}(b_*) = h^*(K_U + B(b_*)|_U)$ for $b_* \in \{b, b'\}$ and the log discrepancy is affine in the boundary [JM12, Corollary 4.8 and Lemma 5.3],

$$a(v, X, B(b_*)) = a\Big(v, X', \sum_i D_i\Big) + \sum_i (1 - \beta_i(b_*)) v(D_i),$$

where $a(v, X', \sum_i D_i) \geq 0$ by [JM12, Proposition 5.1 and Corollary 5.4]. If $a(v, X', \sum_i D_i) = +\infty$, then $a(v, X, B(b)) = a(v, X, B(b')) = +\infty$. Otherwise

$$\begin{aligned} |a(v, X, B(b')) - a(v, X, B(b))| &= \Big| \sum_i (\beta_i(b) - \beta_i(b')) v(D_i) \Big| \leq \epsilon\epsilon_0 \sum_i v(D_i) \\ &\leq \epsilon \sum_i (1 - \beta_i(b)) v(D_i) \leq \epsilon a(v, X, B(b)). \end{aligned}$$

This proves (b). For divisorial valuations, (b) implies that $(U, B(b')|_U)$ is klt, which, as $K_U + B(b')|_U$ is $\mathbb{R}$-Cartier, proves (a). Since $\mathrm{vol}(v)$ does not depend on the boundary, raising the inequalities in (b) to the r-th power, multiplying by $\mathrm{vol}(v)$, and taking the infimum over valuations v with $\operatorname{center}_X v = \eta$ proves (c). Taking the infimum of the inequalities in (b) over prime divisors over $X \ni x$ proves (d).

(3) Since W is an affine subspace of $\mathbb{R}^m$ by (1), possibly shrinking X to a neighborhood of x, we may assume that $K_X + B(b)$ is $\mathbb{R}$-Cartier for any $b \in W$. Let E be the exceptional divisor of f and $g : X' \to Y$ a log resolution of $(Y, E + f_*^{-1} \sum_j P_j)$. Then $h := f \circ g$ is a log resolution

of $(X, \sum_j P_j)$. Let D_i be the irreducible components of $\operatorname{Supp} h_*^{-1}(\sum_j P_j) \cup \operatorname{Ex}(h)$. Then $\sum_i D_i$ is simple normal crossing and $g_*^{-1}E$ is one of the D_i. Since $K_X + B(b)$ is an affine function of $b \in W$ with values in $\mathbb{R}$-Cartier divisors and h^* is $\mathbb{R}$-linear, $a(D_i, X, B(b))$ is an affine function of $b \in W$ for any i. For $b \in W$ let $B_Y(b) := f_*^{-1}B(b)$. Since f is an isomorphism over $X\backslash\{x\}$,

$$K_Y + E + B_Y(b) = f^*(K_X + B(b)) + a(E, X, B(b))E,$$

which is $\mathbb{R}$-Cartier as E is $\mathbb{Q}$-Cartier, and

$$a(D_i, Y, E + B_Y(b)) = a(D_i, X, B(b)) - a(E, X, B(b)) \operatorname{mult}_{D_i} g^*E.$$

Hence $a(D_i, Y, E + B_Y(b))$ is also an affine function of $b \in W$.

Let $b \in W$ with $B(b) \geq 0$. The conditions that f is an isomorphism over $X\backslash\{x\}$, that $\operatorname{Ex}(f) = E$ with $f(E) = x$, that E is $\mathbb{Q}$-Cartier, and that $-E$ is ample over X do not depend on b. Since $\sum_i D_i$ is simple normal crossing, $(X \ni x, B(b))$ is klt if and only if

$$a(D_i, X, B(b)) > 0 \text{ for any } i \text{ with } x \in h(D_i),$$

and $(Y, E + B_Y(b))$ is plt near E if and only if

$$a(D_i, Y, E + B_Y(b)) > 0 \text{ for any } i \text{ with } D_i \neq g_*^{-1}E \text{ and } g(D_i) \cap E \neq \emptyset.$$

Thus W_f is the subset of $\{b \in W \mid B(b) \geq 0\}$ defined by finitely many strict inequalities between continuous functions of b, hence relatively open in $\{b \in W \mid B(b) \geq 0\}$.

Write $b_0 = (b_{0,1}, \ldots, b_{0,m})$, and let

$$W' := \{b \in W \mid b_j = 0 \text{ for any } j \text{ with } b_{0,j} = 0\},$$
$$W'' := \{b \in W' \mid b_j > 0 \text{ for any } j \text{ with } b_{0,j} > 0\} \cap W_f.$$

Since W' is the intersection of W with a coordinate subspace, by (1), W' is an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$, so $W' \cap \mathbb{Q}^m$ is dense in W'. Since $W'' \subset \{b \in W \mid B(b) \geq 0\}$ and W_f is relatively open in $\{b \in W \mid B(b) \geq 0\}$, W'' is an open neighborhood of b_0 in W'. Hence b_0 is a convex combination of finitely many points of $W'' \cap \mathbb{Q}^m$. Any $b \in W'' \cap \mathbb{Q}^m$ satisfies $b \in W_f \cap \mathbb{Q}^m$ and $\operatorname{Supp} B(b) \subset \operatorname{Supp} B(b_0)$. □

Lemma 2.10. *Let $(X \ni x, B)$ be a klt germ of dimension n.*

(1) If $g : (X' \ni x') \to (X \ni x)$ is a finite Galois morphism from a normal variety X' such that $g^{-1}(x) = \{x'\}$ and $K_{X'} + B' = g^*(K_X + B)$ for some $\mathbb{R}$-divisor $B' \geq 0$, then $\widehat{\operatorname{vol}}(X' \ni x', B') = \deg(g) \cdot \widehat{\operatorname{vol}}(X \ni x, B)$.

(2) If $g : (X' \ni x') \to (X \ni x)$ is an étale morphism with $g(x') = x$, then

$$\widehat{\operatorname{vol}}(X' \ni x', g^*B) = \widehat{\operatorname{vol}}(X \ni x, B).$$

(3) If $h : Y \to X$ is a proper birational morphism and (Y, B_Y) is a klt pair such that

$$K_Y + B_Y \leq h^*(K_X + B),$$

then $\widehat{\operatorname{vol}}(Y \ni y, B_Y) \geq \widehat{\operatorname{vol}}(X \ni x, B)$ for any closed point $y \in h^{-1}(x)$.

(4) $a(F, X, B) \geq \frac{\widehat{\operatorname{vol}}(X \ni x, B)}{n^n}$ for any prime divisor F over X with $x \in \operatorname{center}_X F$.

Proof. (1) This is [XZ21, Theorem 1.3] (for real coefficients, see [Han26, Lemma 2.10]).

(2) We argue as in [HLQ23, Proposition 2.24], which treats the case $B = 0$. Possibly shrinking X' to a neighborhood of x', we may assume that $g^{-1}(x) = \{x'\}$. Since g is étale, $(X' \ni x', g^*B)$ is a klt germ and

$$\mathcal{O}_{X,x}/\mathfrak{m}_x^k \cong \mathcal{O}_{X',x'}/\mathfrak{m}_{x'}^k \text{ for any } k \in \mathbb{N}^+,$$

so $\mathfrak{b} \to \mathfrak{b}\mathcal{O}_{X',x'}$ is a bijection between $\mathfrak{m}_x$-primary and $\mathfrak{m}_{x'}$-primary ideals which preserves Hilbert–Samuel multiplicities and log canonical thresholds with respect to $(X \ni x, B)$ and $(X' \ni x', g^*B)$ [HLQ23, Definition 2.2]. Thus (2) follows from [HLQ23, Lemma 2.13].

(3) This is [HLQ24, Lemma 2.4].

(4) First assume that B is a $\mathbb{Q}$-divisor. Then $K_X + B$ is $\mathbb{Q}$-Cartier near x, since the natural map

$$\mathrm{Cl}(\mathcal{O}_{X,x}) \otimes_{\mathbb{Z}} \mathbb{Q} \to \mathrm{Cl}(\mathcal{O}_{X,x}) \otimes_{\mathbb{Z}} \mathbb{R}$$

is injective. By [LLX20, Theorem 6.13], there exists an open neighborhood U of x such that $(U, B|_U)$ is $\frac{\widehat{\mathrm{vol}}(X \ni x, B)}{n^n}$-lc. Since $x \in \mathrm{center}_X F$, F is a prime divisor over U, so

$$a(F, X, B) = a(F, U, B|_U) \geq \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{n^n}.$$

In general, possibly shrinking X to a neighborhood of x, we may write $B = \sum_{j=1}^m b_j P_j$, where $P_1, \dots, P_m$ are distinct prime divisors through x and $b_j > 0$ for any j. For $t \in \mathbb{R}^m$ let $B(t) := \sum_{j=1}^m t_j P_j$, so $B = B(b)$, and let W be the set of $t \in \mathbb{R}^m$ such that $K_X + B(t)$ is $\mathbb{R}$-Cartier near x, so $b \in W$. Fix a real number $0 < \epsilon < 1$. By Lemma 2.9(2) for b and ϵ, there exist an open neighborhood U of x and a real number $c > 0$ such that for any $b' \in W$ with $B(b') \geq 0$ and $\|B(b') - B(b)\| < c$,

$$a(v, X, B(b')) \leq (1+\epsilon) a(v, X, B(b)) \quad \text{for any valuation } v \text{ on } X \text{ with } \mathrm{center}_X v \in U, \tag{2.2}$$

$$\widehat{\mathrm{vol}}(X \ni x, B(b')) \geq (1-\epsilon)^n \widehat{\mathrm{vol}}(X \ni x, B(b)). \tag{2.3}$$

By Lemma 2.9(1), $W \cap \mathbb{Q}^m$ is dense in W, so, since $b_j > 0$ for any j, there exists $b' \in W \cap \mathbb{Q}^m$ with $b'_j > 0$ for any j and $\|B(b') - B(b)\| < c$. By Lemma 2.9(2), $(X \ni x, B(b'))$ is a klt germ. Since the generic point of $\mathrm{center}_X F$ is in U, the rational case, (2.2), and (2.3) imply

$$(1+\epsilon) a(F, X, B) \geq a(F, X, B(b')) \geq \frac{\widehat{\mathrm{vol}}(X \ni x, B(b'))}{n^n} \geq (1-\epsilon)^n \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{n^n}.$$

Letting ϵ tend to 0, we obtain $a(F, X, B) \geq \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{n^n}$. □

2.4. **Log Fano pairs and index one covers.**

Definition 2.11. A pair (S, B) with S projective is a *klt log Fano pair* (resp. a *klt weak log Fano pair*) if (S, B) is klt and $-(K_S + B)$ is ample (resp. nef and big). A normal projective variety S is *of Fano type* if (S, B) is a klt log Fano pair for some $\mathbb{R}$-divisor $B \geq 0$.

Definition 2.12. Let (S, B) be a klt weak log Fano pair of dimension $d \geq 1$ and $L := -(K_S + B)$. For a prime divisor F over S, let

$$S_L(F) := \frac{1}{L^d} \int_0^{+\infty} \mathrm{vol}(h^* L - tF) dt,$$

where $h : S' \to S$ is a resolution such that F is a divisor on S' and $\mathrm{vol}(h^* L - tF)$ is the volume of the $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $h^* L - tF$ [Laz04, Section 2.2.C]. We let

$$\delta(S, B) := \inf \left\{ \frac{a(F, S, B)}{S_L(F)} \;\middle|\; F \text{ is a prime divisor over } S \right\}$$

and

$$\alpha(S, B) := \inf \left\{ \sup\{t \geq 0 \mid (S, B + tG) \text{ is lc}\} \;\middle|\; G \geq 0,\ G \sim_{\mathbb{R}} L \right\}.$$

Lemma 2.13 ([Han26, Lemma 2.7])**.** *Let (S, B) be a klt log Fano pair of dimension $d \geq 1$ and $L := -(K_S + B)$. Then*

$$\delta(S, B)^d L^d \leq \left(\frac{d+1}{d}\right)^d \widehat{\mathrm{vol}}(S \ni y, B)$$

for any closed point $y \in S$.

Definition 2.14. A set of normal projective varieties is *bounded* if there exists a projective morphism $\mathcal{V} \to T$ of schemes of finite type such that any member S of the set satisfies $S \cong \mathcal{V}_t$ for some closed point $t \in T$. A set of couples (S, D), where S is a normal projective variety and D is a reduced divisor on S, is *log bounded* if there exist a projective morphism $\mathcal{V} \to T$ of schemes of finite type and a reduced divisor $\mathcal{D}$ on $\mathcal{V}$ such that for any couple (S, D) in the set, there exist a closed point $t \in T$ and an isomorphism $S \cong \mathcal{V}_t$ mapping D onto $\mathcal{D}|_{\mathcal{V}_t}$.

Theorem 2.15 ([Bir21, Theorems 1.1 and 1.6])**.** *Let d be a positive integer and $\epsilon > 0$ a real number.*

(1) The set of normal projective varieties S of dimension d such that (S, B) is ϵ-lc and $-(K_S + B)$ is nef and big for some $\mathbb{R}$-divisor $B \geq 0$ is bounded.
(2) There exists a real number $t > 0$ depending only on d and ϵ such that $\alpha(S, B) \geq t$ for any projective ϵ-lc pair (S, B) of dimension d such that $-(K_S + B)$ is nef and big.

Definition-Lemma 2.16. For a $\mathbb{Q}$-Cartier Weil divisor D on a normal variety X and a closed point $y \in X$, the *Cartier index* of D at y is

$$\operatorname{ind}_y(X, D) := \min\{r \in \mathbb{N}^+ \mid rD \text{ is Cartier near } y\}.$$

Let X be a normal variety, $x \in X$ a closed point, and S a $\mathbb{Q}$-Cartier prime divisor on X through x. An *index 1 cover* of $(X \ni x, S)$ is a finite cyclic Galois morphism $g : (X' \ni x') \to (X \ni x)$ of degree $\operatorname{ind}_x(X, S)$ from a normal variety X', étale in codimension 1, such that $g^{-1}(x) = \{x'\}$ and g^*S is Cartier. Possibly shrinking X near x, an index 1 cover of $(X \ni x, S)$ exists. Let $B \geq 0$ be an $\mathbb{R}$-divisor with no component S such that $(X, S + B)$ is a pair which is plt near x, g an index 1 cover of $(X \ni x, S)$, $r := \operatorname{ind}_x(X, S)$, $S' := g^*S$, and $B' := g^*B$. Then, possibly shrinking X near x and replacing X' with the preimage:

(1) S' is a normal prime divisor and $(X', S' + B')$ is plt near x'.
(2) $g|_{S'} : S' \to S$ is finite of degree r, $(g|_{S'})^{-1}(x) = \{x'\}$, and

$$K_{S'} + \operatorname{Diff}_{S'}(B') = (g|_{S'})^*(K_S + \operatorname{Diff}_S(B)).$$

(3) We have

$$\widehat{\operatorname{vol}}(S' \ni x', \operatorname{Diff}_{S'}(B')) = r \cdot \widehat{\operatorname{vol}}(S \ni x, \operatorname{Diff}_S(B)).$$

Proof. The existence of an index 1 cover of $(X \ni x, S)$ after shrinking X near x is [Han26, Lemma 2.9].

Possibly shrinking X near x, we may assume that $(X, S + B)$ is plt. Let G be the Galois group of g, so $\#G = r$ and g is the quotient morphism $X' \to X'/G = X$.

(1) Since g is étale in codimension 1, $K_{X'} = g^*K_X$, S' is reduced, and for any prime divisor P' on X' the coefficient of P' in B' equals the coefficient of $g(P')$ in B. Thus $B' \geq 0$, no component of S' is a component of B', and

$$K_{X'} + S' + B' = g^*(K_X + S + B). \tag{2.4}$$

By (2.4), $(X', S' + B')$ is plt, so the irreducible components of S' are normal and pairwise disjoint. Since each of them contains the unique point x' over x, S' is a normal prime divisor.

(2) The morphism $g|_{S'}$ is finite, and $(g|_{S'})^{-1}(x) \subset g^{-1}(x) = \{x'\}$ with $x' \in S'$. Since $S' = g^*S$ is prime, $\deg(g|_{S'}) = \deg(g) = r$. Restricting (2.4) to S' and using the definition of the different (Definition 2.4) on both sides, we obtain

$$K_{S'} + \operatorname{Diff}_{S'}(B') = (K_{X'} + S' + B')|_{S'} = (g|_{S'})^*\big((K_X + S + B)|_S\big) = (g|_{S'})^*(K_S + \operatorname{Diff}_S(B)).$$

(3) Since $S' = g^*S$ is G-invariant and prime, G acts on S'. Since g is étale over the generic point of S, the inertia group of S' in G is trivial, so G acts faithfully on S'. Thus the finite morphism $S'/G \to S$ induced by $g|_{S'}$ has degree $\deg(g|_{S'})/\#G = r/r = 1$. Since S is normal, $S'/G \to S$ is an isomorphism, so $g|_{S'}$ is the quotient morphism $S' \to S'/G \cong S$.

By adjunction, $(S \ni x, \mathrm{Diff}_S(B))$ is a klt germ, and $\mathrm{Diff}_{S'}(B') \geq 0$ since $B' \geq 0$. By (2), $(g|_{S'})^{-1}(x) = \{x'\}$ and $K_{S'} + \mathrm{Diff}_{S'}(B') = (g|_{S'})^*(K_S + \mathrm{Diff}_S(B))$. Thus by Lemma 2.10(1),

$$\widehat{\mathrm{vol}}(S' \ni x', \mathrm{Diff}_{S'}(B')) = \deg(g|_{S'}) \cdot \widehat{\mathrm{vol}}(S \ni x, \mathrm{Diff}_S(B)) = r \cdot \widehat{\mathrm{vol}}(S \ni x, \mathrm{Diff}_S(B)).$$

□

2.5. **Germs with local volume bounded from below.**

Theorem 2.17 ([HLQ24, Lemma 2.6]). *Let d be a positive integer and $\epsilon > 0$ a real number. Then there exists a positive integer I depending only on d and ϵ satisfying the following. Let (X, B) be a pair of dimension d and $V \subsetneq X$ a closed subvariety such that (X, B) is klt near V and $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$ for any closed point $x \in V$. Then ID is Cartier near η for any $\mathbb{Q}$-Cartier Weil divisor D on X and any (not necessarily closed) point $\eta \in V$.*

Theorem 2.18 ([HLQ24, Corollary 2.7]). *Let d be a positive integer and $\epsilon > 0$ a real number. Then there exists a positive integer M depending only on d and ϵ such that $\mathrm{mld}(X \ni x, B) \leq M$ for any klt germ $(X \ni x, B)$ of dimension d with $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$.*

Theorem 2.19 ([HLQ24, Proposition 4.1(1)(5)(6)]). *Let d be a positive integer, $\epsilon, M > 0$ real numbers, and $\mathcal{I} \subset [0, 1]$ a DCC set such that $\overline{\mathcal{I}} = \mathcal{I}$. Let $(X_i \ni x_i, B_i)$ be a sequence of klt germs of dimension d. Assume that for any i the following hold.*

- *$B_i = \sum_j b_{i,j} B_{i,j}$, where $B_{i,j}$ are the irreducible components of B_i, $b_{i,j} \in \mathcal{I}$, and $x_i \in \mathrm{Supp}\, B_{i,j}$ for any j.*
- *$\widehat{\mathrm{vol}}(X_i \ni x_i, B_i) \geq \epsilon$.*
- *$\mathrm{mld}(X_i \ni x_i, B_i) \leq M$.*

Then there exists a real number $c > 0$ such that for any real number $0 < t < c$, possibly passing to a subsequence depending on t, there exist a finite set $\mathcal{I}_t \subset \mathcal{I}$ and a map $p_t : \mathcal{I} \to \mathcal{I}_t$ satisfying the following, where $p_t(B_i) := \sum_j p_t(b_{i,j}) B_{i,j}$.

(1) $\gamma \leq p_t(\gamma) \leq \gamma + t$ for any $\gamma \in \mathcal{I}$.
(2) $K_{X_i} + p_t(B_i)$ is $\mathbb{R}$-Cartier for any i.
(3) $\lim_{i \to +\infty} \|p_t(B_i) - B_i\| = 0$.

Theorem 2.20 ([HLQ24, Proposition 5.1]). *Let d be a positive integer and $\epsilon > 0$ a real number. Then there exist real numbers $\epsilon_0, c > 0$ and a positive integer N depending only on d and ϵ satisfying the following. Let $(X \ni x, B)$ be a klt germ of dimension d such that the coefficients of B are $\geq \epsilon$ and $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$. Then there exists a $\mathbb{Q}$-divisor $B^+ \geq (1 + c)B$ such that NB^+ is a Weil divisor and $(X \ni x, B^+)$ is a klt germ with $\widehat{\mathrm{vol}}(X \ni x, B^+) \geq \epsilon_0$.*

3. Monotonicity of the ratio

Proposition 3.1. *Let n be a positive integer, let $(X \ni x, B_1)$ be a klt germ of dimension n, and let B_0 be an $\mathbb{R}$-divisor on X such that $0 \leq B_0 \leq B_1$ and $K_X + B_0$ is $\mathbb{R}$-Cartier near x. Then $(X \ni x, B_0)$ is a klt germ and*

$$\frac{\widehat{\mathrm{vol}}(X \ni x, B_1)}{\mathrm{mld}(X \ni x, B_1)} \leq \frac{\widehat{\mathrm{vol}}(X \ni x, B_0)}{\mathrm{mld}(X \ni x, B_0)}.$$

Proof. Possibly shrinking X to a neighborhood of x, we may assume that $K_X + B_0$ is $\mathbb{R}$-Cartier, (X, B_1) is klt, and any component of $\mathrm{Supp}\, B_1$ contains x. Since $0 \leq B_0 \leq B_1$, we have $\mathrm{Supp}\, B_0 \subset \mathrm{Supp}\, B_1$, and $B_1 - B_0 \geq 0$ is $\mathbb{R}$-Cartier. For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X and a prime divisor E over X, we let $\mathrm{mult}_E D$ be the coefficient of E in h^*D, where $h : X' \to X$ is a projective birational morphism from a normal variety X' on which E is a divisor. Thus for any prime divisor E over X,

$$a(E, X, B_0) = a(E, X, B_1) + \mathrm{mult}_E(B_1 - B_0) \geq a(E, X, B_1) > 0,$$

so $(X \ni x, B_0)$ is a klt germ.

Step 1. In this step we compare multiplicities along prime divisors over $X \ni x$ with values at a minimizer of the normalized volume. Let B be a $\mathbb{Q}$-divisor on X such that $(X \ni x, B)$ is a klt germ and $K_X + B$ is $\mathbb{Q}$-Cartier, and let $D \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. By [Blu18, Main Theorem and Section 7], there exists a valuation v on X with $\operatorname{center}_X v = x$ and

$$\widehat{\operatorname{vol}}(v, X, B) = \widehat{\operatorname{vol}}(X \ni x, B).$$

Since $\widehat{\operatorname{vol}}(v, X, B) < +\infty$ and $(X \ni x, B)$ is klt, $0 < a(v, X, B) < +\infty$. Let

$$\mu := \frac{v(D)}{a(v, X, B)} \geq 0.$$

We show that for any prime divisor E over $X \ni x$,

$$\operatorname{mult}_E D \leq n\mu a(E, X, B). \tag{3.1}$$

Let c be a positive rational number such that $nc\mu < 1$, that is, $v(cD) < a(v, X, B)/n$. By [XZ25, Lemmas 3.2 and 3.3], possibly shrinking X to a neighborhood of x, there exists a $\mathbb{Q}$-divisor $G \geq cD$ such that $(X, B + G)$ is lc. Since $(X, B + G)$ is a pair and $K_X + B$ is $\mathbb{Q}$-Cartier, $G - cD \geq 0$ is $\mathbb{Q}$-Cartier, so for any prime divisor E over $X \ni x$,

$$a(E, X, B) - c \operatorname{mult}_E D = a(E, X, B + cD) \geq a(E, X, B + G) \geq 0.$$

If $\mu > 0$, then letting c increase to $1/(n\mu)$ implies (3.1). If $\mu = 0$, then c is an arbitrary positive rational number, so $\operatorname{mult}_E D \leq 0$, which is (3.1).

Step 2. In this step we prove the proposition when B_0 and B_1 are $\mathbb{Q}$-divisors. Since an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier, $K_X + B_0$ and $K_X + B_1$ are $\mathbb{Q}$-Cartier. Thus

$$D := B_1 - B_0$$

is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor and $D \geq 0$. Since (X, B_1) is klt and D is $\mathbb{Q}$-Cartier, there exists a rational number $0 < \epsilon < 1$ such that $(X, B_1 + \epsilon D)$ is klt. For any real number $t \in [0, 1 + \epsilon]$ we let

$$B_t := B_0 + tD.$$

Then $0 \leq B_t \leq B_{1+\epsilon}$ and $B_{1+\epsilon} - B_t = (1 + \epsilon - t)D$ is $\mathbb{R}$-Cartier, so the argument for B_0 at the beginning of the proof shows that $(X \ni x, B_t)$ is a klt germ. In particular, $\operatorname{mld}(X \ni x, B_t)$ is positive and finite, and we let

$$\phi(t) := \frac{\widehat{\operatorname{vol}}(X \ni x, B_t)}{\operatorname{mld}(X \ni x, B_t)}.$$

By (2.1), for any $t, t' \in [0, 1 + \epsilon]$ and any valuation w on X with $\operatorname{center}_X w = x$,

$$a(w, X, B_{t'}) = a(w, X, B_t) - (t' - t)w(D). \tag{3.2}$$

Let $q \in [0, 1]$ be a rational number. Then B_q is a $\mathbb{Q}$-divisor and

$$K_X + B_q = (1 - q)(K_X + B_0) + q(K_X + B_1)$$

is $\mathbb{Q}$-Cartier. Let v_q be the valuation v of **Step 1** for $B = B_q$, and let

$$\mu_q := \frac{v_q(D)}{a(v_q, X, B_q)}.$$

Since $(X \ni x, B_{1+\epsilon})$ is klt, $a(v_q, X, B_{1+\epsilon}) > 0$. Since $q \leq 1$ and $v_q(D) \geq 0$, (3.2) implies

$$a(v_q, X, B_q) = a(v_q, X, B_{1+\epsilon}) + (1 + \epsilon - q)v_q(D) \geq \epsilon v_q(D),$$

so

$$0 \leq \mu_q \leq \frac{1}{\epsilon}. \tag{3.3}$$

Let N be a positive integer such that $n/(\epsilon N) \le 1/2$, and let $q = j/N$ for some $j \in \{0, \dots, N-1\}$. By (3.3), $u_q := \mu_q/N$ satisfies $0 \le nu_q \le 1/2$. For any prime divisor E over $X \ni x$, by (3.1) for $B = B_q$,

$$a(E, X, B_{q+1/N}) = a(E, X, B_q) - \frac{1}{N}\operatorname{mult}_E D \ge (1 - nu_q)a(E, X, B_q).$$

Since $1 - nu_q > 0$, taking the infimum over all prime divisors E over $X \ni x$ implies

$$\operatorname{mld}(X \ni x, B_{q+1/N}) \ge (1 - nu_q)\operatorname{mld}(X \ni x, B_q). \tag{3.4}$$

By (3.2),

$$a(v_q, X, B_{q+1/N}) = a(v_q, X, B_q) - v_q(D)/N = (1 - u_q)a(v_q, X, B_q).$$

Thus

$$\begin{aligned}\widehat{\operatorname{vol}}(X \ni x, B_{q+1/N}) &\le \widehat{\operatorname{vol}}(v_q, X, B_{q+1/N}) = (1 - u_q)^n \widehat{\operatorname{vol}}(v_q, X, B_q) \\ &= (1 - u_q)^n \widehat{\operatorname{vol}}(X \ni x, B_q).\end{aligned} \tag{3.5}$$

By (3.4) and (3.5),

$$\phi(q + 1/N) \le \frac{(1 - u_q)^n}{1 - nu_q}\phi(q). \tag{3.6}$$

Since $0 \le nu_q \le \frac{1}{2}$, $(1 - u_q)^n \le 1 - nu_q + \frac{n(n-1)}{2}u_q^2$, so by (3.3),

$$\frac{(1 - u_q)^n}{1 - nu_q} \le 1 + n(n - 1)u_q^2 \le \exp\Big(\frac{n(n - 1)}{\epsilon^2 N^2}\Big).$$

Multiplying (3.6) for $q = 0, 1/N, \dots, (N - 1)/N$, we obtain

$$\phi(1) \le \phi(0) \exp\left(\frac{n(n - 1)}{\epsilon^2 N}\right).$$

Since $\phi(0)$, $\phi(1)$ and ϵ do not depend on N, letting N tend to $+\infty$ implies $\phi(1) \le \phi(0)$, which is the inequality of the proposition.

Step 3. In this step we prove the proposition in general. Let $P_1, \dots, P_m$ be the components of $\operatorname{Supp} B_1$. If $m = 0$, then $B_0 = B_1 = 0$ and there is nothing to prove. Otherwise for $b \in \mathbb{R}^m$ let $B(b) := \sum_{j=1}^m b_j P_j$, let W be the set of $b \in \mathbb{R}^m$ such that $K_X + B(b)$ is $\mathbb{R}$-Cartier near x, and for $i \in \{0, 1\}$ let $b^{(i)} \in W$ be the vector with $B_i = B(b^{(i)})$. Then $(b^{(0)}, b^{(1)})$ belongs to

$$\Pi := \{(c, c') \in W \times W \mid 0 \le c_j \le c'_j \text{ for any } 1 \le j \le m\}.$$

By Lemma 2.9(1), Π is a polyhedron in $\mathbb{R}^{2m}$ defined by linear equations and inequalities with rational coefficients, so rational points are dense in Π. Thus there exist $(b_k^{(0)}, b_k^{(1)}) \in \Pi \cap \mathbb{Q}^{2m}$ for $k \in \mathbb{N}^+$ such that

$$\lim_{k \to +\infty} (b_k^{(0)}, b_k^{(1)}) = (b^{(0)}, b^{(1)}).$$

Let $B_{i,k} := B(b_k^{(i)})$ for $i \in \{0, 1\}$. Then $0 \le B_{0,k} \le B_{1,k}$, and $K_X + B_{i,k}$ is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor near x, hence $\mathbb{Q}$-Cartier near x. By Lemma 2.9(2) for $b = b^{(i)}$ and any $\epsilon \in (0, 1)$, for $i \in \{0, 1\}$ and any $k \gg 0$, the germ $(X \ni x, B_{i,k})$ is klt, and letting ϵ tend to 0, we obtain

$$\lim_{k \to +\infty} \operatorname{mld}(X \ni x, B_{i,k}) = \operatorname{mld}(X \ni x, B_i), \quad \lim_{k \to +\infty} \widehat{\operatorname{vol}}(X \ni x, B_{i,k}) = \widehat{\operatorname{vol}}(X \ni x, B_i).$$

By **Step 2** for $B_{0,k} \le B_{1,k}$, for any $k \gg 0$,

$$\frac{\widehat{\operatorname{vol}}(X \ni x, B_{1,k})}{\operatorname{mld}(X \ni x, B_{1,k})} \le \frac{\widehat{\operatorname{vol}}(X \ni x, B_{0,k})}{\operatorname{mld}(X \ni x, B_{0,k})}.$$

Since $\operatorname{mld}(X \ni x, B_i) > 0$ for $i \in \{0, 1\}$, letting k tend to $+\infty$ implies the inequality of the proposition. The proposition follows. □

4. Kollár components with bounded log discrepancy

4.1. Lc blow-ups.

Lemma 4.1 (cf. [Zhu24a, Lemma 2.2]). *Let $(X \ni x, B)$ be a klt germ such that $\dim X \geq 2$. Let E be a prime divisor over $X \ni x$ and $B^+ \geq B$ an $\mathbb{R}$-divisor on X such that (X, B^+) is an lc pair near x and $a(E, X, B^+) = 0$. Then $(X \ni x, B)$ has an lc blow-up whose exceptional divisor is E.*

Proof. Possibly shrinking X to a neighborhood of x, we may assume that (X, B) is klt, (X, B^+) is lc, and any component of $\operatorname{Supp} B^+$ contains x. Let $P_1, \dots, P_m$ be the components of $\operatorname{Supp} B^+$. For $b'' \in \mathbb{R}^m$ let $B(b'') := \sum_{j=1}^m b''_j P_j$, and let W be the set of $b'' \in \mathbb{R}^m$ such that $K_X + B(b'')$ is $\mathbb{R}$-Cartier near x. Since $0 \leq B \leq B^+$, there exist $b, b^+ \in W$ such that $B(b) = B$ and $B(b^+) = B^+$.

By Lemma 2.9(1), W is an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$. Thus $W \cap \mathbb{Q}^m$ is dense in W, and for any $b'' \in W \cap \mathbb{Q}^m$, $K_X + B(b'')$ is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor near x, hence $\mathbb{Q}$-Cartier near x. Let $w_1, \dots, w_N \in W \cap \mathbb{Q}^m$ be points whose affine span is W. Possibly shrinking X to a neighborhood of x, we may assume that $K_X + B(w_i)$ is $\mathbb{Q}$-Cartier for any i. Since any $b'' \in W$ is an affine combination of $w_1, \dots, w_N$, $K_X + B(b'')$ is $\mathbb{R}$-Cartier for any $b'' \in W$.

Let $g : \widetilde{X} \to X$ be a log resolution of $(X, \operatorname{Supp} B^+)$ such that E is a divisor on $\widetilde{X}$, and let $\Sigma := \operatorname{Exc}(g) \cup g_*^{-1} \operatorname{Supp} B^+$, a simple normal crossing divisor. For $b'' \in W$ define $B_{\widetilde{X}}(b'')$ by

$$K_{\widetilde{X}} + B_{\widetilde{X}}(b'') = g^*(K_X + B(b'')).$$

Then $\operatorname{Supp} B_{\widetilde{X}}(b'') \subset \Sigma$, and the coefficient of any prime divisor in $B_{\widetilde{X}}(b'')$ is an affine function of $b'' \in W$ which takes rational values on $W \cap \mathbb{Q}^m$. Let $\mathcal{Q}$ be the set of $(b', b'') \in W \times W$ satisfying the following.

(1) $0 \leq b'_j \leq b''_j$ for any j.
(2) $\operatorname{mult}_F B_{\widetilde{X}}(b'') \leq 1$ for any component F of Σ.
(3) $\operatorname{mult}_E B_{\widetilde{X}}(b'') = 1$.

Since Σ is a simple normal crossing divisor, (2) holds if and only if $(X, B(b''))$ is lc, and (3) holds if and only if $a(E, X, B(b'')) = 0$. In particular, $(b, b^+) \in \mathcal{Q}$. Moreover, $\mathcal{Q}$ is a polyhedron defined over $\mathbb{Q}$, so $\mathcal{Q} \cap (\mathbb{Q}^m \times \mathbb{Q}^m)$ is dense in $\mathcal{Q}$.

By Lemma 2.9(2) with $\epsilon = \frac{1}{2}$, there exist an open neighborhood U of x and a real number $c > 0$ such that for any $b' \in W$ with $B(b') \geq 0$ and $\|B(b') - B\| < c$,

$$a(v, X, B(b')) \geq \frac{1}{2} a(v, X, B)$$

for any valuation v on X with $\operatorname{center}_X v \in U$, and $a(v, X, B) > 0$ if v is divisorial. In particular, $(U, B(b')|_U)$ is klt. Possibly shrinking X to the neighborhood U of x, we may assume that $U = X$. Pick $(b', b'') \in \mathcal{Q} \cap (\mathbb{Q}^m \times \mathbb{Q}^m)$ such that $\|B(b') - B\| < c$, and let $B' := B(b')$ and $D' := B(b'') - B(b')$. Then:

- B' is a $\mathbb{Q}$-divisor and (X, B') is a klt pair.
- $D' \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, since $K_X + B(b'')$ and $K_X + B'$ are $\mathbb{Q}$-Cartier.
- $(X, B' + D') = (X, B(b''))$ is lc and $a(E, X, B' + D') = 0$.

By [Zhu24a, Lemma 2.2], E is of lc type in the sense of [Zhu24a, Definition 2.1]. In particular, there exists a projective birational morphism $f : Y \to X$ from a normal variety Y such that E is the only exceptional prime divisor of f, E is $\mathbb{Q}$-Cartier, and $-E$ is ample over X.

Since $\operatorname{center}_X E = x$ and $-E$ is ample over X, f is an isomorphism over $X \backslash \{x\}$ with exceptional locus E and $f(E) = x$.

Let $B_Y := f_*^{-1} B$ and $B_Y^+ := f_*^{-1} B^+$. Since $a(E, X, B^+) = 0$ and E is the only exceptional prime divisor of f,

$$K_Y + E + B_Y^+ = f^*(K_X + B^+),$$

so $(Y, E + B_Y^+)$ is lc. Since E is $\mathbb{Q}$-Cartier,

$$K_Y + E + B_Y = f^*(K_X + B) + a(E, X, B)E$$

is $\mathbb{R}$-Cartier. Since $E + B_Y \leq E + B_Y^+$, $(Y, E + B_Y)$ is lc. Thus f is an lc blow-up of $(X \ni x, B)$ with exceptional divisor E. The lemma follows. □

Lemma 4.2. *Let $n \geq 2$ be an integer and $\Gamma \subset [0,1]$ a DCC set. Then there exists a real number $\tau > 0$ depending only on n and Γ satisfying the following. Let $(X \ni x, B)$ be a klt germ of dimension n such that $B \in \Gamma$. Let E be a prime divisor over $X \ni x$ such that $a(E, X, B) < \tau$. Then $(X \ni x, B)$ has an lc blow-up whose exceptional divisor is E.*

Proof. By [HLS26, Theorem 1.5], there exist a positive integer N and a finite set $\Lambda \subset (0,1]$, depending only on n and Γ, satisfying the following: for any contraction $X \to Z$ such that X is of Fano type over Z [HLS26, Definition 3.2], any point $z \in Z$, and any $\mathbb{R}$-complementary pair $(X/Z \ni z, B)$ of dimension n with $B \in \Gamma$, $(X/Z \ni z, B)$ has an (N, Λ)-decomposable $\mathbb{R}$-complement [HLS26, Definitions 1.4 and 3.11]. Let

$$\tau := \frac{\min \Lambda}{N}.$$

Let $(X \ni x, B)$ and E be as in the lemma. Possibly shrinking X to a neighborhood of x, we may assume that (X, B) is klt and $K_X + B \sim_{\mathbb{R}} 0$, since $K_X + B$ is an $\mathbb{R}$-linear combination of Cartier divisors, which are principal near x. Thus, for the identity morphism $X \to X$, X is of Fano type over X and $(X/X \ni x, B)$ is an $\mathbb{R}$-complement of itself. By the choice of N and Λ and by [HLS26, Definitions 1.4 and 3.11], possibly shrinking X to a neighborhood of x, there exist $m \in \mathbb{N}^+$, real numbers $\lambda_1, \dots, \lambda_m \in \Lambda$ with $\sum_{j=1}^m \lambda_j = 1$, and $\mathbb{Q}$-divisors $B_1^+, \dots, B_m^+ \geq 0$ satisfying the following.

- (X, B_j^+) is lc and $N(K_X + B_j^+) \sim 0$ for any j.
- $B^+ := \sum_{j=1}^m \lambda_j B_j^+ \geq B$.

Since $K_X + B^+ = \sum_{j=1}^m \lambda_j (K_X + B_j^+)$, (X, B^+) is lc and

$$a(F, X, B^+) = \sum_{j=1}^m \lambda_j a(F, X, B_j^+)$$

for any prime divisor F over X.

Since (X, B_j^+) is lc and $N(K_X + B_j^+)$ is Cartier, $a(F, X, B_j^+) \in \{0\} \cup [\frac{1}{N}, +\infty)$ for any prime divisor F over X and any j.

Since $B^+ - B \geq 0$ is $\mathbb{R}$-Cartier, $\operatorname{mult}_E(B^+ - B) \geq 0$. Thus

$$\sum_{j=1}^m \lambda_j a(E, X, B_j^+) = a(E, X, B^+) = a(E, X, B) - \operatorname{mult}_E(B^+ - B) \leq a(E, X, B) < \tau.$$

If $a(E, X, B_k^+) > 0$ for some k, then $a(E, X, B_k^+) \geq \frac{1}{N}$, so $a(E, X, B^+) \geq \frac{\lambda_k}{N} \geq \tau$, a contradiction. Thus $a(E, X, B_j^+) = 0$ for any j, so $a(E, X, B^+) = 0$. Since $B^+ \geq B$ and (X, B^+) is an lc pair, Lemma 4.1 implies that $(X \ni x, B)$ has an lc blow-up whose exceptional divisor is E. □

4.2. **Transverse local volumes.**

Lemma 4.3 (cf. [Kim26, Theorem 1.1])**.** *Let (X, B) be a klt pair of dimension n and $V \subset X$ a closed subvariety with generic point η and $\operatorname{codim}_X V = r \geq 1$. Then*

$$\widehat{\operatorname{vol}}(X \ni x, B) \leq \frac{n^n}{r^r} \widehat{\operatorname{vol}}(X \ni \eta, B)$$

for any closed point $x \in V$.

Proof. First assume that B is a $\mathbb{Q}$-divisor. Let $\nu : V^\nu \to V$ be the normalization. Consider the projection $X \times V^\nu \to V^\nu$ with boundary $B \times V^\nu$ and section $(\nu, \mathrm{id}_{V^\nu}) : V^\nu \to X \times V^\nu$. Since $K_{X \times V^\nu / V^\nu} + B \times V^\nu$ is the pullback of $K_X + B$, it is $\mathbb{Q}$-Cartier, and any fiber is the klt pair (X, B). By [BL21, Theorem 1], the function $y \to \widehat{\operatorname{vol}}(X \ni \nu(y), B)$ on the closed points of V^ν is

lower semicontinuous. Thus there exists a closed subset $\Sigma \subset V^\nu$ whose closed points are exactly the closed points $y \in V^\nu$ such that

$$\widehat{\mathrm{vol}}(X \ni \nu(y), B) \leq \frac{n^n}{r^r}\widehat{\mathrm{vol}}(X \ni \eta, B).$$

Since $\dim V = n - r$, [Kim26, Theorem 1.1] implies that

$$\widehat{\mathrm{vol}}(X \ni x', B) = \frac{n^n}{r^r}\widehat{\mathrm{vol}}(X \ni \eta, B)$$

for a general closed point $x' \in V$. Since Σ is closed and contains $\nu^{-1}(x')$ for a general closed point $x' \in V$, $\Sigma = V^\nu$, and the lemma holds when B is a $\mathbb{Q}$-divisor since ν is surjective.

Now let B be an $\mathbb{R}$-divisor and $x \in V$ a closed point. Since $x \in V$, any neighborhood of x contains η. Possibly shrinking X to a neighborhood of x, we may assume that any component of $\operatorname{Supp} B$ contains x. Let $P_1, \dots, P_m$ be the components of $\operatorname{Supp} B$. If $m = 0$, then $B = 0$ and there is nothing to prove. For $b' \in \mathbb{R}^m$ let $B(b') := \sum_{j=1}^m b'_j P_j$, let W be the set of $b' \in \mathbb{R}^m$ such that $K_X + B(b')$ is $\mathbb{R}$-Cartier near x, and let $b \in W$ be the vector with $B(b) = B$. Fix a real number $0 < \epsilon < 1$. By Lemma 2.9(2), there exist an open neighborhood U of x and a real number $c > 0$ satisfying the conclusion of Lemma 2.9(2) for b and ϵ. By Lemma 2.9(1), W is defined over $\mathbb{Q}$, so $W \cap \mathbb{Q}^m$ is dense in W. Since $b_j > 0$ for any j, there exists $b' \in W \cap \mathbb{Q}^m$ such that $B(b') \geq 0$ and $\|B(b') - B\| < c$. Then $K_U + B(b')|_U$ is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor, hence $\mathbb{Q}$-Cartier, and $(U, B(b')|_U)$ is klt since

$$a(v, X, B(b')) \geq (1 - \epsilon)a(v, X, B) > 0$$

for any divisorial valuation v on X with $\operatorname{center}_X v \in U$. Since $x, \eta \in U$ and $\operatorname{codim}_U(V \cap U) = r$, Lemma 2.9(2) and the $\mathbb{Q}$-divisor case for $(U, B(b')|_U)$ and $V \cap U$ imply that

$$(1-\epsilon)^n\widehat{\mathrm{vol}}(X \ni x, B) \leq \widehat{\mathrm{vol}}(X \ni x, B(b')) \leq \frac{n^n}{r^r}\widehat{\mathrm{vol}}(X \ni \eta, B(b')) \leq \frac{n^n}{r^r}(1+\epsilon)^r\widehat{\mathrm{vol}}(X \ni \eta, B).$$

Since ϵ is arbitrary, the lemma follows. □

Proposition 4.4. *Let $n \geq 2$ be an integer and $\Gamma \subset [0, 1]$ a DCC set. Then there exists a real number $t_* \in (0, \frac{1}{2})$ depending only on n and Γ satisfying the following. Let $(X \ni x, B)$ be a klt germ of dimension n such that $B \in \Gamma$. Let $f : Y \to X$ be an lc blow-up of $(X \ni x, B)$ with exceptional divisor E such that $0 < a(E, X, B) \leq t_*$ and $(Y, E + f_*^{-1}B)$ is not plt near E. Then*

$$\widehat{\mathrm{vol}}(X \ni x, B) \leq \frac{n^n}{t_*^2}a(E, X, B)^2.$$

Proof. For any $r \in \{2, \dots, n\}$, let LCT_r be the set of real numbers

$$\sup\{s \geq 0 \mid (X', \Delta' + sM') \text{ is lc}\},$$

where (X', Δ') is an lc pair of dimension r such that $\Delta' \in \Gamma \cup \{1\}$, and $M' \neq 0$ is a reduced $\mathbb{Q}$-Cartier divisor on X'. Since $\Gamma \cup \{1\}$ and $\{1\}$ are DCC sets, LCT_r satisfies the ACC by [HMX14, Theorem 1.1]. Thus $\bigcup_{r=2}^n \mathrm{LCT}_r$ satisfies the ACC, so $\bigcup_{r=2}^n \mathrm{LCT}_r \cap (0, 1)$ is empty or has a maximum. Hence there exists a real number $t_* \in (0, \frac{1}{2})$ such that

$$[1 - t_*, 1) \cap \mathrm{LCT}_r = \emptyset \text{ for any } r \in \{2, \dots, n\}.$$

Step 1. Let $r \in \{2, \dots, n\}$, let $(T \ni u, \Delta)$ be a klt germ of dimension r such that $\Delta \in \Gamma$, and let E_T be a reduced $\mathbb{Q}$-Cartier divisor on T satisfying the following.

- $(T, \Delta + E_T)$ is lc near u.
- There exists a prime divisor G_T over $T \ni u$ such that $a(G_T, T, \Delta + E_T) = 0$.

For any $t \in (0, t_*]$ let $\Delta_t := \Delta + (1 - t)E_T$. In this step we show that $(T \ni u, \Delta_t)$ is a klt germ and

$$a(S, T, \Delta + E_T) = 0 \text{ for any Kollár component } S \text{ of } (T \ni u, \Delta_t).$$

Possibly shrinking T to a neighborhood of u, we may assume that (T, Δ) is klt and $(T, \Delta + E_T)$ is lc. Since $K_T + \Delta_t = t(K_T + \Delta) + (1-t)(K_T + \Delta + E_T)$,

$$a(F, T, \Delta_t) = ta(F, T, \Delta) + (1-t)a(F, T, \Delta + E_T) > 0$$

for any prime divisor F over T, so $(T \ni u, \Delta_t)$ is a klt germ.

Let S be a Kollár component of $(T \ni u, \Delta_t)$, $h : T' \to T$ the plt blow-up with exceptional divisor S, $\Delta_{T'} := h_*^{-1}\Delta$, and $E_{T'} := h_*^{-1}E_T$. Since S is $\mathbb{Q}$-Cartier, $E_{T'} = h^*E_T - (\operatorname{mult}_S E_T)S$ is $\mathbb{Q}$-Cartier. Since $(T', S + \Delta_{T'} + (1-t)E_{T'})$ is plt near S,

$$K_{T'} + S + \Delta_{T'} = (K_{T'} + S + \Delta_{T'} + (1-t)E_{T'}) - (1-t)E_{T'}$$

is $\mathbb{R}$-Cartier, and there exists an open neighborhood T'° of S on which $(T', S + \Delta_{T'} + (1-t)E_{T'})$ is lc. Assume that $E_{T'}|_{T'^{\circ}} \neq 0$. Since $(T'^{\circ}, (S + \Delta_{T'})|_{T'^{\circ}})$ is an lc pair of dimension r with $S + \Delta_{T'} \in \Gamma \cup \{1\}$ and $E_{T'}$ is reduced,

$$\beta := \sup\{s \geq 0 \mid (T'^{\circ}, (S + \Delta_{T'} + sE_{T'})|_{T'^{\circ}}) \text{ is lc}\} \in \mathrm{LCT}_r\,.$$

Since $\beta \geq 1 - t \geq 1 - t_*$, the choice of t_* implies that $\beta \geq 1$. Thus

$$(T', S + \Delta_{T'} + E_{T'}) \text{ is lc near } S. \tag{4.1}$$

If $E_{T'}|_{T'^{\circ}} = 0$, then (4.1) also holds.

Let $e := a(S, T, \Delta + E_T)$. Then $e \geq 0$, and since h extracts only S,

$$K_{T'} + S + \Delta_{T'} + E_{T'} = h^*(K_T + \Delta + E_T) + eS.$$

Suppose that $e > 0$. Then $G_T \neq S$. Since $\operatorname{center}_T G_T = u$ and $h^{-1}(u) = S$, $\operatorname{center}_{T'} G_T \subset S$, so $\operatorname{mult}_{G_T} S > 0$. Thus

$$a(G_T, T', S + \Delta_{T'} + E_{T'}) = a(G_T, T, \Delta + E_T) - e \operatorname{mult}_{G_T} S = -e \operatorname{mult}_{G_T} S < 0,$$

which contradicts (4.1). Thus $e = 0$.

Step 2. Notation as in **Step 1**. For any $t \in (0, t_*]$ let $\phi(t) := \widehat{\mathrm{vol}}(T \ni u, \Delta_t)$, and let

$$c_0 := \inf\{(\operatorname{mult}_F E_T)^r \operatorname{vol}(\operatorname{ord}_F) \mid F \text{ is a prime divisor over } T \ni u \text{ such that } a(F, T, \Delta + E_T) = 0\}.$$

The set in the infimum contains G_T. In this step we show that

$$\phi(t) = t^r c_0 \leq r^r \left(\frac{t}{t_*}\right)^r \text{ for any } t \in (0, t_*]. \tag{4.2}$$

Let F be a prime divisor over $T \ni u$ such that $a(F, T, \Delta + E_T) = 0$. Since $K_T + \Delta_t + tE_T = K_T + \Delta + E_T$,

$$a(F, T, \Delta_t) = a(F, T, \Delta + E_T) + t \operatorname{mult}_F E_T = t \operatorname{mult}_F E_T,$$

so $\widehat{\mathrm{vol}}(F, T, \Delta_t) = t^r (\operatorname{mult}_F E_T)^r \operatorname{vol}(\operatorname{ord}_F)$. Thus $\phi(t) \leq t^r c_0$. Since $r \geq 2$, Lemma 2.8(2) implies that $\phi(t) = \inf_S \widehat{\mathrm{vol}}(S, T, \Delta_t)$, the infimum taken over Kollár components S of $(T \ni u, \Delta_t)$. By **Step 1**, any such S satisfies $a(S, T, \Delta + E_T) = 0$, so $\widehat{\mathrm{vol}}(S, T, \Delta_t) \geq t^r c_0$. Thus $\phi(t) = t^r c_0$. By Lemma 2.8(1), $t_*^r c_0 = \phi(t_*) \leq r^r$, which implies (4.2).

Step 3. Let $(X \ni x, B)$, f, and E be as in the proposition, and let $B_Y := f_*^{-1}B$ and $c := a(E, X, B)$. In this step we construct a transverse germ satisfying the assumptions of **Step 1**.

Possibly shrinking X to a neighborhood of x, we may assume that (X, B) is klt and $(Y, E + B_Y)$ is lc. Since $(Y, E + B_Y)$ is lc and not plt near E, there exists an exceptional prime divisor G over Y such that $a(G, Y, E + B_Y) = 0$ and $\operatorname{center}_Y G \cap E \neq \emptyset$. Since f is an isomorphism over $X \backslash \{x\}$ and (X, B) is klt, $(Y, E + B_Y)$ is klt over $Y \backslash E$. Thus

$$V := \operatorname{center}_Y G \subset E,$$

and since G is exceptional over Y, $r := \operatorname{codim}_Y V \geq 2$. Let η_V be the generic point of V. By Definition 2.5,

$$K_Y + E + B_Y = f^*(K_X + B) + cE.$$

Let $B_c := B_Y + (1-c)E$. Then $K_Y + B_c = f^*(K_X + B)$. Since (X, B) is klt and $B_c \geq 0$ ($c \leq t_* < 1$), (Y, B_c) is klt. Moreover, $K_Y + B_Y$ is $\mathbb{R}$-Cartier since E is $\mathbb{Q}$-Cartier. Since f is an isomorphism over $X\backslash\{x\}$, (X, B) is klt, and

$$a(F, Y, B_Y) = a(F, Y, E + B_Y) + \operatorname{mult}_F E > 0$$

for any prime divisor F over Y with $\operatorname{center}_Y F \subset E$, (Y, B_Y) is klt.

Let $Y^\circ \subset Y$ be an affine open subset containing η_V. By Noether normalization, there exists a morphism $q : Y^\circ \to Z := \mathbb{A}^{\dim V}$ whose restriction to $V \cap Y^\circ$ is finite and surjective. Let $\Bbbk := \mathbb{C}(Z)$, let $Y_\Bbbk$ be the generic fiber of q, and for any field extension $\Bbbk \subset \Bbbk'$ let $Y_{\Bbbk'} := Y_\Bbbk \times_\Bbbk \Bbbk'$. For an $\mathbb{R}$-divisor D on Y°, let $D_{\Bbbk'}$ be the pullback of D to $Y_{\Bbbk'}$. Then $Y_\Bbbk$ is a normal variety over $\Bbbk$ with $\dim Y_\Bbbk = n - \dim V = r$, and η_V is a closed point of $Y_\Bbbk$ whose residue field is a finite extension of $\Bbbk$. Let $\overline{\Bbbk}$ be an algebraic closure of $\Bbbk$.

Let $\mu : \widetilde{Y} \to Y^\circ$ be a log resolution of $(Y^\circ, \operatorname{Supp}(E + B_Y))$ such that G is a divisor on $\widetilde{Y}$. Since $\overline{\Bbbk}$ has characteristic 0, the following hold.

(1) $Y_{\overline{\Bbbk}}$ is normal.
(2) The pullback of K_{Y° to $Y_{\overline{\Bbbk}}$ is a canonical divisor of $Y_{\overline{\Bbbk}}$.
(3) The base change
$$\mu_{\overline{\Bbbk}} : \widetilde{Y}_{\overline{\Bbbk}} \to Y_{\overline{\Bbbk}}$$
is a log resolution of $(Y_{\overline{\Bbbk}}, \operatorname{Supp}(E_{\overline{\Bbbk}} + (B_Y)_{\overline{\Bbbk}}))$.
(4) The pullback of any prime divisor P on Y° or on $\widetilde{Y}$ is a reduced divisor, which is 0 if P does not dominate Z.

Thus, for $D \in \{B_Y, E + B_Y, B_c\}$ and any prime divisor P on $\widetilde{Y}$ dominating Z, any component $\overline{P}$ of the pullback of P to $\widetilde{Y}_{\overline{\Bbbk}}$ satisfies

$$a(\overline{P}, Y_{\overline{\Bbbk}}, D_{\overline{\Bbbk}}) = a(P, Y, D).$$

Since $\mu(G) = V$ dominates Z, we may let $\overline{G}$ be a component of the pullback of G to $\widetilde{Y}_{\overline{\Bbbk}}$. Since $\overline{\Bbbk}$ is an algebraically closed field of characteristic 0 whose cardinality is that of $\mathbb{C}$, we may identify $\overline{\Bbbk}$ with $\mathbb{C}$ by a field isomorphism. Let $u := \mu_{\overline{\Bbbk}}(\overline{G})$, a closed point of $Y_{\overline{\Bbbk}}$ over η_V, let T be the connected component of $Y_{\overline{\Bbbk}}$ containing u, and let

$$\Delta := (B_Y)_{\overline{\Bbbk}}|_T \quad \text{and} \quad E_T := E_{\overline{\Bbbk}}|_T.$$

Since (Y, B_Y) is klt, $(Y, E + B_Y)$ is lc, $B_Y \in \Gamma$, E is a $\mathbb{Q}$-Cartier prime divisor, and $a(G, Y, E + B_Y) = 0$, we obtain the following.

- T is a normal variety of dimension r over $\overline{\Bbbk}$, and $(T \ni u, \Delta)$ is a klt germ such that $\Delta \in \Gamma$.
- E_T is a reduced $\mathbb{Q}$-Cartier divisor, and $(T, \Delta + E_T)$ is lc.
- $\overline{G}$ is a prime divisor over $T \ni u$ such that $a(\overline{G}, T, \Delta + E_T) = 0$.
- $\Delta_c := \Delta + (1-c)E_T$ is the pullback of B_c to T.

Since $0 < c \leq t_*$, **Step 2** for $(T \ni u, \Delta)$, E_T, and $G_T := \overline{G}$ implies that

$$\phi(c) = \widehat{\operatorname{vol}}(T \ni u, \Delta_c) \leq r^r \left(\frac{c}{t_*}\right)^r. \tag{4.3}$$

Step 4. Notation as in **Step 3**. In this step we spread out a prime divisor computing $\phi(c)$ up to ϵ and conclude the proof.

Fix a real number $\epsilon > 0$. By Lemma 2.8(2), there exists a Kollár component $\overline{F}$ of $(T \ni u, \Delta_c)$ such that $\widehat{\operatorname{vol}}(\overline{F}, T, \Delta_c) < \phi(c) + \epsilon$. Since u, $\overline{F}$, and a higher model of T on which $\overline{F}$ is a divisor are defined by finitely many equations, by [Gro66, Théorèmes 8.8.2 and 8.10.5] there exist a

finite extension $\Bbbk \subset \Bbbk' \subset \overline{\Bbbk}$, a $\Bbbk'$-rational point $u_{\Bbbk'}$ of $Y_{\Bbbk'}$, a projective birational morphism $\mu_1 : Y_1 \to Y_{\Bbbk'}$, and a prime divisor F_1 on Y_1 such that u is the base change of $u_{\Bbbk'}$ and $\overline{F} = F_1 \times_{\Bbbk'} \overline{\Bbbk}$.

Let Z' be the normalization of Z in $\Bbbk'$. Since $\Bbbk'$ has characteristic 0, there exists a dense open subset $Z'^{\circ} \subset Z'$ which is étale over Z. Let Y' be the connected component of $Y^{\circ} \times_Z Z'^{\circ}$ containing $u_{\Bbbk'}$, and $g : Y' \to Y^{\circ} \subset Y$ the induced morphism. Then g is étale, Y' is a normal variety, and (Y', g^*B_c) is klt. Let V' be the closure of $u_{\Bbbk'}$ in Y', so that $u_{\Bbbk'}$ is the generic point of V'. Then $g(V') \subset V$ since $g(u_{\Bbbk'}) = \eta_V$, and

$$\operatorname{codim}_{Y'} V' = n - \dim Z' = n - \dim V = r.$$

The generic fiber of $Y' \to Z'^{\circ}$ is an open and closed subset of $Y_{\Bbbk'}$ containing $u_{\Bbbk'}$. Possibly shrinking Z'°, by [Gro66, Théorèmes 8.8.2 and 8.10.5] there exist a projective birational morphism $\mu' : \widetilde{Y}' \to Y'$ and a prime divisor F' on $\widetilde{Y}'$ whose restrictions to the generic fiber of $Y' \to Z'^{\circ}$ are the restrictions of μ_1 and F_1. Then $\operatorname{center}_{Y'} F' = V'$.

We show that $\widehat{\operatorname{vol}}(F', Y', g^*B_c) = \widehat{\operatorname{vol}}(\overline{F}, T, \Delta_c)$. Since g is étale, $K_{Y'} + g^*B_c = g^*(K_Y + B_c)$. The log discrepancies of F' and $\overline{F}$ are computed by the coefficients of $K_{\widetilde{Y}'} - \mu'^*(K_{Y'} + g^*B_c)$ and of the corresponding divisor on $Y_1 \times_{\Bbbk'} \overline{\Bbbk}$, which do not change under restriction to the generic fiber over Z' and under the separable base change $\Bbbk' \to \overline{\Bbbk}$. Thus

$$a(F', Y', g^*B_c) = a(\overline{F}, T, \Delta_c).$$

The local ring $\mathcal{O}_{Y', u_{\Bbbk'}}$ is the local ring of $Y_{\Bbbk'}$ at $u_{\Bbbk'}$, and $\mathcal{O}_{T,u}$ is the localization of $\mathcal{O}_{Y', u_{\Bbbk'}} \otimes_{\Bbbk'} \overline{\Bbbk}$ at u. For any real number $\lambda \geq 0$, the ideal $\mathfrak{a}_\lambda(\operatorname{ord}_{F'})$ is the stalk at $u_{\Bbbk'}$ of $\mu'_*\mathcal{O}_{\widetilde{Y}'}(-\lceil \lambda F' \rceil)$, and pushforward commutes with the flat base change $\Bbbk' \to \overline{\Bbbk}$, so

$$\mathfrak{a}_\lambda(\operatorname{ord}_{\overline{F}}) = \mathfrak{a}_\lambda(\operatorname{ord}_{F'})\mathcal{O}_{T,u}.$$

Since $\mathcal{O}_{Y', u_{\Bbbk'}}/\mathfrak{a}_\lambda(\operatorname{ord}_{F'})$ is an Artinian local ring with residue field $\Bbbk'$,

$$\ell\big(\mathcal{O}_{T,u}/\mathfrak{a}_\lambda(\operatorname{ord}_{\overline{F}})\big) = \dim_{\overline{\Bbbk}}\big(\mathcal{O}_{Y', u_{\Bbbk'}}/\mathfrak{a}_\lambda(\operatorname{ord}_{F'}) \otimes_{\Bbbk'} \overline{\Bbbk}\big) = \ell\big(\mathcal{O}_{Y', u_{\Bbbk'}}/\mathfrak{a}_\lambda(\operatorname{ord}_{F'})\big).$$

Since $\operatorname{codim}_{Y'} V' = \dim T = r$, $\operatorname{vol}(\operatorname{ord}_{F'}) = \operatorname{vol}(\operatorname{ord}_{\overline{F}})$. Thus

$$\widehat{\operatorname{vol}}(Y' \ni u_{\Bbbk'}, g^*B_c) \leq \widehat{\operatorname{vol}}(F', Y', g^*B_c) = \widehat{\operatorname{vol}}(\overline{F}, T, \Delta_c) < \phi(c) + \epsilon.$$

Let $y' \in V'$ be a closed point and $y := g(y')$. Then $y \in V \subset E = f^{-1}(x)$ is a closed point. We have

$$\widehat{\operatorname{vol}}(X \ni x, B) \leq \widehat{\operatorname{vol}}(Y \ni y, B_c) = \widehat{\operatorname{vol}}(Y' \ni y', g^*B_c) \leq \frac{n^n}{r^r}\widehat{\operatorname{vol}}(Y' \ni u_{\Bbbk'}, g^*B_c) < \frac{n^n}{r^r}(\phi(c) + \epsilon),$$

where the first inequality follows from Lemma 2.10(3) since $K_Y + B_c = f^*(K_X + B)$, the equality follows from Lemma 2.10(2) since g is étale, and the second inequality follows from Lemma 4.3 for the klt pair (Y', g^*B_c) and V'. Since ϵ is arbitrary, (4.3) implies that

$$\widehat{\operatorname{vol}}(X \ni x, B) \leq \frac{n^n}{r^r}\phi(c) \leq n^n\Big(\frac{c}{t_*}\Big)^r \leq \frac{n^n}{t_*^2}c^2,$$

where the last inequality follows from $0 < \frac{c}{t_*} \leq 1$ and $r \geq 2$. The proposition follows. □

4.3. **Local volume and adjunction.**

Proposition 4.5. *Let $n \geq 2$ be an integer, $\Gamma \subset [0,1]$ a DCC set, and $\beta > 0$ a real number. Then there exists a real number $t_1 > 0$ depending only on n, Γ, and β satisfying the following. Let $(X \ni x, S + B)$ be a germ of dimension n and $t \in (0, t_1)$ a real number. Assume that*

(1) S is a $\mathbb{Q}$-Cartier prime divisor through x, $B \geq 0$ has no component S, and $B \in \Gamma$,
(2) $(X, S + B)$ is plt near x, and
(3) $\widehat{\operatorname{vol}}(X \ni x, B + (1-t)S) \geq \beta t$.

Then

$$\widehat{\operatorname{vol}}(S \ni x, \operatorname{Diff}_S(B)) \geq \frac{(n-1)^{n-1}}{2^{n+2}n^n}\beta.$$

Proof. **Step 1.** In this step we construct a cyclic cover of X branched along S whose degree tends to $+\infty$ as t tends to 0.

Let $(X \ni x, S+B)$ be a germ satisfying the conditions of Proposition 4.5, and let $t \in (0, \frac{1}{2})$ be a real number such that $\widehat{\mathrm{vol}}(X \ni x, B+(1-t)S) \geq \beta t$. Since $(X, S+B)$ is plt near x and $x \in S$, (X, B) is klt near x. Let $r := \mathrm{ind}_x(X, S)$, let $g : (X' \ni x') \to (X \ni x)$ be an index 1 cover of $(X \ni x, S)$, $S' := g^*S$, and $B' := g^*B$. Possibly shrinking X to an affine neighborhood of x and replacing X' with the preimage, by Definition-Lemma 2.16 we may assume that S' is a normal prime divisor, $(X', S'+B')$ is plt, $S' = \mathrm{div}(\psi)$ for some $\psi \in \mathcal{O}(X')$, and

$$\widehat{\mathrm{vol}}(S' \ni x', \mathrm{Diff}_{S'}(B')) = r \cdot \widehat{\mathrm{vol}}(S \ni x, \mathrm{Diff}_S(B)). \tag{4.4}$$

Since g is étale in codimension 1, $B' \in \Gamma$, S' is not a component of B', and $K_{X'} + B' + bS' = g^*(K_X + B + bS)$ for any $b \in \mathbb{R}$.

Let $k := \lfloor 1/t \rfloor$. Then $k \geq 2$, $\frac{1}{k} \geq t$, and $kt > 1 - t > \frac{1}{2}$. Let

$$Z := \mathrm{Spec}\, \mathcal{O}(X')[w]/(w^k - \psi),$$

let $q : Z \to X'$ be the projection, $S_Z := \mathrm{div}(w)$, and $B_Z := q^*B'$. Since g is étale in codimension 1 and (X, B) is klt near x, (X', B') is klt. Since $S' = \mathrm{div}(\psi)$ is a normal prime divisor and X' has rational singularities, Z is a normal variety, q is the quotient by the action $\zeta \cdot w = \zeta w$ of the group μ_k of k-th roots of unity, q is étale over $X' \backslash S'$, $q^{-1}(x')$ is a single point z, S_Z is a prime divisor with $q|_{S_Z} : S_Z \to S'$ an isomorphism, $q^*S' = kS_Z$, and μ_k acts trivially on $\mathcal{O}(Z)/(w)$. Since q is étale over the generic points of the components of B', $B_Z \in \Gamma$ and S_Z is not a component of B_Z. Since q is étale over $X' \backslash S'$ and $q^*S' = kS_Z$, $K_Z = q^*K_{X'} + (k-1)S_Z$, so

$$K_Z + B_Z = q^*\Big(K_{X'} + B' + \Big(1 - \frac{1}{k}\Big)S'\Big) \quad \text{and} \quad K_Z + S_Z + B_Z = q^*(K_{X'} + S' + B').$$

Thus (Z, B_Z) is klt and $(Z, S_Z + B_Z)$ is plt. Since $q|_{S_Z}$ is an isomorphism, the second equality and (4.4) imply

$$\widehat{\mathrm{vol}}(S_Z \ni z, \mathrm{Diff}_{S_Z}(B_Z)) = \widehat{\mathrm{vol}}(S' \ni x', \mathrm{Diff}_{S'}(B')) = r \cdot \widehat{\mathrm{vol}}(S \ni x, \mathrm{Diff}_S(B)). \tag{4.5}$$

By Lemma 2.10(1) for q and for g, and since $\frac{1}{k} \geq t$ and $kt > \frac{1}{2}$,

$$\begin{aligned} Q := \widehat{\mathrm{vol}}(Z \ni z, B_Z) &= kr \cdot \widehat{\mathrm{vol}}\Big(X \ni x, B + \Big(1 - \frac{1}{k}\Big)S\Big) \\ &\geq kr \cdot \widehat{\mathrm{vol}}(X \ni x, B + (1-t)S) \geq kr\beta t \geq \frac{r\beta}{2}. \end{aligned} \tag{4.6}$$

Step 2. In this step we take a sequence of counterexamples and move the coefficients of the boundaries into a finite set.

Suppose that Proposition 4.5 does not hold. Then for any $i \in \mathbb{N}^+$ there exist a germ $(X_i \ni x_i, S_i + B_i)$ satisfying the conditions of Proposition 4.5 and a real number $\tau_i \in (0, \min\{\frac{1}{2}, \frac{1}{i}\})$ such that $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i + (1-\tau_i)S_i) \geq \beta\tau_i$ and

$$\widehat{\mathrm{vol}}(S_i \ni x_i, \mathrm{Diff}_{S_i}(B_i)) < \frac{(n-1)^{n-1}}{2^{n+2}n^n}\beta. \tag{4.7}$$

Let r_i, k_i, $(Z_i \ni z_i, S_{Z_i} + B_{Z_i})$, w_i, and Q_i be the objects r, k, $(Z \ni z, S_Z + B_Z)$, w, and Q of **Step 1** constructed from $(X_i \ni x_i, S_i + B_i)$ and $t = \tau_i$. Then $\lim_{i\to+\infty} k_i = +\infty$ and $Q_i \geq \frac{r_i\beta}{2} \geq \frac{\beta}{2}$. The group μ_{k_i} acts on Z_i, preserves B_{Z_i}, and acts trivially on $\mathcal{O}(Z_i)/(w_i)$. Possibly replacing Z_i with a μ_{k_i}-invariant affine open neighborhood of z_i, we may assume that any component of B_{Z_i} passes through z_i.

Let $\gamma := \min((\Gamma \backslash \{0\}) \cup \{1\})$, which exists as Γ is a DCC set. The closure $\overline{\Gamma}$ is a DCC set. By Theorem 2.18 with $\epsilon = \frac{\beta}{2}$, there exists $M > 0$ such that $\mathrm{mld}(Z_i \ni z_i, B_{Z_i}) \leq M$ for any i. Let $c_0 > 0$ be the real number of Theorem 2.19 for $\mathcal{I} = \overline{\Gamma}$, $\epsilon = \frac{\beta}{2}$, M, and the germs $(Z_i \ni z_i, B_{Z_i})$,

fix a real number $t_0 \in (0, c_0)$, and let $\mathcal{I}_0 := \mathcal{I}_{t_0}$ and $p := p_{t_0}$. Possibly passing to a subsequence, $b \le p(b)$ for any $b \in \overline{\Gamma}$, $K_{Z_i} + p(B_{Z_i})$ is $\mathbb{R}$-Cartier for any i, and

$$\epsilon_i := \|p(B_{Z_i}) - B_{Z_i}\|$$

satisfies $\lim_{i\to+\infty} \epsilon_i = 0$. Let $B^+_{Z_i} := p(B_{Z_i})$. Since μ_{k_i} permutes the components of B_{Z_i} and preserves their coefficients, $B^+_{Z_i}$ is μ_{k_i}-invariant. Moreover, $\operatorname{Supp} B^+_{Z_i} = \operatorname{Supp} B_{Z_i}$ and any coefficient of $B^+_{Z_i}$ is at least γ. By Theorem 2.20 with $\epsilon = \min\{\gamma, \frac{\beta}{2}\}$, there exist a real number $c > 0$ and, for any i, an $\mathbb{R}$-divisor $\widehat{B}_{Z_i} \ge (1+c)B_{Z_i}$ such that $(Z_i \ni z_i, \widehat{B}_{Z_i})$ is klt. Let $\rho_i := \frac{\epsilon_i}{c\gamma}$. Since any coefficient of $\widehat{B}_{Z_i} - B_{Z_i}$ along a component of B_{Z_i} is at least $c\gamma$,

$$0 \le B^+_{Z_i} - B_{Z_i} \le \rho_i(\widehat{B}_{Z_i} - B_{Z_i}),$$

where $B^+_{Z_i} - B_{Z_i}$ and $\widehat{B}_{Z_i} - B_{Z_i}$ are $\mathbb{R}$-Cartier. Thus, for any valuation v on Z_i whose center contains z_i, since $v(\rho_i(\widehat{B}_{Z_i} - B_{Z_i}) - (B^+_{Z_i} - B_{Z_i})) \ge 0$ and $a(v, Z_i, \widehat{B}_{Z_i}) > 0$, by (2.1),

$$a(v, Z_i, B^+_{Z_i}) \ge a(v, Z_i, B_{Z_i}) - \rho_i\big(a(v, Z_i, B_{Z_i}) - a(v, Z_i, \widehat{B}_{Z_i})\big) \ge (1-\rho_i)a(v, Z_i, B_{Z_i}).$$

Possibly passing to a subsequence, we may assume that $\rho_i \le \frac{1}{2}$ for any i. Then $(Z_i \ni z_i, B^+_{Z_i})$ is klt, and by (4.6),

$$Q^+_i := \widehat{\operatorname{vol}}(Z_i \ni z_i, B^+_{Z_i}) \ge 2^{-n} Q_i \ge 2^{-n-1} r_i \beta. \tag{4.8}$$

Then the following hold for any i.

(1) $B^+_{Z_i} \ge B_{Z_i}$,
(2) $B^+_{Z_i} - B_{Z_i}$ is $\mathbb{R}$-Cartier,
(3) S_{Z_i} is not a component of $B^+_{Z_i}$,
(4) $B^+_{Z_i}$ is μ_{k_i}-invariant, and
(5) $B^+_{Z_i} \in \mathcal{I}_0$.

Step 3. In this step we degenerate $(Z_i \ni z_i, B^+_{Z_i})$ to K-semistable log Fano cone singularities and bound the degrees of their equations independently of i.

By [XZ24, Theorem 2.6] (see also [XZ21, Theorem 1.1] and, for real coefficients, [Zhu24b, Theorem 2.19]), there exists a unique valuation v_i on Z_i with center z_i such that $a(v_i, Z_i, B^+_{Z_i}) = 1$ and v_i minimizes $\widehat{\operatorname{vol}}(\cdot, Z_i, B^+_{Z_i})$ among the valuations with center z_i. For a valuation v on Z_i with center z_i, let

$$\operatorname{gr}_v \mathcal{O}_{Z_i,z_i} := \bigoplus_{\lambda\ge 0} \mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v), \quad \text{where } \mathfrak{a}_{>\lambda}(v) := \bigcup_{\lambda'>\lambda} \mathfrak{a}_{\lambda'}(v).$$

Vertices, Reeb vectors, the Reeb cone, the valuations wt_ξ, log Fano cone singularities, and their K-semistability are as in [XZ24, Subsection 2.3], and the Reeb vector induced by v is as in [XZ24, Subsection 4.2]. Moreover, v_i is quasi-monomial ([Xu20, Theorem 1.2], and [Han26, Theorem 2.4] for real coefficients), the ring

$$R_i := \operatorname{gr}_{v_i} \mathcal{O}_{Z_i,z_i}$$

is finitely generated, and, letting ξ_i be the Reeb vector induced by v_i, o_i the vertex of $\operatorname{Spec} R_i$, and

$$B_{i,0} := \sum_P b_P \cdot \big[\operatorname{Spec} \operatorname{gr}_{v_i}(\mathcal{O}_{Z_i,z_i}/I_P)\big],$$

where P runs over the components of $B^+_{Z_i}$, b_P is the coefficient of P in $B^+_{Z_i}$, I_P is the ideal of P, $\mathcal{O}_{Z_i,z_i}/I_P$ carries the quotient filtration, and $[\cdot]$ is the associated cycle of dimension $n-1$, the triple $(\operatorname{Spec} R_i, B_{i,0}; \xi_i)$ is a K-semistable log Fano cone singularity with

$$\widehat{\operatorname{vol}}(\operatorname{Spec} R_i \ni o_i, B_{i,0}) = Q^+_i.$$

For any $\zeta \in \mu_{k_i}$, as ζ fixes z_i and preserves $B^+_{Z_i}$, the valuation $v_i \circ \zeta$ has center z_i, satisfies $a(v_i \circ \zeta, Z_i, B^+_{Z_i}) = 1$, and minimizes $\widehat{\mathrm{vol}}(\cdot, Z_i, B^+_{Z_i})$ among the valuations with center z_i, so $v_i \circ \zeta = v_i$. Thus μ_{k_i} acts on R_i by graded automorphisms and preserves $B_{i,0}$.

Let $\mathcal{I}_1 := \mathbb{N}\mathcal{I}_0 \cap [0,1)$, where $\mathbb{N}\mathcal{I}_0$ is the additive monoid generated by $\mathcal{I}_0$. The set $\mathcal{I}_1$ is finite as any element of $\mathcal{I}_0 \backslash \{0\}$ is at least γ. Any coefficient of $B_{i,0}$ belongs to $\mathbb{N}\mathcal{I}_0$ and is less than 1 as $(\operatorname{Spec} R_i, B_{i,0})$ is klt, so $B_{i,0} \in \mathcal{I}_1$.

Since $B_{i,0} \in \mathcal{I}_1$ and $Q_i^+ \geq 2^{-n-1}\beta$ by (4.8), by [XZ24, Theorem 1.1 and Definition 2.16], possibly passing to a subsequence, there exist a variety T, a torus $\mathbb{T}$ with lattice of characters $\mathcal{M}$, and an $\mathbb{R}$-Gorenstein family $(\mathcal{X}, \mathcal{D}) \to T$ of log Fano cone singularities with a fiberwise $\mathbb{T}$-action such that, for any i, there exist a closed point $\mathfrak{t}_i \in T$ and an isomorphism

$$(\operatorname{Spec} R_i, B_{i,0}) \cong (\mathcal{X}_{\mathfrak{t}_i}, \mathcal{D}_{\mathfrak{t}_i}) \tag{4.9}$$

under which ξ_i is a Reeb vector in the Reeb cone of $\mathbb{T}$ on $\mathcal{X}_{\mathfrak{t}_i}$. By [LiuJ26, Theorem 1.2], n-dimensional K-semistable log Fano cone singularities whose local volumes and boundary coefficients are bounded from below by fixed positive numbers form a log bounded family, without moving the coefficients into a finite set. We use [XZ24, Theorem 1.1] because we need the family together with its torus action. Possibly passing to a subsequence and replacing T with a stratum of a finite stratification, by generic freeness we may assume the following.

(1) T is irreducible, and there exists a function $\mathfrak{h} : \mathcal{M} \to \mathbb{N}$ such that, for any $\chi \in \mathcal{M}$ and any fiber, $\mathfrak{h}(\chi)$ is the dimension of the χ-weight space of the coordinate ring of the fiber.
(2) There exists $\alpha \in \mathcal{M} \otimes \mathbb{R}$ such that
$$a(\mathrm{wt}_\xi, \mathcal{X}_\mathfrak{t}, \mathcal{D}_\mathfrak{t}) = \langle \xi, \alpha \rangle$$
for any closed point $\mathfrak{t} \in T$ and any ξ in the Reeb cone.
(3) The coordinate ring of $\mathcal{X}$ has a finite presentation over $\mathcal{O}(T)$ by $\mathbb{T}$-homogeneous generators and relations, the ideal of $\operatorname{Supp} \mathcal{D}_\mathfrak{t}$ for $\mathfrak{t} \in T$ is generated by the restrictions of finitely many $\mathbb{T}$-homogeneous elements, and the minimal generators of the coordinate rings of the fibers have $\mathbb{T}$-weights in a fixed finite set $\mathcal{M}_0 \subset \mathcal{M}$. For the second property we replace T by a finite cover over which the components of $\operatorname{Supp} \mathcal{D}$ are ordered and stratify further, so that the reduced supports of the fibers of $\mathcal{D}$ form a flat family.

Let $\mathcal{D}_j$ be the components of $\mathcal{D}$. By Lemma 2.9(1) at every closed point of $\mathcal{X}$, the set of $(e_j)_j$ such that $K_{\mathcal{X}/T} + \sum_j e_j \mathcal{D}_j$ is $\mathbb{R}$-Cartier is an intersection of affine subspaces defined over $\mathbb{Q}$, hence an affine subspace defined over $\mathbb{Q}$, and it contains the coefficients of $\mathcal{D}$. Thus

$$K_{\mathcal{X}/T} + \mathcal{D} = \theta_1(K_{\mathcal{X}/T} + \mathcal{D}^{(1)}) + \cdots + \theta_N(K_{\mathcal{X}/T} + \mathcal{D}^{(N)}), \quad \theta_1 + \cdots + \theta_N = 1,$$

where $\theta_1, \ldots, \theta_N \in \mathbb{R}$, each of $\mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(N)}$ is supported in $\operatorname{Supp} \mathcal{D}$, the divisors in parentheses are $\mathbb{T}$-invariant and $\mathbb{Q}$-Cartier, and α is the corresponding combination of the weights of $\mathbb{T}$-semi-invariant local generators of their Cartier multiples, divided by the corresponding Cartier indices, since, for $\mathcal{D}' = \mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(N)}$,

$$a(\mathrm{wt}_\xi, \mathcal{X}_\mathfrak{t}, \mathcal{D}'_\mathfrak{t}) = \frac{1}{s}\langle \xi, \chi \rangle$$

whenever $s(K_{\mathcal{X}/T} + \mathcal{D}')$ is Cartier near the vertex section and χ is the weight of a $\mathbb{T}$-semi-invariant local generator of $\mathcal{O}_\mathcal{X}(s(K_{\mathcal{X}/T} + \mathcal{D}'))$, viewed as an s-fold relative form [LX18, Theorem 2.15(3) and Lemma 2.18]. Such weights χ are locally constant on T.

By (1), $\mathrm{vol}(\mathrm{wt}_\xi)$ is determined by $\mathfrak{h}$ and ξ, so by (2), the function

$$\xi \to \widehat{\mathrm{vol}}(\mathrm{wt}_\xi, \mathcal{X}_\mathfrak{t}, \mathcal{D}_\mathfrak{t})$$

on the Reeb cone does not depend on $\mathfrak{t}$. By [XZ24, Definition 2.5], wt_{ξ_i} minimizes $\widehat{\mathrm{vol}}$ among the valuations with center o_i, and by [XZ24, Theorem 2.6] such minimizers are unique up to scaling. Thus, under the isomorphisms (4.9), there exists ξ_* in the Reeb cone such that $\xi_i \in \mathbb{R}_{>0}\xi_*$ for any i. Rescaling ξ_*, by (3) we may assume that $\langle \xi_*, \chi \rangle \geq 1$ for any $\chi \in \mathcal{M}_0$, and there exists an

integer $L > 0$ such that the relations and the generators of the ideals of the reduced boundary supports in (3) have ξ_*-degree at most L.

Any homogeneous presentation becomes a minimal presentation after eliminating redundant generators, which does not raise the degrees of relations, and two minimal systems of ξ_i-homogeneous generators of R_i differ by a degree-preserving automorphism of the polynomial ring. Since $\xi_i \in \mathbb{R}_{>0}\xi_*$, degrees measured by ξ_i and by ξ_* are proportional. For a minimal system of ξ_i-homogeneous generators of R_i which are μ_{k_i}-eigenvectors, the kernel of the corresponding surjection from the polynomial ring onto R_i and the preimage of the ideal of $\operatorname{Supp} B_{i,0}$ are μ_{k_i}-invariant and homogeneous, their minimal homogeneous generators have ξ_*-degree at most L while any generator of R_i in the system has ξ_*-degree at least 1, and, for a μ_{k_i}-invariant homogeneous ideal $\mathfrak{b}$ of the polynomial ring, generators of $\mathfrak{b}$ which are μ_{k_i}-eigenvectors are obtained by lifting eigenbases of the graded pieces of $\mathfrak{b}/\mathfrak{n}\mathfrak{b}$, where $\mathfrak{n}$ is the irrelevant ideal. We conclude:

(†) For any i, any minimal system $\phi_1, \dots, \phi_m$ of ξ_i-homogeneous generators of R_i which are μ_{k_i}-eigenvectors, and the surjection $\mathbb{C}[Y_1, \dots, Y_m] \to R_i$ given by $Y_j \to \phi_j$, on which μ_{k_i} acts on Y_j by the character of ϕ_j, the kernel I and the preimage J of the ideal of $\operatorname{Supp} B_{i,0}$ are generated by μ_{k_i}-eigenvectors which are polynomials of total degree at most L in $Y_1, \dots, Y_m$.

Step 4. In this step we show that, for $i \gg 0$, R_i is a polynomial ring in one variable over a graded subring.

Let $\overline{w}_i$ be the image of w_i in the graded piece $(R_i)_{v_i(w_i)}$ (the initial form of w_i). Since $v_i(w_i\varphi) = v_i(w_i) + v_i(\varphi)$ for any $\varphi \in \mathcal{O}_{Z_i,z_i}$, we have

$$\mathfrak{a}_\lambda(v_i) \cap (w_i) = w_i \mathfrak{a}_{\lambda - v_i(w_i)}(v_i) \quad \text{for any } \lambda \geq 0,$$

where $\mathfrak{a}_\lambda(v_i) := \mathcal{O}_{Z_i,z_i}$ for $\lambda < 0$, so

$$\operatorname{gr}_{v_i}\left(\mathcal{O}_{Z_i,z_i}/(w_i)\right) = R_i/(\overline{w}_i),$$

where $\mathcal{O}_{Z_i,z_i}/(w_i)$ carries the quotient filtration. Since μ_{k_i} acts trivially on $\mathcal{O}_{Z_i,z_i}/(w_i)$, μ_{k_i} acts trivially on $R_i/(\overline{w}_i)$, and $\zeta \cdot \overline{w}_i = \zeta \overline{w}_i$ for any $\zeta \in \mu_{k_i}$.

Let $\mathfrak{m} := \bigoplus_{\lambda>0}(R_i)_\lambda$ be the irrelevant ideal. Then μ_{k_i} acts trivially on the cotangent space $\mathfrak{m}/(\mathfrak{m}^2 + \overline{w}_i R_i)$ of $R_i/(\overline{w}_i)$, which is the quotient of $\mathfrak{m}/\mathfrak{m}^2$ by the span of the class $[\overline{w}_i]$ of $\overline{w}_i$. If $\overline{w}_i \in \mathfrak{m}^2$, then μ_{k_i} acts trivially on $\mathfrak{m}/\mathfrak{m}^2$, so R_i is generated by μ_{k_i}-invariant homogeneous lifts of a basis of $\mathfrak{m}/\mathfrak{m}^2$, and μ_{k_i} acts trivially on R_i, which contradicts $\zeta \cdot \overline{w}_i = \zeta \overline{w}_i$, $\overline{w}_i \neq 0$, and $k_i \geq 2$. Thus $[\overline{w}_i] \neq 0$, and

$$\mathfrak{m}/\mathfrak{m}^2 = \mathbb{C} \cdot [\overline{w}_i] \oplus (\mathfrak{m}/\mathfrak{m}^2)^{\mu_{k_i}}.$$

Let $y_1, \dots, y_{m_i} \in R_i$ be μ_{k_i}-invariant homogeneous lifts of a homogeneous basis of $(\mathfrak{m}/\mathfrak{m}^2)^{\mu_{k_i}}$. Then $\overline{w}_i, y_1, \dots, y_{m_i}$ is a minimal system of homogeneous generators of R_i consisting of μ_{k_i}-eigenvectors. We write

$$R_i = \mathbb{C}[Y_0, Y_1, \dots, Y_{m_i}]/I_i$$

with $Y_0 \to \overline{w}_i$ and $Y_j \to y_j$ for $j \geq 1$, where μ_{k_i} fixes $Y_1, \dots, Y_{m_i}$ and acts on Y_0 by $\zeta \cdot Y_0 = \zeta Y_0$, and let $J_i \supset I_i$ be the preimage of the ideal of $\operatorname{Supp} B_{i,0}$. By (†), I_i and J_i are generated by μ_{k_i}-eigenvectors of total degree at most L. Possibly passing to a subsequence, we may assume that $k_i > L$ for any i. Write a μ_{k_i}-eigenvector of total degree at most L as $\sum_e h_e(Y_1, \dots, Y_{m_i}) Y_0^e$ with $h_e \in \mathbb{C}[Y_1, \dots, Y_{m_i}]$. Then the exponents e with $h_e \neq 0$ are congruent modulo k_i and belong to $[0, L]$, so there exists exactly one such e, and the eigenvector is $h_e(Y_1, \dots, Y_{m_i}) Y_0^e$. Since I_i is prime and $Y_0 \notin I_i$, $h_e(Y_1, \dots, Y_{m_i}) Y_0^e \in I_i$ implies $h_e(Y_1, \dots, Y_{m_i}) \in I_i$. Thus I_i is generated by $I_i' := I_i \cap \mathbb{C}[Y_1, \dots, Y_{m_i}]$, and

$$R_i = R_i'[\overline{w}_i], \quad \text{where } R_i' := \mathbb{C}[Y_1, \dots, Y_{m_i}]/I_i'. \tag{4.10}$$

The ring R_i' is a normal domain of dimension $n-1$, as $R_i' = R_i'[\overline{w}_i] \cap \mathrm{Frac}(R_i')$, and R_i' is positively graded by the grading of R_i. Let o_i' be the vertex of $\operatorname{Spec} R_i'$. Then $\operatorname{Spec} R_i = \operatorname{Spec} R_i' \times \mathbb{A}^1$ and $o_i = (o_i', 0)$.

Step 5. In this step we show that $\operatorname{Spec} R_i' \times \{0\}$ is not a component of $B_{i,0}$ for $i \gg 0$, and deduce that $(\operatorname{Spec} R_i, B_{i,0})$ is a product.

Claim 4.6. *For any i there exists a μ_{k_i}-invariant valuation v_i' on Z_i with center z_i and values in $\mathbb{N}$ on $\mathcal{O}_{Z_i,z_i}\backslash\{0\}$, together with an isomorphism $\operatorname{gr}_{v_i'} \mathcal{O}_{Z_i,z_i} \cong R_i$ which maps*

$$\ker\left(\operatorname{gr}_{v_i'} \mathcal{O}_{Z_i,z_i} \to \operatorname{gr}_{v_i'} \mathcal{A}\right) \quad \textit{onto} \quad \ker\left(R_i \to \operatorname{gr}_{v_i} \mathcal{A}\right)$$

for $\mathcal{A} = \mathcal{O}_{Z_i,z_i}/(w_i)$ and for $\mathcal{A} = \mathcal{O}_{Z_i,z_i}/I_P$, where P runs over the components of $B_{Z_i}^+$ and $\mathcal{A}$ carries the quotient filtrations.

Proof. Since μ_{k_i} preserves the ξ_i-weight spaces of R_i, μ_{k_i} commutes with the torus $\mathbb{T}_i \subset \mathrm{Aut}(R_i)$ generated by ξ_i. Choose generators $\phi_1, \dots, \phi_m$ of R_i which are $\mathbb{T}_i$-homogeneous μ_{k_i}-eigenvectors, lifts of them to $\mathcal{O}_{Z_i,z_i}$ which are μ_{k_i}-eigenvectors with the characters of the corresponding generators, and, for R_i and for each of the finitely many modules $\operatorname{gr}_{v_i} \mathcal{A}$, a $\mathbb{T}_i$-graded μ_{k_i}-equivariant free resolution up to the first syzygies over $\mathbb{C}[Y_1, \dots, Y_m]$, where Y_j maps to ϕ_j.

Since the rings in [XZ24, Lemma 4.6] are complete local rings, let

$$\widehat{S} := \mathbb{C}[[Y_1, \dots, Y_m]]$$

act on $\widehat{\mathcal{O}}_{Z_i,z_i}$ by sending Y_j to the lift of ϕ_j. By [XZ24, Lemma 4.6], the resolutions lift to filtered free complexes over $\widehat{S}$, with μ_{k_i}-eigenvectors as matrix entries, since at each step a lift can be replaced by its projection to the corresponding eigencomponent (the average over μ_{k_i}), which changes neither its filtration level nor its initial term.

By [XZ24, Lemma 4.5], the lifted complexes are exact.

By [XZ24, Lemma 4.8], there exists an open neighborhood U of ξ_i in the Reeb cone of $\mathbb{T}_i$ such that, for any $\xi \in U$, the initial terms with respect to ξ of the finitely many matrix entries are those with respect to ξ_i. Then the shifted orders of the entries (the ξ-order of an entry minus the ξ-degree of the corresponding basis element of the domain plus that of the codomain) with respect to $\xi \in U$ are linear functions of ξ, so possibly shrinking U, any strict inequality between the shifted orders of two entries, zero entries included, holds also with respect to ξ.

Let $\xi \in U$ be rational, and let v_ξ be the quasi-monomial valuation of [XZ24, Lemma 4.9], defined on $\widehat{\mathcal{O}}_{Z_i,z_i}$ and restricted to $\mathcal{O}_{Z_i,z_i}$, with $\operatorname{gr}_{v_\xi} \mathcal{O}_{Z_i,z_i} = R_i$ and Reeb vector ξ. The initial matrices of the lifted complexes for v_ξ coincide with those for v_i.

For each lifted complex $M_2 \to M_1 \to \widehat{S}$, the map $M_1 \to \widehat{S}$ is strict for the ξ-filtration: if a preimage r of maximal ξ-order of an element φ of the image had order less than that of φ, then the initial form of r would be in the kernel of the initial map $M_1 \to \widehat{S}$, which equals the initial map for ξ_i, so by the exactness of the graded resolution it would be the initial form of the image of some $s \in M_2$, and r minus the image of s would be a preimage of φ of larger order, a contradiction. Thus $\operatorname{gr}_{v_\xi} \mathcal{A}$ is the quotient of R_i by the image of the initial matrix, so

$$\operatorname{gr}_{v_\xi} \mathcal{A} = \operatorname{gr}_{v_i} \mathcal{A} \quad \text{as quotients of } R_i.$$

The group μ_{k_i} acts on $\widehat{S}$ through the characters of $\phi_1, \dots, \phi_m$ and preserves the filtration by ξ-weights of monomials, and $\widehat{S} \to \widehat{\mathcal{O}}_{Z_i,z_i}$ is μ_{k_i}-equivariant, so the filtration defined by v_ξ is μ_{k_i}-invariant. Since ξ is rational, a positive integer multiple v_i' of v_ξ takes values in $\mathbb{N}$ on $\mathcal{O}_{Z_i,z_i}\backslash\{0\}$. □

Possibly replacing Z_i with a μ_{k_i}-invariant affine open neighborhood of z_i, we may assume that $K_{Z_i} + B_{Z_i}^+ = \sum_j \kappa_j \operatorname{div}(\psi_j)$ for some real numbers κ_j and rational functions ψ_j on Z_i. Let

$$\mathfrak{a}_\lambda := \{\varphi \in \mathcal{O}(Z_i) \mid v_i'(\varphi) \geq \lambda\} \quad \text{for } \lambda \in \mathbb{Z},$$

let u be a variable, and let

$$\mathcal{Z}_i := \operatorname{Spec} \bigoplus_{\lambda\in\mathbb{Z}} \mathfrak{a}_\lambda u^{-\lambda} \longrightarrow \mathbb{A}^1_u.$$

We show that $\mathcal{O}(\mathcal{Z}_i)$ is finitely generated. We have

$$\operatorname{gr}_{v'_i} \mathcal{O}(Z_i) = \operatorname{gr}_{v'_i} \mathcal{O}_{Z_i,z_i} \cong R_i,$$

and, until D is defined, we grade R_i by v'_i through the isomorphism with $\operatorname{gr}_{v'_i} \mathcal{O}_{Z_i,z_i}$. As v'_i takes values in $\mathbb{N}$ on $\mathcal{O}_{Z_i,z_i}\backslash\{0\}$, the rational functions φ on Z_i with $\varphi = 0$ or $v'_i(\varphi) \geq 0$ form a discrete valuation ring $\mathcal{R}$. Any non-zero element of the residue field κ of $\mathcal{R}$ is the class $[f/g]$ of f/g for some $f, g \in \mathcal{O}_{Z_i,z_i}$ with $v'_i(f) = v'_i(g)$, and $[f/g]$ depends only on the initial forms $\overline{f}, \overline{g} \in R_i$, so $\overline{f}/\overline{g} \to [f/g]$ is an isomorphism onto κ from the field K_0 of quotients of homogeneous elements of R_i of equal degree. Let ϑ be a quotient of homogeneous elements of R_i whose degree $d > 0$ generates the group generated by the degrees of the homogeneous elements of R_i. Then $\operatorname{Frac}(R_i) = K_0(\vartheta)$ and ϑ is transcendental over K_0 (non-zero quotients of homogeneous elements of distinct degrees are linearly independent over $\mathbb{C}$), so, as R_i is a finitely generated domain of dimension n, κ has transcendence degree $n-1$ over $\mathbb{C}$. By [KM98, Lemma 2.45], $\mathcal{R}$ is the local ring of a variety projective and birational over Z_i at the generic point of a prime divisor D, so $v'_i = c \operatorname{ord}_D$ for a positive integer c, and D is a divisor over $z_i \in Z_i$ as v'_i has center z_i.

Since $v'_i = c\operatorname{ord}_D$ takes values in $c\mathbb{N}$ on $\mathcal{O}(Z_i)\backslash\{0\}$, the ring $\operatorname{gr}_{\operatorname{ord}_D} \mathcal{O}(Z_i)$ is $\operatorname{gr}_{v'_i} \mathcal{O}(Z_i)$ with degrees divided by c, hence finitely generated. By [XZ25, Lemma 2.14], D is primitive, that is, there exists a projective birational morphism $\pi : Y \to Z_i$ from a normal variety Y on which D is a prime divisor and $-D$ is a π-ample $\mathbb{Q}$-Cartier divisor. As Z_i is normal, $\pi_*\mathcal{O}_Y(-mD) = \mathfrak{a}_{cm}$ for $m \in \mathbb{N}$, and $\bigoplus_{m\geq 0} \pi_*\mathcal{O}_Y(-mD)$ is a finitely generated $\mathcal{O}(Z_i)$-algebra as $-D$ is π-ample.

For $\lambda > 0$, $m := \lceil \lambda/c \rceil$, and $g \in \mathfrak{a}_\lambda = \mathfrak{a}_{cm}$, we have $gu^{-\lambda} = u^{cm-\lambda} \cdot gu^{-cm}$ with $cm - \lambda \geq 0$, and $\mathfrak{a}_\lambda u^{-\lambda} = \mathcal{O}(Z_i)u^{-\lambda}$ for $\lambda \leq 0$. Hence $\mathcal{O}(\mathcal{Z}_i)$ is generated over the finitely generated $\mathbb{C}$-algebra $\mathcal{O}(Z_i)$ by u and the elements gu^{-cm}, where g runs over a finite set of homogeneous generators of $\bigoplus_{m\geq 0} \pi_*\mathcal{O}_Y(-mD)$ and $g \in \pi_*\mathcal{O}_Y(-mD)$.

Thus $\mathcal{O}(\mathcal{Z}_i)$ is finitely generated, u is a nonzerodivisor, $\mathcal{Z}_{i,0} := \operatorname{div}(u) \cong \operatorname{Spec} R_i$ by Claim 4.6, and $\mathcal{Z}_i\backslash\mathcal{Z}_{i,0} = Z_i \times \mathbb{G}_m$. Since R_i and $\mathcal{O}(\mathcal{Z}_i)[u^{-1}] = \mathcal{O}(Z_i)[u, u^{-1}]$ are normal, $\mathcal{Z}_i$ is normal. The group μ_{k_i} acts on $\mathcal{Z}_i$ over $\mathbb{A}^1_u$. For a prime divisor P on Z_i, the closure $\mathcal{P}$ of $P \times \mathbb{G}_m$ has coordinate ring

$$\bigoplus_\lambda \big((\mathfrak{a}_\lambda + I_P)/I_P\big)u^{-\lambda},$$

which has no u-torsion and whose quotient by u is $\operatorname{gr}_{v'_i}(\mathcal{O}(Z_i)/I_P)$. Let $\mathcal{B}_i$ be the closure of $B^+_{Z_i} \times \mathbb{G}_m$, defined componentwise. By Claim 4.6, the restriction of $\mathcal{B}_i$ to $\mathcal{Z}_{i,0}$ is $B_{i,0}$. Let $\widetilde{w}_i := w_i u^{-v'_i(w_i)}$ and $\mathcal{H}_i := \operatorname{div}(\widetilde{w}_i)$. As in **Step 4**,

$$\mathfrak{a}_\lambda \cap (w_i) = w_i \mathfrak{a}_{\lambda - v'_i(w_i)},$$

so $\mathcal{H}_i$ is the closure of $S_{Z_i} \times \mathbb{G}_m$, $\mathcal{O}(\mathcal{H}_i)$ is the Rees algebra of the quotient filtration on $\mathcal{O}(S_{Z_i})$, and by Claim 4.6 and **Step 4**,

$$\mathcal{H}_{i,0} := \mathcal{H}_i \cap \mathcal{Z}_{i,0} = \operatorname{Spec} R'_i \times \{0\}.$$

Moreover, $\zeta \cdot \widetilde{w}_i = \zeta\widetilde{w}_i$ for any $\zeta \in \mu_{k_i}$, and μ_{k_i} acts trivially on $\mathcal{O}(\mathcal{H}_i)$, as μ_{k_i} acts trivially on $\mathcal{O}(S_{Z_i})$.

We choose $K_{\mathcal{Z}_i}$ so that the restriction of $K_{\mathcal{Z}_i} + \mathcal{B}_i$ to $Z_i \times \mathbb{G}_m$ is the pullback of $K_{Z_i} + B^+_{Z_i}$. Then the $\mathbb{R}$-divisor

$$K_{\mathcal{Z}_i} + \mathcal{B}_i - \sum_j \kappa_j \operatorname{div}(\psi_j)$$

is supported on the principal prime divisor $\mathcal{Z}_{i,0} = \operatorname{div}(u)$, so $K_{\mathcal{Z}_i} + \mathcal{B}_i \sim_{\mathbb{R}} 0$. In particular, $K_{\mathcal{Z}_i} + \mathcal{B}_i$ is $\mathbb{R}$-Cartier.

Let η be the generic point of $\mathcal{H}_{i,0}$. Since $\mathcal{O}_{\mathcal{Z}_i,\eta}/(u)$ is the localization of $R_i = R_i'[\overline{w}_i]$ at the prime ideal $(\overline{w}_i)$, which is a discrete valuation ring with uniformizer $\overline{w}_i$, the ring $\mathcal{O}_{\mathcal{Z}_i,\eta}$ is a regular local ring of dimension 2 with parameters u and $\widetilde{w}_i$. Let P be a component of $B^+_{Z_i}$ and $\mathcal{P}^\mu$ the closure of $\sum_{P'} P' \times \mathbb{G}_m$, where P' runs over the distinct components of $B^+_{Z_i}$ in the μ_{k_i}-orbit of P. Since $\mathcal{P}^\mu$ is μ_{k_i}-invariant, μ_{k_i} fixes η, and $\mathcal{O}_{\mathcal{Z}_i,\eta}$ is regular, the ideal of $\mathcal{P}^\mu$ in $\mathcal{O}_{\mathcal{Z}_i,\eta}$ is generated by a μ_{k_i}-eigenvector F. Since S_{Z_i} is not a component of $B^+_{Z_i}$, $\widetilde{w}_i$ does not divide F, so the image of F in the domain $\mathcal{O}_{\mathcal{H}_i,\eta}$ is non-zero, and μ_{k_i} acts trivially on $\mathcal{O}_{\mathcal{H}_i,\eta}$, so F is μ_{k_i}-invariant. Write $F \equiv \widetilde{w}_i^e F_0$ modulo u, where $e \in \mathbb{N}$ and F_0 is a unit of $\mathcal{O}_{\mathcal{Z}_i,\eta}$. Then e is the multiplicity of $\operatorname{Spec} R_i' \times \{0\}$ in the restriction of $\mathcal{P}^\mu$ to $\mathcal{Z}_{i,0}$, and, as F is μ_{k_i}-invariant, $\zeta \cdot F_0 \equiv \zeta^{-e} F_0$ modulo u for any $\zeta \in \mu_{k_i}$. Since μ_{k_i} acts trivially on the residue field $\mathcal{O}_{\mathcal{H}_{i,0},\eta} = \operatorname{Frac}(R_i')$ and the residue of F_0 is non-zero, $\zeta^e = 1$ for any $\zeta \in \mu_{k_i}$, that is, k_i divides e. Since any coefficient of $B^+_{Z_i}$ is at least γ, the coefficient of $\operatorname{Spec} R_i' \times \{0\}$ in $B_{i,0}$ is either 0 or at least $k_i\gamma$. As $(\operatorname{Spec} R_i, B_{i,0})$ is klt, possibly passing to a subsequence such that $k_i\gamma > 1$, $\operatorname{Spec} R_i' \times \{0\}$ is not a component of $B_{i,0}$.

Let $h_e(Y_1, \dots, Y_{m_i})Y_0^e$ be one of the generators of J_i given in **Step 4**. Since J_i is the intersection of the preimages of the prime ideals of the components of $\operatorname{Supp} B_{i,0}$, none of which contains Y_0, we have $h_e(Y_1, \dots, Y_{m_i}) \in J_i$. Thus J_i is generated by $J_i \cap \mathbb{C}[Y_1, \dots, Y_{m_i}]$, so any component of $\operatorname{Supp} B_{i,0}$ is of the form $V \times \mathbb{A}^1$ for a prime divisor V on $\operatorname{Spec} R_i'$. Since $\operatorname{Spec} R_i = \operatorname{Spec} R_i' \times \mathbb{A}^1$, there exists an $\mathbb{R}$-divisor $\Delta_i \geq 0$ on $\operatorname{Spec} R_i'$ such that

$$(\operatorname{Spec} R_i, B_{i,0}) = (\operatorname{Spec} R_i' \times \mathbb{A}^1, \Delta_i \times \mathbb{A}^1), \quad \mathcal{H}_{i,0} = \operatorname{Spec} R_i' \times \{0\}. \tag{4.11}$$

Step 6. In this step we bound the local volume of $(S_{Z_i} \ni z_i, \operatorname{Diff}_{S_{Z_i}}(B^+_{Z_i}))$ from below and conclude the proof.

By (4.11),

$$a(V \times \mathbb{A}^1, \operatorname{Spec} R_i, B_{i,0}) = a(V, \operatorname{Spec} R_i', \Delta_i)$$

for any prime divisor V over $\operatorname{Spec} R_i'$, and $K_{\operatorname{Spec} R_i} + B_{i,0}$ is the pullback of $K_{\operatorname{Spec} R_i'} + \Delta_i$. Thus $(\operatorname{Spec} R_i' \ni o_i', \Delta_i)$ is a klt germ. We show

$$\widehat{\operatorname{vol}}(\operatorname{Spec} R_i' \ni o_i', \Delta_i) \geq \frac{(n-1)^{n-1}}{n^n} Q_i^+. \tag{4.12}$$

Let E be a prime divisor over $\operatorname{Spec} R_i'$ with center o_i', $a_E := a(E, \operatorname{Spec} R_i', \Delta_i)$, and $\omega > 0$ a real number. Let $\widetilde{v}_\omega$ be the valuation on $\operatorname{Spec} R_i$ given by

$$\widetilde{v}_\omega\Big(\sum_e h_e \overline{w}_i^e\Big) := \min_e \big(\operatorname{ord}_E(h_e) + e\omega\big), \quad h_e \in R_i'.$$

On the product of $\mathbb{A}^1$ with a higher model of $\operatorname{Spec} R_i'$ on which E is a prime divisor, $\widetilde{v}_\omega$ is the quasi-monomial valuation with weights 1 and ω along $E \times \mathbb{A}^1$ and along the zero locus of $\overline{w}_i$, and the center of $\widetilde{v}_\omega$ on $\operatorname{Spec} R_i$ is o_i. Near the generic point of $E \times \{0\}$, the crepant pullback of $\Delta_i \times \mathbb{A}^1$ to the product of $\mathbb{A}^1$ with the higher model is $(1 - a_E)E \times \mathbb{A}^1$, so $a(\widetilde{v}_\omega, \operatorname{Spec} R_i, B_{i,0}) = a_E + \omega$ by [JM12, Proposition 5.1]. For $\lambda > 0$,

$$\mathfrak{a}_\lambda(\widetilde{v}_\omega) \cap \mathcal{O}_{\operatorname{Spec} R_i', o_i'}[\overline{w}_i] = \bigoplus_{e \geq 0} \mathfrak{a}_{\lambda - e\omega}(\operatorname{ord}_E)\overline{w}_i^e,$$

where $\mathfrak{a}_{\lambda'}(\operatorname{ord}_E) = \mathcal{O}_{\operatorname{Spec} R_i', o_i'}$ for $\lambda' \leq 0$. Thus

$$\ell\big(\mathcal{O}_{\operatorname{Spec} R_i, o_i}/\mathfrak{a}_\lambda(\widetilde{v}_\omega)\big) = \sum_{0 \leq e < \lambda/\omega} \ell\big(\mathcal{O}_{\operatorname{Spec} R_i', o_i'}/\mathfrak{a}_{\lambda - e\omega}(\operatorname{ord}_E)\big).$$

Since

$$\ell\big(\mathcal{O}_{\operatorname{Spec} R_i', o_i'}/\mathfrak{a}_{\lambda'}(\operatorname{ord}_E)\big) = \frac{\operatorname{vol}(\operatorname{ord}_E)}{(n-1)!}\lambda'^{n-1} + o(\lambda'^{n-1}) \quad \text{as } \lambda' \to +\infty,$$

the right-hand side is

$$\frac{\operatorname{vol}(\operatorname{ord}_E)}{\omega}\frac{\lambda^n}{n!} + o(\lambda^n) \quad \text{as } \lambda \to +\infty,$$

so $\mathrm{vol}(\widetilde{v}_\omega) = \mathrm{vol}(\mathrm{ord}_E)/\omega$. Taking $\omega := \frac{a_E}{n-1}$, we obtain

$$\begin{aligned} Q_i^+ \le \widehat{\mathrm{vol}}(\widetilde{v}_\omega, \operatorname{Spec} R_i, B_{i,0}) &= \Big(\frac{na_E}{n-1}\Big)^n \frac{(n-1)\,\mathrm{vol}(\mathrm{ord}_E)}{a_E} \\ &= \frac{n^n}{(n-1)^{n-1}} \widehat{\mathrm{vol}}(E, \operatorname{Spec} R_i', \Delta_i). \end{aligned}$$

The local volume $\widehat{\mathrm{vol}}(\operatorname{Spec} R_i' \ni o_i', \Delta_i)$ is the infimum of $\widehat{\mathrm{vol}}(E, \operatorname{Spec} R_i', \Delta_i)$ over the prime divisors E over $\operatorname{Spec} R_i'$ with center o_i'. If $n-1 \ge 2$, this holds by Lemma 2.8(2), as any Kollár component of $(\operatorname{Spec} R_i' \ni o_i', \Delta_i)$ is such a prime divisor. If $n = 2$, then o_i' is a prime divisor on the curve $\operatorname{Spec} R_i'$, any valuation with center o_i' is a positive multiple of the divisorial valuation of o_i', and $\widehat{\mathrm{vol}}$ is invariant under scaling. Thus (4.12) follows.

We compare the fiber $\mathcal{H}_{i,0}$ of $\mathcal{H}_i \to \mathbb{A}^1_u$ over 0 with the fibers over $u \ne 0$. Since u is a nonzerodivisor on $\mathcal{O}(\mathcal{H}_i)$ with normal quotient R_i' and

$$\mathcal{H}_i \backslash \mathcal{H}_{i,0} = S_{Z_i} \times \mathbb{G}_m$$

is normal, $\mathcal{H}_i$ is normal. Since $\mathcal{H}_i$ is Cartier, $K_{\mathcal{Z}_i} + \mathcal{H}_i + \mathcal{B}_i$ is $\mathbb{R}$-Cartier. Let

$$\mathcal{B}_{\mathcal{H}_i} := \mathrm{Diff}_{\mathcal{H}_i}(\mathcal{B}_i).$$

Then

$$(\mathcal{H}_i \backslash \mathcal{H}_{i,0}, \mathcal{B}_{\mathcal{H}_i}|_{\mathcal{H}_i \backslash \mathcal{H}_{i,0}}) = (S_{Z_i}, \mathrm{Diff}_{S_{Z_i}}(B_{Z_i}^+)) \times \mathbb{G}_m,$$

and the closure of $\{z_i\} \times \mathbb{G}_m$, whose coordinate ring is the Rees algebra $\mathbb{C}[u]$ of the quotient filtration on $\mathcal{O}_{Z_i,z_i}/\mathfrak{m}_{z_i} = \mathbb{C}$, is the image of a section σ_i of $\mathcal{H}_i \to \mathbb{A}^1_u$ with $\sigma_i(0) = o_i$. Since $\mathcal{O}_{\mathcal{Z}_i,\eta}$ is regular and no component of $\mathcal{B}_i$ passes through η by **Step 5**, $\mathcal{H}_{i,0}$ is not a component of $\mathcal{B}_{\mathcal{H}_i}$. At the generic point of any prime divisor on $\mathcal{H}_{i,0}$, the scheme $\mathcal{Z}_{i,0} = \operatorname{Spec} R_i' \times \mathbb{A}^1$ is regular as R_i' is normal, so $\mathcal{Z}_i$ and $\mathcal{H}_i$ are regular there, and the differents are restrictions of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors. Thus, by (4.11),

$$\mathrm{Diff}_{\mathcal{H}_{i,0}}(\mathcal{B}_{\mathcal{H}_i}) = \big(\mathcal{B}_i|_{\mathcal{Z}_{i,0}}\big)\big|_{\mathcal{H}_{i,0}} = \Delta_i.$$

Since $(\operatorname{Spec} R_i' \ni o_i', \Delta_i)$ is klt, by [Kol23, Theorem 11.17.1], there exists an open neighborhood $\mathcal{U}$ of o_i in $\mathcal{H}_i$ such that

$$\big(\mathcal{U}, (\mathcal{H}_{i,0} + \mathcal{B}_{\mathcal{H}_i})|_{\mathcal{U}}\big)$$

is plt, so $(\mathcal{U}, \mathcal{B}_{\mathcal{H}_i}|_{\mathcal{U}})$ is klt. As $\mathcal{U}$ contains $\sigma_i(u_0) = (z_i, u_0)$ for some $u_0 \ne 0$, $(S_{Z_i} \ni z_i, \mathrm{Diff}_{S_{Z_i}}(B_{Z_i}^+))$ is klt.

We show

$$\widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \mathrm{Diff}_{S_{Z_i}}(B_{Z_i}^+)) \ge \widehat{\mathrm{vol}}(\operatorname{Spec} R_i' \ni o_i', \Delta_i). \tag{4.13}$$

Write $\mathcal{B}_{\mathcal{H}_i} = \sum_{j=1}^m b_j \mathcal{P}_j$, where $\mathcal{P}_j$ are distinct prime divisors and $b_j > 0$, and let $P_j := \mathcal{P}_j \cap (S_{Z_i} \times \{1\})$, so that $\mathrm{Diff}_{S_{Z_i}}(B_{Z_i}^+) = \sum_j b_j P_j$. For $b' \in \mathbb{R}^m$ let $\mathcal{B}(b') := \sum_j b_j' \mathcal{P}_j$, and let W be the set of $b' \in \mathbb{R}^m$ such that $K_{\mathcal{H}_i} + \mathcal{B}(b')$ is $\mathbb{R}$-Cartier near $\sigma_i(0)$. By Lemma 2.9(1), W is an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$ which contains $b = (b_1, \dots, b_m)$, so $W \cap \mathbb{Q}^m$ is dense in W. The $\mathbb{G}_m$-action on $\mathcal{H}_i$ given by the grading preserves the $\mathcal{P}_j$, so for $b' \in W$ the locus where $K_{\mathcal{H}_i} + \mathcal{B}(b')$ is not $\mathbb{R}$-Cartier is a closed $\mathbb{G}_m$-invariant subset not containing $\sigma_i(0)$, hence disjoint from $\sigma_i(\mathbb{A}^1)$. For $b' \in W$, the restrictions of $K_{\mathcal{H}_i} + \mathcal{B}(b')$ to $\mathcal{H}_{i,0}$ and to $S_{Z_i} \times \{1\}$ are $K_{\operatorname{Spec} R_i'} + \Delta(b')$ and $K_{S_{Z_i}} + \sum_j b_j' P_j$, where $\Delta(b') := \sum_j b_j' \mathcal{P}_j|_{\mathcal{H}_{i,0}}$ depends linearly on b', $\Delta(b) = \Delta_i$, and $\operatorname{Supp} \Delta(b') \subset \operatorname{Supp} \Delta_i$. Write $\Delta(b') = \sum_V \mathrm{mult}_V(\Delta(b'))V$, where V runs over the distinct prime components of $\operatorname{Supp} \Delta_i$. The coefficients $\mathrm{mult}_V(\Delta(b'))$ are linear functions of b'. Fix a real number $\epsilon \in (0,1)$. By Lemma 2.9(2) on $\operatorname{Spec} R_i'$ for the prime divisors V and

the vector $(\mathrm{mult}_V \Delta_i)_V$, and on S_{Z_i} for the prime divisors P_j and the vector b, there exists $b' \in W \cap \mathbb{Q}^m_{>0}$ such that both $(\operatorname{Spec} R'_i \ni o'_i, \Delta(b'))$ and $(S_{Z_i} \ni z_i, \sum_j b'_j P_j)$ are klt and

$$\frac{\widehat{\mathrm{vol}}(\operatorname{Spec} R'_i \ni o'_i, \Delta(b'))}{\widehat{\mathrm{vol}}(\operatorname{Spec} R'_i \ni o'_i, \Delta_i)} \geq (1-\epsilon)^{n-1}, \quad \frac{\widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \sum_j b'_j P_j)}{\widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \sum_j b_j P_j)} \leq (1+\epsilon)^{n-1}.$$

By [Kol23, Theorem 11.17.1], $(\mathcal{H}_i, \mathcal{H}_{i,0} + \mathcal{B}(b'))$ is plt near $\sigma_i(0)$, so possibly shrinking $\mathcal{H}_i$ to an open neighborhood of $\sigma_i(\mathbb{A}^1)$, $(\mathcal{H}_i, \mathcal{B}(b')) \to \mathbb{A}^1_u$ with the section σ_i is a $\mathbb{Q}$-Gorenstein flat family of klt singularities over $\mathbb{A}^1_u$ in the sense of [BL21, Definition 7]. The fibers of $(\mathcal{H}_i, \mathcal{B}(b')) \to \mathbb{A}^1_u$ over $u \neq 0$ are $\mathbb{G}_m$-translates of $(S_{Z_i} \ni z_i, \sum_j b'_j P_j)$, which is klt by the choice of b'. By the lower semicontinuity of local volumes [BL21, Theorem 1],

$$\widehat{\mathrm{vol}}\Big(S_{Z_i} \ni z_i, \sum_j b'_j P_j\Big) \geq \widehat{\mathrm{vol}}(\operatorname{Spec} R'_i \ni o'_i, \Delta(b')).$$

Thus

$$(1+\epsilon)^{n-1}\widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \mathrm{Diff}_{S_{Z_i}}(B^+_{Z_i})) \geq (1-\epsilon)^{n-1}\widehat{\mathrm{vol}}(\operatorname{Spec} R'_i \ni o'_i, \Delta_i).$$

Since ϵ is arbitrary, (4.13) holds.

Since $B^+_{Z_i} - B_{Z_i} \geq 0$ is $\mathbb{R}$-Cartier and S_{Z_i} is not a component of $B^+_{Z_i} - B_{Z_i}$,

$$\mathrm{Diff}_{S_{Z_i}}(B_{Z_i}) = \mathrm{Diff}_{S_{Z_i}}(B^+_{Z_i}) - (B^+_{Z_i} - B_{Z_i})|_{S_{Z_i}} \leq \mathrm{Diff}_{S_{Z_i}}(B^+_{Z_i}),$$

and $(B^+_{Z_i} - B_{Z_i})|_{S_{Z_i}}$ is $\mathbb{R}$-Cartier. By (4.5), (4.13), (4.12), and (4.8),

$$\begin{aligned} r_i\widehat{\mathrm{vol}}(S_i \ni x_i, \mathrm{Diff}_{S_i}(B_i)) &= \widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \mathrm{Diff}_{S_{Z_i}}(B_{Z_i})) \\ &\geq \widehat{\mathrm{vol}}(S_{Z_i} \ni z_i, \mathrm{Diff}_{S_{Z_i}}(B^+_{Z_i})) \geq \widehat{\mathrm{vol}}(\operatorname{Spec} R'_i \ni o'_i, \Delta_i) \\ &\geq \frac{(n-1)^{n-1}}{n^n} Q^+_i \geq \frac{(n-1)^{n-1}}{2^{n+1}n^n} r_i \beta. \end{aligned}$$

Since $r_i > 0$ and $2^{-n-1} > 2^{-n-2}$, this contradicts (4.7). The proposition follows. □

4.4. **Existence and bounds.**

Theorem 4.7. *Let $n \geq 2$ be an integer, $\Gamma \subset [0,1]$ a DCC set, and $\epsilon > 0$ a real number. Then there exist real numbers $C \geq 1$ and $\kappa > 0$ depending only on n, Γ, and ϵ satisfying the following. Let $(X \ni x, B)$ be a klt germ of dimension n such that $B \in \Gamma$ and $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon \mathrm{mld}(X \ni x, B)$. Then $(X \ni x, B)$ has a Kollár component E such that*

$$a(E, X, B) \leq C\mathrm{mld}(X \ni x, B) \quad \textit{and} \quad \delta(E, B_E) \geq \kappa.$$

Proof. Let $\tau > 0$ be the real number of Lemma 4.2 for n and Γ, let $t_* \in (0, \frac{1}{2})$ be the real number of Proposition 4.4 for n and Γ, and let $t_1 > 0$ be the real number of Proposition 4.5 for n, Γ, and $\beta = \epsilon/2$. We let

$$m_0 := \min\Big\{\frac{\tau}{2}, \frac{t_*}{2}, \frac{t_1}{2}, \frac{\epsilon t_*^2}{4n^n}\Big\} \quad \text{and} \quad \eta := \frac{\epsilon}{2^{n+3}n^n}.$$

We let $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma\backslash\{0\} \neq \emptyset$, and $\gamma := 1$ otherwise. Let $\gamma_0 := \min\{\gamma, \epsilon m_0\}$. By [HLQ24, Theorem 7.5], there exists a real number $\eta' > 0$ depending only on n and γ_0 satisfying the following. Let $(X \ni x, B)$ be a klt germ of dimension n such that

(1) any non-zero coefficient of B is $\geq \gamma_0$, and
(2) $\widehat{\mathrm{vol}}(X \ni x, B) \geq \gamma_0$.

Then $(X \ni x, B)$ has a plt blow-up $f : Y \to X$ with exceptional divisor E such that $(Y, E + f_*^{-1}B)$ is η'-plt near E. By [HLS26, Proposition 4.3(1)], there exists a real number $c_0 > 0$ depending only on n and η' such that $a(E, X, B) \leq c_0$ for any such $(X \ni x, B)$ and E. Let $\kappa_1 > 0$ (resp. $\kappa_2 > 0$) be the real number of Theorem 2.15(2) for $n-1$ and η (resp. for $n-1$ and η'). We let

$$C := \max\Big\{2, \frac{c_0}{m_0}\Big\} \quad \text{and} \quad \kappa := \min\{\kappa_1, \kappa_2\}.$$

Let $(X \ni x, B)$ be a klt germ of dimension n such that $B \in \Gamma$ and $V \geq \epsilon m$, where

$$m := \mathrm{mld}(X \ni x, B) \quad \text{and} \quad V := \widehat{\mathrm{vol}}(X \ni x, B).$$

Case 1: $m < m_0$. Since $m > 0$, by the definition of $\mathrm{mld}(X \ni x, B)$, there exists a prime divisor E over $X \ni x$ such that $a := a(E, X, B) < 2m$. Since $2m < 2m_0$,

$$0 < a < \min\{\tau, t_*, t_1\} \quad \text{and} \quad a < \frac{\epsilon t_*^2}{2n^n}.$$

Since $a < \tau$, by Lemma 4.2, $(X \ni x, B)$ has an lc blow-up $f : Y \to X$ with exceptional divisor E. Let $B_Y := f_*^{-1}B$. Suppose that $(Y, E + B_Y)$ is not plt near E. Since $0 < a \leq t_*$, by Proposition 4.4 and since $a < \epsilon t_*^2/(2n^n)$ and $a < 2m$,

$$V \leq \frac{n^n}{t_*^2}a^2 < \frac{\epsilon a}{2} < \epsilon m \leq V,$$

a contradiction. Thus f is a plt blow-up and E is a Kollár component of $(X \ni x, B)$ with $a < 2m \leq Cm$.

Possibly shrinking X to a neighborhood of x, we may assume that $(Y, E + B_Y)$ is plt. By Definition 2.5,

$$K_Y + B_Y + (1 - a)E = f^*(K_X + B),$$

and $(Y, B_Y + (1 - a)E)$ is klt since (X, B) is klt and $0 < a < 1$. Let F be a prime divisor over E, and let y be a closed point of $\mathrm{center}_E F$. Since $f(y) = x$, by Lemma 2.10(3) and since $a < 2m$,

$$\widehat{\mathrm{vol}}(Y \ni y, B_Y + (1 - a)E) \geq V \geq \epsilon m > \frac{\epsilon}{2}a.$$

By Proposition 4.5 for $\beta = \epsilon/2$ and $t = a$, and since $B_E = \mathrm{Diff}_E(B_Y)$,

$$\widehat{\mathrm{vol}}(E \ni y, B_E) \geq \frac{(n-1)^{n-1}}{2^{n+2}n^n} \cdot \frac{\epsilon}{2} = (n-1)^{n-1}\eta.$$

Since $(E \ni y, B_E)$ is a klt germ of dimension $n - 1$, by Lemma 2.10(4),

$$a(F, E, B_E) \geq \frac{\widehat{\mathrm{vol}}(E \ni y, B_E)}{(n-1)^{n-1}} \geq \eta.$$

Thus (E, B_E) is η-lc. Since $B_E \geq 0$ and $-(K_E + B_E)$ is ample (Definition 2.5), by Theorem 2.15(2) and [BJ20, Theorem A],

$$\delta(E, B_E) \geq \alpha(E, B_E) \geq \kappa_1 \geq \kappa.$$

Case 2: $m \geq m_0$. Then

$$V \geq \epsilon m \geq \epsilon m_0 \geq \gamma_0.$$

Since $B \in \Gamma$, any non-zero coefficient b of B satisfies $b \geq \gamma \geq \gamma_0$. By the choice of η', $(X \ni x, B)$ has a plt blow-up $f : Y \to X$ with exceptional divisor E such that $(Y, E + B_Y)$ is η'-plt near E, where $B_Y := f_*^{-1}B$. Then E is a Kollár component of $(X \ni x, B)$, and by the choice of c_0,

$$a(E, X, B) \leq c_0 = \frac{c_0}{m_0} \cdot m_0 \leq \frac{c_0}{m_0} \cdot m \leq Cm.$$

Possibly shrinking X to a neighborhood of x, we may assume that $(Y, E + B_Y)$ is η'-plt. By [HLS19, Theorem 3.10(2)], $a(F, E, B_E) > \eta'$ for any prime divisor F over E, so (E, B_E) is η'-lc. Since $B_E \geq 0$ and $-(K_E + B_E)$ is ample, by Theorem 2.15(2) and [BJ20, Theorem A],

$$\delta(E, B_E) \geq \alpha(E, B_E) \geq \kappa_2 \geq \kappa.$$

The theorem follows. □

Set-up 4.8. Let n, ϵ, κ, C, θ, $(X \ni x, B)$, E, f, and Y be as follows.

(1) $n \geq 2$ is an integer, $\epsilon, \kappa > 0$ and $C \geq 1$ are real numbers, and $\theta := \frac{\kappa^{n-1}\epsilon}{Cn^{n-1}}$.

(2) $(X \ni x, B)$ is a klt germ of dimension n such that $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon \mathrm{mld}(X \ni x, B)$.

(3) E is a Kollár component of $(X \ni x, B)$ with plt blow-up $f : Y \to X$ such that $a(E, X, B) \le C\mathrm{mld}(X \ni x, B)$ and $\delta(E, B_E) \ge \kappa$.

Proposition 4.9. *Notation and conditions as in Set-up 4.8. Then:*

(1) $(-(K_E + B_E))^{n-1} \ge \epsilon/C$.
(2) $a(F, E, B_E) \ge \theta$ *for any prime divisor* F *over* E.
(3) $\mathrm{ind}_y(Y, E) \le \theta^{-1}$ *for any closed point* $y \in E$.
(4) The variety E *belongs to a bounded family depending only on* n, ϵ, C, *and* κ.
(5) If $a(E, X, B) \le \theta/2$, *then* $\mathrm{mld}(X \ni x, B) = a(E, X, B)$.

Proof. We let

$$m := \mathrm{mld}(X \ni x, B), \quad V := \widehat{\mathrm{vol}}(X \ni x, B), \quad a := a(E, X, B), \quad \text{and} \quad L := -(K_E + B_E).$$

By Definition 2.5, (E, B_E) is a klt pair of dimension $n - 1 \ge 1$ and L is ample.

(1) By Lemma 2.8(2)(3),

$$\epsilon m \le V \le \widehat{\mathrm{vol}}(E, X, B) = aL^{n-1} \le CmL^{n-1}.$$

Since $m > 0$, $L^{n-1} \ge \epsilon/C$.

(2) Let F be a prime divisor over E, and let y be a closed point of $\mathrm{center}_E F$. By Lemma 2.13 with $d = n - 1$, since $\delta(E, B_E) \ge \kappa$, and by (1),

$$\widehat{\mathrm{vol}}(E \ni y, B_E) \ge \left(\frac{n-1}{n}\right)^{n-1} \delta(E, B_E)^{n-1} L^{n-1} \ge \left(\frac{n-1}{n}\right)^{n-1} \cdot \frac{\kappa^{n-1}\epsilon}{C} = (n-1)^{n-1}\theta.$$

Since $(E \ni y, B_E)$ is a klt germ of dimension $n - 1$, by Lemma 2.10(4),

$$a(F, E, B_E) \ge \frac{\widehat{\mathrm{vol}}(E \ni y, B_E)}{(n-1)^{n-1}} \ge \theta.$$

(3) Let $I(E) := \max_{y \in E} \mathrm{ind}_y(Y, E)$ [Han26, Definition 2.5]. By [Han26, Proposition 3.3],

$$\delta(E, B_E)^{n-1}\widehat{\mathrm{vol}}(E, X, B) \le \frac{n^{n-1}a}{I(E)}.$$

Since $V \le \widehat{\mathrm{vol}}(E, X, B)$ (Lemma 2.8(2)), $\delta(E, B_E) \ge \kappa$, $V \ge \epsilon m$, and $a \le Cm$,

$$\kappa^{n-1}\epsilon m \le \kappa^{n-1} V \le \frac{n^{n-1}a}{I(E)} \le \frac{n^{n-1}Cm}{I(E)}.$$

Since $m > 0$, for any closed point $y \in E$,

$$\mathrm{ind}_y(Y, E) \le I(E) \le \frac{n^{n-1}C}{\kappa^{n-1}\epsilon} = \theta^{-1}.$$

(4) By (2), (E, B_E) is θ-lc. Since E is projective of dimension $n - 1$, $B_E \ge 0$, and L is ample, by Theorem 2.15(1), E belongs to a bounded family depending only on $n - 1$ and θ, hence only on n, ϵ, C, and κ.

(5) By (2), $a(F', E, B_E) \ge \theta > \theta/2$ for any prime divisor F' over E. Possibly shrinking X to a neighborhood of x, we may assume that $(Y, E + B_Y)$ is plt. By [HLS19, Theorem 3.10(2)],

$$a(F, Y, E + B_Y) > \frac{\theta}{2}$$

for any exceptional prime divisor F over Y with $\mathrm{center}_Y F \subset E$. Let $F \ne E$ be a prime divisor over X with $\mathrm{center}_X F = x$. Then $\mathrm{center}_Y F \subset f^{-1}(x) = E$, F is exceptional over Y since E is the only prime divisor on Y with center x on X, and $\mathrm{mult}_F E > 0$ since E is $\mathbb{Q}$-Cartier. By Definition 2.5,

$$a(F, X, B) = a(F, Y, E + B_Y) + a\,\mathrm{mult}_F E > \frac{\theta}{2} \ge a.$$

Since $\mathrm{center}_X E = x$, $m = a$. □

5. The invariant Φ and its ACC

Let (S, B) be a klt log Fano pair, P an ample Cartier divisor on S, and $r > 0$ a real number such that $-(K_S + B) \sim_{\mathbb{R}} rP$. Then $\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) = r\Phi(S,B)$, where $C(S,P)$, o, and $B_{C(S,P)}$ are as in Definition 5.1 and $\Phi(S,B)$ is an infimum over a set of prime divisors over S independent of B (see Definition 5.6 and Theorem 5.9), and $\Phi(S,B_1) \le \Phi(S,B_0)$ for any klt weak log Fano pairs (S,B_0) and (S,B_1) with $B_0 \le B_1$, and Φ is continuous in the coefficients of B (see Proposition 5.7).

5.1. Cones and Gauss valuations.

Definition 5.1. Let S be a normal projective variety, P a semi-ample and big Cartier divisor on S, and $B \ge 0$ an $\mathbb{R}$-divisor on S. Let

$$C(S,P) := \operatorname{Spec} \bigoplus_{k \ge 0} H^0(S, \mathcal{O}_S(kP)),$$

and let $o \in C(S,P)$ be the closed point given by the ideal $\bigoplus_{k>0} H^0(S,\mathcal{O}_S(kP))$, called the *vertex* of $C(S,P)$. Since P is semi-ample, $\bigoplus_{k\ge 0} H^0(S,\mathcal{O}_S(kP))$ is finitely generated [Laz04, Example 2.1.30], so $C(S,P)$ is a normal affine variety of dimension $\dim S+1$. If P is ample, let E_0 be the zero section of $\mathrm{Tot}(\mathcal{O}_S(-P))$. In this case, the natural morphism $\mathrm{Tot}(\mathcal{O}_S(-P)) \to C(S,P)$ contracts E_0 to o and is an isomorphism over $C(S,P)\backslash\{o\}$ [Kol13, 3.8], and we let $B_{C(S,P)}$ be the $\mathbb{R}$-divisor on $C(S,P)$ whose pullback to $\mathrm{Tot}(\mathcal{O}_S(-P))\backslash E_0$ equals the pullback of B. For a prime divisor F over S and a real number $c > 0$, the *Gauss valuation* $v_{F,c}$ on $C(S,P)$ is

$$v_{F,c}\Big(\sum_{k\ge 0} s_k\Big) := \min\{\mathrm{mult}_F(h^* \operatorname{div}(s_k)) + kc \mid s_k \ne 0\},$$

where $s_k \in H^0(S,\mathcal{O}_S(kP))$, $\operatorname{div}(s_k) \ge 0$ is the divisor of zeros of s_k, and $h : S' \to S$ is a resolution such that F is a divisor on S'.

Lemma 5.2. *Let S be a normal projective variety of dimension $n-1 \ge 1$, (S,B) a klt pair, P an ample Cartier divisor on S, and $r > 0$ a real number such that $-(K_S+B) \sim_{\mathbb{R}} rP$. Let $Y := \mathrm{Tot}(\mathcal{O}_S(-P))$, let $g : Y \to S$ be the bundle projection, and let $f : Y \to C(S,P)$ be the natural morphism. Then:*

(1) With $K_{C(S,P)} := f_* K_Y$, the divisor $K_{C(S,P)} + B_{C(S,P)}$ is $\mathbb{R}$-Cartier and

$$K_Y + E_0 + g^*B = f^*(K_{C(S,P)} + B_{C(S,P)}) + rE_0.$$

(2) $(C(S,P) \ni o, B_{C(S,P)})$ is a klt germ, and E_0 is a Kollár component of $(C(S,P) \ni o, B_{C(S,P)})$.

(3) We have

$$a(E_0, C(S,P), B_{C(S,P)}) = \mathrm{mld}(C(S,P) \ni o, B_{C(S,P)}) = r.$$

(4) We have

$$\widehat{\mathrm{vol}}(E_0, C(S,P), B_{C(S,P)}) = r \cdot (-(K_S+B))^{n-1}.$$

Proof. We give the argument for $\mathbb{R}$-divisors (cf. [Kol13, Lemma 3.1] for $\mathbb{Q}$-divisors). We let $X := C(S,P)$ and $B_X := B_{C(S,P)}$. Since Y is the total space of the line bundle $\mathcal{O}_S(-P)$, we have $\omega_{Y/S} \cong g^*\mathcal{O}_S(P)$ and $\mathcal{O}_Y(E_0) \cong g^*\mathcal{O}_S(-P)$. Since $\dim g^{-1}(\operatorname{Sing} S) \le \dim Y - 2$, we may choose K_Y and a rational function ζ on Y such that

$$K_Y + E_0 = g^*K_S \quad \text{and} \quad E_0 + g^*P = \operatorname{div}(\zeta).$$

Write $K_S + B + rP = \sum_{j=1}^m c_j \operatorname{div}(\varphi_j)$, where $c_j \in \mathbb{R}$ and φ_j are rational functions on S. Then

$$K_Y + E_0 + g^*B - rE_0 = g^*(K_S + B + rP) - r\operatorname{div}(\zeta) = \sum_{j=1}^m c_j \operatorname{div}(\varphi_j \circ g) - r\operatorname{div}(\zeta).$$

Since f is birational, $f(E_0) = o$, and $f_*g^*B = B_X$, applying f_* we obtain

$$K_X + B_X = \sum_{j=1}^{m} c_j \operatorname{div}(\varphi_j \circ g) - r \operatorname{div}(\zeta),$$

where $\varphi_j \circ g$ and ζ are regarded as rational functions on X via f. Thus $K_X + B_X$ is $\mathbb{R}$-Cartier and $f^*(K_X + B_X) = K_Y + E_0 + g^*B - rE_0$, which is (1).

For any open subset $U \subset S$ on which $\mathcal{O}_S(-P)$ is trivial, we have

$$(g^{-1}U, E_0 + g^*B) \cong (U \times \mathbb{A}^1, U \times \{0\} + B|_U \times \mathbb{A}^1),$$

the product of the klt pair $(U, B|_U)$ and $(\mathbb{A}^1, \{0\})$. Thus $(Y, E_0 + g^*B)$ is plt, $(Y\backslash E_0, g^*B)$ is klt, and, under the isomorphism $g|_{E_0} : E_0 \to S$, $\operatorname{Diff}_{E_0}(g^*B) = B$.

If E is a prime divisor over X whose center is not o, then $\operatorname{center}_X E$ intersects $X\backslash\{o\} \cong Y\backslash E_0$, so $a(E, X, B_X) = a(E, Y\backslash E_0, g^*B) > 0$. Let E be a prime divisor over $X \ni o$. Since $\operatorname{center}_Y E \subset f^{-1}(o) = E_0$ and E_0 is Cartier, $\operatorname{mult}_E E_0$ is a positive integer. Since $(Y, E_0 + g^*B)$ is lc, (1) implies

$$a(E, X, B_X) = a(E, Y, E_0 + g^*B) + r \operatorname{mult}_E E_0 \geq r.$$

Thus $(X \ni o, B_X)$ is a klt germ. Moreover, $a(E_0, X, B_X) = a(E_0, Y, E_0 + g^*B - rE_0) = r$. Thus (3) holds.

The morphism f is an isomorphism over $X\backslash\{o\}$, the exceptional locus of f is the Cartier prime divisor E_0 with $f(E_0) = o$, $-E_0 \sim g^*P$ is ample over X since $-E_0|_{E_0} \sim P$ is ample and f is an isomorphism outside E_0, $f_*^{-1}B_X = g^*B$, and $(Y, E_0 + g^*B)$ is plt. Thus f is a plt blow-up of $(X \ni o, B_X)$ and E_0 is a Kollár component of $(X \ni o, B_X)$, which proves (2). Since $B_{E_0} = \operatorname{Diff}_{E_0}(g^*B) = B$ under $E_0 \cong S$, Lemma 2.8(3) and $a(E_0, X, B_X) = r$ imply

$$\widehat{\operatorname{vol}}(E_0, X, B_X) = a(E_0, X, B_X) \cdot (-(K_{E_0} + B_{E_0}))^{n-1} = r \cdot (-(K_S + B))^{n-1},$$

which is (4). □

Definition 5.3. Let (S, B) be a klt weak log Fano pair of dimension $n-1 \geq 1$ and $L := -(K_S+B)$. For a prime divisor F over S and a real number $u > 0$, let

$$\Psi_{S,B}(F, u) := (a(F, S, B) + u)^n \Big(\frac{L^{n-1}}{u^n} - n \int_0^{+\infty} \frac{\operatorname{vol}(h^*L - tF)}{(t+u)^{n+1}} dt \Big),$$

where $h : S' \to S$ is a resolution such that F is a divisor on S'. Since $\operatorname{vol}(h^*L - tF)$ does not depend on h and $0 \leq \operatorname{vol}(h^*L - tF) \leq \operatorname{vol}(h^*L) = L^{n-1}$ for any $t \geq 0$, $\Psi_{S,B}(F, u)$ is well defined and

$$0 \leq \Psi_{S,B}(F, u) \leq (a(F, S, B) + u)^n \frac{L^{n-1}}{u^n}.$$

Lemma 5.4. *Let S be a normal projective variety of dimension $n - 1 \geq 1$, P a semi-ample and big Cartier divisor on S, F a prime divisor over S, and $c > 0$ a real number. Then $v_{F,c}$ is a valuation on $C(S, P)$ with center o, and*

$$\operatorname{vol}(v_{F,c}) = \frac{P^{n-1}}{c^n} - n \int_0^{+\infty} \frac{\operatorname{vol}(h^*P - tF)}{(t+c)^{n+1}} dt,$$

where $h : S' \to S$ is a resolution such that F is a divisor on S'. If (S, B) is a klt weak log Fano pair and $-(K_S + B) \sim_{\mathbb{R}} rP$ for some real number $r > 0$, then

$$\operatorname{vol}(v_{F,u/r}) = \frac{r\Psi_{S,B}(F, u)}{(a(F, S, B) + u)^n}$$

for any real number $u > 0$. If moreover P is ample, then $a(v_{F,c}, C(S, P), B_{C(S,P)}) = a(F, S, B) + rc$, and $\widehat{\operatorname{vol}}(v_{F,u/r}, C(S, P), B_{C(S,P)}) = r\Psi_{S,B}(F, u)$ for any real number $u > 0$.

Proof. We let $X := C(S,P)$, $d := n-1$, $v := v_{F,c}$, and $R_k := H^0(S, \mathcal{O}_S(kP))$ for any integer $k \geq 0$, so that $X = \operatorname{Spec} R$ with $R := \bigoplus_{k\geq 0} R_k$. For $s \in R_k\backslash\{0\}$ we let $w(s) := \operatorname{mult}_F(h^* \operatorname{div}(s))$. Since $h_*\mathcal{O}_{S'} = \mathcal{O}_S$, we have $R_k = H^0(S', \mathcal{O}_{S'}(kh^*P))$, and for any real number $\mu \geq 0$,

$$\{s \in R_k\backslash\{0\} \mid w(s) \geq \mu\} \cup \{0\} = H^0(S', \mathcal{O}_{S'}(kh^*P - \lceil \mu \rceil F)).$$

Step 1. We show that v is a valuation with center o. Let $Y := \operatorname{Tot}(\mathcal{O}_S(-P))$, let $Y' := \operatorname{Tot}(\mathcal{O}_{S'}(-h^*P)) = S' \times_S Y$ with bundle projection $g' : Y' \to S'$, let E'_0 be the zero section of Y', let $F_{Y'} := g'^*F$, and let ξ be the generic point of $F_{Y'} \cap E'_0$. Let y_1 be a local equation of $F_{Y'}$ and z a fiber coordinate of g' near ξ, so that $E'_0 = (z = 0)$. Since $R_k = H^0(S', \mathcal{O}_{S'}(kh^*P))$, we regard R as a subring of the ring of regular functions on Y', and the fraction field of R is the function field of Y' since P is big. Any $s \in R_k\backslash\{0\}$ is σz^k near ξ for a regular function σ on S' near the generic point of F with $\operatorname{mult}_F \operatorname{div}(\sigma) = w(s)$. Thus v is the restriction to R of the monomial valuation at ξ in the coordinates (y_1, z) with weights $(1, c)$. Thus v is a valuation. Since $R_0 = \mathbb{C}$ and $w = 0$ on $R_0\backslash\{0\}$, v is trivial on $\mathbb{C}^\times$. Since $v \geq 0$ on R and

$$\{\theta \in R\backslash\{0\} \mid v(\theta) > 0\} = \Big(\bigoplus_{k>0} R_k\Big)\backslash\{0\},$$

the center of v is o.

Step 2. We compute $\operatorname{vol}(v)$. For a real number $\lambda > 0$ let

$$I_\lambda := \{\theta \in R \mid \theta = 0 \text{ or } v(\theta) \geq \lambda\}.$$

A non-zero $\theta = \sum_k \theta_k$ belongs to I_λ if and only if $w(\theta_k) + kc \geq \lambda$ for any k with $\theta_k \neq 0$, so $I_\lambda = \bigoplus_{k\geq 0}(I_\lambda \cap R_k)$ with

$$I_\lambda \cap R_k = H^0(S', \mathcal{O}_{S'}(kh^*P - \lceil \lambda - kc \rceil F)) \text{ if } kc < \lambda, \quad I_\lambda \cap R_k = R_k \text{ if } kc \geq \lambda.$$

Since P is semi-ample, R is generated over $R_0 = \mathbb{C}$ by finitely many homogeneous elements of positive degree, and each of them has v-value at least c. Thus $\mathfrak{m}^N \subset I_\lambda$ for any integer $N \geq \lambda/c$, where $\mathfrak{m} := \bigoplus_{k>0} R_k$. Hence $\mathfrak{a}_\lambda(v) = I_\lambda \mathcal{O}_{X,o}$, $\mathcal{O}_{X,o}/\mathfrak{a}_\lambda(v) \cong R/I_\lambda$, and, since $I_\lambda \cap R_0 = 0$,

$$\ell(\mathcal{O}_{X,o}/\mathfrak{a}_\lambda(v)) = 1 + \sum_{1\leq k<\lambda/c} \Big(h^0(S', kh^*P) - h^0(S', kh^*P - \lceil \lambda - kc \rceil F)\Big).$$

For any integer $k \geq 1$ and real number $t \geq 0$ let

$$\phi_k(t) := \frac{d!}{k^d} h^0(S', kh^*P - \lceil kt \rceil F).$$

Then ϕ_k is decreasing, $0 \leq \phi_k(t) \leq \phi_k(0)$, and

$$\lim_{k\to+\infty} \phi_k(t) = \operatorname{vol}(h^*P - tF)$$

for any $t \geq 0$. For rational t such that $h^*P - tF$ is big, the graded linear series $\bigoplus_{k\geq 0} H^0(S', kh^*P - \lceil kt \rceil F)$ of h^*P [Laz04, Definition 2.4.1] contains an ample series [BC11, Definition 1.1 and Lemma 1.6], so the limit exists [LM09, Theorem 2.13 and Remark 2.14], and it is $\operatorname{vol}(h^*P - tF)$ by taking k divisible by the denominator of t. For the other $t \geq 0$, the limit follows since ϕ_k is decreasing and vol is continuous [Laz04, Corollary 2.2.45], where $\operatorname{vol}(h^*P - tF) = 0$ if $h^*P - tF$ is not big. In particular, since P is nef, $\lim_{k\to+\infty} \phi_k(0) = \operatorname{vol}(P) = P^d$, so there exists a real number M such that $\phi_k(0) \leq M$ for any $k \geq 1$.

For real numbers $\lambda \geq 1$ and $\rho \geq 0$, let $k_\rho := \lfloor \lambda\rho \rfloor$, let

$$\psi_\lambda(\rho) := \Big(\frac{k_\rho}{\lambda}\Big)^d \Big(\phi_{k_\rho}(0) - \phi_{k_\rho}\Big(\frac{\lambda}{k_\rho} - c\Big)\Big) \quad \text{if } 1 \leq k_\rho < \frac{\lambda}{c},$$

and let $\psi_\lambda(\rho) := 0$ otherwise. Since $\lceil \lambda - kc \rceil = \lceil k(\lambda/k - c) \rceil$ and ψ_λ is constant on $[k/\lambda, (k+1)/\lambda)$ for any integer $k \geq 0$, we have

$$\frac{n!}{\lambda^n} \ell(\mathcal{O}_{X,o}/\mathfrak{a}_\lambda(v)) = \frac{n!}{\lambda^n} + n \int_0^{+\infty} \psi_\lambda(\rho) d\rho.$$

Moreover, $0 \leq \psi_\lambda \leq Mc^{-d}$, and $\psi_\lambda(\rho) = 0$ for any $\rho \geq 1/c + 1$, since $k_\rho < \lambda/c$ implies $\rho < (k_\rho + 1)/\lambda \leq 1/c + 1$.

Since the volume is continuous on $N^1(S')_{\mathbb{R}}$ [Laz04, Corollary 2.2.45], $t \to \mathrm{vol}(h^*P - tF)$ is continuous. Let $\rho \in (0, 1/c)$ and $t := 1/\rho - c$. As $\lambda \to +\infty$, we have $k_\rho \to +\infty$, $k_\rho/\lambda \to \rho$, and $\lambda/k_\rho - c \to t$, so for any $\epsilon \in (0, t)$ and any $\lambda \gg 0$,

$$\phi_{k_\rho}(t + \epsilon) \leq \phi_{k_\rho}\Big(\frac{\lambda}{k_\rho} - c\Big) \leq \phi_{k_\rho}(t - \epsilon).$$

Letting $\lambda \to +\infty$ and then $\epsilon \to 0$, we obtain

$$\lim_{\lambda \to +\infty} \psi_\lambda(\rho) = \rho^d \Big(P^d - \mathrm{vol}\left(h^*P - (1/\rho - c)F\right)\Big).$$

If $\rho > 1/c$, then $k_\rho \geq \lambda\rho - 1 \geq \lambda/c$ for any $\lambda \geq (\rho - 1/c)^{-1}$, so $\lim_{\lambda \to +\infty} \psi_\lambda(\rho) = 0$. Thus ψ_λ converges pointwise on $[0, +\infty) \backslash \{1/c\}$. By the dominated convergence theorem, and substituting $t = 1/\rho - c$,

$$\mathrm{vol}(v) = n \int_0^{1/c} \rho^d \Big(P^d - \mathrm{vol}\left(h^*P - (1/\rho - c)F\right)\Big) d\rho = \frac{P^{n-1}}{c^n} - n \int_0^{+\infty} \frac{\mathrm{vol}(h^*P - tF)}{(t+c)^{n+1}} dt.$$

Step 3. We compute $a(v, X, B_X)$, where $B_X := B_{C(S,P)}$, assuming that P is ample, (S, B) is klt, and $-(K_S + B) \sim_{\mathbb{R}} rP$. Let $g : Y \to S$ be the bundle projection, let $f : Y \to X$ be the natural morphism, let E_0 be the zero section of Y, and let $h_Y : Y' \to Y$ be the induced morphism. Let $B_{S'}$ be the $\mathbb{R}$-divisor on S' such that $K_{S'} + B_{S'} = h^*(K_S + B)$. Since $Y' = S' \times_S Y$, we have $h_Y^* E_0 = E_0'$ and $K_{Y'} + E_0' = g'^* K_{S'}$, so $K_{Y'} + E_0' + g'^* B_{S'} = g'^* h^*(K_S + B) = h_Y^*(K_Y + E_0 + g^*B)$, and Lemma 5.2(1) implies

$$K_{Y'} + g'^* B_{S'} + (1 - r)E_0' = (f \circ h_Y)^*(K_X + B_X).$$

Near ξ, Y' is smooth, $F_{Y'}$ and E_0' are smooth and intersect transversally, and the support of $g'^* B_{S'} + (1 - r)E_0'$ is contained in $F_{Y'} \cup E_0'$, the coefficients of $F_{Y'}$ and E_0' being $1 - a(F, S, B)$ and $1 - r$. By the description of v in **Step 1** and [JM12, Proposition 5.1],

$$a(v, X, B_X) = 1 \cdot a(F_{Y'}, X, B_X) + c \cdot a(E_0', X, B_X) = a(F, S, B) + rc.$$

Finally, assume that (S, B) is a klt weak log Fano pair and $-(K_S + B) \sim_{\mathbb{R}} rP$, let $L := -(K_S + B)$, let $u > 0$ be a real number, and let $c := u/r$. Since $h^*L \equiv rh^*P$, we have $L^{n-1} = r^{n-1}P^{n-1}$ and $\mathrm{vol}(h^*L - tF) = r^{n-1}\,\mathrm{vol}(h^*P - (t/r)F)$ for any $t \geq 0$. Substituting $t = rt'$,

$$\frac{L^{n-1}}{u^n} - n \int_0^{+\infty} \frac{\mathrm{vol}(h^*L - tF)}{(t+u)^{n+1}} dt = \frac{1}{r}\Big(\frac{P^{n-1}}{c^n} - n \int_0^{+\infty} \frac{\mathrm{vol}(h^*P - t'F)}{(t'+c)^{n+1}} dt' \Big) = \frac{\mathrm{vol}(v_{F,c})}{r}.$$

Thus, by Definition 5.3, $\mathrm{vol}(v_{F,u/r}) = r(a(F, S, B) + u)^{-n} \Psi_{S,B}(F, u)$. If moreover P is ample, then $a(v_{F,u/r}, X, B_X) = a(F, S, B) + u$ by **Step 3**, so

$$\widehat{\mathrm{vol}}(v_{F,u/r}, X, B_X) = (a(F, S, B) + u)^n \,\mathrm{vol}(v_{F,u/r}) = r\Psi_{S,B}(F, u).$$

The lemma follows. □

To obtain a set of divisors over S independent of B in Definition 5.6, we need the following lemma.

Lemma 5.5. *Let d be a positive integer. Then there exists a positive integer N depending only on d satisfying the following. Let (S, B) be a klt log Fano pair of dimension d, let $B^+ \geq B$ be a $\mathbb{Q}$-divisor such that (S, B^+) is lc and $K_S + B^+ \sim_{\mathbb{Q}} 0$, and let F be a prime divisor over S such that $a(F, S, B^+) = 0$. Then there exists a $\mathbb{Q}$-divisor $G \geq 0$ on S such that (S, G) is lc, $N(K_S + G) \sim 0$, and $a(F, S, G) = 0$.*

Proof. By [HLS19, Theorem 1.10], there exists a positive integer N depending only on d satisfying the following: for any lc pair (Y, D) of dimension d such that Y is of Fano type, D is a reduced divisor, and (Y, D) is $\mathbb{R}$-complementary (that is, there exists an $\mathbb{R}$-divisor $D^+ \geq D$ such that (Y, D^+) is lc and $K_Y + D^+ \sim_{\mathbb{R}} 0$), there exists a $\mathbb{Q}$-divisor $G_Y \geq D$ such that (Y, G_Y) is lc and $N(K_Y + G_Y) \sim 0$.

Let $t \in (0, 1)$ be a rational number such that $ta(F, S, B) < 1$. Let $B_t := tB + (1-t)B^+$. Then

$$a(E, S, B_t) = ta(E, S, B) + (1-t)a(E, S, B^+) > 0$$

for any prime divisor E over S, $K_S + B_t \sim_{\mathbb{Q}} t(K_S + B)$, and $a(F, S, B_t) = ta(F, S, B) < 1$. Thus (S, B_t) is a klt log Fano pair. Let $\mathcal{E} := \{F\}$ if F is exceptional over S, and $\mathcal{E} := \emptyset$ otherwise. By [BCHM10, Corollary 1.4.3], there exists a birational morphism $g : Y \to S$ such that Y is $\mathbb{Q}$-factorial and the exceptional divisors of g are exactly the elements of $\mathcal{E}$. Let $F_Y := \operatorname{center}_Y F$, which is a prime divisor on Y, and $F_Y = g_*^{-1}F$ if F is a divisor on S. Let D_Y and B_Y^+ be the $\mathbb{R}$-divisors on Y such that

$$K_Y + D_Y = g^*(K_S + B_t) \quad \text{and} \quad K_Y + B_Y^+ = g^*(K_S + B^+).$$

The only possible exceptional divisor of g is F_Y. Since $g_*^{-1}B_t \geq 0$ and $\operatorname{mult}_{F_Y} D_Y = 1 - ta(F, S, B) > 0$, we have $D_Y \geq 0$ and (Y, D_Y) is a klt weak log Fano pair. Since $g_*^{-1}B^+ \geq 0$ and $\operatorname{mult}_{F_Y} B_Y^+ = 1 - a(F, S, B^+) = 1$, we have $B_Y^+ \geq F_Y$, (Y, B_Y^+) is lc, and $K_Y + B_Y^+ \sim_{\mathbb{Q}} 0$.

Since (Y, D_Y) is a klt weak log Fano pair, Y is of Fano type.

Since Y is $\mathbb{Q}$-factorial, (Y, F_Y) is a pair, and since $F_Y \leq B_Y^+$, (Y, F_Y) is lc and (Y, B_Y^+) is an $\mathbb{R}$-complement of (Y, F_Y). By the choice of N with $(Y, D) = (Y, F_Y)$, there exists a $\mathbb{Q}$-divisor $G_Y \geq F_Y$ such that (Y, G_Y) is lc and $N(K_Y + G_Y) \sim 0$. Since (Y, G_Y) is lc, $\operatorname{mult}_{F_Y} G_Y = 1$, that is, $a(F, Y, G_Y) = 0$.

Let $G := g_*G_Y \geq 0$ and $K_S := g_*K_Y$, and write $N(K_Y + G_Y) = \operatorname{div}(\varphi)$ for a rational function φ on Y. Then $N(K_S + G) = \operatorname{div}(\varphi)$ on S. Thus $N(K_S + G) \sim 0$, $K_S + G$ is $\mathbb{Q}$-Cartier, and

$$g^*(K_S + G) = \frac{1}{N}\operatorname{div}(\varphi) = K_Y + G_Y.$$

Hence $a(E, S, G) = a(E, Y, G_Y)$ for any prime divisor E over S. Thus (S, G) is lc and $a(F, S, G) = 0$. The lemma follows. □

Definition 5.6. Let d be a positive integer, and let

$$N_d := \min\{N \in \mathbb{N}^+ \mid N \text{ satisfies the property in Lemma 5.5}\}.$$

For a normal projective variety S of dimension d, let $\mathcal{C}(S)$ be the set of prime divisors F over S such that $a(F, S, G) = 0$ for some $\mathbb{Q}$-divisor $G \geq 0$ on S such that (S, G) is lc and $N_d(K_S + G) \sim 0$. For a klt weak log Fano pair (S, B) of dimension d, let

$$\Phi(S, B) := \inf\left(\{(-(K_S + B))^d\} \cup \{\Psi_{S,B}(F, u) \mid F \in \mathcal{C}(S),\ u \in \mathbb{R}_{>0}\}\right),$$

where $\Psi_{S,B}$ is as in Definition 5.3 with $n = d + 1$.

5.2. **The local volume of the cone.**

Proposition 5.7. *Let S be a normal projective variety of dimension $n - 1 \geq 1$ and $D_1, \ldots, D_m$ distinct prime divisors on S. For $b \in \mathbb{R}^m$ let $B(b) := \sum_{j=1}^m b_j D_j$, and let $\mathcal{P}$ be the set of $b \in \mathbb{R}^m$ such that $(S, B(b))$ is a klt weak log Fano pair. Then:*

(1) $\Phi(S, B(b')) \leq \Phi(S, B(b))$ for any $b, b' \in \mathcal{P}$ with $b' - b \in \mathbb{R}^m_{\geq 0}$.

(2) For any prime divisor F over S, $\Psi_{S,B(b)}(F, u)$ is a continuous function of $(b, u) \in \mathcal{P} \times \mathbb{R}_{>0}$.

(3) The number $\Phi(S, B(b))$ is a positive continuous function of $b \in \mathcal{P}$.

Proof. We may assume that $\mathcal{P} \neq \emptyset$, and we fix a canonical divisor K_S. Let W be the set of $b \in \mathbb{R}^m$ such that $K_S + B(b)$ is $\mathbb{R}$-Cartier, and for $b \in W$ let $L(b) := -(K_S + B(b))$. If $b = \sum_k \mu_k c_k$ for finitely many $c_k \in W$ and real numbers μ_k with $\sum_k \mu_k = 1$, then $L(b) = \sum_k \mu_k L(c_k)$. Thus W is an affine subspace of $\mathbb{R}^m$ containing $\mathcal{P}$, and for any resolution $h : S' \to S$, the $\mathbb{R}$-divisor $h^*L(b)$ is an affine function of $b \in W$.

For any prime divisor F over S and any $\mathbb{R}$-divisors B, B' on S such that $K_S + B$ and $K_S + B'$ are $\mathbb{R}$-Cartier,

$$a(F, S, B') = a(F, S, B) - \mathrm{mult}_F(B' - B). \tag{5.1}$$

Step 1. We prove (2). Let $(c_i, u_i)_{i\geq 1}$ be a sequence in $\mathcal{P} \times \mathbb{R}_{>0}$ converging to $(b, u) \in \mathcal{P} \times \mathbb{R}_{>0}$. Let F be a prime divisor over S and $h : S' \to S$ a resolution such that F is a divisor on S'. Since $h^*L(b)$ is affine in $b \in W$, (5.1) implies $a(F, S, B(c_i)) \to a(F, S, B(b))$, and $L(c_i)^{n-1} = (h^*L(c_i))^{n-1} \to L(b)^{n-1}$. Since the volume is continuous on $N^1(S')_{\mathbb{R}}$ [Laz04, Corollary 2.2.45], $\mathrm{vol}(h^*L(c_i) - yF) \to \mathrm{vol}(h^*L(b) - yF)$ for any $y \geq 0$. Since $L(c_i)$ is nef and $F \geq 0$,

$$0 \leq \mathrm{vol}(h^*L(c_i) - yF) \leq \mathrm{vol}(h^*L(c_i)) = L(c_i)^{n-1} \quad \text{for any } y \geq 0.$$

For $i \gg 0$, $u_i > u/2$ and $L(c_i)^{n-1} < L(b)^{n-1} + 1$, so the integrand $\mathrm{vol}(h^*L(c_i) - yF)/(y + u_i)^{n+1}$ of Definition 5.3 is bounded by the integrable function $(L(b)^{n-1} + 1)(y + u/2)^{-n-1}$ of $y \in \mathbb{R}_{\geq 0}$. By the dominated convergence theorem, $\Psi_{S,B(c_i)}(F, u_i) \to \Psi_{S,B(b)}(F, u)$, and (2) follows.

Step 2. Let B_0, B_1 be $\mathbb{Q}$-divisors on S such that (S, B_0) and (S, B_1) are klt weak log Fano pairs, and let $L_0 := -(K_S + B_0)$ and $L_1 := -(K_S + B_1)$. Let $t \in [0, 1)$ be a rational number and $G \geq 0$ a $\mathbb{Q}$-divisor on S such that

$$L_1 \sim_{\mathbb{Q}} (1 - t)L_0 + G.$$

In this step we show that for any prime divisor F over S and any real number $u > 0$, letting $u_0 := (u + \mathrm{mult}_F G)/(1 - t)$, we have

$$\frac{\Psi_{S,B_0}(F, u_0)}{(a(F, S, B_0) + u_0)^n} \leq (1 - t)\frac{\Psi_{S,B_1}(F, u)}{(a(F, S, B_1) + u)^n} \quad \text{and} \quad L_0^{n-1} \leq (1 - t)^{1-n} L_1^{n-1}. \tag{5.2}$$

Here L_0, L_1, and G are $\mathbb{Q}$-Cartier, and $u_0 > 0$.

First, let (S, B) be a klt weak log Fano pair such that B is a $\mathbb{Q}$-divisor, $L := -(K_S + B)$, and q any positive rational number such that qL is a semi-ample Cartier divisor. By Lemma 5.4 for $P := qL$ and $r := 1/q$,

$$\Psi_{S,B}(F, u) = q(a(F, S, B) + u)^n \,\mathrm{vol}(v_{F,qu}) \tag{5.3}$$

for any prime divisor F over S and any real number $u > 0$, where $v_{F,qu}$ is the Gauss valuation on $C(S, qL)$.

Since L_0 and L_1 are nef and big, by the base point free theorem there exists a positive integer q such that $P_0 := q(1 - t)L_0$ and $P_1 := qL_1$ are semi-ample Cartier divisors, qG is Cartier, and $P_1 \sim P_0 + qG$. Let $\sigma \in H^0(S, \mathcal{O}_S(P_1 - P_0))$ be a section whose divisor of zeros is qG. For any $k \geq 0$, multiplication by σ^k is an injective map

$$H^0(S, \mathcal{O}_S(kP_0)) \to H^0(S, \mathcal{O}_S(kP_1))$$

which adds kqG to the divisor of zeros of any non-zero section. Thus multiplication by σ^k in degree k defines an injective homomorphism of graded rings

$$\iota : \bigoplus_{k\geq 0} H^0(S, \mathcal{O}_S(kP_0)) \to \bigoplus_{k\geq 0} H^0(S, \mathcal{O}_S(kP_1)).$$

Let $v_0 := v_{F,q(1-t)u_0}$ be the Gauss valuation on $C(S, P_0)$ and $v_1 := v_{F,qu}$ the Gauss valuation on $C(S, P_1)$. Since $q(1 - t)u_0 = qu + \mathrm{mult}_F(qG)$, Definition 5.1 implies

$$v_1 \circ \iota = v_0.$$

Thus for any real number $\lambda > 0$, the ideal $\{g \mid v_0(g) \geq \lambda\}$ is the preimage under ι of the ideal $\{g \mid v_1(g) \geq \lambda\}$, and ι induces an injection from the quotient by the first ideal into the quotient by the second ideal. By Lemma 5.4, v_0 and v_1 have centers the vertices of $C(S, P_0)$ and $C(S, P_1)$ respectively, so the ideals are primary to the maximal ideals of the vertices, and the dimensions of the quotients are the lengths of $\mathcal{O}_{C(S,P_i),o}/\mathfrak{a}_\lambda(v_i)$ for $i \in \{0, 1\}$. Since $\dim C(S, P_0) = \dim C(S, P_1) = n$,

$$\mathrm{vol}(v_0) \leq \mathrm{vol}(v_1).$$

By (5.3), applied to $(B_0, q(1-t))$ and to (B_1, q), the first inequality of (5.2) follows. Since $G \geq 0$ and L_0, L_1 are nef,

$$L_1^{n-1} = \mathrm{vol}((1-t)L_0 + G) \geq \mathrm{vol}((1-t)L_0) = (1-t)^{n-1}L_0^{n-1}.$$

Step 3. Let B_0, B_1 be $\mathbb{Q}$-divisors on S such that $B_0 \geq B_1$ and $(S, B_0), (S, B_1)$ are klt weak log Fano pairs. In this step we show that

$$\Psi_{S,B_0}(F, u + \mathrm{mult}_F(B_0 - B_1)) \leq \Psi_{S,B_1}(F, u) \quad \text{and} \quad L_0^{n-1} \leq L_1^{n-1} \tag{5.4}$$

for any prime divisor F over S and any real number $u > 0$, where $L_i := -(K_S + B_i)$, and that

$$\Phi(S, B_0) \leq \Phi(S, B_1). \tag{5.5}$$

We apply **Step 2** with $t := 0$ and $G := B_0 - B_1$, so that $L_1 = L_0 + G$ and $u_0 = u + \mathrm{mult}_F(B_0 - B_1)$. By (5.1), $a(F, S, B_0) + u_0 = a(F, S, B_1) + u$, so (5.2) implies (5.4). Since $\mathcal{C}(S)$ does not depend on the boundary and $u + \mathrm{mult}_F(B_0 - B_1) > 0$, taking the infimum of (5.4) over $F \in \mathcal{C}(S)$ and $u > 0$ implies (5.5).

Step 4. Let $b^* \in \mathcal{P}$. In this step we construct a rational polytope $\Pi \subset \mathcal{P}$ which contains a neighborhood of b^* in $\mathcal{P}$, and a real number $\alpha_0 > 0$ such that $\alpha(S, B(b)) \geq \alpha_0$ for any $b \in \Pi$.

By [LZ26, Proposition 3.3], the closure $\overline{\mathcal{P}}$ of $\mathcal{P}$ is a rational polytope. We choose rational numbers $e_j < b_j^* < e_j'$ for $1 \leq j \leq m$ and let $\Pi := \overline{\mathcal{P}} \cap \prod_{j=1}^m [e_j, e_j']$. Then Π is a rational polytope containing the neighborhood $\mathcal{P} \cap \prod_{j=1}^m (e_j, e_j')$ of b^* in $\mathcal{P}$. Since klt and bigness are open conditions and effectivity and nefness are closed conditions in $b \in W$, possibly shrinking the intervals $[e_j, e_j']$, we have $\Pi \subset \mathcal{P}$, and since Π is compact, there exists a real number $\epsilon \in (0, 1)$ such that $(S, B(b))$ is ϵ-lc for any $b \in \Pi$. By Theorem 2.15(2), there exists a real number $\alpha_0 > 0$ such that $\alpha(S, B(b)) \geq \alpha_0$ for any $b \in \Pi$.

Step 5. We prove (1). Let $b, b' \in \mathcal{P}$ with $b' - b \in \mathbb{R}^m_{\geq 0}$, and let $\Pi \ni b$ and $\Pi' \ni b'$ be rational polytopes as in **Step 4**, applied to $b^* := b$ and $b^* := b'$ respectively. The set of $(c, c') \in \Pi \times \Pi'$ with $c' - c \in \mathbb{R}^m_{\geq 0}$ is a rational polytope containing (b, b'), so it contains a sequence of rational points (c_i, c_i') converging to (b, b'). Let F be a prime divisor over S and $u > 0$ a real number. By (5.4) applied to $B_0 := B(c_i')$ and $B_1 := B(c_i)$,

$$\Psi_{S,B(c_i')}\big(F, u + \mathrm{mult}_F(B(c_i') - B(c_i))\big) \leq \Psi_{S,B(c_i)}(F, u) \quad \text{and} \quad L(c_i')^{n-1} \leq L(c_i)^{n-1}.$$

Since $B(c_i') - B(c_i) = L(c_i) - L(c_i')$ and $h^*L(c)$ is affine in $c \in W$ for a resolution $h : S' \to S$ such that F is a divisor on S', $\mathrm{mult}_F(B(c_i') - B(c_i)) \to \mathrm{mult}_F(B(b') - B(b))$. By (2), letting $i \to +\infty$, we obtain

$$\Psi_{S,B(b')}\big(F, u + \mathrm{mult}_F(B(b') - B(b))\big) \leq \Psi_{S,B(b)}(F, u) \quad \text{and} \quad L(b')^{n-1} \leq L(b)^{n-1}.$$

Taking the infimum over $F \in \mathcal{C}(S)$ and $u > 0$ implies (1).

Step 6. We prove that $b \to \Phi(S, B(b))$ is continuous on $\mathcal{P}$. Let $b^* \in \mathcal{P}$, let Π and α_0 be as in **Step 4**, and let $V \subset \mathbb{Q}^m$ be the set of vertices of Π. By Serre's theorem, there exists an ample Cartier divisor $M \geq \sum_{j=1}^m D_j$ on S. For any $w \in V$, $L(w)$ is a big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Since the big cone of S is open, there exists a rational number $\gamma > 0$ such that $L(w) - \gamma M$ is big for any

$w \in V$, and we choose $\mathbb{Q}$-divisors $R_w \geq 0$ such that $R_w \sim_{\mathbb{Q}} L(w) - \gamma M$. For $\mu = (\mu_w)_{w\in V} \in \mathbb{R}^V_{\geq 0}$ with $\sum_w \mu_w = 1$, let

$$b(\mu) := \sum_{w\in V} \mu_w w \in \Pi \quad \text{and} \quad H(\mu) := \sum_{w\in V} \mu_w R_w + \gamma M.$$

Then $H(\mu) \geq \gamma \sum_{j=1}^m D_j$ and $H(\mu) \sim_{\mathbb{R}} L(b(\mu))$, and $H(\mu) \sim_{\mathbb{Q}} L(b(\mu))$ if $\mu \in \mathbb{Q}^V$. For $t \in [0,1)$ let

$$\beta_t := (1-t)\Big(1 + \frac{t}{1-t}\max\{1, \alpha_0^{-1}\}\Big)^n.$$

Then $\beta_t \geq (1-t)^{1-n}$, and $\beta_t \to 1$ as $t \to 0$. We show that for any $\mu, \mu' \in \mathbb{R}^V_{\geq 0}$ with $\sum_w \mu_w = \sum_w \mu'_w = 1$ and any rational number $t \in [0,1)$ such that $\|B(b(\mu')) - B(b(\mu))\| < \gamma t$,

$$\Phi(S, B(b(\mu))) \leq \beta_t \Phi(S, B(b(\mu'))). \tag{5.6}$$

First assume that $\mu, \mu' \in \mathbb{Q}^V$. Let $B_0 := B(b(\mu))$, $B_1 := B(b(\mu'))$, $L_i := -(K_S + B_i)$, $H := H(\mu')$, and

$$G := tH - (1-t)(B_1 - B_0).$$

Since

$$(1-t)(B_1 - B_0) \leq \|B_1 - B_0\| \sum_j D_j \leq \gamma t \sum_j D_j \leq tH,$$

we have $G \geq 0$. Since $(1-t)L_0 + G = (1-t)L_1 + tH \sim_{\mathbb{Q}} L_1$, we may apply **Step 2**. Let F be a prime divisor over S, $u > 0$ a real number, and $u_0 := (u + \operatorname{mult}_F G)/(1-t)$. By (5.1), $a(F, S, B_0) = a(F, S, B_1) + \operatorname{mult}_F(B_1 - B_0)$, so

$$a(F, S, B_0) + u_0 = a(F, S, B_1) + u + \frac{t}{1-t}(u + \operatorname{mult}_F H).$$

Since $\alpha(S, B_1) \geq \alpha_0$ and $H \sim_{\mathbb{Q}} L_1$, $(S, B_1 + \alpha_0 H)$ is lc, so $\alpha_0 \operatorname{mult}_F H \leq a(F, S, B_1)$. Thus

$$a(F, S, B_0) + u_0 \leq \Big(1 + \frac{t}{1-t}\max\{1, \alpha_0^{-1}\}\Big)(a(F, S, B_1) + u).$$

By (5.2) and since $\beta_t \geq (1-t)^{1-n}$,

$$\Psi_{S,B_0}(F, u_0) \leq \beta_t \Psi_{S,B_1}(F, u) \quad \text{and} \quad L_0^{n-1} \leq \beta_t L_1^{n-1}. \tag{5.7}$$

Since $\Phi(S, B_0) \leq \min\{L_0^{n-1}, \Psi_{S,B_0}(F, u_0)\}$ for any $F \in \mathcal{C}(S)$ and $u > 0$, (5.7) implies (5.6).

For arbitrary μ, μ', we choose rational weights $\mu_i \to \mu$ and $\mu'_i \to \mu'$. For $i \gg 0$, $\|B(b(\mu'_i)) - B(b(\mu_i))\| < \gamma t$. Let F be a prime divisor over S, $u > 0$ a real number,

$$G_i := tH(\mu'_i) - (1-t)\big(B(b(\mu'_i)) - B(b(\mu_i))\big),$$

and $u_{0,i} := (u + \operatorname{mult}_F G_i)/(1-t)$. Since

$$G_i = t\gamma M + \sum_{w\in V} \big(t\mu'_{i,w} R_w + (1-t)(\mu'_{i,w} - \mu_{i,w})L(w)\big),$$

$\operatorname{mult}_F G_i$ converges, and since $G_i \geq 0$, $u_{0,i}$ converges to a real number $u_0 \geq u/(1-t) > 0$. By (5.7) for (μ_i, μ'_i) and by (2), letting $i \to +\infty$,

$$\Psi_{S,B(b(\mu))}(F, u_0) \leq \beta_t \Psi_{S,B(b(\mu'))}(F, u) \quad \text{and} \quad L(b(\mu))^{n-1} \leq \beta_t L(b(\mu'))^{n-1}.$$

Taking the infimum over $F \in \mathcal{C}(S)$ and $u > 0$ implies (5.6).

Now let $c \in \mathcal{P}\backslash\{b^*\}$ be a point close to b^*. By **Step 4**, we may assume that $c \in \Pi$ and $\|B(c) - B(b^*)\| < \gamma/2$. Write $c = b(\mu)$ and $b^* = b(\mu')$, and let t be a rational number such that

$$\frac{\|B(c) - B(b^*)\|}{\gamma} < t < \frac{2\|B(c) - B(b^*)\|}{\gamma}.$$

By (5.6) for (μ, μ') and for (μ', μ),

$$\beta_t^{-1}\Phi(S, B(b^*)) \leq \Phi(S, B(c)) \leq \beta_t \Phi(S, B(b^*)).$$

Since $\Phi(S,B(b^*)) \le L(b^*)^{n-1} < +\infty$ and $\beta_t \to 1$ as $c \to b^*$, the function $b \to \Phi(S,B(b))$ is continuous at b^*.

Step 7. We prove that $\Phi(S,B) > 0$ for any klt weak log Fano pair (S,B) such that B is a $\mathbb{Q}$-divisor, and that $\Phi(S,B(b)) > 0$ for any $b \in \mathcal{P}$. This, together with **Step 6**, implies (3).

Let (S,B) be a klt weak log Fano pair such that B is a $\mathbb{Q}$-divisor, and $L := -(K_S+B)$. First assume that L is ample. Let q be a positive integer such that $P := qL$ is Cartier, and let $r := 1/q$. By Lemma 5.2, $(C(S,P) \ni o, B_{C(S,P)})$ is a klt germ and $\widehat{\mathrm{vol}}(E_0, C(S,P), B_{C(S,P)}) = rL^{n-1}$. By Lemma 5.4, for any $F \in \mathcal{C}(S)$ and $u > 0$, $v_{F,u/r}$ has center o and

$$\widehat{\mathrm{vol}}(v_{F,u/r}, C(S,P), B_{C(S,P)}) = r\Psi_{S,B}(F,u).$$

Since the local volume is at most the normalized volume of any valuation with center o, Lemma 2.8(1) implies

$$r\Phi(S,B) \ge \widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) > 0.$$

In general, by the Kodaira lemma, there exists a $\mathbb{Q}$-divisor $B' \ge B$ such that (S,B') is a klt log Fano pair. By the ample case and (5.5) for $B_0 := B'$ and $B_1 := B$,

$$\Phi(S,B) \ge \Phi(S,B') > 0.$$

Let $c \in \mathcal{P}$, and let Π, γ, and $b(\mu)$ be as in **Steps 4 and 6** for $b^* := c$. Write $c = b(\mu')$, choose $\mu \in \mathbb{Q}^V$ such that $\|B(b(\mu)) - B(c)\| < \gamma$, and let t be a rational number such that $\|B(b(\mu)) - B(c)\|/\gamma < t < 1$. Since $b(\mu) \in \mathcal{P} \cap \mathbb{Q}^m$, by (5.6) and the previous paragraph,

$$\Phi(S,B(c)) \ge \beta_t^{-1}\Phi(S,B(b(\mu))) > 0. \quad \square$$

To obtain the lower bound in Theorem 5.9, we need the following lemma.

Lemma 5.8 (cf. [LX20, Theorem 4.5] and [LW24, Corollary 3.27])**.** *Let S be a normal projective variety of dimension $n-1 \ge 1$ and (S,B) a klt log Fano pair such that B is a $\mathbb{Q}$-divisor. Let P be an ample Cartier divisor on S and r a positive rational number such that $-(K_S+B) \sim_{\mathbb{Q}} rP$. Let $\mathbb{G}_m$ act on $C(S,P)$ by the grading, and let $E \ne E_0$ be a $\mathbb{G}_m$-invariant Kollár component of $(C(S,P) \ni o, B_{C(S,P)})$. Then there exist a prime divisor F over S and positive rational numbers c, c' such that the following hold.*

1. *The valuation $v_{F,c}$ is c' times the divisorial valuation of E.*
2. *There exists a $\mathbb{Q}$-divisor $B^+ \ge B$ such that (S,B^+) is lc, $K_S+B^+ \sim_{\mathbb{Q}} 0$, and $a(F,S,B^+) = 0$.*

Proof. For $k \ge 0$ let $R_k := H^0(S, \mathcal{O}_S(kP))$, so that $C(S,P) = \operatorname{Spec} \bigoplus_{k\ge 0} R_k$ and $\lambda \in \mathbb{G}_m$ acts on R_k by multiplication by λ^k.

Step 1. We show that for any $g = \sum_k g_k$ with $g_k \in R_k$,

$$\mathrm{ord}_E(g) = \min\{\mathrm{ord}_E(g_k) \mid g_k \ne 0\}. \tag{5.8}$$

Let $I := \{k \mid g_k \ne 0\}$ and $N := \#I$. The determinant of the matrix $(\lambda_i^k)_{1\le i\le N, k\in I}$ is a non-zero polynomial in $\lambda_1, \dots, \lambda_N$, so there exist $\lambda_1, \dots, \lambda_N \in \mathbb{C}^*$ such that the matrix is invertible. Then any g_k with $k \in I$ is a $\mathbb{C}$-linear combination of $\lambda_1 \cdot g, \dots, \lambda_N \cdot g$, where $\lambda_i \cdot g = \sum_{k\in I} \lambda_i^k g_k$. Since E is $\mathbb{G}_m$-invariant, $\mathrm{ord}_E(\lambda_i \cdot g) = \mathrm{ord}_E(g)$, so $\mathrm{ord}_E(g_k) \ge \mathrm{ord}_E(g)$ for any $k \in I$. Since $\mathrm{ord}_E(g) \ge \min_{k\in I} \mathrm{ord}_E(g_k)$, (5.8) follows.

Step 2. We construct F, c, and c'. We regard R_k as the subspace

$$\{\varphi T^k \mid \varphi \in \mathbb{C}(S)^*,\ \mathrm{div}(\varphi) + kP \ge 0\} \cup \{0\}$$

of $\mathbb{C}(S)[T]$, so that the divisor of zeros of $\varphi T^k \in R_k$ is $\mathrm{div}(\varphi) + kP$, and the function field of $C(S,P)$ is $\mathbb{C}(S)(T)$. Let w be the restriction of ord_E to $\mathbb{C}(S)$. Then

$$\mathrm{ord}_E(\varphi T^k) = w(\varphi) + k\,\mathrm{ord}_E(T) \quad \text{for any non-zero } \varphi T^k \in R_k.$$

Since $\mathbb{C}(S)(T)$ has transcendence degree 1 over $\mathbb{C}(S)$, if w is non-trivial, then its residue field has transcendence degree $n-2=\dim S-1$ over $\mathbb{C}$: at least $n-2$ by [ZS60, Chapter VI, §6, Lemma 2, Corollary 1], since the residue field of ord_E has transcendence degree $n-1$, and less than $n-1$ since w is non-trivial. Thus either w is trivial, or $w=m\,\mathrm{ord}_F$ for a prime divisor F over S and a real number $m>0$. In the second case, $m\in\mathbb{N}^+$ since w takes integer values and $\mathrm{ord}_F(\mathbb{C}(S)^*)=\mathbb{Z}$.

Assume that w is trivial. Since E_0 is the zero section of $\mathrm{Tot}(\mathcal{O}_S(-P))$, $\mathrm{ord}_{E_0}(g)=\min\{k\mid g_k\neq 0\}$ for any $g=\sum_k g_k$ with $g_k\in R_k$. Then $\mathrm{ord}_E(g_k)=k\,\mathrm{ord}_E(T)$ for any non-zero $g_k\in R_k$, and $\mathrm{ord}_E(g_k)>0$ if $k>0$ since $\mathrm{center}_{C(S,P)}E=o$, so $\mathrm{ord}_E(T)>0$. Thus (5.8) implies $\mathrm{ord}_E(g)=\mathrm{ord}_E(T)\,\mathrm{ord}_{E_0}(g)$ for any $g\in\bigoplus_k R_k$, hence on $\mathbb{C}(S)(T)$. Since ord_E and ord_{E_0} both map $\mathbb{C}(S)(T)^*$ onto $\mathbb{Z}$, $\mathrm{ord}_E(T)=1$ and $E=E_0$, a contradiction. Thus $w=m\,\mathrm{ord}_F$.

Let $h:S'\to S$ be a resolution such that F is a divisor on S', and let

$$c:=\frac{\mathrm{ord}_E(T)}{m}-\mathrm{mult}_F(h^*P)\in\mathbb{Q}.$$

Then for any non-zero $\varphi T^k\in R_k$,

$$\frac{1}{m}\mathrm{ord}_E(\varphi T^k)=\mathrm{ord}_F(\varphi)+\frac{k\,\mathrm{ord}_E(T)}{m}=\mathrm{mult}_F\left(h^*(\mathrm{div}(\varphi)+kP)\right)+kc.$$

Let $k>0$ be an integer such that kP is base point free, and let $\varphi T^k\in R_k$ be a section whose divisor of zeros does not contain $\mathrm{center}_S F$. Then $\mathrm{mult}_F(h^*(\mathrm{div}(\varphi)+kP))=0$, and $\mathrm{ord}_E(\varphi T^k)>0$ since $\mathrm{center}_{C(S,P)}E=o$. Thus $c>0$. By (5.8) and Definition 5.1, $\frac{1}{m}\mathrm{ord}_E=v_{F,c}$ on $\bigoplus_k R_k$, hence on $\mathbb{C}(S)(T)$. Let $c':=1/m$.

Step 3. We construct B^+. By Lemma 5.2, $(C(S,P)\ni o,B_{C(S,P)})$ is a klt germ, and $B_{C(S,P)}$ is a $\mathbb{Q}$-divisor. Let ξ_0 be the generator of the grading action of $\mathbb{G}_m$. Then $(C(S,P),B_{C(S,P)},\mathbb{G}_m;\xi_0)$ is a log Fano cone singularity [LW24, Definition 2.2], with base (S,B) polarized by P. Let $p:(\mathcal{X},\Delta_{\mathcal{X}})\to\mathbb{A}^1$ be the $\mathbb{G}_m$-equivariant family induced by the Rees construction of ord_E, that is,

$$\mathcal{X}:=\mathrm{Spec}\bigoplus_{\lambda\in\mathbb{Z}}t^{-\lambda}\{g\in\bigoplus_k R_k\mid \mathrm{ord}_E(g)\geq\lambda\},$$

and $\Delta_{\mathcal{X}}$ is the closure of $B_{C(S,P)}\times(\mathbb{A}^1\backslash\{0\})$ in $\mathcal{X}$, as in [XZ25, (2.1)]. By [XZ25, Lemma 2.17], $(\mathcal{X},\Delta_{\mathcal{X}}+p^{-1}(0))$ is plt. By (5.8), the ideals $\{g\mid\mathrm{ord}_E(g)\geq\lambda\}$ are homogeneous, so the grading action of $\mathbb{G}_m$ extends to $\mathcal{X}$ and commutes with the $\mathbb{G}_m$-action of the Rees construction. Thus p is a normal test configuration of $(C(S,P),B_{C(S,P)};\xi_0)$ [LW24, Definition 3.9]. Since $p^{-1}(0)$ is a Cartier divisor and $(\mathcal{X},\Delta_{\mathcal{X}}+p^{-1}(0))$ is plt, p is a special test configuration [LW24, Definition 3.11]. By [LW24, Theorem 3.17], p corresponds to a special test configuration of (S,B) polarized by P, which has integral central fiber. Since

$$\{g\in R_k\mid\mathrm{ord}_E(g)\geq\lambda\}=\{\varphi T^k\in R_k\mid m\,\mathrm{ord}_F(\varphi)\geq\lambda-k\,\mathrm{ord}_E(T)\}\quad\text{for any } k\geq 0\text{ and }\lambda\in\mathbb{R},$$

the special test configuration of (S,B) corresponding to p is induced by the restriction $w=m\,\mathrm{ord}_F$ of ord_E to $\mathbb{C}(S)$. Since the coefficients of B form a finite subset of $[0,1]\cap\mathbb{Q}$, by [LW24, Theorem 3.24] ((1) implies (2)) for the trivial torus, there exists a $\mathbb{Q}$-divisor $B^+\geq B$ such that (S,B^+) is lc, $K_S+B^+\sim_{\mathbb{Q}}0$, and $a(F,S,B^+)=0$. □

Theorem 5.9. *Let S be a normal projective variety of dimension $n-1\geq 1$, (S,B) a klt pair, P an ample Cartier divisor on S, and $r>0$ a real number such that $-(K_S+B)\sim_{\mathbb{R}}rP$. Then*

$$\widehat{\mathrm{vol}}(C(S,P)\ni o,B_{C(S,P)})=r\Phi(S,B).$$

In particular,

$$\Phi(S,B)=\frac{\widehat{\mathrm{vol}}(C(S,P)\ni o,B_{C(S,P)})}{\mathrm{mld}(C(S,P)\ni o,B_{C(S,P)})}.$$

Proof. By Lemma 5.2, $(C(S,P) \ni o, B_{C(S,P)})$ is a klt germ and $\mathrm{mld}(C(S,P) \ni o, B_{C(S,P)}) = r$, so the second equality follows from the first. Let $L := -(K_S + B)$. Then (S,B) is a klt log Fano pair.

First assume that B is a $\mathbb{Q}$-divisor, $r \in \mathbb{Q}$, and $L \sim_{\mathbb{Q}} rP$. By Lemma 5.4, for any $F \in \mathcal{C}(S)$ and $u > 0$, $v_{F,u/r}$ has center o and $\widehat{\mathrm{vol}}(v_{F,u/r}, C(S,P), B_{C(S,P)}) = r\Psi_{S,B}(F,u)$, and by Lemma 5.2, $\widehat{\mathrm{vol}}(E_0, C(S,P), B_{C(S,P)}) = rL^{n-1}$. Thus

$$\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) \le r\Phi(S,B).$$

By [LX20, Proposition 4.4] for the $\mathbb{G}_m$-action given by the grading,

$$\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) = \inf_E \widehat{\mathrm{vol}}(E, C(S,P), B_{C(S,P)}),$$

where E runs over all $\mathbb{G}_m$-invariant Kollár components of $(C(S,P) \ni o, B_{C(S,P)})$. We have $\widehat{\mathrm{vol}}(E_0, C(S,P), B_{C(S,P)}) = rL^{n-1} \ge r\Phi(S,B)$. Let $E \ne E_0$ be a $\mathbb{G}_m$-invariant Kollár component. By Lemma 5.8, there exist a prime divisor F over S, positive rational numbers c, c', and a $\mathbb{Q}$-divisor $B^+ \ge B$ such that c' times the divisorial valuation of E equals $v_{F,c}$, (S, B^+) is lc, $K_S + B^+ \sim_{\mathbb{Q}} 0$, and $a(F, S, B^+) = 0$. By Lemma 5.5 and Definition 5.6 with $d = n-1$, there exists a $\mathbb{Q}$-divisor $G \ge 0$ on S such that (S,G) is lc, $N_{n-1}(K_S + G) \sim 0$, and $a(F,S,G) = 0$, so $F \in \mathcal{C}(S)$. Since the normalized volume is invariant under scaling of valuations, Lemma 5.4 implies

$$\widehat{\mathrm{vol}}(E, C(S,P), B_{C(S,P)}) = \widehat{\mathrm{vol}}(v_{F,c}, C(S,P), B_{C(S,P)}) = r\Psi_{S,B}(F, rc) \ge r\Phi(S,B).$$

Thus $\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) = r\Phi(S,B)$.

Now we prove the theorem in general. We fix a canonical divisor K_S and write $B = \sum_{j=1}^m b_j D_j$, where $D_1, \dots, D_m$ are distinct prime divisors and $b_j > 0$, and

$$K_S + B + rP = \sum_{k=1}^N z_k \,\mathrm{div}(\varphi_k)$$

for real numbers z_k and rational functions φ_k on S. The set of $(b', r', z') \in \mathbb{R}^m \times \mathbb{R} \times \mathbb{R}^N$ such that

$$K_S + \sum_j b'_j D_j + r'P = \sum_k z'_k \,\mathrm{div}(\varphi_k)$$

is the solution set of finitely many linear equations with integer coefficients and contains (b, r, z), so it contains a sequence of rational points $(b^{(i)}, r_i, z^{(i)})$ converging to (b, r, z). Let $B_i := \sum_j b_j^{(i)} D_j$. For $i \gg 0$, $b_j^{(i)} > 0$ for any j and $r_i > 0$. Let $h : S' \to S$ be a log resolution of $(S, \mathrm{Supp}\, B)$. Since

$$h^*(K_S + B_i) = \sum_k z_k^{(i)} h^* \,\mathrm{div}(\varphi_k) - r_i h^* P$$

converges to $h^*(K_S + B)$ coefficientwise, (S, B_i) is klt for $i \gg 0$. Thus for $i \gg 0$, B_i is a $\mathbb{Q}$-divisor, $r_i \in \mathbb{Q}_{>0}$, $-(K_S + B_i) \sim_{\mathbb{Q}} r_i P$, and (S, B_i) is klt, so the previous paragraph implies

$$\widehat{\mathrm{vol}}(C(S,P) \ni o, (B_i)_{C(S,P)}) = r_i \Phi(S, B_i).$$

Let $\mathcal{P}$ be the set of $b' \in \mathbb{R}^m$ such that $(S, \sum_j b'_j D_j)$ is a klt weak log Fano pair. Then $b \in \mathcal{P}$ and $b^{(i)} \in \mathcal{P}$ for $i \gg 0$, so $\Phi(S, B_i) \to \Phi(S, B)$ by Proposition 5.7(3) for $D_1, \dots, D_m$.

For $1 \le j \le m$, let $\widetilde{D}_j$ be the prime divisor on $C(S,P)$ whose pullback to $\mathrm{Tot}(\mathcal{O}_S(-P)) \backslash E_0$ is the pullback of D_j. Then $\widetilde{D}_1, \dots, \widetilde{D}_m$ are distinct prime divisors through o, $B_{C(S,P)} = \sum_j b_j \widetilde{D}_j$, and $(B_i)_{C(S,P)} = \sum_j b_j^{(i)} \widetilde{D}_j$. By Lemma 5.2, $(C(S,P) \ni o, (B_i)_{C(S,P)})$ is a klt germ for $i \gg 0$.

Let $0 < \epsilon < 1$ be a real number. By Lemma 2.9(2) for the germ $C(S,P) \ni o$ and the prime divisors $\widetilde{D}_1, \dots, \widetilde{D}_m$, for $i \gg 0$,

$$(1-\epsilon)^n \widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) \le \widehat{\mathrm{vol}}(C(S,P) \ni o, (B_i)_{C(S,P)}) \le (1+\epsilon)^n \widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}).$$

Thus for $i \gg 0$,

$$(1-\epsilon)^n \widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) \le r_i \Phi(S, B_i) \le (1+\epsilon)^n \widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}).$$

Letting $i \to +\infty$ and then $\epsilon \to 0$, we obtain $\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) = r\Phi(S,B)$. □

5.3. **Families and the ACC.**

Lemma 5.10. *Let d and N be positive integers and $\gamma > 0$ a real number. Then the set of couples $(S, \operatorname{Supp} B)$, where $S \subset \mathbb{P}^N$ is a normal closed subvariety of dimension d and degree at most N, and $B \ge 0$ is an $\mathbb{R}$-divisor on S such that $B \ge \gamma \operatorname{Supp} B$ and $-(K_S + B)$ is ample, is log bounded.*

Proof. Let H be a hyperplane section of S. Then $H^d \le N$. If $d \ge 2$, let Σ be the intersection of $d-1$ general members of $|H|$. Since S is normal, Σ is a smooth curve contained in the non-singular locus of S, and intersection numbers with H^{d-1} below are computed on Σ. We show

$$-K_S \cdot H^{d-1} \le (d-1)N + 2.$$

If $d = 1$, then $S \cong \mathbb{P}^1$ since $-K_S = -(K_S + B) + B$ is ample, so $-\deg K_S = 2$. If $d \ge 2$, then by adjunction,

$$2g(\Sigma) - 2 = (K_S + (d-1)H) \cdot \Sigma = K_S \cdot H^{d-1} + (d-1)H^d,$$

hence $-K_S \cdot H^{d-1} \le (d-1)H^d + 2 \le (d-1)N + 2$. Since $B \ge \gamma \operatorname{Supp} B$ and $-(K_S + B)$ is ample,

$$\gamma \deg_H(\operatorname{Supp} B) \le B \cdot H^{d-1} < -K_S \cdot H^{d-1} \le (d-1)N + 2.$$

Thus $\operatorname{Supp} B$ is an effective cycle of dimension $d-1$ in $\mathbb{P}^N$ of degree less than $((d-1)N+2)/\gamma$. Since such cycles and the closed subvarieties of $\mathbb{P}^N$ of dimension d and degree at most N are parametrized by schemes of finite type [Kol96, Theorem I.3.21], the set of couples $(S, \operatorname{Supp} B)$ is log bounded. □

Lemma 5.11. *Let $n \ge 2$ be an integer, (S,B) a klt log Fano pair of dimension $n-1$, P an ample Cartier divisor on S, and $r > 0$ a real number such that $-(K_S + B) \sim_{\mathbb{R}} rP$. Then*

$$\Phi(S,B) \le \frac{n^n}{(n-1)^{n-1}} \widehat{\mathrm{vol}}(S \ni y, B)$$

for any closed point $y \in S$. If moreover $\Phi(S,B) \ge \epsilon$ for some real number $\epsilon > 0$, then:

1. *$a(F,S,B) \ge \epsilon/n^n$ for any prime divisor F over S.*
2. *The variety S belongs to a bounded family depending only on n and ϵ.*
3. *For any real number $\gamma > 0$, if every coefficient of B is $\ge \gamma$, then $(S, \operatorname{Supp} B)$ belongs to a log bounded family depending only on n, ϵ, and γ.*

Proof. Let $y \in S$ be a closed point, let F be a prime divisor over S such that $\operatorname{center}_S F = y$, and let $c > 0$ be a real number. We first show

$$\mathrm{vol}(v_{F,c}) \le \frac{\mathrm{vol}(\mathrm{ord}_F)}{c}. \tag{5.9}$$

Let $U \ni y$ be an open subset of S and e a nowhere vanishing section of $\mathcal{O}_S(P)|_U$. For any integer $k \ge 0$ and any $s \in H^0(S, \mathcal{O}_S(kP))$, let $f_s \in \mathcal{O}_S(U)$ be the function such that $s|_U = f_s e^k$. Since $\operatorname{center}_S F = y \in U$ and $\operatorname{div}(s)|_U = \operatorname{div}(f_s)$, we have

$$v_{F,c}(s) = \operatorname{mult}_F \operatorname{div}(f_s) + kc.$$

The length $\ell(\mathcal{O}_{S,y}/\mathfrak{a}_\mu(\mathrm{ord}_F))$ is increasing in $\mu \ge 0$ and is 0 for $\mu = 0$. Let $\lambda > 0$ be a real number. By Lemma 5.4, the center of $v_{F,c}$ is o, so $\mathfrak{a}_\lambda(v_{F,c})$ is primary to the maximal ideal of o.

Since $v_{F,c}(\sum_k s_k) = \min_k v_{F,c}(s_k)$ for homogeneous elements s_k of degree k, and since sending $s \in H^0(S, \mathcal{O}_S(kP))$ to the class of f_s induces, for $0 \le k < \lambda/c$, an injection

$$H^0(S, \mathcal{O}_S(kP))/\{s \mid \mathrm{mult}_F \operatorname{div}(f_s) \ge \lambda - kc\} \hookrightarrow \mathcal{O}_{S,y}/\mathfrak{a}_{\lambda-kc}(\mathrm{ord}_F),$$

and

$$\{s \mid \mathrm{mult}_F \operatorname{div}(f_s) \ge \lambda - kc\} = H^0(S, \mathcal{O}_S(kP)) \quad \text{if } kc \ge \lambda,$$

we have

$$\dim_{\mathbb{C}} \mathcal{O}_{C(S,P),o}/\mathfrak{a}_\lambda(v_{F,c}) = \sum_{k\ge 0} \dim_{\mathbb{C}} \frac{H^0(S, \mathcal{O}_S(kP))}{\{s \mid \mathrm{mult}_F \operatorname{div}(f_s) \ge \lambda - kc\}} \le \sum_{0\le k<\lambda/c} \ell(\mathcal{O}_{S,y}/\mathfrak{a}_{\lambda-kc}(\mathrm{ord}_F)).$$

Let $\tau > 0$ be a real number. By Definition 2.7, there exists a real number M_τ such that

$$\ell(\mathcal{O}_{S,y}/\mathfrak{a}_\mu(\mathrm{ord}_F)) \le \Big(\frac{\mathrm{vol}(\mathrm{ord}_F)}{(n-1)!} + \tau\Big)\mu^{n-1} + M_\tau$$

for any $\mu \ge 0$. Since

$$\sum_{0\le k<\lambda/c} (\lambda - kc)^{n-1} \le \lambda^{n-1} + \int_0^{\lambda/c} (\lambda - xc)^{n-1} dx = \lambda^{n-1} + \frac{\lambda^n}{nc},$$

we obtain

$$\dim_{\mathbb{C}} \mathcal{O}_{C(S,P),o}/\mathfrak{a}_\lambda(v_{F,c}) \le \Big(\frac{\mathrm{vol}(\mathrm{ord}_F)}{(n-1)!} + \tau\Big)\Big(\lambda^{n-1} + \frac{\lambda^n}{nc}\Big) + M_\tau\Big(\frac{\lambda}{c} + 1\Big).$$

Multiplying by $n!/\lambda^n$ and letting $\lambda \to +\infty$, we obtain $\mathrm{vol}(v_{F,c}) \le (\mathrm{vol}(\mathrm{ord}_F) + (n-1)!\tau)/c$. Letting $\tau \to 0$, we obtain (5.9).

Let $a := a(F, S, B)$. Then $a > 0$ since (S, B) is klt, and $a(v_{F,c}, C(S,P), B_{C(S,P)}) = a + rc$ by Lemma 5.4. Taking $c := a/((n-1)r)$, by (5.9) we have

$$\begin{aligned}\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) &\le (a+rc)^n \,\mathrm{vol}(v_{F,c}) \le \frac{(a+rc)^n}{c}\,\mathrm{vol}(\mathrm{ord}_F)\\ &= \frac{rn^n}{(n-1)^{n-1}} a^{n-1}\,\mathrm{vol}(\mathrm{ord}_F) = \frac{rn^n}{(n-1)^{n-1}}\widehat{\mathrm{vol}}(F, S, B).\end{aligned}$$

Moreover, $\widehat{\mathrm{vol}}(S \ni y, B)$ is the infimum of $\widehat{\mathrm{vol}}(F, S, B)$ over all prime divisors F over S such that $\mathrm{center}_S F = y$: if $n \ge 3$, this follows from Lemma 2.8(2) for the klt germ $(S \ni y, B)$ of dimension $n - 1 \ge 2$, since Kollár components of $(S \ni y, B)$ are such prime divisors. If $n = 2$, then S is a smooth curve, F is the prime divisor y on S, and any valuation v on S with $\mathrm{center}_S v = y$ is $t\,\mathrm{ord}_F$ for a real number $t > 0$, so $\widehat{\mathrm{vol}}(v, S, B) = ta(F, S, B) \cdot \mathrm{vol}(\mathrm{ord}_F)/t = \widehat{\mathrm{vol}}(F, S, B)$. Thus

$$\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) \le \frac{rn^n}{(n-1)^{n-1}}\widehat{\mathrm{vol}}(S \ni y, B).$$

Since $\widehat{\mathrm{vol}}(C(S,P) \ni o, B_{C(S,P)}) = r\Phi(S, B)$ by Theorem 5.9, the first inequality of the lemma holds.

From now on assume that $\Phi(S, B) \ge \epsilon$.

(1) Let F be a prime divisor over S and $y \in \mathrm{center}_S F$ a closed point. By the first inequality,

$$\widehat{\mathrm{vol}}(S \ni y, B) \ge \frac{(n-1)^{n-1}}{n^n}\epsilon.$$

Since $(S \ni y, B)$ is a klt germ of dimension $n - 1$, Lemma 2.10(4) implies

$$a(F, S, B) \ge \frac{\widehat{\mathrm{vol}}(S \ni y, B)}{(n-1)^{n-1}} \ge \frac{\epsilon}{n^n}.$$

(2) By (1), (S, B) is ϵ/n^n-lc, so (2) follows from Theorem 2.15(1) with $d = n - 1$.

(3) By (2), there exists a positive integer N depending only on n and ϵ such that S is isomorphic to a closed subvariety of $\mathbb{P}^N$ of degree at most N. Thus (3) follows from Lemma 5.10 with $d = n - 1$. □

To obtain Theorem 5.14, we need the following result.

Construction-Theorem 5.12 (cf. [LZ26, Proposition 5.2, Remark 5.3, and Theorem 5.4])**.** Let d, T, $\mathcal{S} \to T$, and $\mathcal{B}_1, \dots, \mathcal{B}_m$ be as follows.

(a) d is a positive integer and T is a variety.
(b) $\mathcal{S} \to T$ is a flat projective morphism whose fibers S_t are normal varieties of dimension d of Fano type.
(c) $\mathcal{B}_1, \dots, \mathcal{B}_m \geq 0$ are relative Mumford divisors on $\mathcal{S} \to T$ [Kol23, Definition 4.68] such that $\mathcal{B}_1|_{S_t}, \dots, \mathcal{B}_m|_{S_t}$ are pairwise distinct prime divisors for any $t \in T$, and $B_t(b) := \sum_{j=1}^m b_j \mathcal{B}_j|_{S_t}$ for $b \in \mathbb{R}^m$ and $t \in T$.

Then, after replacing T with the members of a finite stratification into locally closed subvarieties and then with finite surjective covers of these, the following hold for each new T.

(1) The following hold.
- The set $\mathcal{P}$ of $b \in \mathbb{R}^m$ such that $(S_t, B_t(b))$ is a klt weak log Fano pair does not depend on $t \in T$.
- For any $b \in \mathcal{P}$, $(-(K_{S_t} + B_t(b)))^d$ does not depend on t.
- The set W of $b \in \mathbb{R}^m$ such that $K_{S_t} + B_t(b)$ is $\mathbb{R}$-Cartier does not depend on $t \in T$, and W is either empty or an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$.
- There exists a projective birational morphism $g : \mathcal{S}' \to \mathcal{S}$ such that $\mathcal{S}' \to T$ is smooth and $g_t := g|_{S'_t} : S'_t \to S_t$ is a log resolution of $(S_t, \sum_{j=1}^m \operatorname{Supp} \mathcal{B}_j|_{S_t})$ for any $t \in T$.
- For any $\mathcal{F}$ in the set $\mathcal{E}$ of prime divisors on $\mathcal{S}'$ which are exceptional over $\mathcal{S}$ or are $g_*^{-1}D$ for a component D of some $\mathcal{B}_j$, every fiber $\mathcal{F}_t$ is a prime divisor and $a(\mathcal{F}_t, S_t, B_t(b))$ is an affine function of $b \in W$ which does not depend on t.
- For any $t \in T$, the exceptional prime divisors of g_t and the strict transforms of $\mathcal{B}_1|_{S_t}, \dots, \mathcal{B}_m|_{S_t}$ under g_t are exactly the fibers $\mathcal{F}_t$, $\mathcal{F} \in \mathcal{E}$.

(2) There exist finitely many surjective morphisms $Z_k \to T$ from varieties Z_k, $\mathbb{Q}$-linear combinations $\mathcal{G}_k \geq 0$ of relative Mumford divisors on $\mathcal{S} \times_T Z_k \to Z_k$, and sets $\mathcal{C}_k$ of prime divisors over $\mathcal{S} \times_T Z_k$ with the following properties.
- Every $F \in \mathcal{C}_k$ is a prime divisor on a variety Y_F with a projective birational morphism $Y_F \to \mathcal{S} \times_T Z_k$ such that $Y_F \to Z_k$ and $F \to Z_k$ are smooth.
- For $z \in Z_k$ over $t \in T$, we identify the fiber of $\mathcal{S} \times_T Z_k \to Z_k$ over z with S_t, and $\mathcal{G}_{k,z}$, $Y_{F,z}$, and F_z denote the fibers over z of $\mathcal{G}_k$, $Y_F \to Z_k$, and $F \to Z_k$.
- For any $F \in \mathcal{C}_k$ and $z \in Z_k$ over $t \in T$, the morphism $Y_{F,z} \to S_t$ is birational and the fiber F_z is a prime divisor on $Y_{F,z}$.
- For any $z \in Z_k$ over $t \in T$, $(S_t, \mathcal{G}_{k,z})$ is lc.
- For any $z \in Z_k$ over $t \in T$, $N_d(K_{S_t} + \mathcal{G}_{k,z}) \sim 0$.
- For any $F \in \mathcal{C}_k$ and $z \in Z_k$ over $t \in T$, $a(F_z, S_t, \mathcal{G}_{k,z}) = 0$.
- For any $t \in T$,

$$\mathcal{C}(S_t) = \{F_z \mid F \in \mathcal{C}_k \text{ for some } k,\ z \in Z_k \text{ over } t\}.$$

(3) For any k, $F \in \mathcal{C}_k$, $b \in \mathcal{P}$, and real number $u > 0$, the numbers $a(F_z, S_t, B_t(b))$ and $\Psi_{S_t, B_t(b)}(F_z, u)$, where $z \in Z_k$ is over $t \in T$, do not depend on z.

(4) For any $b \in \mathcal{P}$ and $t \in T$,

$$\Phi(S_t, B_t(b)) = \inf\left(\{(-(K_{S_t} + B_t(b)))^d\} \cup \{\Psi_{S_t, B_t(b)}(F_z, u) \mid k,\ F \in \mathcal{C}_k,\ u \in \mathbb{R}_{>0}\}\right),$$

where $z \in Z_k$ is any point over t, and $\Phi(S_t, B_t(b))$ does not depend on t.

Proof. Each property proved below is preserved when T is replaced by a locally closed subvariety of T or by a variety finite and surjective over T, and all families over T are replaced by their base changes.

(†) Let U be a smooth variety, $\mathcal{V} \to U$ a flat projective morphism whose fibers $\mathcal{V}_u$ are normal varieties, and $\mathcal{D} \geq 0$ a relative Mumford divisor on $\mathcal{V} \to U$. Then, after replacing U with the members of a finite stratification into smooth locally closed subvarieties and then with smooth varieties finite and surjective over these, there exists a projective birational morphism $g : \mathcal{V}' \to \mathcal{V}$ satisfying the following, where $\mathcal{E}$ is the set of prime divisors on $\mathcal{V}'$ which are exceptional over $\mathcal{V}$ or are of the form $g_*^{-1}D$ for a component D of $\mathcal{D}$:

(i) $\mathcal{V}' \to U$ is smooth, and for any $u \in U$, $g_u := g|_{\mathcal{V}'_u}$ is a log resolution of $(\mathcal{V}_u, \operatorname{Supp} \mathcal{D}|_{\mathcal{V}_u})$ whose exceptional prime divisors and divisors $g_{u*}^{-1}D$ for components D of $\operatorname{Supp} \mathcal{D}|_{\mathcal{V}_u}$ are the fibers E_u, $E \in \mathcal{E}$.

(ii) For any non-empty subset $\mathcal{E}' \subset \mathcal{E}$, any irreducible component of $\bigcap_{E \in \mathcal{E}'} E$ is smooth over U with irreducible fibers.

To prove (†), let g be a log resolution of $(\mathcal{V}, \operatorname{Supp} \mathcal{D})$. Since U is smooth and the fibers $\mathcal{V}_u$ are normal, $\mathcal{V}$ is normal by [Gro66, Théorème 12.1.6(iv)] and [Mat89, Corollary to Theorem 23.9], so $\operatorname{codim}_{\mathcal{V}} g(\operatorname{Exc}(g)) \geq 2$. By generic smoothness, there exists a dense open subset $U^\circ \subset U$ over which $\mathcal{V}'$ and all irreducible components of all intersections of members of $\mathcal{E}$ are smooth, and over which $\operatorname{codim}_{\mathcal{V}_u}(g(\operatorname{Exc}(g)) \cap \mathcal{V}_u) \geq 2$. Then (i) holds over U°, except that the fibers E_u with $E \in \mathcal{E}$ may be reducible. Taking the Stein factorizations of the irreducible components above over U°, after a base change by a smooth variety finite and surjective over a dense open subset of U°, all of them have connected, hence irreducible, fibers, so (ii) holds there, and so does (i), since by (ii) for $\mathcal{E}' = \{E\}$ every fiber E_u is a prime divisor. By Noetherian induction on U, (†) holds.

Proof of (1). By [LZ26, Propositions 3.5 and 4.1], possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties, we may assume that T is smooth and that $\mathcal{P}$ does not depend on $t \in T$. Let A be a Cartier divisor on $\mathcal{S}$ which is ample over T. Let $N_A \gg 0$ be an integer, and for any j let $\mathcal{B}'_j \geq 0$ be the divisor of a general section of $\mathcal{O}_{\mathcal{S}}(N_A A - \mathcal{B}_j)$, so that $\mathcal{B}_j + \mathcal{B}'_j \sim N_A A$. By generic flatness and Noetherian induction, possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties, we may assume that every $\mathcal{B}'_j$ is a relative Mumford divisor on $\mathcal{S} \to T$. For $b \in \mathbb{R}^m$, let $b_j^{\pm} := \max\{\pm b_j, 0\}$ and $b^{\pm} := (b_1^{\pm}, \dots, b_m^{\pm})$. Since

$$\Big(K_{S_t} + \sum_{j=1}^m b_j^+ \mathcal{B}_j|_{S_t} + \sum_{j=1}^m b_j^- \mathcal{B}'_j|_{S_t}\Big) - (K_{S_t} + B_t(b)) = \sum_{j=1}^m b_j^-(\mathcal{B}_j + \mathcal{B}'_j)|_{S_t}$$

is an $\mathbb{R}$-linear combination of Cartier divisors, $K_{S_t} + B_t(b)$ is $\mathbb{R}$-Cartier if and only if $(b^+, b^-) \in W'_t$, where W'_t is the set of $c = (c_1, \dots, c_{2m}) \in \mathbb{R}^{2m}_{\geq 0}$ such that

$$K_{S_t} + \sum_{j=1}^m (c_j \mathcal{B}_j|_{S_t} + c_{m+j} \mathcal{B}'_j|_{S_t})$$

is $\mathbb{R}$-Cartier. By [LZ26, Proposition 3.4], possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties, we may assume that W'_t does not depend on $t \in T$, so $W = \{b \in \mathbb{R}^m \mid (b^+, b^-) \in W'_t\}$ does not depend on t. By (†) for $\mathcal{S} \to T$ and $\sum_{j=1}^m \mathcal{B}_j$, we may assume that g as in the fourth item of (1) exists, that $\mathcal{F}_t$ is a prime divisor for any $\mathcal{F} \in \mathcal{E}$ and $t \in T$, and that the sixth item of (1) holds.

We may assume that $W \neq \emptyset$. Since the $\mathbb{R}$-Cartier $\mathbb{R}$-divisors on S_t are the $\mathbb{R}$-linear combinations of Cartier divisors, W is an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$. Let $b^{(0)}, \dots, b^{(\dim W)} \in W \cap \mathbb{Q}^m$ be affinely independent, and for $b \in W$, let $\lambda_0(b), \dots, \lambda_{\dim W}(b)$ be the real numbers such that

$$\sum_i \lambda_i(b) = 1 \quad \text{and} \quad b = \sum_i \lambda_i(b) b^{(i)};$$

they are affine functions of b with rational coefficients. Let $\mathcal{B}(b) := \sum_{j=1}^m b_j \mathcal{B}_j$. For any i and t, $K_{S_t} + B_t(b^{(i)})$ is $\mathbb{Q}$-Cartier. Let $r_0 > 0$ be an integer such that $r_0 b^{(i)} \in \mathbb{Z}^m$ for any i. By [Kol23, Corollary 3.30], for any i and integer $r > 0$, the closed points $t \in T$ such that $r! r_0(K_{S_t} + B_t(b^{(i)}))$ is Cartier are the closed points of a constructible subset $T_{i,r} \subset T$, and if $T_{i,r}$ is dense in T, then $r! r_0(K_{\mathcal{S}/T} + \mathcal{B}(b^{(i)}))$ is Cartier over a dense open subset of T. Since the union of $T_{i,1} \subset T_{i,2} \subset \cdots$ contains all closed points of T and $\mathbb{C}$ is uncountable, $T_{i,r}$ is dense in T for $r \gg 0$. Thus, by Noetherian induction on T, possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties, we may assume that

$$\mathcal{K}_i := K_{\mathcal{S}/T} + \mathcal{B}(b^{(i)})$$

is $\mathbb{Q}$-Cartier for any i. For any t, $\mathcal{K}_i|_{S_t} = K_{S_t} + B_t(b^{(i)})$, since both sides are $\mathbb{Q}$-Cartier on the normal variety S_t and are equal on the non-singular locus of S_t.

Let $\mathcal{F} \in \mathcal{E}$. Since $\mathcal{S}' \to T$ is smooth, $K_{\mathcal{S}'/T}|_{S'_t} = K_{S'_t}$, and since $\mathcal{F}_t$ is a prime divisor,

$$a(\mathcal{F}_t, S_t, B_t(b^{(i)})) = 1 + \mathrm{mult}_{\mathcal{F}}(K_{\mathcal{S}'/T} - g^*\mathcal{K}_i)$$

for any i and t, which does not depend on t. For any $b \in W$, we have $K_{S_t} + B_t(b) = \sum_i \lambda_i(b)\mathcal{K}_i|_{S_t}$, so

$$a(\mathcal{F}_t, S_t, B_t(b)) = \sum_i \lambda_i(b) a(\mathcal{F}_t, S_t, B_t(b^{(i)})),$$

which is an affine function of $b \in W$ and does not depend on t. For any $b \in \mathcal{P}$,

$$(-(K_{S_t} + B_t(b)))^d = (-1)^d \sum_{i_1,\dots,i_d} \lambda_{i_1}(b) \cdots \lambda_{i_d}(b) \mathcal{K}_{i_1}|_{S_t} \cdots \mathcal{K}_{i_d}|_{S_t},$$

and the intersection numbers $\mathcal{K}_{i_1}|_{S_t} \cdots \mathcal{K}_{i_d}|_{S_t}$ do not depend on t since $\mathcal{S} \to T$ is flat, T is connected, and some multiple of each $\mathcal{K}_i$ is Cartier. This proves (1).

Proof of (2). Let $\pi : \mathcal{S} \to T$ be the morphism and $N := N_d$. The sheaf $\mathcal{M} := \mathcal{O}_{\mathcal{S}}(-NK_{\mathcal{S}/T})$ is reflexive of rank 1 on the normal variety $\mathcal{S}$, so the restriction of $\mathcal{M}$ to the generic fiber of π is reflexive, hence S_2. By generic flatness and Noetherian induction, possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties, we may assume the following.

- $\mathcal{M}$ is flat over T.
- $\mathcal{M}|_{S_t} \cong \mathcal{O}_{S_t}(-NK_{S_t})$ for any t.
- $h^0(S_t, \mathcal{O}_{S_t}(-NK_{S_t}))$ does not depend on t.

The first item holds over a dense open subset of T by generic flatness. For the second, the set of points of $\mathcal{S}$ at which the fiber of $\mathcal{M}$ is equidimensional and S_2 is open [Gro66, Théorème 12.1.1(iv)] and contains the generic fiber of π, hence contains S_t for any t in a dense open subset of T, and $\mathcal{M}|_{S_t}$ is isomorphic to $\mathcal{O}_{S_t}(-NK_{S_t})$ on the smooth locus of π, whose complement in S_t has codimension ≥ 2 since S_t is normal. The third holds over a dense open subset of T by upper semicontinuity. By the three items, $\pi_*\mathcal{M}$ is locally free and commutes with base change. Let $Z \to T$ be the projective bundle whose fiber over t is the space of 1-dimensional subspaces of $H^0(S_t, \mathcal{O}_{S_t}(-NK_{S_t}))$, and let $\mathcal{S}_Z := \mathcal{S} \times_T Z$. Let $p : \mathcal{S}_Z \to \mathcal{S}$ and $q : \mathcal{S}_Z \to Z$ be the projections,

$$\sigma \in H^0(\mathcal{S}_Z, p^*\mathcal{M} \otimes q^*\mathcal{O}_Z(1))$$

the universal section, and $\mathcal{H} := (\sigma = 0)$. Then $\mathcal{H}$ is a relative Mumford divisor on $\mathcal{S}_Z \to Z$ whose fiber H_z over $z \in Z$ over $t \in T$ is the member of $|-NK_{S_t}|$ given by z. Let $\mathcal{G} := \mathcal{H}/N$. Then $N(K_{\mathcal{S}_Z/Z} + \mathcal{G})$ is linearly equivalent to the pullback of a Cartier divisor on Z, since $\mathcal{H}$ is the zero divisor of σ and

$$p^*\mathcal{M} \cong \mathcal{O}_{\mathcal{S}_Z}(-NK_{\mathcal{S}_Z/Z}),$$

so $K_{\mathcal{S}_Z/Z} + \mathcal{G}$ is $\mathbb{Q}$-Cartier and $K_{\mathcal{S}_Z/Z} + \mathcal{G} \sim_{\mathbb{Q}} 0/Z$. For any $t \in T$ and any $\mathbb{Q}$-divisor $G \geq 0$ on S_t such that $N(K_{S_t} + G) \sim 0$, NG is a member of $|-NK_{S_t}|$, so $G = \mathcal{G}_z$ for exactly one $z \in Z$ over t.

Applying (†) to the restrictions of $\mathcal{S}_Z \to Z$ and $\mathcal{H} + \sum_{j=1}^m \mathcal{B}_j$ over the members of a finite stratification of Z into smooth locally closed subvarieties, we obtain finitely many smooth varieties Z_k with morphisms $Z_k \to Z$ whose images cover Z, and projective birational morphisms $h_k : Y_k \to \mathcal{S}_{Z_k} := \mathcal{S} \times_T Z_k$ satisfying (i) and (ii) of (†) with the set $\mathcal{E}_k$ of prime divisors on Y_k which are exceptional over $\mathcal{S}_{Z_k}$ or are of the form $(h_k)_*^{-1}D$ for a component D of $\mathcal{H}$ or of some $\mathcal{B}_j$. Let $\mathcal{G}_k$ be the pullback of $\mathcal{G}$ to $\mathcal{S}_{Z_k}$, and let Δ_k be the $\mathbb{Q}$-divisor such that

$$K_{Y_k/Z_k} + \Delta_k = h_k^*(K_{\mathcal{S}_{Z_k}/Z_k} + \mathcal{G}_k).$$

For any $z \in Z_k$ over $t \in T$, the fiber $\Delta_{k,z}$ satisfies

$$K_{Y_{k,z}} + \Delta_{k,z} = h_{k,z}^*(K_{S_t} + \mathcal{G}_{k,z}),$$

and $\operatorname{mult}_{E_z} \Delta_{k,z} = \operatorname{mult}_E \Delta_k$ for any $E \in \mathcal{E}_k$ since E_z is a prime divisor. Since $h_{k,z}$ is a log resolution of $(S_t, \operatorname{Supp} \mathcal{G}_{k,z})$ and $\operatorname{Supp} \Delta_{k,z} \subset \sum_{E\in\mathcal{E}_k} E_z$, $(S_t, \mathcal{G}_{k,z})$ is lc if and only if $\operatorname{mult}_E \Delta_k \le 1$ for any $E \in \mathcal{E}_k$. Thus either $(S_t, \mathcal{G}_{k,z})$ is lc for all $z \in Z_k$, or it is lc for no $z \in Z_k$. We discard the Z_k of the second kind. Then for any $z \in Z_k$ over t, $(S_t, \mathcal{G}_{k,z})$ is lc and $N(K_{S_t} + \mathcal{G}_{k,z}) \sim 0$, and any $z' \in Z$ over t such that $(S_t, \mathcal{G}_{z'})$ is lc is the image of a point of some Z_k.

Let

$$\mathcal{E}_k^{=1} := \{E \in \mathcal{E}_k \mid \operatorname{mult}_E \Delta_k = 1\}.$$

For any non-empty subset $\mathcal{E}' \subset \mathcal{E}_k^{=1}$, any irreducible component V of $\bigcap_{E\in\mathcal{E}'} E$, and any primitive vector $w \in (\mathbb{N}^+)^{\mathcal{E}'}$, let F be the toroidal divisor over V given by w, and let $Y_F \to Y_k$ be a projective toroidal modification on which F is a divisor, given by the star subdivision along w followed by a regular refinement [KKMS73, Chapter II, §2, Theorems 6*, 10*, and 11*]. Since $Y_k \to Z_k$ is smooth and $\sum_{E\in\mathcal{E}_k} E$ is a divisor with simple normal crossings relative to Z_k by (i) and (ii), $Y_F \to Y_k$ is locally the product of a toric modification of an affine space and Z_k-smooth factors. Thus $Y_F \to Z_k$ and $F \to Z_k$ are smooth, and for any $z \in Z_k$, $Y_{F,z} \to Y_{k,z}$ is the toroidal modification of $(Y_{k,z}, \sum_{E\in\mathcal{E}_k} E_z)$ defined by the star subdivision along the ray of w followed by the regular refinement above, and F_z is the divisor over the irreducible variety V_z corresponding to w, which is a prime divisor. Let $\mathcal{C}_k$ be the set of all prime divisors F obtained in this way. For any $F \in \mathcal{C}_k$ and $z \in Z_k$ over t, since $\operatorname{Supp} \Delta_{k,z} \subset \sum_{E\in\mathcal{E}_k} E_z$ and $a(E_z, S_t, \mathcal{G}_{k,z}) = 1 - \operatorname{mult}_E \Delta_k = 0$ for any $E \in \mathcal{E}'$, we have

$$a(F_z, S_t, \mathcal{G}_{k,z}) = \sum_{E\in\mathcal{E}'} w_E a(E_z, S_t, \mathcal{G}_{k,z}) = 0.$$

Conversely, let $z \in Z_k$ be a point over t and F_1 a prime divisor over S_t such that

$$a(F_1, S_t, \mathcal{G}_{k,z}) = 0.$$

Then $a(F_1, Y_{k,z}, \Delta_{k,z}) = 0$, where $\Delta_{k,z}$ is a sub-boundary, that is, all coefficients of $\Delta_{k,z}$ are ≤ 1, and $\operatorname{Supp} \Delta_{k,z} \subset \sum_{E\in\mathcal{E}_k} E_z$, which is a divisor with simple normal crossings. Let η be the generic point of $\operatorname{center}_{Y_{k,z}}(F_1)$, let $\mathcal{E}'$ be the set of $E \in \mathcal{E}_k$ such that $\eta \in E_z$, and let $w_E := \operatorname{mult}_{F_1} E_z \ge 1$ for $E \in \mathcal{E}'$. Let v_w be the quasi-monomial valuation which is monomial, at the generic point ξ of the irreducible component of $\bigcap_{E\in\mathcal{E}'} E_z$ containing η, which is unique since $\bigcap_{E\in\mathcal{E}'} E_z$ is smooth, in local equations of the E_z with weights w_E (with v_w trivial if $\mathcal{E}' = \emptyset$). Let v_{F_1} be the divisorial valuation of F_1. Since $v_w(E_z) = w_E = v_{F_1}(E_z)$ for any $E \in \mathcal{E}'$, v_w is the retraction of v_{F_1} to the log smooth pair $(Y_{k,z}, \sum_{E\in\mathcal{E}_k} E_z)$ [JM12, Section 4.3], so by [JM12, Proposition 5.1 and Corollary 5.4],

$$a(F_1, Y_{k,z}, 0) \ge a(v_w, Y_{k,z}, 0) = \sum_{E\in\mathcal{E}'} w_E,$$

with equality if and only if $v_w = v_{F_1}$. Since $\operatorname{mult}_{E_z} \Delta_{k,z} = \operatorname{mult}_E \Delta_k$ and $\operatorname{mult}_E \Delta_k \le 1$, we get

$$0 = a(F_1, Y_{k,z}, \Delta_{k,z}) = a(F_1, Y_{k,z}, 0) - \sum_{E\in\mathcal{E}'} w_E \operatorname{mult}_E \Delta_k \ge \sum_{E\in\mathcal{E}'} w_E(1 - \operatorname{mult}_E \Delta_k) \ge 0.$$

Thus $v_{F_1} = v_w$, so $\mathcal{E}' \neq \emptyset$ and $\eta = \mathrm{center}_{Y_{k,z}}(v_w) = \xi$, and $\mathrm{mult}_E \Delta_k = 1$ for any $E \in \mathcal{E}'$, that is, $\mathcal{E}' \subset \mathcal{E}_k^{=1}$. Since the value group of v_{F_1} is $\mathbb{Z}$, the vector $w := (w_E)_{E\in\mathcal{E}'} \in (\mathbb{N}^+)^{\mathcal{E}'}$ is primitive. Therefore $\mathrm{center}_{Y_{k,z}}(F_1)$ is an irreducible component of $\bigcap_{E\in\mathcal{E}'} E_z$ and F_1 is the toroidal divisor over $\mathrm{center}_{Y_{k,z}}(F_1)$ given by w [KKMS73, Chapter II, §2, Theorem 6*]. By (ii) of (†), $\mathrm{center}_{Y_{k,z}}(F_1) = V_z$ for an irreducible component V of $\bigcap_{E\in\mathcal{E}'} E$, so $F_1 = F_z$ for the prime divisor $F \in \mathcal{C}_k$ defined by $\mathcal{E}'$, V, and w.

Let $t \in T$ and $F_1 \in \mathcal{C}(S_t)$. By Definition 5.6, $a(F_1, S_t, G) = 0$ for some $\mathbb{Q}$-divisor $G \geq 0$ on S_t such that (S_t, G) is lc and $N(K_{S_t} + G) \sim 0$. Then $G = \mathcal{G}_{z'}$ for some $z' \in Z$ over t, z' is the image of a point z of some Z_k, and $\mathcal{G}_{k,z} = G$, so $F_1 = F_z$ for some $F \in \mathcal{C}_k$ by the previous paragraph. Conversely, $F_z \in \mathcal{C}(S_t)$ for any $F \in \mathcal{C}_k$ and $z \in Z_k$ over t, since $(S_t, \mathcal{G}_{k,z})$ is lc, $N(K_{S_t} + \mathcal{G}_{k,z}) \sim 0$, and $a(F_z, S_t, \mathcal{G}_{k,z}) = 0$. Thus

$$\mathcal{C}(S_t) = \{F_z \mid F \in \mathcal{C}_k \text{ for some } k,\ z \in Z_k \text{ over } t\}.$$

Finally, if T is irreducible, there exists a dense open subset $T^\circ \subset T$ such that for any k, the open subset $Z_k \times_T T^\circ$ of Z_k is either empty or surjective onto T°: if $Z_k \to T$ is dominant, then the image contains a dense open subset by Chevalley's theorem, and otherwise the closure of the image is a proper closed subset. By Noetherian induction on T, possibly replacing T with the members T_1 of a finite stratification into locally closed subvarieties and, for each T_1, each Z_k with the members of a finite stratification of $Z_k \times_T T_1$ into smooth locally closed subvarieties, and reindexing, we may assume that each Z_k is smooth and $Z_k \to T$ is surjective. This proves (2).

Proof of (3). Let k be an index and $F \in \mathcal{C}_k$. For $z \in Z_k$ over t, let $h_{F,z} : Y_{F,z} \to S_t$ be the induced morphism. Then $Y_{F,z}$ is a resolution of S_t on which F_z is a divisor, since $Y_F \to Z_k$ is smooth and $Y_{F,z} \to S_t$ is birational. Let $\mathcal{B}_k(b)$ be the pullback of $\mathcal{B}(b)$ to $\mathcal{S}_{Z_k}$.

First let $b \in \mathcal{P} \cap \mathbb{Q}^m$. Then the numbers $\lambda_i(b)$ defined above are rational, so

$$K_{\mathcal{S}/T} + \mathcal{B}(b) = \sum_i \lambda_i(b)\mathcal{K}_i$$

is $\mathbb{Q}$-Cartier, hence so is $K_{\mathcal{S}_{Z_k}/Z_k} + \mathcal{B}_k(b)$. Let $L_b := -(K_{\mathcal{S}_{Z_k}/Z_k} + \mathcal{B}_k(b))$. The prime divisor F is a toroidal divisor with respect to

$$\mathrm{Exc}(h_k) + \mathrm{Supp}\left((h_k)_*^{-1}(\mathcal{B}_k(b) + \mathcal{G}_k)\right),$$

since every member of the set $\mathcal{E}' \subset \mathcal{E}_k^{=1}$ defining F is exceptional over $\mathcal{S}_{Z_k}$ or a component of $(h_k)_*^{-1}\mathcal{G}_k$. By [LZ26, Proposition 5.2 and Remark 5.3] with $D = \mathcal{G}_k$ and the fiberwise log resolution h_k, the numbers $a(F_z, S_t, B_t(b))$ and $\mathrm{vol}(h_{F,z}^* L_b|_{S_t} - xF_z)$ for any rational number $x \geq 0$ do not depend on $z \in Z_k$, where $L_b|_{S_t} = -(K_{S_t} + B_t(b))$. Thus, by the continuity of the volume function, $\mathrm{vol}(h_{F,z}^* L_b|_{S_t} - xF_z)$ does not depend on $z \in Z_k$ for any real number $x \geq 0$. By Definition 5.3, $\Psi_{S_t,B_t(b)}(F_z, u)$ is determined by $a(F_z, S_t, B_t(b))$, $(-(K_{S_t} + B_t(b)))^d$, and $\mathrm{vol}(h_{F,z}^* L_b|_{S_t} - xF_z)$ for $x \geq 0$, so it does not depend on z for any real number $u > 0$ by (1).

Now let $b \in \mathcal{P}$ be arbitrary, and fix $z \in Z_k$ over t and a real number $u > 0$. By Proposition 5.7(2) for $S = S_t$ and $D_j = \mathcal{B}_j|_{S_t}$, $\Psi_{S_t,B_t(b')}(F_z, u)$ is a continuous function of $b' \in \mathcal{P}$, and so is the affine function

$$a(F_z, S_t, B_t(b')) = \sum_i \lambda_i(b') a(F_z, S_t, B_t(b^{(i)}))$$

of b'. Since $\mathcal{P}$ is convex and the closure of $\mathcal{P}$ is a rational polytope [LZ26, Proposition 3.3], $\mathcal{P} \cap \mathbb{Q}^m$ is dense in $\mathcal{P}$. Since $a(F_z, S_t, B_t(b'))$ and $\Psi_{S_t,B_t(b')}(F_z, u)$ do not depend on z for $b' \in \mathcal{P} \cap \mathbb{Q}^m$, they do not depend on z for $b' = b$. This proves (3).

Proof of (4). Let $b \in \mathcal{P}$ and $t \in T$. By Definition 5.6, (2), and the surjectivity of each $Z_k \to T$, the formula in (4) holds for any choice of the points $z \in Z_k$ over t, and by (3) the right-hand side does not depend on the choice of z. By (1) and (3), the right-hand side does not depend on t. This proves (4). □

Set-up 5.13. Let d, T, $\mathcal{S} \to T$, $\mathcal{B}_1, \ldots, \mathcal{B}_m$, $\mathcal{P}$, W, Z_k, $\mathcal{G}_k$, and $\mathcal{C}_k$ be as follows.

(1) d, T, $\mathcal{S} \to T$, and $\mathcal{B}_1, \dots, \mathcal{B}_m$ satisfy conditions (a)–(c) of Construction-Theorem 5.12.
(2) Conclusions (1)–(4) of Construction-Theorem 5.12 hold for T.
(3) $\mathcal{P}$ and W are the sets in conclusion (1), and Z_k, $\mathcal{G}_k$, and $\mathcal{C}_k$ are the data in conclusion (2).

Theorem 5.14. *Let $n \geq 2$ be an integer and $\Gamma \subset [0,1]$ a DCC set. Then the set*

$$\left\{ \Phi(S,B) \;\middle|\; \begin{array}{l} (S,B) \text{ is a klt log Fano pair of dimension } n-1,\ B \in \Gamma, \text{ and} \\ -(K_S+B) \sim_{\mathbb{R}} rP \text{ for some ample Cartier divisor } P \text{ on } S \\ \text{and some } r \in \mathbb{R}_{>0} \end{array} \right\}$$

satisfies the ACC.

Proof. Suppose that there exist klt log Fano pairs (S_i, B_i) of dimension $n-1$ with $B_i \in \Gamma$, ample Cartier divisors P_i on S_i, and real numbers $r_i > 0$ such that for any $i \geq 1$, $-(K_{S_i}+B_i) \sim_{\mathbb{R}} r_i P_i$ and

$$\Phi(S_i, B_i) < \Phi(S_{i+1}, B_{i+1}).$$

Let $\epsilon := \Phi(S_1, B_1)$. Then $\epsilon > 0$ by Proposition 5.7(3), and $\Phi(S_i, B_i) \geq \epsilon$ for any i. Let $\gamma := \min(\Gamma \setminus \{0\})$ if $\Gamma \not\subset \{0\}$, which exists since Γ is a DCC set, and $\gamma := 1$ otherwise. Then every coefficient of B_i is $\geq \gamma$. By Lemma 5.11(3), there exist a projective morphism $\mathcal{S} \to T$ of schemes of finite type and a reduced divisor $\mathcal{B}$ on $\mathcal{S}$ such that for any i, $(S_i, \operatorname{Supp} B_i) \cong (S_{t_i}, \mathcal{B}|_{S_{t_i}})$ for some $t_i \in T$, where S_t denotes the fiber of $\mathcal{S} \to T$ over t.

Possibly replacing T with finitely many locally closed subvarieties and finite surjective covers of these, and passing to a subsequence, we may assume the following:

- T is a smooth variety, $\mathcal{S} \to T$ is flat, and every fiber S_t is a normal variety.
- $\mathcal{B} = \sum_{j=1}^m \mathcal{B}_j$, where each $\mathcal{B}_j$ is a relative Mumford divisor on $\mathcal{S} \to T$ [Kol23, Definition 4.68], each $\mathcal{B}_j|_{S_t}$ is a prime divisor, and $\mathcal{B}_1|_{S_t}, \dots, \mathcal{B}_m|_{S_t}$ are pairwise distinct for any $t \in T$.
- For any $t \in T$, there exist non-negative real numbers $c_1, \dots, c_m$ such that $(S_t, \sum_{j=1}^m c_j \mathcal{B}_j|_{S_t})$ is a klt log Fano pair. This is possible since every t_i belongs to the set of such t, which is constructible [LZ26, Proposition 4.1]. In particular, every S_t is of Fano type.
- $S_i = S_{t_i}$ and

$$B_i = B_{t_i}(b_i) := \sum_{j=1}^m b_{i,j} \mathcal{B}_j|_{S_{t_i}}$$

for some $b_i = (b_{i,1}, \dots, b_{i,m}) \in (\Gamma \cup \{0\})^m$.

Then the conditions (a)–(c) of Construction-Theorem 5.12 hold with $d = n-1$, since $\dim S_{t_1} = n-1$ and $\mathcal{S} \to T$ is flat over the variety T. By Construction-Theorem 5.12, possibly replacing T with one of the new T there which has points over infinitely many of the t_i, replacing the t_i by such points, and passing to a subsequence, we may assume that $d = n-1$, T, $\mathcal{S} \to T$, $\mathcal{B}_1, \dots, \mathcal{B}_m$, together with $\mathcal{P}$, W, Z_k, $\mathcal{G}_k$, and $\mathcal{C}_k$ of conclusions (1) and (2), satisfy Set-up 5.13.

Since $\Gamma \cup \{0\}$ is a DCC set, any sequence in $\Gamma \cup \{0\}$ has an increasing subsequence. Passing to a subsequence m times, we may assume that $b_{i,j} \leq b_{i+1,j}$ for any i and j. Since $(S_{t_i}, B_{t_i}(b_i))$ is a klt log Fano pair, $b_i \in \mathcal{P}$ for any i. Fix $t_* \in T$ and let $S_* := S_{t_*}$. By Construction-Theorem 5.12(4),

$$\Phi(S_{t_i}, B_{t_i}(b_i)) = \Phi(S_*, B_{t_*}(b_i))$$

for any i. Since $\mathcal{B}_1|_{S_*}, \dots, \mathcal{B}_m|_{S_*}$ are pairwise distinct prime divisors and $\mathcal{P}$ is the set of $b \in \mathbb{R}^m$ such that $(S_*, B_{t_*}(b))$ is a klt weak log Fano pair, Proposition 5.7(1) and $b_{i+1} - b_i \in \mathbb{R}^m_{\geq 0}$ imply

$$\Phi(S_{t_{i+1}}, B_{t_{i+1}}(b_{i+1})) = \Phi(S_*, B_{t_*}(b_{i+1})) \leq \Phi(S_*, B_{t_*}(b_i)) = \Phi(S_{t_i}, B_{t_i}(b_i))$$

for any i, which contradicts $\Phi(S_i, B_i) < \Phi(S_{i+1}, B_{i+1})$. □

6. Degeneration to the cone of a Kollár component

Let $(X \ni x, B)$ be a klt germ and E a Kollár component of $(X \ni x, B)$. In this section, $(X \ni x, B)$ degenerates along E to the cone C_E of E, a finite cyclic cover of a cone of Definition 5.1 over E (see Construction-Theorem 6.2).

Theorem 6.1. *Let $n \geq 2$ be an integer.*

(1) For any klt germ $(X \ni x, B)$ of dimension n and any Kollár component E of $(X \ni x, B)$,

$$\Phi(E, B_E) \leq \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{a(E, X, B)} \leq (-(K_E + B_E))^{n-1}.$$

(2) For any DCC set $\Gamma \subset [0, 1]$, the set of $\Phi(E, B_E)$, where E is a Kollár component of a klt germ $(X \ni x, B)$ of dimension n such that $B \in \Gamma$, satisfies the ACC.

In the proofs of Construction-Theorem 6.2 and Theorem 6.1(1), we write $B = \sum_i b_i B_i$, where the B_i are the distinct components of B and $b_i > 0$ for any i, let $b := (b_i)_i$, and let W be the set of real vectors $b' = (b'_i)_i$ such that $K_X + \sum_i b'_i B_i$ is $\mathbb{R}$-Cartier near x.

The left inequality of Theorem 6.1(1) depends on the following construction.

Construction-Theorem 6.2. Let n, $(X \ni x, B)$, E, f, and a be as follows.

(a) $n \geq 2$ is an integer and $(X \ni x, B)$ is a klt germ of dimension n.
(b) E is a Kollár component of $(X \ni x, B)$ with plt blow-up $f : Y \to X$, and $a := a(E, X, B)$.

Possibly shrinking X to an affine neighborhood of x, for $j \in \mathbb{Z}$ let

$$\mathfrak{a}_j := H^0(Y, \mathcal{O}_Y(-jE)) \subset H^0(X, \mathcal{O}_X),$$

let

$$\mathrm{gr}_E \mathcal{O}_{X,x} := \bigoplus_{j \geq 0} \mathfrak{a}_j / \mathfrak{a}_{j+1},$$

let $C_E := \operatorname{Spec} \mathrm{gr}_E \mathcal{O}_{X,x}$, and let $o \in C_E$ be the closed point given by $\bigoplus_{j>0} \mathfrak{a}_j/\mathfrak{a}_{j+1}$. Let

$$\mathcal{X} := \operatorname{Spec} \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j t^{-j} \to \mathbb{A}^1_t,$$

so that C_E is the fiber over 0, let $B_{\mathcal{X}}$ be the closure in $\mathcal{X}$ of $B \times (\mathbb{A}^1 \backslash \{0\})$, and let $B_{C_E} := B_{\mathcal{X}}|_{C_E}$ be the restriction of $B_{\mathcal{X}}$ to the Cartier divisor $C_E = (t = 0)$. Then:

(1) The ring $\mathrm{gr}_E \mathcal{O}_{X,x}$ is a finitely generated normal $\mathbb{C}$-algebra whose degree 0 part is $\mathbb{C}$, and the degrees of the non-zero homogeneous elements of $\mathrm{gr}_E \mathcal{O}_{X,x}$ generate $\mathbb{Z}$.
(2) The divisor $K_{\mathcal{X}} + B_{\mathcal{X}} + C_E$ is $\mathbb{R}$-Cartier, the pair $(\mathcal{X}, B_{\mathcal{X}} + C_E)$ is plt near C_E,

$$K_{C_E} + B_{C_E} = (K_{\mathcal{X}} + B_{\mathcal{X}} + C_E)|_{C_E},$$

and $(C_E \ni o, B_{C_E})$ is a klt germ.
(3) If q is a sufficiently divisible positive integer, then qE is Cartier near E, and $P_q := -qE|_E$ is an ample Cartier divisor on E with

$$-(K_E + B_E) \sim_{\mathbb{R}} \frac{a}{q} P_q.$$

For such q, the maps

$$\mathfrak{a}_j/\mathfrak{a}_{j+1} \to H^0(E, \mathcal{O}_E(\tfrac{j}{q} P_q)) \quad (j \in q\mathbb{N})$$

sending the class of ϕ to the restriction to E of $f^*\phi \in H^0(Y, \mathcal{O}_Y(-jE))$ form an isomorphism of graded $\mathbb{C}$-algebras

$$\bigoplus_{j \in q\mathbb{N}} \mathfrak{a}_j/\mathfrak{a}_{j+1} \cong \bigoplus_{j \in q\mathbb{N}} H^0(E, \mathcal{O}_E(\tfrac{j}{q} P_q)),$$

and the induced morphism $g : C_E \to C(E, P_q)$ is a finite cyclic Galois morphism of degree q such that g^{-1} of the vertex of $C(E, P_q)$ is $\{o\}$ and

$$K_{C_E} + B_{C_E} = g^*(K_{C(E,P_q)} + (B_E)_{C(E,P_q)}).$$

(4) $\widehat{\mathrm{vol}}(C_E \ni o, B_{C_E}) = a\Phi(E, B_E)$.

Proof. By Lemma 2.9(3), there exist rational vectors $b^{(1)}, \dots, b^{(N)} \in W$ and real numbers $\lambda_1, \dots, \lambda_N > 0$ satisfying the following.

(i) $\sum_{k=1}^N \lambda_k = 1$ and $\sum_{k=1}^N \lambda_k b^{(k)} = b$.

(ii) For any k, $B^{(k)} := \sum_i b_i^{(k)} B_i \geq 0$, $K_X + B^{(k)}$ is $\mathbb{Q}$-Cartier near x, and f is a plt blow-up of $(X \ni x, B^{(k)})$.

For any k, let $B_{\mathcal{X}}^{(k)}$ be the closure in $\mathcal{X}$ of $B^{(k)} \times (\mathbb{A}^1\backslash\{0\})$. Taking closures is linear in the coefficients, so

$$B = \sum_{k=1}^N \lambda_k B^{(k)} \quad \text{and} \quad B_{\mathcal{X}} = \sum_{k=1}^N \lambda_k B_{\mathcal{X}}^{(k)}. \tag{6.1}$$

(1) We have

$$\mathfrak{a}_j = \{\phi \in H^0(X, \mathcal{O}_X) \mid \mathrm{ord}_E \phi \geq j\}$$

for any $j \in \mathbb{Z}$. Let $\mathfrak{m}_x \subset H^0(X, \mathcal{O}_X)$ be the maximal ideal of x. Then $\mathfrak{a}_j = H^0(X, \mathcal{O}_X)$ for any $j \leq 0$, $\mathfrak{a}_1 = \mathfrak{m}_x$, and $\mathfrak{m}_x^j \subset \mathfrak{a}_j \subset \mathfrak{m}_x$ for any $j \geq 1$. Thus for any $j \geq 1$, $\mathfrak{a}_j$ is $\mathfrak{m}_x$-primary, $\mathfrak{a}_j \mathcal{O}_{X,x} = \mathfrak{a}_j(\mathrm{ord}_E)$, and

$$\mathfrak{a}_j/\mathfrak{a}_{j+1} = \mathfrak{a}_j(\mathrm{ord}_E)/\mathfrak{a}_{j+1}(\mathrm{ord}_E),$$

so $\mathrm{gr}_E \mathcal{O}_{X,x}$ depends only on the germ $X \ni x$ and on E. Moreover,

$$\mathfrak{a}_0/\mathfrak{a}_1 = H^0(X, \mathcal{O}_X)/\mathfrak{m}_x = \mathbb{C}.$$

For any $j \geq 0$, $\mathfrak{a}_j \neq \mathfrak{a}_{j+1}$ if and only if $j = \mathrm{ord}_E \phi$ for some $\phi \in H^0(X, \mathcal{O}_X)\backslash\{0\}$. Since X is affine, any non-zero rational function on X is ϕ_1/ϕ_2 for some $\phi_1, \phi_2 \in H^0(X, \mathcal{O}_X)\backslash\{0\}$, and ord_E maps the non-zero rational functions on X onto $\mathbb{Z}$. Thus the degrees j with $\mathfrak{a}_j \neq \mathfrak{a}_{j+1}$ generate $\mathbb{Z}$.

Let q_0 be a positive integer such that $q_0 E$ is Cartier. Since $\mathcal{O}_Y(-q_0 E)$ is an f-ample line bundle and X is affine, $\bigoplus_{j \in q_0\mathbb{N}} \mathfrak{a}_j$ is a finitely generated $H^0(X, \mathcal{O}_X)$-algebra, and $\bigoplus_{j \in j_0 + q_0\mathbb{N}} \mathfrak{a}_j$ is a finite module over it for any integer $0 \leq j_0 < q_0$. Thus $\bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j t^{-j}$ is a finitely generated $\mathbb{C}[t]$-algebra, which is torsion free, hence flat, over $\mathbb{C}[t]$. Since $t \cdot \mathfrak{a}_{j+1} t^{-j-1} = \mathfrak{a}_{j+1} t^{-j}$ for any j and $\mathfrak{a}_j = \mathfrak{a}_{j+1}$ for any $j < 0$, the fiber of $\mathcal{X}$ over 0 is C_E. In particular, $\mathrm{gr}_E \mathcal{O}_{X,x}$ is finitely generated. Since $\mathfrak{a}_j = H^0(X, \mathcal{O}_X)$ for any $j \leq 0$, the fiber of $\mathcal{X}$ over $\mathbb{A}^1\backslash\{0\}$ is $X \times (\mathbb{A}^1\backslash\{0\})$.

For any k, $(X \ni x, B^{(k)})$ is a klt germ with $\mathbb{Q}$-Cartier $K_X + B^{(k)}$ and Kollár component E, so $(\mathcal{X}, B_{\mathcal{X}}^{(k)} + C_E)$ is plt near C_E by [XZ25, Lemma 2.17]. Thus C_E is normal. Since C_E is normal and $\mathfrak{a}_0/\mathfrak{a}_1 = \mathbb{C}$, C_E is connected and $\mathrm{gr}_E \mathcal{O}_{X,x}$ is a normal domain. This proves (1).

(2) For any k, $K_{\mathcal{X}} + B_{\mathcal{X}}^{(k)} + C_E$ is $\mathbb{Q}$-Cartier near C_E and $a(F, \mathcal{X}, B_{\mathcal{X}}^{(k)} + C_E) > 0$ for any exceptional prime divisor F over $\mathcal{X}$ whose center on $\mathcal{X}$ intersects C_E. By (6.1),

$$K_{\mathcal{X}} + B_{\mathcal{X}} + C_E = \sum_{k=1}^N \lambda_k (K_{\mathcal{X}} + B_{\mathcal{X}}^{(k)} + C_E)$$

is $\mathbb{R}$-Cartier near C_E, hence on $\mathcal{X}$, since $\mathcal{X}\backslash C_E$ is $X \times (\mathbb{A}^1\backslash\{0\})$, on which $K_{\mathcal{X}} + B_{\mathcal{X}} + C_E$ is the pullback of $K_X + B$, and $K_X + B$ is $\mathbb{R}$-Cartier after shrinking X. For any such F,

$$a(F, \mathcal{X}, B_{\mathcal{X}} + C_E) = \sum_{k=1}^N \lambda_k a(F, \mathcal{X}, B_{\mathcal{X}}^{(k)} + C_E) > 0.$$

Since the coefficients of $B_{\mathcal{X}}$ are the $b_i \in (0,1)$, $(\mathcal{X}, B_{\mathcal{X}} + C_E)$ is plt near C_E. Since C_E is a normal Cartier divisor, $\mathcal{X}$ is smooth at the generic point of any prime divisor on C_E, so

$$(K_{\mathcal{X}} + B_{\mathcal{X}} + C_E)|_{C_E} = K_{C_E} + B_{\mathcal{X}}|_{C_E} = K_{C_E} + B_{C_E}.$$

Thus (C_E, B_{C_E}) is klt by [HLS19, Theorem 3.10(2)], and $(C_E \ni o, B_{C_E})$ is a klt germ.

(3) Since $\mathcal{O}_Y(-q_0E)$ is f-ample and X is affine, by Serre vanishing, for any sufficiently divisible positive integer q and any $j \in q\mathbb{N}$, restriction to E induces an isomorphism

$$\mathfrak{a}_j/\mathfrak{a}_{j+1} \cong H^0(E, \mathcal{O}_E(\tfrac{j}{q}P_q)), \tag{6.2}$$

where $P_q := -qE|_E$ (for $j = 0$, both sides are $\mathbb{C}$). The restriction to E is multiplicative, so the isomorphisms (6.2) for $j \in q\mathbb{N}$ form the isomorphism of graded $\mathbb{C}$-algebras in (3). Since $-E$ is f-ample and $f(E) = x$, P_q is an ample Cartier divisor on E, and $C(E, P_q)$ is the cone of Definition 5.1. Possibly shrinking X to an affine neighborhood of x, we may assume that $K_X + B \sim_{\mathbb{R}} 0$. Since

$$K_Y + E + B_Y = f^*(K_X + B) + aE,$$

restricting to E we obtain

$$-(K_E + B_E) \sim_{\mathbb{R}} -aE|_E = \frac{a}{q}P_q.$$

Let o_q be the vertex of $C(E, P_q)$.

Let μ_q act on $\operatorname{gr}_E \mathcal{O}_{X,x}$ by $\xi \cdot \overline{\phi} = \xi^j \overline{\phi}$ for any $\xi \in \mu_q$ and any homogeneous $\overline{\phi}$ of degree j. The ring of invariants is

$$\bigoplus_{j \in q\mathbb{N}} \mathfrak{a}_j/\mathfrak{a}_{j+1},$$

so g is the quotient morphism by μ_q, and g is finite. If $\xi \in \mu_q$ acts trivially, then $\xi^j = 1$ for any j with $\mathfrak{a}_j \neq \mathfrak{a}_{j+1}$, so $\xi = 1$ by (1). Thus μ_q acts faithfully, and g is a finite cyclic Galois morphism of degree q. If $z \in g^{-1}(o_q)$, then $\overline{\phi}^q(z) = 0$ for any homogeneous $\overline{\phi}$ of positive degree, so $\overline{\phi}(z) = 0$ and $z = o$. Thus $g^{-1}(o_q) = \{o\}$.

By the Hurwitz formula, $K_{C_E} = g^*K_{C(E,P_q)} + R(g)$, where $R(g)$ is the ramification divisor of g. The gradings define $\mathbb{G}_m$-actions on C_E and on $C(E, P_q)$ for which g is equivariant, and $B_{\mathcal{X}}$, hence B_{C_E}, is $\mathbb{G}_m$-invariant. Since $\mathbb{G}_m$ is connected, any component of $R(g)$, of B_{C_E}, or of $(B_E)_{C(E,P_q)}$ is $\mathbb{G}_m$-invariant. Since $C(E, P_q)\backslash\{o_q\}$ is the complement of the zero section in $\operatorname{Tot}(\mathcal{O}_E(-P_q))$, any $\mathbb{G}_m$-invariant prime divisor on $C(E, P_q)$ is $D_{C(E,P_q)}$ for a prime divisor D on E, and $\operatorname{mult}_{D_{C(E,P_q)}}(B_E)_{C(E,P_q)} = \operatorname{mult}_D B_E$. For a prime divisor Q on C_E, let e_Q be the ramification index of g along Q. It suffices to show that

$$\operatorname{mult}_D B_E = 1 - \frac{1 - \operatorname{mult}_Q B_{C_E}}{e_Q} \tag{6.3}$$

for any prime divisor D on E and any prime divisor Q on C_E with $g(Q) = D_{C(E,P_q)}$: then the coefficients of $g^*(B_E)_{C(E,P_q)}$ and of $R(g) + B_{C_E}$ along Q are $e_Q \operatorname{mult}_D B_E$ and $e_Q - 1 + \operatorname{mult}_Q B_{C_E}$, which are equal.

Fix D, and let $m := |\operatorname{Cl}(\mathcal{O}_{Y,D})|$, where $\mathcal{O}_{Y,D}$ is the local ring of Y at the generic point of D. Étale locally near the generic point of D, (Y, E) is $(\mathbb{A}^2_{u,y}/\mu_m, (y = 0))$ times a smooth factor, where we work over the generic point of D, so that the smooth factor is the spectrum of a field. Here $\zeta \cdot (u, y) = (\zeta u, \zeta^c y)$ for any $\zeta \in \mu_m$ and some integer c coprime to m, μ_m acts freely outside the origin, and D is the image of the origin. Since E generates the group $\operatorname{Cl}(\mathcal{O}_{Y,D})$ of order m and qE is Cartier, $m \mid q$. We have $B_Y = \sum_i b_i f_*^{-1} B_i$. For any i, let $(h_i = 0)$ be the inverse image of $f_*^{-1}B_i$ in $\mathbb{A}^2_{u,y}$. Since $E \not\subset f_*^{-1}B_i$, $h_i(u, 0) \neq 0$, and

$$(h_i(u, 0) = 0) = w_i \cdot (u = 0)$$

near the origin of $\mathbb{A}^1_u$ for some $w_i \in \mathbb{N}$. The morphism $\mathbb{A}^2_{u,y} \to \mathbb{A}^2_{u,y}/\mu_m$ is étale outside the origin, and $(y=0) = \mathbb{A}^1_u \to E$ has ramification index m along $(u=0)$. Restricting the pullback of $K_Y + E + B_Y$ to $(y=0)$, we obtain

$$\operatorname{mult}_D B_E = 1 - \frac{1 - \sum_i w_i b_i}{m} = \frac{m - 1 + \sum_i w_i b_i}{m}.$$

By [LX20, Section 2.4], $C_E \backslash \{o\}$ is the complement of the zero section in the relative Spec over E of $\bigoplus_{j\geq 0} \mathcal{O}_Y(-jE)/\mathcal{O}_Y(-(j+1)E)$, and B_{C_E} is the intersection of C_E with the closure of $B_Y \times (\mathbb{A}^1 \backslash \{0\})$ in the deformation to the normal cone. Étale locally near the generic point of D, the algebra $\bigoplus_{j\geq 0} \mathcal{O}_Y(-jE)/\mathcal{O}_Y(-(j+1)E)$ is $\mathbb{C}[u,y]^{\mu_m}$ graded by the degree in y, and the part of h_i of degree 0 in y is $h_i(u,0) \neq 0$. Thus étale locally near the generic point of any prime divisor Q on C_E with $g(Q) = D_{C(E,P_q)}$, C_E is $\mathbb{A}^2_{u,y}/\mu_m$ times a smooth factor, where y has degree 1 and u has degree 0, Q is the image of $(u=0)$, and B_{C_E} is the image of $\sum_i b_i (h_i(u,0) = 0)$. Thus $\operatorname{mult}_Q B_{C_E} = \sum_i w_i b_i$. The group μ_q acts by $\xi \cdot (u,y) = (u, \xi y)$. Since $m \mid q$, the group generated by μ_m and μ_q is $\mu_m \times \mu_q$ acting by $(\zeta, \xi) \cdot (u,y) = (\zeta u, \xi y)$, and the quotient is $\mathbb{A}^2_{u^m, y^q}$. Thus the composite morphism $\mathbb{A}^2_{u,y} \to C_E \to C(E, P_q)$ has ramification index m along $(u=0)$, and since $\mathbb{A}^2_{u,y} \to C_E$ is étale in codimension 1, $e_Q = m$. Thus

$$1 - \frac{1 - \operatorname{mult}_Q B_{C_E}}{e_Q} = 1 - \frac{1 - \sum_i w_i b_i}{m} = \operatorname{mult}_D B_E,$$

which is (6.3). This proves (3).

(4) Since $a > 0$, by (3) and Lemma 5.2, $(C(E,P_q) \ni o_q, (B_E)_{C(E,P_q)})$ is a klt germ. By (2), (3), and Lemma 2.10(1),

$$\widehat{\operatorname{vol}}(C_E \ni o, B_{C_E}) = q \widehat{\operatorname{vol}}(C(E,P_q) \ni o_q, (B_E)_{C(E,P_q)}).$$

By (3) and Theorem 5.9,

$$\widehat{\operatorname{vol}}(C(E,P_q) \ni o_q, (B_E)_{C(E,P_q)}) = \frac{a}{q} \Phi(E, B_E).$$

Thus $\widehat{\operatorname{vol}}(C_E \ni o, B_{C_E}) = a\Phi(E, B_E)$. □

Definition-Lemma 6.3. For a set $\Gamma \subset [0,1]$, let

$$D(\Gamma) := \left\{ \frac{m - 1 + \sum_{i=1}^k \gamma_i}{m} \;\middle|\; m \in \mathbb{N}^+,\ k \in \mathbb{N},\ \gamma_1, \dots, \gamma_k \in \Gamma \right\} \cap [0,1].$$

If Γ is a DCC set, then $D(\Gamma)$ is a DCC set. Moreover, if E is a Kollár component of a klt germ $(X \ni x, B)$ with $B \in \Gamma$, then $B_E \in D(\Gamma)$.

Proof. If Γ is a DCC set, then $D(\Gamma)$ is a DCC set by [HMX14, Proposition 3.4.1(3)].

Let $f : Y \to X$ be the plt blow-up with exceptional divisor E, and write $B = \sum_i b_i B_i$, where the B_i are distinct prime divisors and $b_i \in \Gamma$, so that $B_Y = \sum_i b_i f_*^{-1} B_i$. Possibly shrinking Y to a neighborhood of E, we may assume that $(Y, E + B_Y)$ is plt. Let P be a prime divisor on E. By [HLS19, Theorem 3.10(3)], there exist $m \in \mathbb{N}^+$ and $e_i \in \mathbb{N}$ such that

$$\operatorname{mult}_P B_E = \frac{m - 1 + \sum_i e_i b_i}{m}.$$

Since (E, B_E) is klt, $\operatorname{mult}_P B_E < 1$. Thus $\operatorname{mult}_P B_E \in D(\Gamma)$, where $\gamma_1, \dots, \gamma_k$ are the b_i, each b_i repeated e_i times. Hence $B_E \in D(\Gamma)$. □

Proof of Theorem 6.1. (1) Let $f : Y \to X$ be the plt blow-up of E. Possibly shrinking X to an affine neighborhood of x, let C_E, o, $\mathcal{X}$, and B_{C_E} be as in Construction-Theorem 6.2. By Construction-Theorem 6.2(4), the left inequality is equivalent to

$$\widehat{\operatorname{vol}}(C_E \ni o, B_{C_E}) \leq \widehat{\operatorname{vol}}(X \ni x, B).$$

Let W_f be the set of $b' \in W$ such that $\sum_i b'_i B_i \geq 0$ and f is a plt blow-up of $(X \ni x, \sum_i b'_i B_i)$. By Lemma 2.9(1)(3), W is an affine subspace defined over $\mathbb{Q}$ and W_f is relatively open in $\{b' \in W \mid \sum_i b'_i B_i \geq 0\}$. Since $b \in W_f$ and $b_i > 0$ for any i, there exist rational vectors $b'_1, b'_2, \cdots \in W_f$ with $\lim_{k\to+\infty} b'_k = b$. For any k, let $B'_k := \sum_i b'_{k,i} B_i$, where $b'_k = (b'_{k,i})_i$, and let $(B'_k)_{C_E}$ be the restriction to C_E of the closure in $\mathcal{X}$ of $B'_k \times (\mathbb{A}^1 \backslash \{0\})$.

For any k, $(X \ni x, B'_k)$ is a klt germ with $\mathbb{Q}$-Cartier $K_X + B'_k$, E is a Kollár component of $(X \ni x, B'_k)$, and the degeneration of $(X \ni x, B'_k)$ induced by E is $(C_E \ni o, (B'_k)_{C_E})$ [LX20, Section 2.4]. Thus by [LX20, Proposition 4.8],

$$\widehat{\mathrm{vol}}(C_E \ni o, (B'_k)_{C_E}) \leq \widehat{\mathrm{vol}}(X \ni x, B'_k).$$

By Construction-Theorem 6.2(2) for $(X \ni x, B'_k)$ and for $(X \ni x, B)$, $K_{C_E} + (B'_k)_{C_E}$ is $\mathbb{R}$-Cartier and $(C_E \ni o, B_{C_E})$ is a klt germ. The coefficients of $(B'_k)_{C_E}$ along the prime divisors in $\operatorname{Supp} B_{C_E}$ are linear functions of b'_k, so they converge to the coefficients of B_{C_E}. Let $0 < \epsilon < 1$ be a real number. By Lemma 2.9(2) for the prime divisors in $\operatorname{Supp} B$ on X and for the prime divisors in $\operatorname{Supp} B_{C_E}$ on C_E, for any $k \gg 0$,

$$(1-\epsilon)^n \widehat{\mathrm{vol}}(C_E \ni o, B_{C_E}) \leq \widehat{\mathrm{vol}}(C_E \ni o, (B'_k)_{C_E}) \leq \widehat{\mathrm{vol}}(X \ni x, B'_k) \leq (1+\epsilon)^n \widehat{\mathrm{vol}}(X \ni x, B).$$

Letting $\epsilon \to 0$, we obtain the left inequality. By Lemma 2.8(2)(3),

$$\widehat{\mathrm{vol}}(X \ni x, B) \leq \widehat{\mathrm{vol}}(E, X, B) = a(E, X, B) \cdot (-(K_E + B_E))^{n-1}.$$

Since $a(E, X, B) > 0$, the right inequality follows.

(2) Let E be a Kollár component of a klt germ $(X \ni x, B)$ of dimension n such that $B \in \Gamma$, let $f : Y \to X$ be the plt blow-up of E, and let q and P_q be as in Construction-Theorem 6.2(3). By Definition-Lemma 6.3, $B_E \in D(\Gamma)$, and $D(\Gamma) \subset [0,1]$ is a DCC set. By Construction-Theorem 6.2(3), $-(K_E + B_E) \sim_{\mathbb{R}} \frac{a(E,X,B)}{q} P_q$, where P_q is an ample Cartier divisor on E and $\frac{a(E,X,B)}{q} > 0$. Thus (E, B_E) is a klt log Fano pair of dimension $n-1$ with $B_E \in D(\Gamma)$, and $\Phi(E, B_E)$ belongs to the set of Theorem 5.14 for the DCC set $D(\Gamma)$. That set satisfies the ACC by Theorem 5.14. □

7. Finite approximation of Φ

Theorem 7.1. *Notation and conditions as in Set-up 5.13. Let $\mathcal{P}_0 \subset \mathcal{P}$ be a compact set and $\epsilon > 0$ a real number. Then, after replacing T with the members of a finite stratification into locally closed subvarieties and then with finite surjective covers of these, there exist the following.*

- *(i) A smooth variety $\mathcal{Z}$ with a projective morphism $\mathcal{Z} \to \mathcal{S}$ such that $\mathcal{Z} \to T$ is smooth and $\mathcal{Z}_t \to S_t$ is birational for any $t \in T$.*
- *(ii) A finite set J.*
- *(iii) For any $j \in J$, a prime divisor $\mathcal{F}_j$ on $\mathcal{Z}$, a $\mathbb{Q}$-linear combination $\mathcal{G}_j \geq 0$ of relative Mumford divisors on $\mathcal{S} \to T$, and a positive rational number u_j.*

Let $F_{j,t} := \mathcal{F}_j|_{\mathcal{Z}_t}$ for $t \in T$ and $j \in J$. Then the following hold for any $t \in T$ and $j \in J$.

- *(1) The divisor $F_{j,t}$ is prime, $(S_t, \mathcal{G}_j|_{S_t})$ is lc, $N_d(K_{S_t} + \mathcal{G}_j|_{S_t}) \sim 0$, and $a(F_{j,t}, S_t, \mathcal{G}_j|_{S_t}) = 0$.*
- *(2) For any $b \in \mathcal{P}$, the numbers $a(F_{j,t}, S_t, B_t(b))$ and $\Psi_{S_t, B_t(b)}(F_{j,t}, u_j)$ do not depend on t.*
- *(3) For any $b \in \mathcal{P}_0$,*

$$0 \leq \min\left(\{(-(K_{S_t} + B_t(b)))^d\} \cup \{\Psi_{S_t,B_t(b)}(F_{j,t}, u_j) \mid j \in J\}\right) - \Phi(S_t, B_t(b)) < \epsilon.$$

Proof. Replacing T as in the theorem does not change the fibers S_t, so it does not change $\mathcal{P}$ or the numbers in conclusions (1)–(4) of Construction-Theorem 5.12. In the proof, we write $\mathcal{G}^0_k$ for $\mathcal{G}_k$ of Set-up 5.13 and $\mathcal{G}^0_{k,z}$ for the fiber of $\mathcal{G}^0_k$ over $z \in Z_k$.

Step 1. We find finitely many candidates which approximate Φ on $\mathcal{P}_0$. For any k, $F \in \mathcal{C}_k$, $b \in \mathcal{P}$, and $u \in \mathbb{R}_{>0}$, let

$$\psi_{k,F,u}(b) := \Psi_{S_t, B_t(b)}(F_z, u),$$

where $z \in Z_k$ is over $t \in T$. By Construction-Theorem 5.12(3), $\psi_{k,F,u}(b)$ does not depend on z. By Construction-Theorem 5.12(1)(4), for any $b \in \mathcal{P}$ the numbers

$$\Psi_0(b) := (-(K_{S_t} + B_t(b)))^d \quad \text{and} \quad \phi(b) := \Phi(S_t, B_t(b))$$

do not depend on $t \in T$. By Construction-Theorem 5.12(4), for any $b \in \mathcal{P}$,

$$\phi(b) = \inf\left(\{\Psi_0(b)\} \cup \{\psi_{k,F,u}(b) \mid k,\ F \in \mathcal{C}_k,\ u \in \mathbb{R}_{>0}\}\right). \tag{7.1}$$

Fix $t_0 \in T$. By Proposition 5.7(2)(3) for $S = S_{t_0}$, $D_j = \mathcal{B}_j|_{S_{t_0}}$, and $n = d+1$, $\psi_{k,F,u}(b)$ is a continuous function of $(b,u) \in \mathcal{P} \times \mathbb{R}_{>0}$ for any k and $F \in \mathcal{C}_k$, and ϕ is a positive continuous function on $\mathcal{P}$. By Construction-Theorem 5.12(1), W is empty or an affine subspace of $\mathbb{R}^m$, and $\mathcal{P} \subset W$. Since intersection numbers of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors are multilinear, Ψ_0 is the restriction to $\mathcal{P}$ of a polynomial function on W. In particular, Ψ_0 is continuous on $\mathcal{P}$.

Let $b \in \mathcal{P}_0$. By (7.1) and the continuity of $\psi_{k,F,u}(b)$ in u, there exists a function

$$g_b \in \{\Psi_0\} \cup \{\psi_{k,F,u} \mid k,\ F \in \mathcal{C}_k,\ u \in \mathbb{Q}_{>0}\}$$

such that $g_b(b) - \phi(b) < \epsilon$. Since g_b and ϕ are continuous on $\mathcal{P}$, the set

$$U_b := \{b' \in \mathcal{P}_0 \mid g_b(b') - \phi(b') < \epsilon\}$$

is an open neighborhood of b in $\mathcal{P}_0$. Since $\mathcal{P}_0$ is compact, there exists a finite set $\mathcal{P}_1 \subset \mathcal{P}_0$ such that $\mathcal{P}_0 = \bigcup_{b \in \mathcal{P}_1} U_b$. Thus there exist a finite set J and, for any $j \in J$, an index k_j, an element $F'_j \in \mathcal{C}_{k_j}$, and $u_j \in \mathbb{Q}_{>0}$, such that

$$\{g_b \mid b \in \mathcal{P}_1\} \subset \{\Psi_0\} \cup \{\psi_j \mid j \in J\}, \quad \text{where } \psi_j := \psi_{k_j,F'_j,u_j}.$$

By (7.1) and $\mathcal{P}_0 = \bigcup_{b \in \mathcal{P}_1} U_b$, for any $b \in \mathcal{P}_0$,

$$0 \le \min\left(\{\Psi_0(b)\} \cup \{\psi_j(b) \mid j \in J\}\right) - \phi(b) < \epsilon. \tag{7.2}$$

Step 2. We reduce the theorem to a statement over a non-empty open subset of any irreducible closed subset of T. For a morphism $T' \to T$ from a variety T', let $\mathcal{S}_{T'} := \mathcal{S} \times_T T'$, and for $t \in T'$ let S_t be the fiber of $\mathcal{S}_{T'} \to T'$ over t, which is the fiber of $\mathcal{S} \to T$ over the image of t. By Noetherian induction, it suffices to prove the following.

($*$) Let $T_1 \subset T$ be an irreducible closed subset. Then there exist

- a non-empty open subset $T_1^\circ \subset T_1$ and a finite surjective morphism $T' \to T_1^\circ$ from a smooth variety T',
- for any $j \in J$, a morphism $\sigma_j : T' \to Z_{k_j}$ over T,
- a smooth variety $\mathcal{Z}$ with a projective morphism $\mathcal{Z} \to \mathcal{S}_{T'}$, and
- for any $j \in J$, a prime divisor $\mathcal{F}_j$ on $\mathcal{Z}$ and a $\mathbb{Q}$-linear combination $\mathcal{G}_j \ge 0$ of relative Mumford divisors on $\mathcal{S}_{T'} \to T'$,

such that the following hold for any $t \in T'$ and $j \in J$.

(a) $\mathcal{Z} \to T'$ is smooth and $\mathcal{Z}_t \to S_t$ is birational.
(b) $F_{j,t} := \mathcal{F}_j|_{\mathcal{Z}_t}$ is a prime divisor, and $F_{j,t} = F'_{j,\sigma_j(t)}$ as prime divisors over S_t.
(c) $\mathcal{G}_j|_{S_t} = \mathcal{G}^0_{k_j,\sigma_j(t)}$.

We show that the data in ($*$) satisfy the conclusions of the theorem for $\mathcal{S}_{T'} \to T'$. Let $t \in T'$, $j \in J$, and $z := \sigma_j(t)$. Since $F'_j \in \mathcal{C}_{k_j}$, by (c) and Construction-Theorem 5.12(2), $(S_t, \mathcal{G}_j|_{S_t})$ is lc, $N_d(K_{S_t} + \mathcal{G}_j|_{S_t}) \sim 0$, and $a(F'_{j,z}, S_t, \mathcal{G}_j|_{S_t}) = 0$. Together with (b), we obtain (1). By (b), for any $b \in \mathcal{P}$,

$$a(F_{j,t}, S_t, B_t(b)) = a(F'_{j,z}, S_t, B_t(b)) \quad \text{and} \quad \Psi_{S_t,B_t(b)}(F_{j,t}, u_j) = \psi_j(b).$$

Thus (2) follows from Construction-Theorem 5.12(3), and (3) follows from (7.2).

Step 3. We construct T_1°, T', and σ_j. Since $Z_{k_j} \to T$ is surjective for any $j \in J$, there exist a non-empty open subset $T_1^\circ \subset T_1$, a finite surjective morphism $T' \to T_1^\circ$ from a smooth variety T',

and morphisms $\sigma_j : T' \to Z_{k_j}$ over T for $j \in J$. Below, shrinking T' means replacing T_1° with a non-empty open subset and T' with its preimage.

Step 4. We construct $\mathcal{Z}$, $\mathcal{F}_j$, and $\mathcal{G}_j$, and prove (a)(b)(c). For $j \in J$, let

$$\mathcal{Y}_j := Y_{F'_j} \times_{Z_{k_j}} T' \quad \text{and} \quad F'_{j,T'} := F'_j \times_{Z_{k_j}} T',$$

where $T' \to Z_{k_j}$ is σ_j, and let $\mathcal{G}_j$ be the pullback of $\mathcal{G}^0_{k_j}$ under the projection

$$\mathcal{S}_{T'} = (\mathcal{S} \times_T Z_{k_j}) \times_{Z_{k_j}} T' \to \mathcal{S} \times_T Z_{k_j}.$$

Since restrictions of relative Mumford divisors to fibers are compatible with base change [Kol23, Definition 4.68], $\mathcal{G}_j$ is a $\mathbb{Q}$-linear combination ≥ 0 of relative Mumford divisors on $\mathcal{S}_{T'} \to T'$ satisfying (c). By Construction-Theorem 5.12(2), $\mathcal{Y}_j \to \mathcal{S}_{T'}$ is projective, $\mathcal{Y}_j \to T'$ and $F'_{j,T'} \to T'$ are smooth, and for any $t \in T'$, the morphism $\mathcal{Y}_{j,t} = Y_{F'_j,\sigma_j(t)} \to S_t$ is birational and the fiber of $F'_{j,T'}$ over t is the prime divisor $F'_{j,\sigma_j(t)}$ on $\mathcal{Y}_{j,t}$.

Since $\mathcal{Y}_{j,t} \to S_t$ is birational for any $t \in T'$, $\mathcal{Y}_j \to \mathcal{S}_{T'}$ is birational. Let $\mathcal{Z}$ be a resolution of the main component of the fiber product of the $\mathcal{Y}_j$, $j \in J$, over $\mathcal{S}_{T'}$ (with $\mathcal{Z} :=$ a resolution of $\mathcal{S}_{T'}$ if $J = \emptyset$), let $h_j : \mathcal{Z} \to \mathcal{Y}_j$ be the induced morphism, and let $\mathcal{F}_j := h_{j*}^{-1} F'_{j,T'}$. By generic smoothness, possibly shrinking T', we may assume that $\mathcal{Z} \to T'$ is smooth, $\mathcal{Z}_t \to S_t$ is birational, and $\mathcal{F}_j|_{\mathcal{Z}_t}$ is the strict transform of the prime divisor $F'_{j,\sigma_j(t)}$ for any $t \in T'$ and $j \in J$, which are (a) and (b). This proves $(*)$, and the theorem follows. □

Lemma 7.2. *Let n, ϵ, κ, and C be as in Set-up 4.8(1), and $\Gamma \subset [0,1]$ a DCC set. For any positive integer i, let $(X_i \ni x_i, B_i)$ and E_i satisfy Set-up 4.8(2)(3) in place of $(X \ni x, B)$ and E. Assume that $B_i \in \Gamma$ for any i. Then, possibly passing to a subsequence, there exist the following.*

(1) $\mathcal{S} \to T$ and $\mathcal{B}_1, \dots, \mathcal{B}_m$ satisfying the conditions of Set-up 5.13 with $d = n - 1$.

(2) For any i, a closed point $t_i \in T$ and a vector $b_i \in \mathcal{P}$ such that $E_i \cong S_{t_i}$ by an isomorphism mapping B_{E_i} onto $B_{t_i}(b_i)$.

Moreover, the limit $b_\infty := \lim_{i \to +\infty} b_i$ exists, and $b_\infty \in \mathcal{P}$.

Proof. Let $d := n - 1$, and let θ be as in Set-up 4.8(1). By Definition 2.5, (E_i, B_{E_i}) is a klt log Fano pair.

We show that the pairs $(E_i, \operatorname{Supp} B_{E_i})$ are log bounded. Let

$$\gamma := \min\left(\{\tfrac{1}{2}\} \cup (\Gamma \setminus \{0\})\right),$$

which is positive since Γ is a DCC set. By Definition-Lemma 6.3, $B_{E_i} \in D(\Gamma)$, and every non-zero element of $D(\Gamma)$ is at least γ, so

$$B_{E_i} \geq \gamma \operatorname{Supp} B_{E_i} \quad \text{for any } i.$$

By Proposition 4.9(4), there exists a positive integer N such that any E_i is isomorphic to a closed subvariety of $\mathbb{P}^N$ of degree at most N. Thus, by Lemma 5.10, the couples $(E_i, \operatorname{Supp} B_{E_i})$ form a log bounded set, that is, there exist a projective morphism $\mathcal{S} \to T$ of schemes of finite type, a reduced divisor $\mathcal{D}$ on $\mathcal{S}$, and, for any i, a closed point $t_i \in T$ and an isomorphism $E_i \cong S_{t_i}$ which maps $\operatorname{Supp} B_{E_i}$ onto $\mathcal{D}|_{S_{t_i}}$.

Possibly replacing T with the members of a finite stratification into smooth locally closed subvarieties and then with finite surjective covers of the strata, and passing to a subsequence, we may assume that T is a smooth variety, $\mathcal{S} \to T$ is flat with normal fibers, and there exist relative Mumford divisors $\mathcal{B}_1, \dots, \mathcal{B}_m \geq 0$ on $\mathcal{S} \to T$ such that for any $t \in T$, the divisors $\mathcal{B}_1|_{S_t}, \dots, \mathcal{B}_m|_{S_t}$ are pairwise distinct prime divisors and $\mathcal{D} \cap S_t = \bigcup_{j=1}^m \operatorname{Supp} \mathcal{B}_j|_{S_t}$. Let $B_t(b) := \sum_{j=1}^m b_j \mathcal{B}_j|_{S_t}$ for $b \in \mathbb{R}^m$ and $t \in T$, and let $b_i \in \mathbb{R}^m$ be the vector such that the isomorphism $E_i \cong S_{t_i}$ maps B_{E_i} onto $B_{t_i}(b_i)$. Thus $b_i \in D(\Gamma)^m$.

By [LZ26, Proposition 4.1], the set T_0 of $t \in T$ such that $(S_t, \sum_{j=1}^m c_j \mathcal{B}_j|_{S_t})$ is a klt weak log Fano pair for some $c \in \mathbb{R}^m_{\geq 0}$ is an algebraic subset of T, and $t_i \in T_0$ for any i by taking $c = b_i$. Possibly passing to a subsequence and replacing T with an irreducible locally closed subset of T_0

which contains infinitely many t_i, we may assume that $T_0 = T$. Then every S_t is of Fano type, and by Construction-Theorem 5.12, possibly replacing T with a stratum and a finite surjective cover of it and passing to a subsequence, we may assume that $\mathcal{S} \to T$ and $\mathcal{B}_1, \dots, \mathcal{B}_m$ satisfy the conditions of Set-up 5.13. Since $(S_{t_i}, B_{t_i}(b_i)) \cong (E_i, B_{E_i})$ is a klt log Fano pair and $\mathcal{P}$ does not depend on t, $b_i \in \mathcal{P}$ for any i.

Since $D(\Gamma)$ is a DCC set by Definition-Lemma 6.3, possibly passing to a subsequence m times, we may assume that for any j, the sequence of the j-th coordinates of the b_i is increasing. Since $D(\Gamma) \subset [0,1]$, the limit $b_\infty := \lim_{i\to+\infty} b_i$ exists and $b_\infty \in [0,1]^m$. In particular, $B_t(b_\infty) \geq 0$ for any $t \in T$.

It remains to show $b_\infty \in \mathcal{P}$. Let W and $g : \mathcal{S}' \to \mathcal{S}$ be as in Construction-Theorem 5.12(1). By Construction-Theorem 5.12(1), W is an affine subspace of $\mathbb{R}^m$ containing $\mathcal{P}$, so $b_\infty \in W$. Let $\mathcal{F} \in \mathcal{E}$, where $\mathcal{E}$ is as in Construction-Theorem 5.12(1), and let

$$a_{\mathcal{F}}(b) := a(\mathcal{F}_t, S_t, B_t(b)) \quad \text{for } b \in W,$$

which is an affine function of b and does not depend on $t \in T$. By Proposition 4.9(2) for $(X_i \ni x_i, B_i)$ and E_i, $a_{\mathcal{F}}(b_i) \geq \theta$ for any i, so $a_{\mathcal{F}}(b_\infty) \geq \theta$. Let $t \in T$. By Construction-Theorem 5.12(1), the log discrepancies of $(S_t, B_t(b_\infty))$ at the prime divisors $\mathcal{F}_t$, $\mathcal{F} \in \mathcal{E}$, which are the exceptional prime divisors of g_t and the strict transforms of $\mathcal{B}_1|_{S_t}, \dots, \mathcal{B}_m|_{S_t}$, are the numbers $a_{\mathcal{F}}(b_\infty) \geq \theta > 0$. Since g_t is a log resolution of $(S_t, \sum_{j=1}^m \operatorname{Supp} \mathcal{B}_j|_{S_t})$ and

$$\operatorname{Supp} B_t(b_\infty) \subset \bigcup_{j=1}^m \operatorname{Supp} \mathcal{B}_j|_{S_t},$$

it is also a log resolution of $(S_t, B_t(b_\infty))$, so $(S_t, B_t(b_\infty))$ is klt.

Let $t \in T$. Since $b_i \in \mathcal{P}$, $-(K_{S_t} + B_t(b_i))$ is nef for any i. Since the class of $K_{S_t} + B_t(b)$ in $N^1(S_t)_{\mathbb{R}}$ is affine in $b \in W$ and the nef cone is closed, $-(K_{S_t} + B_t(b_\infty))$ is nef. Moreover, since intersection numbers of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors are multilinear, $(-(K_{S_t} + B_t(b)))^d$ is a polynomial function of $b \in W$ for each t, and its values at $b \in \mathcal{P}$ do not depend on t by Construction-Theorem 5.12(1). By Proposition 4.9(1),

$$(-(K_{S_t} + B_t(b_\infty)))^d = \lim_{i\to+\infty} (-(K_{S_{t_i}} + B_{t_i}(b_i)))^d = \lim_{i\to+\infty} (-(K_{E_i} + B_{E_i}))^d \geq \frac{\epsilon}{C} > 0.$$

Thus $-(K_{S_t} + B_t(b_\infty))$ is nef and big [Laz04, Theorem 2.2.16]. Hence $(S_t, B_t(b_\infty))$ is a klt weak log Fano pair, and $b_\infty \in \mathcal{P}$. The lemma follows. □

8. Finitely many families of stacks with faithful line bundles

Definition 8.1. Let $\mathfrak{S}$ be a tame Deligne–Mumford stack [AOV08, Definition 3.1] with coarse map $\rho : \mathfrak{S} \to S$. A line bundle $\mathcal{D}$ on $\mathfrak{S}$ is *faithful on stabilizers* if for any geometric point $\overline{x}$ of $\mathfrak{S}$, the stabilizer group of $\overline{x}$ acts faithfully on the fiber of $\mathcal{D}$ at $\overline{x}$.

Let T be a variety, $\mathcal{E} \to T$ a flat projective morphism between varieties, and E_t the fiber of $\mathcal{E} \to T$ over $t \in T$. A *family over T of normal tame Deligne–Mumford stacks with coarse spaces E_t* is a tame Deligne–Mumford stack $\mathfrak{S}$ flat over T with a coarse map $\rho : \mathfrak{S} \to \mathcal{E}$ over T such that for any $t \in T$, the fiber $\mathfrak{S}_t$ is normal. We let $\rho_t : \mathfrak{S}_t \to E_t$ be the restriction of ρ, which is a coarse map [AOV08, Corollary 3.3].

For line bundles $\mathcal{D}_1, \dots, \mathcal{D}_m$ on $\mathfrak{S}$ or on a normal variety and $\lambda = (\lambda_1, \dots, \lambda_m) \in \mathbb{Z}^m$, we let

$$\mathcal{D}(\lambda) := \bigotimes_{i=1}^m \mathcal{D}_i^{\otimes \lambda_i},$$

and similarly, for line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$,

$$\mathcal{H}(\lambda) := \bigotimes_{i=1}^m \mathcal{H}_i^{\otimes \lambda_i}.$$

Lemma 8.2 (cf. [CLZ25, Theorem 1.3]). *Let $\mathcal{E} \to T$ be a flat projective morphism between varieties whose fibers E_t are normal varieties of Fano type. Then, after replacing T with the members of a finite stratification into locally closed subvarieties and then with finite surjective covers of these, there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{E}$ and a rational polyhedral cone $\sigma \subset \mathbb{R}^m$ such that for any $t \in T$, the following hold.*

(1) $\operatorname{Pic}(E_t)$ is a free abelian group with basis $\mathcal{H}_1|_{E_t}, \dots, \mathcal{H}_m|_{E_t}$.
(2) For any $\lambda \in \mathbb{Z}^m$, $\mathcal{H}(\lambda)|_{E_t}$ is nef if and only if $\lambda \in \sigma$.

Proof. Let $T^\circ \subset T$ be a non-empty open subset and $T' \to T^\circ$ a finite surjective morphism from a variety T'. The lemma for $\mathcal{E} \times_T T' \to T'$ and for the restrictions of $\mathcal{E}$ over the irreducible components of $T \backslash T^\circ$ implies the lemma for $\mathcal{E} \to T$. Since $\dim(T \backslash T^\circ) < \dim T$, by Noetherian induction, we may replace T by T' at any point of the proof. Since an étale morphism from a variety to T with dense image is finite over a non-empty open subset of T, we may also replace T by a variety étale over T with dense image. In particular, we may assume that T is smooth. The set of points of T over which the fiber of $\mathcal{E} \to T$ is geometrically integral is constructible [Gro66, Théorème 9.7.7(iv)] and contains all closed points of T, so all geometric fibers of $\mathcal{E} \to T$ are integral.

Step 1. We show that for any $t \in T$, $\operatorname{Pic}(E_t)$ is a free abelian group of finite rank, and that $\operatorname{Pic}(E_{\overline{\eta}})$ is finitely generated, where $\overline{\eta}$ is a geometric generic point of T.

Let $t \in T$ and $d := \dim E_t$. If $d = 0$, then $\operatorname{Pic}(E_t) = 0$, so we may assume that $d \geq 1$. Since E_t is of Fano type, (E_t, B) is a klt log Fano pair for some $\mathbb{R}$-divisor $B \geq 0$, and since the coefficient vectors of the $\mathbb{R}$-divisors B' with $\operatorname{Supp} B' \subset \operatorname{Supp} B$ and $K_{E_t} + B'$ $\mathbb{R}$-Cartier form an affine subspace defined over $\mathbb{Q}$ (Lemma 2.9(1) at every closed point of E_t) and klt and ampleness are open conditions on the coefficients in the affine subspace, there exists a $\mathbb{Q}$-divisor $B_t \geq 0$ on E_t such that (E_t, B_t) is klt and $-(K_{E_t} + B_t)$ is ample. For any Cartier divisor $M \equiv 0$ on E_t, $M - (K_{E_t} + B_t)$ is ample, so by the Kawamata–Viehweg vanishing theorem,

$$H^i(E_t, \mathcal{O}_{E_t}(M)) = 0 \text{ for any } i \geq 1. \tag{8.1}$$

For $M = 0$, $H^1(E_t, \mathcal{O}_{E_t}) = H^2(E_t, \mathcal{O}_{E_t}) = 0$, so by the exponential sequence and GAGA, the first Chern class is an isomorphism $\operatorname{Pic}(E_t) \to H^2(E_t, \mathbb{Z})$. Thus $\operatorname{Pic}(E_t)$ is finitely generated. Let M be a Cartier divisor on E_t such that $nM \sim 0$ for some $n \in \mathbb{N}^+$. The function $j \to \chi(E_t, \mathcal{O}_{E_t}(jM))$ on $\mathbb{Z}$ is a polynomial [Laz04, Theorem 1.1.24] and is periodic with period n, so the function is constant. By (8.1) for M and for 0,

$$h^0(E_t, \mathcal{O}_{E_t}(M)) = \chi(E_t, \mathcal{O}_{E_t}(M)) = \chi(E_t, \mathcal{O}_{E_t}) = 1.$$

Let H be an ample divisor on E_t, and $M' \geq 0$ a divisor such that $M' \sim M$. Since $nM' \sim 0$, $M' \cdot H^{d-1} = 0$, so $M' = 0$. Thus $M \sim 0$, and $\operatorname{Pic}(E_t)$ is torsion-free.

Let $k(\overline{\eta})$ be the residue field of $\overline{\eta}$. By semicontinuity,

$$H^1(E_{\overline{\eta}}, \mathcal{O}_{E_{\overline{\eta}}}) = H^2(E_{\overline{\eta}}, \mathcal{O}_{E_{\overline{\eta}}}) = 0.$$

Since $k(\overline{\eta})$ and $\mathbb{C}$ are algebraically closed fields of characteristic 0 whose transcendence degrees over $\mathbb{Q}$ are equal, there exists a field isomorphism $k(\overline{\eta}) \to \mathbb{C}$. Through such an isomorphism, $E_{\overline{\eta}}$ becomes a projective variety, and the exponential sequence and GAGA imply that $\operatorname{Pic}(E_{\overline{\eta}})$ is finitely generated.

Step 2. We show that the Picard scheme $\operatorname{Pic}_{\mathcal{E}/T}$ is étale and separated over T.

By [Kle05, Theorem 9.4.8], $\operatorname{Pic}_{\mathcal{E}/T}$ exists, is separated and locally of finite type over T, and represents the étale Picard functor. In particular, for any algebraically closed field k and any morphism $\operatorname{Spec} k \to T$, the k-points of $\operatorname{Pic}_{\mathcal{E}/T}$ over $\operatorname{Spec} k$ form the group $\operatorname{Pic}(\mathcal{E} \times_T \operatorname{Spec} k)$. For any $t \in T$, the tangent space at 0 of the fiber of $\operatorname{Pic}_{\mathcal{E}/T}$ over t is $H^1(E_t, \mathcal{O}_{E_t}) = 0$ [Kle05, Theorem 9.5.11], and since $H^2(E_t, \mathcal{O}_{E_t}) = 0$, $\operatorname{Pic}_{\mathcal{E}/T}$ is smooth over a neighborhood of t [Kle05, Proposition 9.5.19]. Since the fibers are group schemes, their tangent spaces at all points are 0, so $\operatorname{Pic}_{\mathcal{E}/T} \to T$ is étale. Thus:

(∗) For a connected scheme T' étale over T and two morphisms $\varsigma_1, \varsigma_2 : T' \to \mathrm{Pic}_{\mathcal{E}/T}$ over T, the subset of T' where ς_1 and ς_2 coincide is open and closed. In particular, if ς_1 and ς_2 coincide at one point or at one geometric point of T', then $\varsigma_1 = \varsigma_2$.

Step 3. We construct $\mathcal{H}_1, \dots, \mathcal{H}_m$ and prove (1).

We first show that $\mathrm{Pic}(E_{\overline{\eta}})$ is torsion-free. Let $x \in \mathrm{Pic}(E_{\overline{\eta}})$ and $n \in \mathbb{N}^+$ such that $nx = 0$. Since x is defined over a finite extension of the function field of T, there exist a connected variety T' étale over T with dense image and a morphism $\varsigma : T' \to \mathrm{Pic}_{\mathcal{E}/T}$ over T whose value at a geometric generic point of T' over $\overline{\eta}$ is x. By (∗), $n\varsigma = 0$. For any closed point $t' \in T'$ over $t \in T$, $\varsigma(t')$ is a torsion element of $\mathrm{Pic}(E_t)$, so $\varsigma(t') = 0$ by **Step 1**. By (∗), $\varsigma = 0$, so $x = 0$.

By **Step 1** and the previous paragraph, $\mathrm{Pic}(E_{\overline{\eta}})$ is a free abelian group with a basis $x_1, \dots, x_m$. Since $x_1, \dots, x_m$ are defined over a finite extension of the function field of T, there exist a connected variety T' étale over T with dense image, a geometric generic point $\overline{\eta}'$ of T', and morphisms $\varsigma_i : T' \to \mathrm{Pic}_{\mathcal{E}/T}$ over T for $i = 1, \dots, m$ such that the value of ς_i at $\overline{\eta}'$ is x_i. Since $\mathrm{Pic}_{\mathcal{E}/T}$ represents the étale sheafification of the Picard functor, possibly replacing T' with a connected variety étale over T' with dense image, ς_i is the class of a line bundle $\mathcal{H}_i$ on $\mathcal{E} \times_T T'$ for any i. Possibly replacing T with T', we may assume that T is a connected normal variety and that ς_i is the class of a line bundle $\mathcal{H}_i$ on $\mathcal{E}$ for any i.

Let $t \in T$. If $u \in \mathbb{Z}^m$ and $\mathcal{H}(u)|_{E_t} \cong \mathcal{O}_{E_t}$, then $\sum_{i=1}^m u_i\varsigma_i$ and 0 coincide at t, so by (∗) they coincide at a geometric generic point of T. Thus $\sum_{i=1}^m u_i x_i = 0$, and $u = 0$. Let $\mathcal{M}$ be a line bundle on E_t, $y_{\mathcal{M}} \in \mathrm{Pic}_{\mathcal{E}/T}$ the point corresponding to $\mathcal{M}$, and T'' the connected component of $\mathrm{Pic}_{\mathcal{E}/T}$ containing $y_{\mathcal{M}}$. Then T'' is étale over T, and the inclusion $T'' \to \mathrm{Pic}_{\mathcal{E}/T}$ is a morphism over T. The value of the inclusion at a geometric generic point of T'' belongs to $\mathrm{Pic}(E_{\overline{\eta}})$, so the value is equal to $\sum_{i=1}^m u_i x_i$ for some $u \in \mathbb{Z}^m$. By (∗), the inclusion $T'' \to \mathrm{Pic}_{\mathcal{E}/T}$ coincides with the composition of $T'' \to T$ and $\sum_{i=1}^m u_i\varsigma_i$. Evaluating both at $y_{\mathcal{M}}$, we have $\mathcal{M} \cong \mathcal{H}(u)|_{E_t}$. Thus (1) holds.

Step 4. We prove (2).

Since any fiber of $\mathcal{E} \to T$ is of Fano type, by [CLZ25, Theorems 1.5 and 1.2(ii)], possibly replacing T with a finite surjective cover of a non-empty open subset of T, there exists a rational polyhedral cone $\sigma \subset \mathbb{R}^m$ such that for any $t \in T$,

$$\sigma = \Big\{\lambda \in \mathbb{R}^m \Big| \sum_{i=1}^m \lambda_i [\mathcal{H}_i|_{E_t}] \in \mathrm{Nef}(E_t)\Big\},$$

where $[\mathcal{H}_i|_{E_t}]$ is the numerical class of $\mathcal{H}_i|_{E_t}$ in $N^1(E_t)_{\mathbb{R}}$ and $\mathrm{Nef}(E_t) \subset N^1(E_t)_{\mathbb{R}}$ is the nef cone. This implies (2). The lemma follows. □

Theorem 8.3. *Let $\mathcal{E} \to T$ be a flat projective morphism between varieties whose fibers E_t are normal varieties of Fano type, $\mathcal{V} \subset \mathcal{E}$ a closed subset such that*

$$\mathrm{Sing}(E_t) \subset \mathcal{V} \cap E_t \neq E_t \text{ for any } t \in T,$$

and q a positive integer. Then, after replacing T with the members of a finite stratification into locally closed subvarieties and then with finite surjective covers of these, there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{E}$ whose restrictions to E_t form a basis of the free abelian group $\mathrm{Pic}(E_t)$ for any $t \in T$, and a finite set $\mathfrak{P}$ of pairs $(\mathfrak{S}', \mathcal{D}')$ with the following properties.

- *(i) $\mathfrak{S}'$, with coarse map $\rho' : \mathfrak{S}' \to \mathcal{E}$, is a family over T of normal tame Deligne–Mumford stacks with coarse spaces E_t.*
- *(ii) $\mathcal{D}'$ is a line bundle on $\mathfrak{S}'$ faithful on stabilizers.*
- *(iii) $\mathcal{D}'^{\otimes q} \cong \rho'^*\mathcal{H}(\lambda')$ for some $\lambda' \in \{0, \dots, q-1\}^m$.*

Moreover, the following hold.

- *(1) Let $t \in T$, and $\mathfrak{S}$ a normal separated tame Deligne–Mumford stack with coarse map $\rho : \mathfrak{S} \to E_t$ such that ρ is an isomorphism over $E_t \backslash \mathcal{V}$ and the orders of the stabilizers*

of $\mathfrak{S}$ divide q. Let $\mathcal{D}$ be a line bundle on $\mathfrak{S}$ faithful on stabilizers. Then there exist $(\mathfrak{S}', \mathcal{D}') \in \mathfrak{P}$, $\lambda \in \mathbb{Z}^m$, and an isomorphism $g : \mathfrak{S} \to \mathfrak{S}'_t$ such that $\rho'_t \circ g = \rho$ and

$$g^*\left(\mathcal{D}'|_{\mathfrak{S}'_t} \otimes \rho'^*\mathcal{H}(\lambda)|_{\mathfrak{S}'_t}\right) \cong \mathcal{D}.$$

(2) For any $(\mathfrak{S}', \mathcal{D}') \in \mathfrak{P}$ and $t \in T$, the stack $\mathfrak{S}'_t$ is separated, the orders of the stabilizers of $\mathfrak{S}'_t$ divide q, and ρ'_t is an isomorphism over $E_t \backslash \mathcal{V}$.

Proof. By Noetherian induction, at any point of the proof we may replace T by a finite surjective cover of a non-empty open subset of T, or by a variety étale over T with dense image.

Step 1. We reduce to line bundles $\mathcal{D}$ whose q-th powers are pullbacks of finitely many line bundles on $\mathcal{E}$.

By Lemma 8.2, we may assume that there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{E}$ whose restrictions to E_t form a basis of $\mathrm{Pic}(E_t)$ for any $t \in T$. Possibly shrinking T, we may assume that T is smooth. Let

$$\mathcal{U} := \mathcal{E} \backslash \mathcal{V} \text{ and } U_t := E_t \backslash \mathcal{V} \text{ for any } t \in T.$$

The set of points $x \in \mathcal{E}$ such that the fiber of $\mathcal{E} \to T$ through x is geometrically normal at x is open [Gro66, Théorème 12.1.6(iv)] and contains the closed points of $\mathcal{E}$, so it is $\mathcal{E}$. Since T is smooth, $\mathcal{E}$ is normal [Mat89, Corollary to Theorem 23.9]. Since U_t is smooth and dense in E_t for any $t \in T$, $\mathcal{U}$ is a non-empty open subset of $\mathcal{E}$ smooth over T. Since $\mathcal{E}$ is irreducible, $\mathcal{U}$ is connected.

Let t, $\mathfrak{S}$, ρ, and $\mathcal{D}$ be as in (1). The stabilizer group of any geometric point of $\mathfrak{S}$ acts on the fiber of $\mathcal{D}$ through a character whose order divides q, so the stabilizer group acts trivially on the fiber of $\mathcal{D}^{\otimes q}$. Since ρ is a good moduli space, $\rho_*\mathcal{D}^{\otimes q}$ is a line bundle on E_t and

$$\mathcal{D}^{\otimes q} \cong \rho^*\rho_*\mathcal{D}^{\otimes q}$$

[Alp13, Example 8.1 and Theorem 10.3]. By Lemma 8.2(1), $\rho_*\mathcal{D}^{\otimes q} \cong \mathcal{H}(u)|_{E_t}$ for a unique $u \in \mathbb{Z}^m$. Write $u = q\lambda + \lambda'$ with $\lambda \in \mathbb{Z}^m$ and $\lambda' \in \{0, \dots, q-1\}^m$, and let

$$\mathcal{D}_0 := \mathcal{D} \otimes \rho^*\mathcal{H}(-\lambda)|_{E_t}.$$

Then

$$\mathcal{D}_0^{\otimes q} \cong \rho^*\mathcal{H}(\lambda')|_{E_t},$$

and $\mathcal{D}_0$ is faithful on stabilizers, since stabilizers act trivially on the fibers of pullbacks of line bundles on E_t. If $g : \mathfrak{S} \to \mathfrak{S}'_t$ is an isomorphism such that $\rho'_t \circ g = \rho$ and $g^*(\mathcal{D}'|_{\mathfrak{S}'_t}) \cong \mathcal{D}_0$, then

$$g^*\left(\mathcal{D}'|_{\mathfrak{S}'_t} \otimes \rho'^*\mathcal{H}(\lambda)|_{\mathfrak{S}'_t}\right) \cong \mathcal{D}_0 \otimes \rho^*\mathcal{H}(\lambda)|_{E_t} \cong \mathcal{D}.$$

Claim 8.4. *Let $\lambda' \in \{0, \dots, q-1\}^m$. Then there exists a finite set $\mathfrak{P}_{\lambda'}$ of pairs $(\mathfrak{S}', \mathcal{D}')$ as in the theorem with $\mathcal{D}'^{\otimes q} \cong \rho'^*\mathcal{H}(\lambda')$ satisfying (2) and the following: for any t, $\mathfrak{S}$, and ρ as in (1) and any line bundle $\mathcal{D}$ on $\mathfrak{S}$ faithful on stabilizers with*

$$\mathcal{D}^{\otimes q} \cong \rho^*\mathcal{H}(\lambda')|_{E_t},$$

there exist $(\mathfrak{S}', \mathcal{D}') \in \mathfrak{P}_{\lambda'}$ and an isomorphism $g : \mathfrak{S} \to \mathfrak{S}'_t$ such that $\rho'_t \circ g = \rho$ and $g^*(\mathcal{D}'|_{\mathfrak{S}'_t}) \cong \mathcal{D}$.

Since $\{0, \dots, q-1\}^m$ is finite, Claim 8.4 for all λ' implies the theorem with $\mathfrak{P} := \bigcup_{\lambda'} \mathfrak{P}_{\lambda'}$. We fix λ' and prove Claim 8.4.

Let $\mathcal{M}_0 := \mathcal{H}(\lambda')$. Let $\mathcal{M}_0^\times$ be the complement of the zero section in the total space of $\mathcal{M}_0$, with the action

$$\alpha : \mathbb{G}_m \times \mathcal{M}_0^\times \to \mathcal{M}_0^\times, \quad \alpha(c, v) = c^q v,$$

where $c^q v$ is the scalar multiplication on the fibers of $\mathcal{M}_0$. For $t \in T$ and $x \in \mathcal{E}$, let $\mathcal{M}_0^\times|_{E_t}$ and $\mathcal{M}_0^\times(x)$ be the fibers of $\mathcal{M}_0^\times$ over t and over x, and let

$$\mathcal{M}_0^\times|_{\mathcal{U}} := \mathcal{M}_0^\times \times_{\mathcal{E}} \mathcal{U} \text{ and } \mathcal{M}_0^\times|_{U_t} := \mathcal{M}_0^\times|_{E_t} \times_{E_t} U_t.$$

Then $\mathcal{M}_0^\times(x) \cong \mathbb{G}_m$, and $\mathcal{M}_0^\times|_{\mathcal{U}}$ is a smooth connected variety. We let $\mu_q \subset \mathbb{G}_m$ act on $\mathbb{G}_m$ by multiplication, and we consider the following condition on a μ_q-torsor W° over $\mathcal{M}_0^\times|_{\mathcal{U}}$ or over $\mathcal{M}_0^\times|_{U_t}$:

(R) For any $x \in \mathcal{U}$ such that $\mathcal{M}_0^\times(x)$ is contained in the base and any $v_0 \in \mathcal{M}_0^\times(x)$, the restriction of W° to $\mathcal{M}_0^\times(x)$ is isomorphic to the μ_q-torsor $\mathbb{G}_m \to \mathcal{M}_0^\times(x)$, $z \to z^q v_0$.

Step 2. We show that a stack with a faithful line bundle is the quotient of a finite normal cover of $\mathcal{M}_0^\times|_{E_t}$.

Let t, $\mathfrak{S}$, ρ, and $\mathcal{D}$ be as in (1) with $\mathcal{D}^{\otimes q} \cong \rho^*\mathcal{M}_0|_{E_t}$, and fix an isomorphism

$$\psi : \mathcal{D}^{\otimes q} \to \rho^*\mathcal{M}_0|_{E_t}.$$

Let W be the complement of the zero section in the total space of $\mathcal{D}$, with $\mathbb{G}_m$ acting by scalar multiplication. Then $W \to \mathfrak{S}$ is a $\mathbb{G}_m$-torsor, so $\mathfrak{S} \cong [W/\mathbb{G}_m]$, and $\mathcal{D}$ is the line bundle associated with the $\mathbb{G}_m$-torsor $W \to \mathfrak{S}$ and the identity character of $\mathbb{G}_m$. Let $w : W \to \mathcal{M}_0^\times|_{E_t}$ be the morphism $\omega \to \psi(\omega^{\otimes q})$. Then w is $\mathbb{G}_m$-equivariant, where $\mathbb{G}_m$ acts on $\mathcal{M}_0^\times|_{E_t}$ through α, and the composition of w and $\mathcal{M}_0^\times|_{E_t} \to E_t$ is the composition of $W \to \mathfrak{S}$ and ρ.

We show that W is a normal scheme and that w is finite. By [Ols16, Theorem 11.3.1], for any geometric point $\overline{x}$ of $\mathfrak{S}$ with stabilizer group G, there exist an étale morphism $E' \to E_t$ whose image contains $\rho(\overline{x})$, a normal affine scheme $\operatorname{Spec} R$ with an action of G, and an isomorphism $\mathfrak{S} \times_{E_t} E' \cong [\operatorname{Spec} R/G]$ with $E' = \operatorname{Spec} R^G$. Possibly shrinking E', $\mathcal{D}$ corresponds to the free R-module Re with G acting on e through a character $\chi : G \to \mathbb{G}_m$. Since $\mathcal{D}$ is faithful on stabilizers, χ is injective. Since $\mathcal{D}$ corresponds to Re,

$$W \times_{E_t} E' \cong [\operatorname{Spec} R[\varpi, \varpi^{-1}]/G],$$

where ϖ is the coordinate dual to e. Since χ is injective, G acts freely on $\operatorname{Spec} R[\varpi, \varpi^{-1}]$, so $W \times_{E_t} E' \cong \operatorname{Spec}(R[\varpi, \varpi^{-1}])^G$ is a normal affine scheme. Moreover,

$$\mathcal{M}_0^\times|_{E_t} \times_{E_t} E' \cong \operatorname{Spec} R^G[y, y^{-1}]$$

with $w^*y = \varpi^q$, up to a unit of R^G given by ψ, which is G-invariant since $\chi^q = 1$, the order of G dividing q. Since R is finite over R^G, $R[\varpi, \varpi^{-1}]$ is finite over $R^G[\varpi^q, \varpi^{-q}]$, so $(R[\varpi, \varpi^{-1}])^G$ is finite over $R^G[\varpi^q, \varpi^{-q}]$. Thus w is finite after the base change $E' \to E_t$, so w is finite and W is a normal scheme.

Let $W^\circ := w^{-1}(\mathcal{M}_0^\times|_{U_t})$. Since ρ is an isomorphism over U_t, W° is the complement of the zero section in the total space of $\mathcal{D}|_{U_t}$, and $w|_{W^\circ} : W^\circ \to \mathcal{M}_0^\times|_{U_t}$ is a μ_q-torsor, where $\mu_q \subset \mathbb{G}_m$ acts on W° by scalar multiplication. For any $x \in U_t$, the restriction of w to the fiber of $\mathcal{D}$ over x minus 0 is $z \to \psi(z^{\otimes q})$, so $W^\circ \to \mathcal{M}_0^\times|_{U_t}$ satisfies (R). Since $\mathfrak{S}$ is normal and the coarse space E_t of $\mathfrak{S}$ is irreducible, $\mathfrak{S}$ is irreducible, so $\rho^{-1}(U_t)$ is dense in $\mathfrak{S}$. Since $W \to \mathfrak{S}$ is smooth, W° is dense in W. Since W is normal and finite over $\mathcal{M}_0^\times|_{E_t}$, W is the normalization of $\mathcal{M}_0^\times|_{E_t}$ in W°.

Step 3. We show that, possibly replacing T with a finite surjective cover of a non-empty open subset of T, there exist finitely many μ_q-torsors $W_1^\circ, \dots, W_N^\circ$ on $\mathcal{M}_0^\times|_{\mathcal{U}}$ such that for any $t \in T$, any μ_q-torsor on $\mathcal{M}_0^\times|_{U_t}$ is isomorphic to $W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$ for some $k \in \{1, \dots, N\}$.

Let

$$\overline{\mathcal{M}}_0 := \mathbb{P}_{\mathcal{E}}(\mathcal{O}_{\mathcal{E}} \oplus \mathcal{M}_0),$$

which contains $\mathcal{M}_0^\times$ as the complement of the zero section and the infinity section. Since $\mathcal{M}_0^\times|_{\mathcal{U}}$ is smooth, there exists a log resolution $f : Q \to \overline{\mathcal{M}}_0$ of

$$(\overline{\mathcal{M}}_0, \overline{\mathcal{M}}_0 \backslash \mathcal{M}_0^\times|_{\mathcal{U}})$$

that is an isomorphism over $\mathcal{M}_0^\times|_{\mathcal{U}}$. Let $D_1, \dots, D_n$ be the irreducible components of $Q \backslash f^{-1}(\mathcal{M}_0^\times|_{\mathcal{U}})$, and $D := \sum_{j=1}^n D_j$, an snc divisor. Possibly shrinking T, by generic smoothness,

$Q \to T$ is smooth and $\bigcap_{j\in J} D_j$ is smooth over T for any $J \subset \{1,\dots,n\}$. Thus for any $t \in T$, D_t is an snc divisor on the smooth variety Q_t and

$$Q_t \backslash D_t \cong \mathcal{M}_0^\times|_{U_t}.$$

Let $\mathfrak{Q}$ be the root stack of Q of order q along $D_1,\dots,D_n$ [Cad07, Definition 2.2.4], and $\pi : \mathfrak{Q} \to T$ the induced morphism. Then $\mathfrak{Q}$ is a smooth tame Deligne–Mumford stack (Deligne–Mumford by [Cad07, Corollary 2.3.4], smooth by the local description below applied to Q and D, and tame by [AOV08, Theorem 3.2] as its stabilizers are finite groups and $\mathbb{C}$ has characteristic 0), proper over T since its coarse map to Q is proper [AOV08, Section 3] and Q is proper over T, isomorphic to Q over $Q\backslash D$ [Cad07, Remark 2.2.3 and Example 2.4.2], and for any $t \in T$, the fiber $\mathfrak{Q}_t$ is the root stack of Q_t of order q along $D_{1,t},\dots,D_{n,t}$ [Cad07, Remark 2.2.3]. Near a point of Q_t with an affine open neighborhood $\operatorname{Spec} B$ on which $D_t = (s_1 \cdots s_r = 0)$ for a part $s_1,\dots,s_r \in B$ of a system of local coordinates at the point, by [Cad07, Remark 2.2.5 and Example 2.4.1], $\mathfrak{Q}_t$ is étale locally the quotient by μ_q^r of the smooth scheme

$$X := \operatorname{Spec} B[z_1,\dots,z_r]/(z_1^q - s_1,\dots,z_r^q - s_r). \tag{8.2}$$

We show that for any $t \in T$, restriction is a bijection from the set of isomorphism classes of μ_q-torsors on $\mathfrak{Q}_t$ to that on $\mathcal{M}_0^\times|_{U_t}$. A finite étale cover of the normal stack $\mathfrak{Q}_t$ is the normalization of $\mathfrak{Q}_t$ in the restriction of the cover to the dense open subset $\mathcal{M}_0^\times|_{U_t}$, so restriction is injective. Let $W^\circ \to \mathcal{M}_0^\times|_{U_t}$ be a μ_q-torsor. The ramification index of W° along any component of D_t divides q. By Abhyankar's lemma [Gro03, Exposé XIII, Proposition 5.2], on any chart X in (8.2), the normalization X' of X in the pullback of W° is étale over X at the generic points of the divisors $(z_j = 0)$, hence étale outside a closed subset $Z \subset X$ with $\dim Z \le \dim X - 2$. Since X is regular, by purity of the branch locus [Gro03, Exposé X, Théorème 3.1], X' is étale over X. Since normalization commutes with étale base change, the normalizations X' on the charts descend to a finite étale cover of $\mathfrak{Q}_t$ with a μ_q-action extending W°, that is, a μ_q-torsor on $\mathfrak{Q}_t$ extending W°. Thus restriction is surjective.

By [Gro03, Exposé XIII, Proposition 6.1 and Corollaire 6.2] and [ACV03, Proposition A.0.1 and Theorem A.0.2], applied to the proper morphism $Q \to T$ through the Leray spectral sequence of the coarse map $p : \mathfrak{Q} \to Q$, where $p_*\mu_q = \mu_q$ and $R^1p_*\mu_q$ is locally constant on each stratum $\bigcap_{j\in J} D_j \backslash \bigcup_{j\notin J} D_j$ with stalks $H^1(\mu_q^J, \mu_q)$ by (8.2) with Q, D, and $\mathfrak{Q}$ in place of Q_t, D_t, and $\mathfrak{Q}_t$, $R^1\pi_*\mu_q$ is a constructible sheaf on T whose stalk at any $t \in T$ is $H^1(\mathfrak{Q}_t, \mu_q)$. Possibly shrinking T, $R^1\pi_*\mu_q$ is locally constant, and possibly replacing T with a connected finite étale cover, $R^1\pi_*\mu_q$ is the constant sheaf associated with a finite group $\{c_1,\dots,c_N\}$. By the definition of $R^1\pi_*\mu_q$, any c_k is, étale locally on T, the class of a μ_q-torsor on $\mathfrak{Q}$. Possibly replacing T with a connected variety étale over T with dense image, for any $k \in \{1,\dots,N\}$ there exists a μ_q-torsor $\mathcal{W}_k$ on $\mathfrak{Q}$ whose class is c_k. For any $t \in T$, the classes of $\mathcal{W}_k|_{\mathfrak{Q}_t}$, $k \in \{1,\dots,N\}$, exhaust $H^1(\mathfrak{Q}_t, \mu_q)$. By the bijection above, the restrictions of the $\mathcal{W}_k$ to $\mathcal{M}_0^\times|_{\mathcal{U}}$ are the desired $W_1^\circ,\dots,W_N^\circ$.

Step 4. We construct $\mathfrak{P}_{\lambda'}$.

We use the Riemann existence theorem [Gro03, Exposé XII, Théorème 5.1]: for a connected variety Y, finite étale covers of Y and morphisms between them are the finite topological covers of Y and morphisms between them. In particular, isomorphism classes of μ_q-torsors on Y correspond to homomorphisms $\pi_1(Y) \to \mu_q$.

Claim 8.5. *Let Λ and Y be connected varieties, $1 \in \Lambda$ a point, $Y' \to Y$ a μ_q-torsor, and $\beta : \Lambda \times Y \to Y$ a morphism such that $\beta(1, y) = y$ for any $y \in Y$. Let $\tau, \tau' : \Lambda \times Y' \to Y'$ be μ_q-equivariant morphisms over β such that $\tau(1, y') = \tau'(1, y') = y'$ for any $y' \in Y'$. Then $\tau = \tau'$.*

Proof. Both $(c, y') \to (c, \tau(c, y'))$ and $(c, y') \to (c, \tau'(c, y'))$ are isomorphisms of μ_q-torsors from the pullback of Y' by the projection $\Lambda \times Y \to Y$ to the pullback of Y' by β. Thus there exists a locally constant function $\zeta : \Lambda \times Y \to \mu_q$ such that $\tau'(c, y') = \zeta(c, y)\tau(c, y')$ for any $y' \in Y'$ over $y \in Y$. Since $\Lambda \times Y$ is connected, ζ is constant, and $\zeta = 1$ on $\{1\} \times Y$. Thus $\zeta = 1$, and $\tau = \tau'$. □

For any $k \in \{1, \dots, N\}$, since $\mathcal{M}_0^\times|_{\mathcal{U}} \to \mathcal{U}$ is locally trivial with fiber $\mathbb{G}_m$ in the analytic topology and $\mathcal{U}$ is connected, the class of $W_k^\circ|_{\mathcal{M}_0^\times(x)}$ in $\mathrm{Hom}(\pi_1(\mathcal{M}_0^\times(x)), \mu_q) \cong \mu_q$ does not depend on $x \in \mathcal{U}$. Let $\mathcal{K}$ be the set of k such that W_k° satisfies (R). Fix $k \in \mathcal{K}$.

We construct a $\mathbb{G}_m$-action on W_k°. Let

$$p_2 : \mathbb{G}_m \times \mathcal{M}_0^\times|_{\mathcal{U}} \to \mathcal{M}_0^\times|_{\mathcal{U}}$$

be the projection. Since

$$\pi_1(\mathbb{G}_m \times \mathcal{M}_0^\times|_{\mathcal{U}}) = \pi_1(\mathbb{G}_m) \times \pi_1(\mathcal{M}_0^\times|_{\mathcal{U}}),$$

we compare the homomorphisms from the left-hand side to μ_q of $\alpha^* W_k^\circ$ and $p_2^* W_k^\circ$ on $\pi_1(\mathcal{M}_0^\times|_{\mathcal{U}})$ and on $\pi_1(\mathbb{G}_m)$. On $\pi_1(\mathcal{M}_0^\times|_{\mathcal{U}})$, through $v \to (1, v)$, both are the homomorphism of W_k°. On $\pi_1(\mathbb{G}_m)$, through $c \to (c, v_0)$ for a fixed $v_0 \in \mathcal{M}_0^\times|_{\mathcal{U}}$, the homomorphism of $p_2^* W_k^\circ$ is trivial, and the homomorphism of $\alpha^* W_k^\circ$ sends the generator to the image of the q-th power of a loop in $\mathcal{M}_0^\times|_{\mathcal{U}}$, which is trivial in μ_q. Thus $\alpha^* W_k^\circ \cong p_2^* W_k^\circ$. Any two such isomorphisms differ by a locally constant function to μ_q, so there exists an isomorphism $p_2^* W_k^\circ \to \alpha^* W_k^\circ$ that is the identity over $\{1\} \times \mathcal{M}_0^\times|_{\mathcal{U}}$. Such an isomorphism corresponds to a μ_q-equivariant morphism

$$a_k : \mathbb{G}_m \times W_k^\circ \to W_k^\circ$$

over α such that $a_k(1, w) = w$ for any $w \in W_k^\circ$. The morphisms

$$\tau_1, \tau_2 : \mathbb{G}_m \times \mathbb{G}_m \times W_k^\circ \to W_k^\circ, \quad \tau_1(c, c', w) = a_k(cc', w), \quad \text{and} \quad \tau_2(c, c', w) = a_k(c, a_k(c', w))$$

are μ_q-equivariant over the morphism

$$\mathbb{G}_m \times \mathbb{G}_m \times \mathcal{M}_0^\times|_{\mathcal{U}} \to \mathcal{M}_0^\times|_{\mathcal{U}}, \quad (c, c', v) \to \alpha(cc', v) = \alpha(c, \alpha(c', v)),$$

and $\tau_1 = \tau_2$ for $c = c' = 1$, so by Claim 8.5 with $\Lambda = \mathbb{G}_m \times \mathbb{G}_m$, $1 = (1, 1)$, and $Y = \mathcal{M}_0^\times|_{\mathcal{U}}$, $\tau_1 = \tau_2$. Thus a_k is a $\mathbb{G}_m$-action on W_k° commuting with the μ_q-action. Since $\mathcal{M}_0^\times|_{U_t}$ is connected for any $t \in T$, by Claim 8.5 with $\Lambda = \mathbb{G}_m$, the restriction of a_k to $\mathbb{G}_m \times W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$ is the unique μ_q-equivariant morphism over $\alpha|_{\mathbb{G}_m \times \mathcal{M}_0^\times|_{U_t}}$ that is the identity for $c = 1$.

We show that $W_k^\circ \to \mathcal{U}$ is a $\mathbb{G}_m$-torsor. Let $x \in \mathcal{U}$ and $v_0 \in \mathcal{M}_0^\times(x)$. By (R), there exists a μ_q-equivariant isomorphism ι_x from $\mathbb{G}_m$ to $W_k^\circ|_{\mathcal{M}_0^\times(x)}$ over $\mathcal{M}_0^\times(x)$, where $\mathbb{G}_m \to \mathcal{M}_0^\times(x)$ is $z \to z^q v_0$. Through ι_x, the action $(c, z) \to cz$ and the restriction of a_k are two μ_q-equivariant morphisms over $\alpha|_{\mathbb{G}_m \times \mathcal{M}_0^\times(x)}$ that are the identity for $c = 1$, so they coincide by Claim 8.5 with $\Lambda = \mathbb{G}_m$. Thus $\mathbb{G}_m$ acts simply transitively on the fiber of W_k° over x. Since $W_k^\circ \to \mathcal{U}$ is smooth and surjective, $W_k^\circ \to \mathcal{U}$ is a $\mathbb{G}_m$-torsor, so

$$[W_k^\circ/\mathbb{G}_m] \cong \mathcal{U} \text{ and } [W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}/\mathbb{G}_m] \cong U_t \text{ for any } t \in T.$$

Let W_k be the normalization of $\mathcal{M}_0^\times$ in W_k°. Then $W_k \to \mathcal{M}_0^\times$ is finite, W_k is normal, and W_k° is the preimage of $\mathcal{M}_0^\times|_{\mathcal{U}}$ in W_k. The automorphism $(c, v) \to (c, \alpha(c, v))$ of $\mathbb{G}_m \times \mathcal{M}_0^\times$ lifts to the automorphism $(c, w) \to (c, a_k(c, w))$ of $\mathbb{G}_m \times W_k^\circ$, and hence to an automorphism of the normalization $\mathbb{G}_m \times W_k$ of $\mathbb{G}_m \times \mathcal{M}_0^\times$ in $\mathbb{G}_m \times W_k^\circ$. Composing with the projection to W_k, we obtain a morphism $\overline{a}_k : \mathbb{G}_m \times W_k \to W_k$ extending a_k. Since W_k is separated and W_k° is dense in W_k, $\overline{a}_k$ is a $\mathbb{G}_m$-action, and $W_k \to \mathcal{M}_0^\times$ is $\mathbb{G}_m$-equivariant. By generic flatness, possibly shrinking T, W_k is flat over T. Since the generic fiber of $W_k \to T$ is normal and the characteristic is 0, possibly shrinking T further, for any $t \in T$, the fiber $W_{k,t}$ is normal and $W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$ is dense in $W_{k,t}$. Since $W_{k,t}$ is finite over $\mathcal{M}_0^\times|_{E_t}$, $W_{k,t}$ is the normalization of $\mathcal{M}_0^\times|_{E_t}$ in $W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$, and the $\mathbb{G}_m$-action on $W_{k,t}$ is the unique extension of the restriction of a_k.

Let $\mathfrak{S}'_k := [W_k/\mathbb{G}_m]$, and $\mathcal{D}'_k$ the line bundle on $\mathfrak{S}'_k$ associated with the $\mathbb{G}_m$-torsor $W_k \to \mathfrak{S}'_k$ and the identity character of $\mathbb{G}_m$. The pair $(\mathfrak{S}'_k, \mathcal{D}'_k)$ has the following properties.

- If $c \in \mathbb{G}_m$ fixes $w \in W_k$, then c^q fixes the image of w in $\mathcal{M}_0^\times$, so $c \in \mu_q$. Thus the stabilizers of $\mathfrak{S}'_k$ are subgroups of μ_q, and their orders divide q. Since the characteristic is 0, $\mathfrak{S}'_k$ is a tame Deligne–Mumford stack [AOV08, Theorem 3.2]. The stabilizer of w acts on the fiber of $\mathcal{D}'_k$ through the inclusion into $\mathbb{G}_m$, so $\mathcal{D}'_k$ is faithful on stabilizers.
- Since W_k is finite over $\mathcal{M}_0^\times$ and $\mathcal{M}_0^\times$ is affine over $\mathcal{E}$, $W_k = \operatorname{Spec}_{\mathcal{E}} \mathcal{A}$ for a $\mathbb{Z}$-graded $\mathcal{O}_{\mathcal{E}}$-algebra $\mathcal{A} = \bigoplus_{j\in\mathbb{Z}} \mathcal{A}_j$, the grading given by the $\mathbb{G}_m$-action. Since $\mathcal{A}$ is finite over the push-forward of $\mathcal{O}_{\mathcal{M}_0^\times}$ to $\mathcal{E}$, whose homogeneous component of degree j is a power of $\mathcal{M}_0$ if $q \mid j$ and 0 otherwise, $\mathcal{A}_0$ is a finite $\mathcal{O}_{\mathcal{E}}$-algebra. Since the stabilizers are finite, all $\mathbb{G}_m$-orbits on W_k have dimension 1, so they are closed, the boundary of an orbit being a union of orbits of smaller dimension. Thus $[W_k/\mathbb{G}_m] \to \operatorname{Spec}_{\mathcal{E}} \mathcal{A}_0$ is a good moduli space with closed orbits, hence a coarse map [Alp13, Theorem 13.2, Proposition 7.7, and Remark 7.3]. Since $W_k^\circ \to \mathcal{U}$ is a $\mathbb{G}_m$-torsor, W_k is a normal variety, so $\mathcal{A}_0$ is a sheaf of normal domains. Since $[W_k^\circ/\mathbb{G}_m] \cong \mathcal{U}$, the finite morphism $\operatorname{Spec}_{\mathcal{E}} \mathcal{A}_0 \to \mathcal{E}$ is an isomorphism over $\mathcal{U}$. Since $\mathcal{E}$ is normal, $\operatorname{Spec}_{\mathcal{E}} \mathcal{A}_0 \to \mathcal{E}$ is an isomorphism. Thus $\mathcal{E}$ is the coarse space of $\mathfrak{S}'_k$. Let $\rho'_k : \mathfrak{S}'_k \to \mathcal{E}$ be the coarse map.
- Since $W_k \to \mathcal{M}_0^\times$ is $\mathbb{G}_m$-equivariant and $\mathbb{G}_m$ acts on $\mathcal{M}_0^\times$ through α, $\mathcal{D}_k'^{\otimes q} \cong \rho_k'^* \mathcal{M}_0$.
- The morphism $\mathfrak{S}'_k \to [\mathcal{M}_0^\times/\mathbb{G}_m]$ induced by $W_k \to \mathcal{M}_0^\times$ is representable and finite, since $W_k \to \mathcal{M}_0^\times$ is the pullback of $\mathfrak{S}'_k \to [\mathcal{M}_0^\times/\mathbb{G}_m]$ along the smooth surjection
$$\mathcal{M}_0^\times \to [\mathcal{M}_0^\times/\mathbb{G}_m].$$
Locally on $\mathcal{E}$,
$$[\mathcal{M}_0^\times/\mathbb{G}_m] \cong \mathcal{E} \times B\mu_q,$$
where $B\mu_q := [\operatorname{Spec} \mathbb{C}/\mu_q]$, so $[\mathcal{M}_0^\times/\mathbb{G}_m]$ is separated over $\mathcal{E}$. Thus $\mathfrak{S}'_k$ is separated, and so is $\mathfrak{S}'_{k,t}$ for any $t \in T$.
- For any $t \in T$, $\mathfrak{S}'_{k,t} = [W_{k,t}/\mathbb{G}_m]$ is normal since $W_{k,t}$ is normal, $\rho'_{k,t} : \mathfrak{S}'_{k,t} \to E_t$ is the coarse map by [AOV08, Corollary 3.3], and $\rho'_{k,t}$ is an isomorphism over U_t since $W_k^\circ \to \mathcal{U}$ is a $\mathbb{G}_m$-torsor.

Thus $\mathfrak{S}'_k$ is a family over T in the sense of Definition 8.1, $(\mathfrak{S}'_k, \mathcal{D}'_k)$ is a pair as in the theorem with $\mathcal{D}_k'^{\otimes q} \cong \rho_k'^* \mathcal{H}(\lambda')$, and $(\mathfrak{S}'_k, \mathcal{D}'_k)$ satisfies (2). Let
$$\mathfrak{P}_{\lambda'} := \{(\mathfrak{S}'_k, \mathcal{D}'_k) \mid k \in \mathcal{K}\}.$$

Step 5. We conclude the proof.

Let t, $\mathfrak{S}$, ρ, and $\mathcal{D}$ be as in (1) with $\mathcal{D}^{\otimes q} \cong \rho^* \mathcal{M}_0|_{E_t}$, and let ψ, W, w, and W° be as in **Step 2**. By **Step 3**, there exist $k \in \{1, \dots, N\}$ and an isomorphism of μ_q-torsors
$$\varphi : W^\circ \to W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$$
over $\mathcal{M}_0^\times|_{U_t}$. Since W° satisfies (R), so does $W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$. Since the class of $W_k^\circ|_{\mathcal{M}_0^\times(x)}$ does not depend on $x \in \mathcal{U}$ (**Step 4**), W_k° satisfies (R), that is, $k \in \mathcal{K}$. The scalar action of $\mathbb{G}_m$ on W°, transported through φ, is a μ_q-equivariant morphism over $\alpha|_{\mathbb{G}_m \times \mathcal{M}_0^\times|_{U_t}}$ that is the identity for $c = 1$, so the transported action coincides with the restriction of a_k by **Step 4**. Thus φ is $\mathbb{G}_m$-equivariant. Since W and $W_{k,t}$ are the normalizations of $\mathcal{M}_0^\times|_{E_t}$ in W° and in $W_k^\circ|_{\mathcal{M}_0^\times|_{U_t}}$ respectively (**Steps 2 and 4**), φ extends to an isomorphism $\widetilde{\varphi} : W \to W_{k,t}$ over $\mathcal{M}_0^\times|_{E_t}$, and $\widetilde{\varphi}$ is $\mathbb{G}_m$-equivariant since W° is dense in W. The isomorphism $\widetilde{\varphi}$ induces an isomorphism
$$g : \mathfrak{S} \cong [W/\mathbb{G}_m] \to [W_{k,t}/\mathbb{G}_m] = \mathfrak{S}'_{k,t}$$
such that $g^*(\mathcal{D}'_k|_{\mathfrak{S}'_{k,t}}) \cong \mathcal{D}$, since both line bundles are associated with the identity character of $\mathbb{G}_m$. The morphism g commutes with the morphisms
$$\mathfrak{S} \to [\mathcal{M}_0^\times|_{E_t}/\mathbb{G}_m] \quad \text{and} \quad \mathfrak{S}'_{k,t} \to [\mathcal{M}_0^\times|_{E_t}/\mathbb{G}_m],$$

whose compositions with $[\mathcal{M}_0^{\times}|_{E_t}/\mathbb{G}_m] \to E_t$ are ρ and $\rho'_{k,t}$ by **Step 2** and **Step 4**. Thus $\rho'_{k,t} \circ g = \rho$. This proves Claim 8.4, and the theorem follows. □

9. Uniform generators of filtered algebras in families

Definition 9.1. Let $\mathfrak{S}$ be an integral tame Deligne–Mumford stack with coarse map $\rho : \mathfrak{S} \to S$, where S is a normal variety. A line bundle $\mathcal{D}$ on $\mathfrak{S}$ is *nef* if $\mathcal{D}^{\otimes q} \cong \rho^*\mathcal{N}$ for some positive integer q and some nef line bundle $\mathcal{N}$ on S. Let F be a prime divisor over S and $h : S' \to S$ a resolution such that F is a divisor on S'. For a line bundle $\mathcal{D}$ on $\mathfrak{S}$ such that $\mathcal{D}^{\otimes q} \cong \rho^*\mathcal{N}$ for a positive integer q and a line bundle $\mathcal{N}$ on S, and a non-zero section s of $\mathcal{D}$, we let

$$\nu_F(s) := \frac{1}{q} \operatorname{mult}_F(h^* \operatorname{div}(s^{\otimes q})),$$

where $s^{\otimes q}$ is regarded as a section of $\mathcal{N}$, and $\nu_F(s)$ does not depend on q. We let $\nu_F(0) := +\infty$. If moreover $\mathcal{D}_1, \dots, \mathcal{D}_m$ are line bundles on $\mathfrak{S}$ and q is a positive integer such that $\mathcal{D}_i^{\otimes q}$ is the pullback of a line bundle on S for any i, then for a submonoid Λ of $\mathbb{Z}^m$ we let

$$\mathcal{A}_F(\Lambda) := \bigoplus_{\lambda \in \Lambda,\ k \in \mathbb{N}} \{s \in H^0(\mathfrak{S}, \mathcal{D}(\lambda)) \mid q\nu_F(s) \geq k\},$$

with the multiplication induced by

$$\mathcal{D}(\lambda) \otimes \mathcal{D}(\lambda') \cong \mathcal{D}(\lambda + \lambda').$$

Let $\mathfrak{S}$, ρ, F, and h be as in Definition 9.1. Since $\rho_*\mathcal{O}_{\mathfrak{S}} = \mathcal{O}_S$ and

$$\operatorname{div}(s^{\otimes qq'}) = q' \operatorname{div}(s^{\otimes q})$$

for any positive integer q', $\nu_F(s)$ and the nefness of $\mathcal{D}$ do not depend on q, $q\nu_F(s) \in \mathbb{N}$, $\nu_F(ss') = \nu_F(s) + \nu_F(s')$ for sections s of $\mathcal{D}(\lambda)$ and s' of $\mathcal{D}(\lambda')$, and $\nu_F(s+s') \geq \min\{\nu_F(s), \nu_F(s')\}$ for sections s, s' of $\mathcal{D}(\lambda)$, so $\mathcal{A}_F(\Lambda)$ is a $\Lambda \times \mathbb{N}$-graded $\mathbb{C}$-algebra.

Theorem 9.2. *Let $\mathcal{E} \to T$, ρ, q, $\mathcal{D}_0$, $\mathcal{M}_0$, h, J, $\mathcal{F}_j$, and $\mathcal{G}_j$ be as follows.*

- *(a) $\mathcal{E} \to T$ is a flat projective morphism between varieties whose fibers E_t are normal varieties of Fano type.*
- *(b) $\rho : \mathfrak{S} \to \mathcal{E}$ is a family over T of normal tame Deligne–Mumford stacks $\mathfrak{S}_t$ with coarse spaces E_t, and ρ_t is an isomorphism over a dense open subset of E_t for any $t \in T$.*
- *(c) q is a positive integer, and $\mathcal{D}_0$ is a line bundle on $\mathfrak{S}$ with $\mathcal{D}_0^{\otimes q} \cong \rho^*\mathcal{M}_0$ for a line bundle $\mathcal{M}_0$ on $\mathcal{E}$.*
- *(d) $h : \mathcal{Z} \to \mathcal{E}$ is a projective morphism such that $\mathcal{Z} \to T$ is smooth and projective and $h_t : \mathcal{Z}_t \to E_t$ is birational for any $t \in T$.*
- *(e) J is a finite set, and for any $j \in J$, $\mathcal{F}_j$ is a prime divisor on $\mathcal{Z}$ whose fibers $F_{j,t}$ are prime divisors, and $\mathcal{G}_j \geq 0$ is a $\mathbb{Q}$-linear combination of relative Mumford divisors on $\mathcal{E} \to T$ such that $(E_t, \mathcal{G}_j|_{E_t})$ is lc, $K_{E_t} + \mathcal{G}_j|_{E_t} \sim_{\mathbb{Q}} 0$, and $a(F_{j,t}, E_t, \mathcal{G}_j|_{E_t}) = 0$.*

Then, after replacing T with the members of a finite stratification into locally closed subvarieties and then with finite surjective covers of these, there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{E}$ whose restrictions to E_t form a basis of $\operatorname{Pic}(E_t)$ for any $t \in T$ and a rational polyhedral cone $\Sigma \subset \mathbb{R}^{1+m}$ such that the following hold, where for $\lambda = (\lambda_0, \dots, \lambda_m) \in \mathbb{Z}^{1+m}$ we let

$$\mathcal{D}(\lambda) := \mathcal{D}_0^{\otimes \lambda_0} \otimes \rho^*\Big(\bigotimes_{i=1}^{m} \mathcal{H}_i^{\otimes \lambda_i}\Big),$$

and $\nu_{F_{j,t}}$ and $\mathcal{A}_{F_{j,t}}$ are as in Definition 9.1 for $\mathfrak{S}_t$ and the restrictions of $\mathcal{D}_0, \rho^*\mathcal{H}_1, \dots, \rho^*\mathcal{H}_m$ to $\mathfrak{S}_t$.

- *(1) For any $t \in T$ and $\lambda \in \mathbb{Z}^{1+m}$, $\mathcal{D}(\lambda)|_{\mathfrak{S}_t}$ is nef if and only if $\lambda \in \Sigma$.*

(2) *For any $j \in J$, there exist a finite set I_j and, for any $\beta \in I_j$, a section $s_{j,\beta}$ of $\mathcal{D}(\lambda_{j,\beta})$ on $\mathfrak{S}$ with $\lambda_{j,\beta} \in \Sigma \cap \mathbb{Z}^{1+m}$ and an integer $c_{j,\beta} \geq 0$, such that for any $t \in T$, the restrictions $s_{j,\beta}|_{\mathfrak{S}_t}$ in the degrees $(\lambda_{j,\beta}, c_{j,\beta})$ are elements of $\mathcal{A}_{F_{j,t}}(\Sigma \cap \mathbb{Z}^{1+m})$ which generate this $\mathbb{C}$-algebra.*

(3) *There exists a real number $\Theta > 0$ such that $\nu_{F_{j,t}}(s_{j,\beta}|_{\mathfrak{S}_t}) \leq \Theta$ for any j, β, and t with $s_{j,\beta}|_{\mathfrak{S}_t} \neq 0$.*

Proof. In the proof, shrinking T means replacing T with a dense open subset and all the data with their restrictions. Since J is finite and the data in (2) and (3) for one j are preserved by base change, we construct them for one $j \in J$ at a time. By noetherian induction on T, possibly shrinking T or replacing T with a finite surjective cover of a dense open subset, we may assume that T is smooth and affine, and we shrink T in this way again after each later replacement of T, so that the normality of $\mathcal{E}$ and $\mathfrak{S}$ in **Step 1** persists. Let $d := \dim E_t$, which does not depend on t since $\mathcal{E} \to T$ is flat.

Step 1. In this step we construct $\mathcal{H}_1, \dots, \mathcal{H}_m$ and Σ, prove (1), and describe the sections of $\mathcal{D}(\lambda)$ by $\mathbb{Q}$-divisors on $\mathcal{E}$ and on E_t.

By [Gro66, Théorème 12.1.6(iv)], the set of points $x \in \mathcal{E}$ such that the fiber of $\mathcal{E} \to T$ through x is geometrically normal at x is open. It contains all closed points of $\mathcal{E}$ since the fibers E_t are normal, so it is $\mathcal{E}$. Since T is smooth, $\mathcal{E}$ is normal by [Mat89, Corollary to Theorem 23.9]. An étale atlas of $\mathfrak{S}$ is flat over T with normal fibers over closed points, so by [Gro66, Théorème 12.1.6(iv)] and [Mat89, Corollary to Theorem 23.9] it is normal, that is, $\mathfrak{S}$ is normal. Since ρ is a homeomorphism and $\mathcal{E}$ is integral, $\mathfrak{S}$ is integral. Since the inertia of the tame Deligne–Mumford stack $\mathfrak{S}$ is finite and unramified over $\mathfrak{S}$, its identity section is open and closed, so the locus of trivial stabilizers, the complement of the image of the rest of the inertia, is an open substack of $\mathfrak{S}$, and it is non-empty since ρ_t is an isomorphism over a dense open subset of E_t. Thus there exists a dense open subset $U_1 \subset \mathcal{E}$ over which ρ is an isomorphism.

By Lemma 8.2, after replacing T with the members of a finite stratification and then with finite surjective covers, there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{E}$ and a rational polyhedral cone $\sigma \subset \mathbb{R}^m$ such that for any $t \in T$, $\mathcal{H}_1|_{E_t}, \dots, \mathcal{H}_m|_{E_t}$ is a basis of $\mathrm{Pic}(E_t)$, and for any $u \in \mathbb{Z}^m$,

$$\mathcal{H}(u)|_{E_t} \text{ is nef} \iff u \in \sigma.$$

For $u \in \mathbb{Z}^m$, let T_u be the set of closed points $t \in T$ with

$$(\mathcal{M}_0 \otimes \mathcal{H}(-u))|_{E_t} \cong \mathcal{O}_{E_t}.$$

Since E_t is an integral projective variety, a line bundle $\mathcal{M}$ on E_t is trivial if and only if $h^0(E_t, \mathcal{M}) > 0$ and $h^0(E_t, \mathcal{M}^{-1}) > 0$. Thus T_u is closed by upper semicontinuity. Since $\mathcal{H}_1|_{E_t}, \dots, \mathcal{H}_m|_{E_t}$ is a basis of $\mathrm{Pic}(E_t)$, the set of closed points of T is the disjoint union of the countably many closed subsets T_u. Since $\mathbb{C}$ is uncountable and T is irreducible, there exists $v \in \mathbb{Z}^m$ such that T_v is the set of all closed points of T, that is,

$$\mathcal{M}_0|_{E_t} \cong \mathcal{H}(v)|_{E_t} \quad \text{for any } t \in T.$$

For $\lambda \in \mathbb{Z}^{1+m}$ we write $\lambda = (\lambda_0, \overline{\lambda})$ with $\lambda_0 \in \mathbb{Z}$ and $\overline{\lambda} = (\lambda_1, \dots, \lambda_m) \in \mathbb{Z}^m$. Let

$$\mathcal{N}_\lambda := \mathcal{M}_0^{\otimes \lambda_0} \otimes \mathcal{H}(q\overline{\lambda}).$$

Then

$$\mathcal{D}(\lambda)^{\otimes q} \cong \rho^* \mathcal{N}_\lambda \quad \text{and} \quad \mathcal{N}_\lambda|_{E_t} \cong \mathcal{H}(\lambda_0 v + q\overline{\lambda})|_{E_t}.$$

Let

$$\Sigma := \{\lambda \in \mathbb{R}^{1+m} \mid \lambda_0 v/q + \overline{\lambda} \in \sigma\},$$

which is a rational polyhedral cone. Let $t \in T$ and $\lambda \in \mathbb{Z}^{1+m}$. Since

$$\mathcal{D}(\lambda)^{\otimes q}|_{\mathfrak{S}_t} \cong \rho_t^*(\mathcal{N}_\lambda|_{E_t}) \quad \text{and} \quad \mathcal{N}_\lambda|_{E_t} \cong \mathcal{H}(\lambda_0 v + q\overline{\lambda})|_{E_t},$$

by Definition 9.1 and since $\rho_{t*}\mathcal{O}_{\mathfrak{S}_t} = \mathcal{O}_{E_t}$, $\mathcal{D}(\lambda)|_{\mathfrak{S}_t}$ is nef if and only if $\mathcal{H}(\lambda_0 v + q\overline{\lambda})|_{E_t}$ is nef, that is, $\lambda \in \Sigma$. This proves (1).

Since $\rho^{-1}(U_1) \cong U_1$, $\mathcal{D}_0$ has a rational section ζ_0. Let $\xi_1, \dots, \xi_m$ be rational sections of $\mathcal{H}_1, \dots, \mathcal{H}_m$. There exists a dense open subset $U \subset U_1$ such that ζ_0 is regular and nowhere vanishing on $\rho^{-1}(U)$ and $\xi_1, \dots, \xi_m$ are regular and nowhere vanishing on U. Possibly shrinking T, we may assume that $U \cap E_t$ is dense in E_t for any $t \in T$. Then $\xi_i|_{E_t}$ is a rational section of $\mathcal{H}_i|_{E_t}$. Let $\xi_0 := \zeta_0^{\otimes q}$, a rational section of $\mathcal{M}_0$. For $\lambda \in \mathbb{Z}^{1+m}$ and $t \in T$, let

$$\zeta(\lambda) := \zeta_0^{\otimes \lambda_0} \otimes \rho^*\Big(\bigotimes_{i=1}^m \xi_i^{\otimes \lambda_i}\Big),$$

$$D_\lambda := \frac{\lambda_0}{q}\operatorname{div}(\xi_0) + \sum_{i=1}^m \lambda_i \operatorname{div}(\xi_i), \quad \text{and}$$

$$D_{\lambda,t} := \frac{\lambda_0}{q}\operatorname{div}(\xi_0|_{E_t}) + \sum_{i=1}^m \lambda_i \operatorname{div}(\xi_i|_{E_t}).$$

Then $\zeta(\lambda)$ is a rational section of $\mathcal{D}(\lambda)$, regular and nowhere vanishing on $\rho^{-1}(U)$. Moreover,

$$\zeta(\lambda)\zeta(\lambda') = \zeta(\lambda+\lambda') \quad \text{and} \quad qD_\lambda = \operatorname{div}(\zeta(\lambda)^{\otimes q}),$$

where $\zeta(\lambda)^{\otimes q}$ is regarded as a rational section of $\mathcal{N}_\lambda$. In particular, D_λ and $D_{\lambda,t}$ are $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on $\mathcal{E}$ and on E_t. Since $\rho_*\mathcal{O}_{\mathfrak{S}} = \mathcal{O}_{\mathcal{E}}$, any rational section of $\mathcal{D}(\lambda)$ is $f\zeta(\lambda)$ for a rational function f on $\mathcal{E}$. We show that for any open subset $E^\circ \subset \mathcal{E}$,

$$f\zeta(\lambda) \in H^0(\rho^{-1}(E^\circ), \mathcal{D}(\lambda)) \iff (\operatorname{div}(f) + D_\lambda)|_{E^\circ} \geq 0, \tag{9.1}$$

that is,

$$\rho_*\mathcal{D}(\lambda) = \mathcal{O}_{\mathcal{E}}(\lfloor D_\lambda \rfloor)\cdot\zeta(\lambda),$$

and that (9.1) holds with $\mathfrak{S}$, $\mathcal{E}$, D_λ, and $\zeta(\lambda)$ replaced by $\mathfrak{S}_t$, E_t, $D_{\lambda,t}$, and $\zeta(\lambda)|_{\mathfrak{S}_t}$, that is,

$$\rho_{t*}(\mathcal{D}(\lambda)|_{\mathfrak{S}_t}) = \mathcal{O}_{E_t}(\lfloor D_{\lambda,t} \rfloor)\cdot\zeta(\lambda)|_{\mathfrak{S}_t}.$$

Let $\varphi : Y \to \rho^{-1}(E^\circ)$ be an étale surjective morphism from a scheme Y of finite type, and $\psi := \rho\circ\varphi : Y \to E^\circ$. Since $\mathfrak{S}$ is normal, Y is normal, so $f\zeta(\lambda)$ is regular on $\rho^{-1}(E^\circ)$ if and only if $\varphi^*(f\zeta(\lambda))$ is regular at any point of codimension 1 of Y. Since ψ is quasi-finite and surjective and any irreducible component of Y is of dimension $\dim \mathcal{E}$, ψ maps the set of points of codimension 1 of Y onto the set of points of codimension 1 of E°. Let Q be a point of codimension 1 of Y, $P := \psi(Q)$, and e_Q the coefficient of the closure of Q in ψ^*P, a positive integer. Let ω be a local generator of $\varphi^*\mathcal{D}(\lambda)$ at Q and ω' a local generator of $\mathcal{N}_\lambda$ at P. Then $\omega^{\otimes q} = \mu\psi^*\omega'$ for a function μ invertible at Q. Write $\varphi^*(f\zeta(\lambda)) = \gamma\omega$ and $\zeta(\lambda)^{\otimes q} = \gamma'\omega'$ near P, where γ, γ' are rational functions on Y and on $\mathcal{E}$ and $\operatorname{mult}_P \operatorname{div}(\gamma') = q \operatorname{mult}_P D_\lambda$. Since $\gamma^q\mu = \psi^*(f^q\gamma')$,

$$\operatorname{mult}_Q \operatorname{div}(\gamma) = \frac{e_Q}{q}\operatorname{mult}_P \operatorname{div}(f^q\gamma') = e_Q \operatorname{mult}_P\big(\operatorname{div}(f) + D_\lambda\big).$$

Thus $\varphi^*(f\zeta(\lambda))$ is regular at Q if and only if $\operatorname{mult}_P(\operatorname{div}(f) + D_\lambda) \geq 0$, which proves (9.1). Since $\mathfrak{S}_t$ is normal with coarse space E_t and

$$qD_{\lambda,t} = \operatorname{div}(\zeta(\lambda)^{\otimes q}|_{E_t}),$$

the argument above starting with the choice of φ, with $\mathfrak{S}_t$, E_t, $D_{\lambda,t}$, and $\zeta(\lambda)|_{\mathfrak{S}_t}$ in place of $\mathfrak{S}$, $\mathcal{E}$, D_λ, and $\zeta(\lambda)$ proves the version of (9.1) on $\mathfrak{S}_t$.

Let $t \in T$, F a prime divisor over E_t, and $h' : S' \to E_t$ a resolution such that F is a divisor on S'. If $s = f\zeta(\lambda)|_{\mathfrak{S}_t}$ is a non-zero section of $\mathcal{D}(\lambda)|_{\mathfrak{S}_t}$, then

$$s^{\otimes q} = f^q\zeta(\lambda)^{\otimes q}|_{\mathfrak{S}_t}$$

is a section of $\mathcal{N}_\lambda|_{E_t}$ with divisor $q(\operatorname{div}(f) + D_{\lambda,t})$, so

$$\nu_F(s) = \operatorname{mult}_F h'^*(\operatorname{div}(f) + D_{\lambda,t}). \tag{9.2}$$

Step 2. In this step we fix $j \in J$ and extract F_j on a $\mathbb{Q}$-factorial variety of Fano type over the geometric generic point of T.

Let $\overline{\eta}$ be a geometric generic point of T, whose residue field $k(\overline{\eta})$ is an algebraic closure of the function field $k(\eta)$ of T. Let $k_0 \subset \mathbb{C}$ be a countable algebraically closed subfield over which T, $\mathcal{E}$, $\mathcal{Z}$, $\mathcal{F}_j$, and $\mathcal{G}_j$ are defined, and T_0 the corresponding k_0-variety with $T = T_0 \times_{k_0} \mathbb{C}$. Let $t \in T$ be a closed point outside the countably many proper closed subsets of T defined over k_0. Then t maps to the generic point of T_0, so the function field of T_0 embeds into $\mathbb{C}$ and into $k(\overline{\eta})$. Since $\mathbb{C}$ and $k(\overline{\eta})$ are algebraically closed fields of transcendence degree $2^{\aleph_0}$ over the function field of T_0, there exists an isomorphism $k(\overline{\eta}) \cong \mathbb{C}$ over it, under which the base changes of $\mathcal{E}$, $\mathcal{Z}$, $\mathcal{F}_j$, $\mathcal{G}_j$ to $\overline{\eta}$ are isomorphic to their restrictions to t. Thus $E_{\overline{\eta}}$ is a normal variety of Fano type, and with $G := \mathcal{G}_j|_{E_{\overline{\eta}}}$, the pair $(E_{\overline{\eta}}, G)$ is lc, $K_{E_{\overline{\eta}}} + G \sim_{\mathbb{Q}} 0$ and $a(F_{j,\overline{\eta}}, E_{\overline{\eta}}, G) = 0$. In the rest of this step we work over $k(\overline{\eta}) \cong \mathbb{C}$ and write $E := E_{\overline{\eta}}$ and $F_j := F_{j,\overline{\eta}}$.

Since E is of Fano type, (E, B) is a klt log Fano pair for some $\mathbb{R}$-divisor $B \geq 0$, and since the coefficient vectors of the $\mathbb{R}$-divisors B' with $\operatorname{Supp} B' \subset \operatorname{Supp} B$ and $K_E + B'$ $\mathbb{R}$-Cartier form an affine subspace W defined over $\mathbb{Q}$ (Lemma 2.9(1) at any closed point of E) and klt and ampleness are open conditions on W, there exists a $\mathbb{Q}$-divisor $B_0 \geq 0$ such that (E, B_0) is a klt log Fano pair. Let $\epsilon \in (0,1)$ be a rational number with $\epsilon a(F_j, E, B_0) < 1$, and $\Delta := (1-\epsilon)G + \epsilon B_0$. Then (E, Δ) is klt, $-(K_E + \Delta) \sim_{\mathbb{Q}} -\epsilon(K_E + B_0)$ is ample, and

$$0 < a(F_j, E, \Delta) = (1-\epsilon)a(F_j, E, G) + \epsilon a(F_j, E, B_0) = \epsilon a(F_j, E, B_0) < 1.$$

By [BCHM10, Corollary 1.4.3], there exists a projective birational morphism $g : Z' \to E$ from a $\mathbb{Q}$-factorial variety Z' whose only exceptional prime divisor is the center of F_j on Z' if F_j is exceptional over E, and which is small otherwise. Let F' be the center of F_j on Z', which is a prime divisor. Let Δ' be the $\mathbb{Q}$-divisor on Z' with $K_{Z'} + \Delta' = g^*(K_E + \Delta)$ and $g_*\Delta' = \Delta$. Since $0 < a(F_j, E, \Delta) < 1$, $\Delta' \geq 0$ and (Z', Δ') is klt. Since $-(K_{Z'} + \Delta') = g^*(-(K_E + \Delta))$ is nef and big, Z' is of Fano type, and Z' is a Mori dream space by [BCHM10, Corollary 1.3.2]. Let $A \geq 0$ be an ample Cartier divisor on Z'.

The data Z', g, F', and A are defined over a finite extension of $k(\eta)$. Possibly replacing T with a finite surjective cover and shrinking T, we may assume that they are the base changes to $\overline{\eta}$ of a projective morphism $g : \mathcal{Z}' \to \mathcal{E}$, a prime divisor $\mathcal{F}'$ on $\mathcal{Z}'$, and a Cartier divisor $\mathcal{A} \geq 0$ on $\mathcal{Z}'$. Let $\mathcal{V}$ be the normalization of the closure of the graph of the birational map $\mathcal{Z} \dashrightarrow \mathcal{Z}'$ over $\mathcal{E}$, $\pi : \mathcal{V} \to \mathcal{Z}$ the projection, and $\mathcal{F}_{\mathcal{V}} := \pi_*^{-1}\mathcal{F}_j$. By generic flatness and [Gro66, Théorèmes 8.10.5(i), 9.7.7, and 12.1.6, and Corollaire 9.6.4], possibly shrinking T, we may assume the following for any closed point $t \in T$:

- $\mathcal{Z}'$ and $\mathcal{V}$ are flat over T with normal fibers Z'_t and V_t, and the morphisms $g_t : Z'_t \to E_t$, $V_t \to \mathcal{Z}_t$, and $V_t \to Z'_t$ are birational.
- $A_t := \mathcal{A}|_{Z'_t} \geq 0$ is an ample Cartier divisor, $F'_t := \mathcal{F}'|_{Z'_t}$ is a prime divisor, and $\mathcal{F}_{\mathcal{V}}|_{V_t}$ is a prime divisor which maps birationally onto $F_{j,t}$ and onto F'_t.

Thus, computing both sides on V_t, for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on E_t,

$$\operatorname{mult}_{F_{j,t}} h_t^* D = \operatorname{mult}_{F'_t} g_t^* D. \tag{9.3}$$

Step 3. In this step we show that the ring $R_{\overline{\eta}}$ defined below is a finitely generated normal domain.

For

$$\mathbf{a} = (\lambda, k, n) \in \mathbb{Z}^{1+m} \times \mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^{m+3},$$

let

$$\Upsilon(\mathbf{a}) := g^* D_\lambda - \frac{k}{q}\mathcal{F}' + n\mathcal{A} \quad \text{and} \quad \Upsilon_t(\mathbf{a}) := g_t^* D_{\lambda,t} - \frac{k}{q}F'_t + nA_t,$$

$\mathbb{Q}$-divisors on $\mathcal{Z}'$ and on Z'_t, where here and below t is a closed point of T, the generic point η or $\overline{\eta}$. The $\mathbb{Q}$-divisors $\Upsilon(\mathbf{a})$ and $\Upsilon_t(\mathbf{a})$ are linear in $\mathbf{a}$. Let

$$R_t := \bigoplus_{\mathbf{a}\in\mathbb{Z}^{m+3}} H^0\big(Z'_t, \mathcal{O}_{Z'_t}(\lfloor \Upsilon_t(\mathbf{a})\rfloor)\big)\chi^{\mathbf{a}},$$

a subring of the Laurent polynomial ring $k(Z'_t)[\chi^{\mathbf{a}} \mid \mathbf{a}\in\mathbb{Z}^{m+3}]$ over the function field $k(Z'_t)$, where for a $\mathbb{Q}$-divisor D,

$$H^0(Z'_t, \mathcal{O}_{Z'_t}(\lfloor D\rfloor)) = \{f \mid \operatorname{div}(f) + D \geq 0\}.$$

We show that the ring R_t is a normal domain. For a prime divisor P on Z'_t, let w_P be the valuation of $k(Z'_t)(\chi^{\mathbf{a}} \mid \mathbf{a})$ with

$$w_P\Big(\sum_{\mathbf{a}} f_{\mathbf{a}}\chi^{\mathbf{a}}\Big) = \min_{\mathbf{a}}\big(\operatorname{mult}_P \operatorname{div}(f_{\mathbf{a}}) + \operatorname{mult}_P \Upsilon_t(\mathbf{a})\big),$$

which exists since $\operatorname{mult}_P \Upsilon_t(\mathbf{a})$ is linear in $\mathbf{a}$. Then R_t is the set of elements of $k(Z'_t)[\chi^{\mathbf{a}} \mid \mathbf{a}]$ on which w_P is non-negative for any P, so R_t is the intersection of the normal domain $k(Z'_t)[\chi^{\mathbf{a}} \mid \mathbf{a}]$ with the valuation rings of the valuations w_P of $k(Z'_t)(\chi^{\mathbf{a}} \mid \mathbf{a})$.

We show that $R_{\overline{\eta}}$ is finitely generated over $k(\overline{\eta}) \cong \mathbb{C}$, and we write $R := R_{\overline{\eta}}$. Recall from **Step 2** that $Z' = Z'_{\overline{\eta}}$ and $A = A_{\overline{\eta}}$. Let $L_1, \dots, L_r$ be Cartier divisors on Z' whose classes form a basis of $\operatorname{Pic}(Z')_{\mathbb{Q}}$ and span a cone Λ_L containing the effective cone of Z', as in [HK00, Definition 2.6], and $L(u) := \sum_{i=1}^r u_i L_i$ for $u \in \mathbb{Z}^r$. Since $H^0(Z', \mathcal{O}_{Z'}(L(u))) \neq 0$ implies $[L(u)] \in \Lambda_L$, that is, $u \in \mathbb{N}^r$, the ring $\bigoplus_{u\in\mathbb{Z}^r} H^0(Z', \mathcal{O}_{Z'}(L(u)))$ is the Cox ring of Z', which is finitely generated since Z' is a Mori dream space [HK00, Definition 2.6 and Proposition 2.9]. Let $z_1, \dots, z_{N_1}$ be homogeneous generators of $\bigoplus_{u\in\mathbb{Z}^r} H^0(Z', \mathcal{O}_{Z'}(L(u)))$ of degrees $u^{(1)}, \dots, u^{(N_1)} \in \mathbb{Z}^r$. Since Z' is $\mathbb{Q}$-factorial and the quotient of $\operatorname{Pic}(Z')$ by the subgroup generated by the classes of $L_1, \dots, L_r$ is a torsion group, there exists a positive integer N such that for any basis vector e of $\mathbb{Z}^{m+3}$, $N\Upsilon_{\overline{\eta}}(e)$ is Cartier and

$$\mathcal{O}_{Z'}(N\Upsilon_{\overline{\eta}}(e)) \cong \mathcal{O}_{Z'}(L(\ell(e))) \quad \text{for some } \ell(e) \in \mathbb{Z}^r.$$

Fixing the isomorphisms for the basis vectors and taking tensor products, we obtain isomorphisms

$$\mathcal{O}_{Z'}(N\Upsilon_{\overline{\eta}}(\mathbf{a})) \cong \mathcal{O}_{Z'}(L(\ell(\mathbf{a}))) \quad \text{for all } \mathbf{a} \in \mathbb{Z}^{m+3},$$

compatible with multiplication, where $\ell : \mathbb{Z}^{m+3} \to \mathbb{Z}^r$ is the induced homomorphism. Thus the subring

$$R^{(N)} := \bigoplus_{\mathbf{a}\in\mathbb{Z}^{m+3}} H^0(Z', \mathcal{O}_{Z'}(N\Upsilon_{\overline{\eta}}(\mathbf{a})))\chi^{N\mathbf{a}}$$

of R is isomorphic to the subring of

$$\bigoplus_u H^0(Z', \mathcal{O}_{Z'}(L(u)))[\chi^{\mathbf{a}} \mid \mathbf{a}]$$

spanned by the monomials $z_1^{\beta_1}\cdots z_{N_1}^{\beta_{N_1}}\chi^{\mathbf{a}}$ with

$$(\beta, \mathbf{a}) \in \mathbb{N}^{N_1} \times \mathbb{Z}^{m+3} \quad \text{and} \quad \sum_{\alpha=1}^{N_1} \beta_\alpha u^{(\alpha)} = \ell(\mathbf{a}).$$

The set of such $(\beta, \mathbf{a})$ is the set of lattice points of a rational polyhedral cone, which is a finitely generated monoid by Gordan's lemma [Ful93, Section 1.2, Proposition 1]. Thus $R^{(N)}$ is finitely generated. For a homogeneous element $f\chi^{\mathbf{a}} \in R$, we have $\operatorname{div}(f) + \Upsilon_{\overline{\eta}}(\mathbf{a}) \geq 0$, so

$$N\operatorname{div}(f) + \Upsilon_{\overline{\eta}}(N\mathbf{a}) \geq 0$$

and $(f\chi^{\mathbf{a}})^N \in R^{(N)}$. Thus R is integral over $R^{(N)}$. Let $e_A \in \mathbb{Z}^{m+3}$ be the vector $(0,0,1)$. Since $A \geq 0$ is ample, $\chi^{Nne_A} \in R^{(N)}$ for $n \geq 0$, the ratios of elements of $H^0(Z', \mathcal{O}_{Z'}(NnA))$ for $n \gg 0$ generate $k(Z')$, and

$$H^0(Z', \mathcal{O}_{Z'}(N\Upsilon_{\overline{\eta}}(\mathbf{a}) + NnA)) \neq 0$$

for any $\mathbf{a}$ and $n \gg 0$. Thus the fraction field of $R^{(N)}$ is $k(Z')(\chi^{N\mathbf{a}} \mid \mathbf{a})$, over which the fraction field of R is finite. Since R is contained in the integral closure of $R^{(N)}$ in the fraction field of R, which is a finite $R^{(N)}$-module, R is a finite $R^{(N)}$-module. Thus $R = R_{\overline{\eta}}$ is finitely generated.

The prime divisors on Z'_η remain reduced after base change to $\overline{\eta}$, and distinct prime divisors have no common components after base change. Thus rounding down commutes with base change to $\overline{\eta}$, and $R_\eta \otimes_{k(\eta)} k(\overline{\eta}) = R_{\overline{\eta}}$ degreewise. Homogeneous elements of R_η which generate $R_{\overline{\eta}}$ generate R_η, so R_η is finitely generated. Since $R_{\overline{\eta}}$ is a normal domain over the algebraically closed field $k(\overline{\eta})$, $\operatorname{Spec} R_\eta$ is geometrically normal and geometrically integral.

Step 4. In this step we spread R_η to a finitely generated $\mathcal{O}(T)$-algebra $\mathcal{R}$ with $\mathcal{R} \otimes k(t) = R_t$ for any closed point $t \in T$.

Let

$$\Xi := \operatorname{Supp}(g^* \operatorname{div}(\xi_0)) \cup \bigcup_{i=1}^{m} \operatorname{Supp}(g^* \operatorname{div}(\xi_i)) \cup \mathcal{F}' \cup \operatorname{Supp} \mathcal{A}.$$

Possibly shrinking T, any prime divisor contained in Ξ dominates T. Let Π be the finite set of prime divisors contained in Ξ. Let $\mathcal{Z}'^\circ \subset \mathcal{Z}'$ be the open subset of points at which $\mathcal{Z}' \to T$ is smooth, which are contained in at most one $P \in \Pi$, and at which such a P is smooth over T. Since $Z'_{\overline{\eta}}$ is normal and the restrictions of the elements of Π to $Z'_{\overline{\eta}}$ are reduced divisors without common components, $\dim(Z'_{\overline{\eta}} \backslash \mathcal{Z}'^\circ) \leq d-2$. By [Gro66, Proposition 9.9.1], possibly shrinking T, $\dim(Z'_t \backslash \mathcal{Z}'^\circ) \leq d-2$ for any t, so

$$\dim(\mathcal{Z}' \backslash \mathcal{Z}'^\circ) \leq \dim \mathcal{Z}' - 2. \tag{9.4}$$

Let $Z'^\circ_t := \mathcal{Z}'^\circ \cap Z'_t$. On $\mathcal{Z}'^\circ$ the elements of Π are pairwise disjoint Cartier divisors smooth over T. Thus restriction to Z'°_t commutes with taking the divisors of the rational sections $g^*\xi_i$ and with rounding down:

$$\lfloor \Upsilon(\mathbf{a}) \rfloor|_{Z'^\circ_t} = \lfloor \Upsilon_t(\mathbf{a}) \rfloor|_{Z'^\circ_t} \quad \text{for any } \mathbf{a} \in \mathbb{Z}^{m+3}.$$

Let

$$\mathcal{B} := \bigoplus_{\mathbf{a} \in \mathbb{Z}^{m+3}} \mathcal{O}_{\mathcal{Z}'^\circ}(\lfloor \Upsilon(\mathbf{a}) \rfloor)\chi^{\mathbf{a}} \quad \text{and} \quad \mathcal{W} := \operatorname{Spec}_{\mathcal{Z}'^\circ} \mathcal{B},$$

with the multiplication of rational functions. Let N' be a positive integer such that $N'\Upsilon(\mathbf{a})$ has integral coefficients for any $\mathbf{a}$. For $\mathbf{a}' \in \{0, \dots, N'-1\}^{m+3}$ and $\mathbf{a} \in \mathbb{Z}^{m+3}$,

$$\lfloor \Upsilon(\mathbf{a}' + N'\mathbf{a}) \rfloor = \lfloor \Upsilon(\mathbf{a}') \rfloor + N'\Upsilon(\mathbf{a}).$$

Thus the subalgebra

$$\mathcal{B}^{(N')} := \bigoplus_{\mathbf{a}} \mathcal{O}_{\mathcal{Z}'^\circ}(N'\Upsilon(\mathbf{a}))\chi^{N'\mathbf{a}}$$

is locally a Laurent polynomial ring in $m+3$ variables over $\mathcal{O}_{\mathcal{Z}'^\circ}$, and $\mathcal{B}$ is the direct sum of the N'^{m+3} invertible $\mathcal{B}^{(N')}$-modules

$$\bigoplus_{\mathbf{a}} \mathcal{O}_{\mathcal{Z}'^\circ}(\lfloor \Upsilon(\mathbf{a}' + N'\mathbf{a}) \rfloor)\chi^{\mathbf{a}'+N'\mathbf{a}}, \quad \mathbf{a}' \in \{0, \dots, N'-1\}^{m+3}.$$

Hence $\mathcal{W}$ is of finite type over T, and

$$\mathcal{W}_t = \operatorname{Spec}_{Z'^\circ_t} \bigoplus_{\mathbf{a}} \mathcal{O}_{Z'^\circ_t}(\lfloor \Upsilon_t(\mathbf{a}) \rfloor)\chi^{\mathbf{a}}.$$

Since Z'_t is normal and $\dim(Z'_t \backslash Z'^\circ_t) \leq d-2$,

$$\Gamma(\mathcal{W}_t, \mathcal{O}_{\mathcal{W}_t}) = \bigoplus_{\mathbf{a}} H^0\big(Z'^\circ_t, \mathcal{O}(\lfloor \Upsilon_t(\mathbf{a}) \rfloor)\big)\chi^{\mathbf{a}} = R_t. \tag{9.5}$$

Let $x_1, \dots, x_{N_2}$ be homogeneous generators of R_η of degrees $\mathbf{a}_1, \dots, \mathbf{a}_{N_2}$ (**Step 3**). Possibly shrinking T, for any α the element x_α extends to a section of

$$\mathcal{O}_{\mathcal{Z}'}(\lfloor \Upsilon(\mathbf{a}_\alpha) \rfloor),$$

hence to an element of $\Gamma(\mathcal{W}, \mathcal{O}_\mathcal{W})$. Let $\mathcal{R} \subset \Gamma(\mathcal{W}, \mathcal{O}_\mathcal{W})$ be the graded $\mathcal{O}(T)$-subalgebra generated by $x_1, \dots, x_{N_2}$, which is of finite presentation and satisfies $\mathcal{R} \otimes_{\mathcal{O}(T)} k(\eta) = R_\eta$. Let $\mathcal{Y} := \operatorname{Spec} \mathcal{R}$ and $\iota : \mathcal{W} \to \mathcal{Y}$ the induced morphism over T.

We show that

$$\iota_\eta : \mathcal{W}_\eta \to \mathcal{Y}_\eta = \operatorname{Spec} R_\eta$$

is an open immersion. Let n_0 be a positive integer such that $n_0 A_\eta$ is globally generated, and take finitely many sections $\tau \in H^0(Z'_\eta, \mathcal{O}(n_0 A_\eta))$ whose non-vanishing loci $Z'_{\eta,\tau}$ cover Z'_η, so that $\tau\chi^{n_0 e_A} \in R_\eta$. The open subset $Z'_{\eta,\tau}$ is affine with complement $\operatorname{Supp}(\operatorname{div}(\tau) + n_0 A_\eta)$. Let

$$\widetilde{\mathcal{B}}_\eta := \bigoplus_{\mathbf{a}} \mathcal{O}_{Z'_\eta}(\lfloor \Upsilon_\eta(\mathbf{a}) \rfloor)\chi^{\mathbf{a}},$$

a sheaf of algebras on Z'_η whose restriction to Z'°_η is $\mathcal{B}|_{Z'^\circ_\eta}$, and let $y\chi^{\mathbf{a}}$ be a section of $\widetilde{\mathcal{B}}_\eta$ over $Z'_{\eta,\tau}$. Then

$$\operatorname{div}(y) + \lfloor \Upsilon_\eta(\mathbf{a}) \rfloor \geq 0 \quad \text{on } Z'_{\eta,\tau},$$

so for $c \gg 0$,

$$\operatorname{div}(y\tau^c) + \lfloor \Upsilon_\eta(\mathbf{a} + cn_0 e_A) \rfloor = \operatorname{div}(y) + \lfloor \Upsilon_\eta(\mathbf{a}) \rfloor + c\big(\operatorname{div}(\tau) + n_0 A_\eta\big) \geq 0,$$

and

$$y\chi^{\mathbf{a}} = (y\tau^c \chi^{\mathbf{a}+cn_0 e_A})(\tau\chi^{n_0 e_A})^{-c}.$$

Thus

$$(R_\eta)_{\tau\chi^{n_0 e_A}} = \Gamma(Z'_{\eta,\tau}, \widetilde{\mathcal{B}}_\eta).$$

The preimage $\iota_\eta^{-1}(\operatorname{Spec}(R_\eta)_{\tau\chi^{n_0 e_A}})$ is the part of $\mathcal{W}_\eta$ over $Z'^\circ_\eta \cap Z'_{\eta,\tau}$, on which ι_η is the open immersion from the relative spectrum of $\widetilde{\mathcal{B}}_\eta$ over $Z'^\circ_\eta \cap Z'_{\eta,\tau}$ to the relative spectrum of $\widetilde{\mathcal{B}}_\eta$ over $Z'_{\eta,\tau}$. Since the open subsets $Z'_{\eta,\tau}$ cover Z'_η, ι_η is an open immersion.

Moreover,

$$\dim(\mathcal{Y}_\eta \backslash \mathcal{W}_\eta) \leq \dim \mathcal{Y}_\eta - 2.$$

Otherwise there exists a prime divisor $Q \subset \mathcal{Y}_\eta \backslash \mathcal{W}_\eta$ on the normal affine scheme $\mathcal{Y}_\eta$. Since $\mathcal{Y}_\eta$ is affine and the coherent sheaf $\mathcal{O}_{\mathcal{Y}_\eta}(Q)$ strictly contains $\mathcal{O}_{\mathcal{Y}_\eta}$,

$$H^0(\mathcal{Y}_\eta, \mathcal{O}_{\mathcal{Y}_\eta}(Q)) \supsetneq R_\eta.$$

The elements of $H^0(\mathcal{Y}_\eta, \mathcal{O}_{\mathcal{Y}_\eta}(Q))$ are regular on $\mathcal{W}_\eta$, which contradicts $\Gamma(\mathcal{W}_\eta, \mathcal{O}_{\mathcal{W}_\eta}) = R_\eta$ (9.5). After base change to $\overline{\eta}$, $\mathcal{Y}_{\overline{\eta}} = \operatorname{Spec} R_{\overline{\eta}}$ is normal and integral, and

$$\dim(\mathcal{Y}_{\overline{\eta}} \backslash \mathcal{W}_{\overline{\eta}}) \leq \dim \mathcal{Y}_{\overline{\eta}} - 2.$$

By [Gro66, Théorème 8.10.5(iii)], possibly shrinking T, ι is an open immersion. By [Gro66, Théorème 9.7.7, Propositions 9.9.1 and 9.9.4, and Théorème 12.1.6], possibly shrinking T, for any closed point $t \in T$ the fiber $\mathcal{Y}_t$ is normal and integral, and

$$\dim(\mathcal{Y}_t \backslash \mathcal{W}_t) \leq \dim \mathcal{Y}_t - 2.$$

Since $\mathcal{Y}_t$ is normal and affine, by (9.5) we obtain, for any closed point $t \in T$,

$$\mathcal{R} \otimes_{\mathcal{O}(T)} k(t) = \Gamma(\mathcal{Y}_t, \mathcal{O}_{\mathcal{Y}_t}) = \Gamma(\mathcal{W}_t, \mathcal{O}_{\mathcal{W}_t}) = R_t, \tag{9.6}$$

an isomorphism of $\mathbb{Z}^{m+3}$-graded rings mapping $x_\alpha \otimes 1$ to $x_\alpha|_{Z'_t}$.

Step 5. In this step we restrict to the degrees in $\Sigma \times \mathbb{R}_{\geq 0} \times \{0\}$ and prove (2) and (3).

Let $t \in T$ be a closed point, $\lambda \in \mathbb{Z}^{1+m}$, $k \in \mathbb{N}$, and f a rational function on E_t, equivalently on Z'_t. By (9.1) on $\mathfrak{S}_t$, (9.2), and (9.3), $f\zeta(\lambda)|_{\mathfrak{S}_t}$ is a section of $\mathcal{D}(\lambda)|_{\mathfrak{S}_t}$ with $q\nu_{F_{j,t}}(f\zeta(\lambda)|_{\mathfrak{S}_t}) \geq k$ if and only if

$$\operatorname{div}(f) + D_{\lambda,t} \geq 0 \quad \text{and} \quad \operatorname{mult}_{F'_t} g_t^*\big(\operatorname{div}(f) + D_{\lambda,t}\big) \geq \frac{k}{q}.$$

Since $g_t^* D \geq 0$ and $g_{t*} g_t^* D = D$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on E_t, this holds if and only if $\operatorname{div}(f) + g_t^* D_{\lambda,t} - (k/q)F'_t \geq 0$ on Z'_t. Since $\zeta(\lambda)\zeta(\lambda') = \zeta(\lambda+\lambda')$, the maps

$$f\chi^{(\lambda,k,0)} \to f\zeta(\lambda)|_{\mathfrak{S}_t}$$

form an isomorphism of graded algebras

$$\bigoplus_{\lambda \in \Sigma \cap \mathbb{Z}^{1+m},\ k \in \mathbb{N}} H^0\Big(Z'_t, \mathcal{O}\big(\lfloor g_t^* D_{\lambda,t} - \tfrac{k}{q} F'_t \rfloor\big)\Big)\chi^{(\lambda,k,0)} \cong \mathcal{A}_{F_{j,t}}(\Sigma \cap \mathbb{Z}^{1+m}). \tag{9.7}$$

Let

$$\Sigma^+ := \Sigma \times \mathbb{R}_{\geq 0} \times \{0\} \subset \mathbb{R}^{1+m} \times \mathbb{R} \times \mathbb{R},$$

a rational polyhedral cone. Then the left hand side of (9.7) is

$$\bigoplus_{\mathbf{a} \in \Sigma^+ \cap \mathbb{Z}^{m+3}} (R_t)_{\mathbf{a}}.$$

The monoid

$$\Big\{\beta \in \mathbb{N}^{N_2} \mid \sum_{\alpha=1}^{N_2} \beta_\alpha \mathbf{a}_\alpha \in \Sigma^+\Big\}$$

is the set of lattice points of a rational polyhedral cone, so it is generated by a finite subset I_j by Gordan's lemma [Ful93, Section 1.2, Proposition 1]. For $\beta \in I_j$, let

$$x^\beta := \prod_{\alpha=1}^{N_2} x_\alpha^{\beta_\alpha} \in \mathcal{R} \quad \text{and} \quad (\lambda_{j,\beta}, c_{j,\beta}, 0) := \sum_{\alpha=1}^{N_2} \beta_\alpha \mathbf{a}_\alpha \in \Sigma^+.$$

Since $\mathcal{R}$ is spanned over $\mathcal{O}(T)$ by the monomials in $x_1, \ldots, x_{N_2}$, the elements x^β with $\beta \in I_j$ generate

$$\bigoplus_{\mathbf{a} \in \Sigma^+ \cap \mathbb{Z}^{m+3}} \mathcal{R}_{\mathbf{a}}.$$

By (9.6), for any closed point $t \in T$, the restrictions $x^\beta|_{Z'_t}$ with $\beta \in I_j$ generate

$$\bigoplus_{\mathbf{a} \in \Sigma^+ \cap \mathbb{Z}^{m+3}} (R_t)_{\mathbf{a}}.$$

Let $\beta \in I_j$ and $\lambda := \lambda_{j,\beta}$. The element x^β is $f_{j,\beta}\chi^{(\lambda,c_{j,\beta},0)}$ for a rational function $f_{j,\beta}$ on $\mathcal{Z}'$ with

$$\operatorname{div}(f_{j,\beta}) + g^* D_\lambda \geq \frac{c_{j,\beta}}{q}\mathcal{F}' \geq 0$$

at the points of codimension 1 of $\mathcal{Z}'^\circ$. By (9.4), any point of codimension 1 of $\mathcal{Z}'$ is contained in $\mathcal{Z}'^\circ$. Since g is birational,

$$\operatorname{div}(f_{j,\beta}) + D_\lambda = g_*(\operatorname{div}(f_{j,\beta}) + g^* D_\lambda) \geq 0 \quad \text{on } \mathcal{E}.$$

By (9.1) with $E^\circ = \mathcal{E}$,

$$s_{j,\beta} := f_{j,\beta}\zeta(\lambda_{j,\beta}) \in H^0(\mathfrak{S}, \mathcal{D}(\lambda_{j,\beta})).$$

Since $\operatorname{div}(f_{j,\beta}) + D_\lambda \geq 0$ and $D_\lambda|_U = 0$, $f_{j,\beta}$ is regular on U. Since $U \cap E_t$ is dense in E_t, the section

$$s_{j,\beta}|_{\mathfrak{S}_t} = f_{j,\beta}|_{E_t}\zeta(\lambda_{j,\beta})|_{\mathfrak{S}_t}$$

corresponds to $x^\beta|_{Z'_t}$ under (9.7). Thus for any $t \in T$, the restrictions $s_{j,\beta}|_{\mathfrak{S}_t}$ in the degrees $(\lambda_{j,\beta}, c_{j,\beta})$ generate $\mathcal{A}_{F_{j,t}}(\Sigma \cap \mathbb{Z}^{1+m})$, with $\lambda_{j,\beta} \in \Sigma \cap \mathbb{Z}^{1+m}$ and $c_{j,\beta} \geq 0$. This proves (2).

Let $\mathcal{Q}$ be a line bundle on $\mathcal{Z}$ very ample over T. Let $\beta \in I_j$, $\lambda := \lambda_{j,\beta}$, and $t \in T$ with $s := s_{j,\beta}|_{\mathfrak{S}_t} \neq 0$. Then $h_t^* \operatorname{div}(s^{\otimes q}) \geq 0$ is a Cartier divisor on $\mathcal{Z}_t$ with line bundle $(h^*\mathcal{N}_\lambda)|_{\mathcal{Z}_t}$, and

$$h_t^* \operatorname{div}(s^{\otimes q}) - q\nu_{F_{j,t}}(s)F_{j,t} \geq 0.$$

Since $\mathcal{Q}|_{\mathcal{Z}_t}$ is ample,

$$q\nu_{F_{j,t}}(s) \cdot \left(F_{j,t} \cdot (\mathcal{Q}|_{\mathcal{Z}_t})^{d-1}\right) \leq (h^*\mathcal{N}_\lambda)|_{\mathcal{Z}_t} \cdot (\mathcal{Q}|_{\mathcal{Z}_t})^{d-1}.$$

The intersection number $F_{j,t} \cdot (\mathcal{Q}|_{\mathcal{Z}_t})^{d-1}$ is a positive integer, and the right hand side does not depend on t since $\mathcal{Z} \to T$ is flat and T is connected. Let Θ be the maximum of 1 and the numbers

$$\frac{1}{q}(h^*\mathcal{N}_{\lambda_{j,\beta}})|_{\mathcal{Z}_t} \cdot (\mathcal{Q}|_{\mathcal{Z}_t})^{d-1}$$

over the finitely many $j \in J$, $\beta \in I_j$, and members of the stratification. Then $\nu_{F_{j,t}}(s_{j,\beta}|_{\mathfrak{S}_t}) \leq \Theta$ whenever $s_{j,\beta}|_{\mathfrak{S}_t} \neq 0$. This proves (3), and the theorem follows. □

10. Lifting sections and an upper bound

In this section, we bound the local volume of a klt germ $(X \ni x, B)$ from above by means of a Kollár component E of $(X \ni x, B)$.

10.1. **Lifting to the formal neighborhood.**

Definition 10.1. Let Y be a normal variety and E a $\mathbb{Q}$-Cartier prime divisor on Y. The *index* 1 *covering stack* of E is the normal tame Deligne–Mumford stack $\mathfrak{Y}$ with coarse space $\rho : \mathfrak{Y} \to Y$ obtained by gluing the local index 1 covers of E (Definition-Lemma 2.16) as in [LX20, Section 2.4]: ρ is an isomorphism over the open subset of Y on which E is Cartier, and $\mathfrak{E} := \rho^*E$ is a Cartier divisor on $\mathfrak{Y}$ whose coarse space is E. We denote the coarse map $\mathfrak{E} \to E$ again by ρ and let $\mathcal{L} := \mathcal{O}_{\mathfrak{E}}(-\mathfrak{E})$.

Proposition 10.2. *Let $(X \ni x, B)$ be a klt germ of dimension $n \geq 2$, E a Kollár component of $(X \ni x, B)$ with plt blow-up $f : Y \to X$, $\rho : \mathfrak{Y} \to Y$ the index 1 covering stack of E, $\mathfrak{E} := \rho^*E$, $\mathcal{L} := \mathcal{O}_{\mathfrak{E}}(-\mathfrak{E})$, and $\widehat{\mathfrak{Y}}$ the formal completion of $\mathfrak{Y}$ along $\mathfrak{E}$. Then:*

(1) Possibly shrinking X to an affine neighborhood of x, the maps sending $\phi \in \mathfrak{a}_j$ $(j \geq 0)$ to the restriction to $\mathfrak{E}$ of

$$\rho^*f^*\phi \in H^0(\mathfrak{Y}, \mathcal{O}_{\mathfrak{Y}}(-j\mathfrak{E}))$$

induce an isomorphism of graded $\mathbb{C}$-algebras

$$\operatorname{gr}_E \mathcal{O}_{X,x} = \bigoplus_{j\geq 0} \mathfrak{a}_j/\mathfrak{a}_{j+1} \cong \bigoplus_{j\geq 0} H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}), \tag{10.1}$$

where $\mathfrak{a}_j$ is as in Construction-Theorem 6.2. Moreover, for q and P_q as in Construction-Theorem 6.2(3) and any $j \in q\mathbb{N}$, $\mathcal{L}^{\otimes j} = \rho^*\mathcal{O}_E(\frac{j}{q}P_q)$, and the isomorphism (10.1) sends the class of any $\phi \in \mathfrak{a}_j$ to $\rho^*((f^*\phi)|_E)$, where $(f^*\phi)|_E \in H^0(E, \mathcal{O}_E(\frac{j}{q}P_q))$ is the image of the class of ϕ under the isomorphism of Construction-Theorem 6.2(3).

(2) (a) Any line bundle $\mathcal{D}$ on $\mathfrak{E}$ extends, uniquely up to isomorphism, to a line bundle $\widehat{\mathcal{D}}$ on $\widehat{\mathfrak{Y}}$.

(b) The extension $\widehat{\mathcal{L}}$ of $\mathcal{L}$ satisfies

$$\widehat{\mathcal{L}} \cong \mathcal{O}_{\widehat{\mathfrak{Y}}}(-\mathfrak{E}).$$

(c) If $\mathcal{D}$ is a nef line bundle on $\mathfrak{E}$, then

$$H^0(\widehat{\mathfrak{Y}}, \widehat{\mathcal{D}}) \to H^0(\mathfrak{E}, \mathcal{D})$$

is surjective.

Proof. Let $a := a(E, X, B)$, and let $V \subset E$ be the closed subset on which E is not Cartier, so that $\operatorname{codim}_Y V \geq 2$ and ρ is an isomorphism over $Y \backslash V$. Since $\mathfrak{Y}$ and $\mathfrak{E}$ are tame, their coarse pushforwards are exact [AOV08, Definition 3.1].

Step 1. We prove (1). Possibly shrinking X to an affine neighborhood of x, we may assume that (X, B) is klt and $(Y, E + B_Y)$ is plt. Fix a real number η with $0 < \eta < \min\{1, a\}$. Then $(Y, B_Y + (1-\eta)E)$ is klt, and by Definition 2.5,

$$K_Y + B_Y + (1-\eta)E = f^*(K_X + B) + (a - \eta)E.$$

Thus for any integer $j \geq 0$,

$$-jE - (K_Y + B_Y + (1-\eta)E) \sim_{\mathbb{R}, X} -(j + a - \eta)E$$

is ample over X. Since $-jE$ is a $\mathbb{Q}$-Cartier Weil divisor, by the Kawamata–Viehweg vanishing theorem, $R^i f_* \mathcal{O}_Y(-jE) = 0$ for any $i > 0$, so

$$H^i(Y, \mathcal{O}_Y(-jE)) = 0 \quad \text{for any } i > 0 \text{ and } j \geq 0,$$

as X is affine. Locally on Y, $\mathcal{F}_j := \rho_* \mathcal{O}_{\mathfrak{Y}}(-j\mathfrak{E})$ is the invariant part of the pushforward of an invertible sheaf under an index 1 cover, hence reflexive of rank 1. Since

$$\mathcal{F}_j|_{Y \backslash V} \cong \mathcal{O}_Y(-jE)|_{Y \backslash V},$$

we have $\mathcal{F}_j = \mathcal{O}_Y(-jE)$. Since $\mathfrak{E}$ is Cartier,

$$\mathcal{O}_{\mathfrak{Y}}(-j\mathfrak{E})|_{\mathfrak{E}} = \mathcal{L}^{\otimes j},$$

and applying the exact functor ρ_* to the restriction sequence of $\mathcal{O}_{\mathfrak{Y}}(-j\mathfrak{E})$ to $\mathfrak{E}$, we obtain the exact sequence

$$0 \to \mathcal{O}_Y(-(j+1)E) \to \mathcal{O}_Y(-jE) \to \rho_* \mathcal{L}^{\otimes j} \to 0,$$

where $\rho_* \mathcal{L}^{\otimes j}$ is regarded as a sheaf on Y supported on E. Since $H^0(Y, \mathcal{O}_Y(-jE)) = \mathfrak{a}_j$, $H^1(Y, \mathcal{O}_Y(-(j+1)E)) = 0$, and

$$H^0(E, \rho_* \mathcal{L}^{\otimes j}) = H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}),$$

the map sending $\phi \in \mathfrak{a}_j$ to $(\rho^* f^* \phi)|_{\mathfrak{E}}$ induces an isomorphism

$$\mathfrak{a}_j / \mathfrak{a}_{j+1} \cong H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}) \quad \text{for any } j \geq 0.$$

Restriction to $\mathfrak{E}$ is compatible with multiplication, so the isomorphism in (1) follows. For q and P_q as in Construction-Theorem 6.2(3) and $j \in q\mathbb{N}$, jE is Cartier near E and $\mathfrak{E} = \rho^* E$, so

$$\mathcal{L}^{\otimes j} = \rho^*(\mathcal{O}_Y(-jE)|_E) = \rho^* \mathcal{O}_E(\tfrac{j}{q} P_q)$$

and $(\rho^* f^* \phi)|_{\mathfrak{E}} = \rho^*((f^*\phi)|_E)$ for any $\phi \in \mathfrak{a}_j$. Thus (1) holds.

Step 2. We show that $H^i(\mathfrak{E}, \mathcal{D} \otimes \mathcal{L}^{\otimes j}) = 0$ for any nef line bundle $\mathcal{D}$ on $\mathfrak{E}$ and any integers $j \geq 0$ and $i > 0$. Let q and P_q be as in Construction-Theorem 6.2(3), and let o_q be the vertex of $C(E, P_q)$. By (1), $\mathcal{L}^{\otimes q} \cong \rho^* \mathcal{O}_E(P_q)$, so by the projection formula,

$$\rho_* \mathcal{L}^{\otimes (j+kq)} \cong \rho_* \mathcal{L}^{\otimes j} \otimes \mathcal{O}_E(kP_q) \quad \text{for any integers } j, k \geq 0. \tag{10.2}$$

Let

$$\tau : \mathfrak{Y}_0 := \operatorname{Spec}_{\mathfrak{E}} \bigoplus_{j \geq 0} \mathcal{L}^{\otimes j} \to \mathfrak{E}$$

be the total space of $\mathcal{L}^{-1}$, and let $\mathfrak{E}_0 \subset \mathfrak{Y}_0$ be the zero section, defined by the ideal $\bigoplus_{j>0} \mathcal{L}^{\otimes j}$. Since ρ_* is exact and, étale locally on E, $\mathfrak{E}$ is the quotient $[U/G]$ of an affine scheme U by a finite group G [Ols16, Theorem 11.3.1],

$$Y_0 := \operatorname{Spec}_E \bigoplus_{j \geq 0} \rho_* \mathcal{L}^{\otimes j}$$

is the coarse space of $\mathfrak{Y}_0$. Let $\rho_0 : \mathfrak{Y}_0 \to Y_0$ be the coarse map, $\tau_0 : Y_0 \to E$ the projection, and $E_{Y_0} := \rho_0(\mathfrak{E}_0)$, so that $\tau_0|_{E_{Y_0}} : E_{Y_0} \to E$ is an isomorphism and ρ_{0*} is exact, since $\tau_{0*}\rho_{0*} = \rho_* \tau_*$

is exact and τ_0 is affine. Since $\mathfrak{E}$ is normal (étale locally, the pullback of $(Y, E+B_Y)$ to an index 1 cover of E is plt, as the cover is étale in codimension 1) and $\mathfrak{Y}_0$ is the total space of a line bundle on $\mathfrak{E}$, $\mathfrak{Y}_0$ is normal, so Y_0 is normal, as étale locally on E it is the quotient of a normal variety by G. Let $y \in E$ be a closed point, $r := \mathrm{ind}_y(Y, E)$, $Y' \to Y$ an index 1 cover of $(Y \ni y, E)$ with Galois group μ_r, and t a μ_r-eigenfunction defining the pullback of E near the point y' over y, with character χ. If $\chi^k = 1$ for some integer $0 < k < r$, then t^k is the pullback of a local equation of kE near y, which contradicts the minimality of r. Thus χ is injective. Hence the stabilizer of any geometric point of $\mathfrak{E}$ acts faithfully on the fiber of $\mathcal{L}$, so any point of $\mathfrak{Y}_0$ with non-trivial stabilizer is contained in $\mathfrak{E}_0$ over V. Thus ρ_0 is an isomorphism over $Y_0 \backslash V_0$, where $V_0 := E_{Y_0} \cap \tau_0^{-1}(V)$ and $\mathrm{codim}_{Y_0} V_0 \geq 2$.

By (10.2), the $\bigoplus_{k\geq 0} \mathcal{O}_E(kP_q)$-module $\bigoplus_{j\geq 0} \rho_* \mathcal{L}^{\otimes j}$ is generated by $\bigoplus_{0\leq j<q} \rho_* \mathcal{L}^{\otimes j}$, so the inclusion of $\mathcal{O}_E$-algebras induces a finite morphism

$$h : Y_0 \to W := \mathrm{Spec}_E \bigoplus_{k\geq 0} \mathcal{O}_E(kP_q) = \mathrm{Tot}(\mathcal{O}_E(-P_q)).$$

Let E_0 be the zero section of W. Near the generic point of E, $\mathfrak{E}$ is a scheme, $\mathcal{L}$ has a local generator u, and the inclusion is $\mathcal{O}_E[u^q] \subset \mathcal{O}_E[u]$. Thus $\deg h = q$ and

$$h^* E_0 = q E_{Y_0}.$$

In particular, E_{Y_0} is $\mathbb{Q}$-Cartier.

By Construction-Theorem 6.2(3), $-(K_E + B_E) \sim_{\mathbb{R}} \frac{a}{q} P_q$. Since $\dim E = n - 1 \geq 1$ and (E, B_E) is klt, by Lemma 5.2, E_0 is a Kollár component of $(C(E, P_q) \ni o_q, (B_E)_{C(E,P_q)})$ and $a(E_0, C(E, P_q), (B_E)_{C(E,P_q)}) = a/q$. Thus, for the natural morphism $\mu : W \to C(E, P_q)$ and $B_W := \mu_*^{-1}(B_E)_{C(E,P_q)}$, $(W, E_0 + B_W)$ is plt near E_0, $-E_0$ is ample over $C(E, P_q)$, and

$$K_W + E_0 + B_W \sim_{\mathbb{R}, C(E,P_q)} \frac{a}{q} E_0.$$

By (1),

$$H^0(Y_0, \mathcal{O}_{Y_0}) = \bigoplus_{j\geq 0} H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}) \cong \mathrm{gr}_E \mathcal{O}_{X,x}.$$

Let $\pi_0 : Y_0 \to C_E$ be the induced morphism. By (1), in the degrees $j \in q\mathbb{N}$, the isomorphism of (1) is ρ^* composed with the isomorphism of Construction-Theorem 6.2(3). Hence for the morphism $g : C_E \to C(E, P_q)$ in Construction-Theorem 6.2(3), $g \circ \pi_0 = \mu \circ h$, where μ is proper, contracts E_0 to o_q, and is an isomorphism over $C(E, P_q)\backslash\{o_q\}$. Since $g^{-1}(o_q) = \{o\}$ and $h^* E_0 = q E_{Y_0}$, π_0 is proper and $\pi_0^{-1}(o) = E_{Y_0}$. Both $Y_0 \backslash E_{Y_0}$ and $C_E \backslash \{o\}$ are normal and finite of degree q over $C(E, P_q)\backslash\{o_q\}$, so $\pi_0|_{Y_0\backslash E_{Y_0}}$ is finite, surjective and of degree 1, hence an isomorphism onto $C_E \backslash \{o\}$. Moreover, $-qE_{Y_0} = h^*(-E_0)$ is ample over $C(E, P_q)$, hence $-E_{Y_0}$ is ample over C_E.

Let $B_{Y_0} := (\pi_0)_*^{-1} B_{C_E}$. By Construction-Theorem 6.2(3), $K_{Y_0} + B_{Y_0} = h^*(K_W + B_W)$ over $Y_0 \backslash E_{Y_0}$. Since the ramification index of h along E_{Y_0} is q, the Hurwitz formula implies

$$K_{Y_0} + B_{Y_0} = h^*(K_W + B_W) + (q-1)E_{Y_0},$$

so

$$K_{Y_0} + E_{Y_0} + B_{Y_0} = h^*(K_W + E_0 + B_W) \sim_{\mathbb{R}, C_E} a E_{Y_0}.$$

Since h is finite, $(Y_0, E_{Y_0} + B_{Y_0})$ is plt near E_{Y_0}. By Construction-Theorem 6.2(2), possibly shrinking C_E to an affine neighborhood of o, we may assume that $(Y_0 \backslash E_{Y_0}, B_{Y_0})$ is klt. Then, for η as in **Step 1**, $(Y_0, B_{Y_0} + (1-\eta)E_{Y_0})$ is klt and

$$K_{Y_0} + B_{Y_0} + (1-\eta)E_{Y_0} \sim_{\mathbb{R}, C_E} (a - \eta)E_{Y_0}.$$

Let $\mathcal{D}$ be a nef line bundle on $\mathfrak{E}$, and let N be a positive integer and $\mathcal{D}'$ a nef line bundle on E such that $\mathcal{D}^{\otimes N} \cong \rho^* \mathcal{D}'$. As in **Step 1**, the pushforward under ρ_0 of a line bundle on $\mathfrak{Y}_0$ is

reflexive of rank 1. Since ρ_0 is an isomorphism over $Y_0\backslash V_0$, there exists a Weil divisor D_0 on Y_0 such that

$$\rho_{0*}(\tau^*\mathcal{D}\otimes\mathcal{O}_{\mathfrak{Y}_0}(-j\mathfrak{E}_0))\cong\mathcal{O}_{Y_0}(D_0-jE_{Y_0})\quad\text{for any } j\geq 0,$$

and $\mathcal{O}_{Y_0}(ND_0)\cong\tau_0^*\mathcal{D}'$. Thus D_0 is $\mathbb{Q}$-Cartier. Since the curves contracted by π_0 are contained in E_{Y_0} and $\tau_0|_{E_{Y_0}}$ is an isomorphism, D_0 is nef over C_E. For any integer $j\geq 0$,

$$(D_0-jE_{Y_0})-(K_{Y_0}+B_{Y_0}+(1-\eta)E_{Y_0})\sim_{\mathbb{R},C_E}D_0-(j+a-\eta)E_{Y_0}$$

is ample over C_E, so by the Kawamata–Viehweg vanishing theorem and since C_E is affine,

$$H^i(Y_0,\mathcal{O}_{Y_0}(D_0-jE_{Y_0}))=0\quad\text{for any } i>0 \text{ and } j\geq 0.$$

The conormal bundle of $\mathfrak{E}_0$ in $\mathfrak{Y}_0$ is $\mathcal{L}$, so under $\mathfrak{E}_0\cong\mathfrak{E}$,

$$(\tau^*\mathcal{D}\otimes\mathcal{O}_{\mathfrak{Y}_0}(-j\mathfrak{E}_0))|_{\mathfrak{E}_0}\cong\mathcal{D}\otimes\mathcal{L}^{\otimes j}.$$

Applying the exact functor ρ_{0*} to the restriction sequence, we obtain the exact sequence

$$0\to\mathcal{O}_{Y_0}(D_0-(j+1)E_{Y_0})\to\mathcal{O}_{Y_0}(D_0-jE_{Y_0})\to\rho_*(\mathcal{D}\otimes\mathcal{L}^{\otimes j})\to 0,$$

where $\rho_*(\mathcal{D}\otimes\mathcal{L}^{\otimes j})$ is regarded as a sheaf on $E_{Y_0}\cong E$. The long exact cohomology sequence and the vanishing above imply that for any $i>0$,

$$H^i(\mathfrak{E},\mathcal{D}\otimes\mathcal{L}^{\otimes j})=H^i(E,\rho_*(\mathcal{D}\otimes\mathcal{L}^{\otimes j}))=0,$$

where the first equality holds as ρ_* is exact.

Step 3. We prove (2). For any integer $k\geq 0$, let $\mathfrak{E}^{[k]}\subset\mathfrak{Y}$ be the closed substack defined by $\mathcal{O}_{\mathfrak{Y}}(-(k+1)\mathfrak{E})$, so that $\mathfrak{E}^{[0]}=\mathfrak{E}$ and $\widehat{\mathfrak{Y}}$ is the limit of the $\mathfrak{E}^{[k]}$. The ideal $\mathcal{I}_k$ of $\mathfrak{E}^{[k]}$ in $\mathfrak{E}^{[k+1]}$ satisfies

$$\mathcal{I}_k=\mathcal{O}_{\mathfrak{Y}}(-(k+1)\mathfrak{E})/\mathcal{O}_{\mathfrak{Y}}(-(k+2)\mathfrak{E})=\mathcal{L}^{\otimes(k+1)}\quad\text{and}\quad\mathcal{I}_k^2=0,$$

so on the étale site of $\mathfrak{E}^{[k+1]}$ we have the exact sequence

$$1\to\mathcal{L}^{\otimes(k+1)}\to\mathcal{O}^*_{\mathfrak{E}^{[k+1]}}\to\mathcal{O}^*_{\mathfrak{E}^{[k]}}\to 1,$$

where the first map sends a local section φ to $1+\varphi$. By **Step 2** for $\mathcal{D}=\mathcal{O}_{\mathfrak{E}}$,

$$H^1(\mathfrak{E},\mathcal{L}^{\otimes(k+1)})=H^2(\mathfrak{E},\mathcal{L}^{\otimes(k+1)})=0.$$

Hence the restriction maps

$$\mathrm{Pic}(\mathfrak{E}^{[k+1]})\to\mathrm{Pic}(\mathfrak{E}^{[k]})\quad\text{and}\quad H^0(\mathfrak{E}^{[k+1]},\mathcal{O}^*_{\mathfrak{E}^{[k+1]}})\to H^0(\mathfrak{E}^{[k]},\mathcal{O}^*_{\mathfrak{E}^{[k]}})$$

are bijective and surjective, respectively. Thus any line bundle on $\mathfrak{E}$ extends to a compatible system of line bundles on the $\mathfrak{E}^{[k]}$. For two such systems $(\mathcal{N}_k)_k$ and $(\mathcal{N}'_k)_k$ and an isomorphism $\alpha_k:\mathcal{N}_k\to\mathcal{N}'_k$, by the bijectivity there exists an isomorphism $\beta:\mathcal{N}_{k+1}\to\mathcal{N}'_{k+1}$, and $\beta|_{\mathfrak{E}^{[k]}}=\upsilon\alpha_k$ for a unit υ on $\mathfrak{E}^{[k]}$. For a lift $\widetilde{\upsilon}$ of υ to $\mathfrak{E}^{[k+1]}$, $\widetilde{\upsilon}^{-1}\beta$ extends α_k. Thus the extension is unique up to isomorphism, and as $\mathcal{L}=\mathcal{O}_{\mathfrak{E}}(-\mathfrak{E})$,

$$\widehat{\mathcal{L}}\cong\mathcal{O}_{\widehat{\mathfrak{Y}}}(-\mathfrak{E}).$$

Let $\mathcal{D}$ be a nef line bundle on $\mathfrak{E}$, $\widehat{\mathcal{D}}$ the extension of $\mathcal{D}$, and $\widehat{\mathcal{D}}_k:=\widehat{\mathcal{D}}|_{\mathfrak{E}^{[k]}}$. From the exact sequence

$$0\to\mathcal{D}\otimes\mathcal{L}^{\otimes(k+1)}\to\widehat{\mathcal{D}}_{k+1}\to\widehat{\mathcal{D}}_k\to 0$$

and the vanishing

$$H^1(\mathfrak{E},\mathcal{D}\otimes\mathcal{L}^{\otimes(k+1)})=0$$

in **Step 2**, any element of $H^0(\mathfrak{E}^{[k]},\widehat{\mathcal{D}}_k)$ lifts to $H^0(\mathfrak{E}^{[k+1]},\widehat{\mathcal{D}}_{k+1})$. By induction on k,

$$H^0(\widehat{\mathfrak{Y}},\widehat{\mathcal{D}})\to H^0(\mathfrak{E},\mathcal{D})$$

is surjective. □

Set-up 10.3. Let n, $(X \ni x, B)$, E, f, a, $\rho : \mathfrak{Y} \to Y$, $\mathfrak{E}$, $\mathcal{L}$, q, $\mathcal{D}_1, \dots, \mathcal{D}_m$, λ_0, Σ, F, s_i $(i \in I)$, (μ_i, k_i) $(i \in I)$, $\widehat{\mathfrak{Y}}$, $\widehat{\mathcal{D}}_1, \dots, \widehat{\mathcal{D}}_m$, ι, $\widetilde{s}_i$ $(i \in I)$, U, g, E', $B_{Y'}$, e, F', $\widehat{Y'}$, $\varpi_1, \dots, \varpi_m$, and ϕ_i $(i \in I)$ be as follows.

(1) $n \geq 2$ is an integer, $(X \ni x, B)$ is a klt germ of dimension n, E is a Kollár component of $(X \ni x, B)$ with plt blow-up $f : Y \to X$, and $a := a(E, X, B)$.

(2) $\rho : \mathfrak{Y} \to Y$ is the index 1 covering stack of E, $\mathfrak{E} := \rho^* E$, $\mathcal{L} := \mathcal{O}_{\mathfrak{E}}(-\mathfrak{E})$, and q is a positive integer such that qE is Cartier near E.

(3) $\mathcal{D}_1, \dots, \mathcal{D}_m$ are line bundles on $\mathfrak{E}$ such that $\mathcal{D}(\lambda)^{\otimes q}$ is the pullback of a line bundle on E for any $\lambda \in \mathbb{Z}^m$, $\lambda_0 \in \mathbb{Z}^m$ satisfies $\mathcal{L} \cong \mathcal{D}(\lambda_0)$, and Σ is the set of $\lambda \in \mathbb{Z}^m$ such that $\mathcal{D}(\lambda)$ is nef.

(4) F is a prime divisor over E, s_i $(i \in I)$ are finitely many homogeneous elements generating the $\mathbb{C}$-algebra $\mathcal{A}_F(\Sigma)$, and $(\mu_i, k_i) \in \Sigma \times \mathbb{N}$ is the degree of s_i in $\mathcal{A}_F(\Sigma)$.

(5) $\widehat{\mathfrak{Y}}$ is the formal completion of $\mathfrak{Y}$ along $\mathfrak{E}$, $\widehat{\mathcal{D}}_1, \dots, \widehat{\mathcal{D}}_m$ are extensions of $\mathcal{D}_1, \dots, \mathcal{D}_m$ to $\widehat{\mathfrak{Y}}$ (Proposition 10.2(2)),
$$\widehat{\mathcal{D}}(\lambda) := \widehat{\mathcal{D}}_1^{\otimes \lambda_1} \otimes \cdots \otimes \widehat{\mathcal{D}}_m^{\otimes \lambda_m}$$
for $\lambda = (\lambda_1, \dots, \lambda_m) \in \mathbb{Z}^m$,
$$\iota : \widehat{\mathcal{D}}(\lambda_0) \to \mathcal{O}_{\widehat{\mathfrak{Y}}}(-\mathfrak{E})$$
is an isomorphism whose restriction to $\mathfrak{E}$ is the given isomorphism $\mathcal{D}(\lambda_0) \cong \mathcal{L}$, and for any $i \in I$, $\widetilde{s}_i$ is a section of $\widehat{\mathcal{D}}(\mu_i)$ restricting to s_i.

(6) $U \subset Y$ is an open neighborhood of the generic point of $\operatorname{center}_E F$, $g : Y' \to U$ is a finite morphism from a normal variety Y' which factors through an étale morphism $Y' \to \rho^{-1}(U)$, $E' := g^* E$, and $B_{Y'} := g^* B_Y$.

(7) e is a positive integer and F' is a prime divisor over E' such that for any Cartier divisor $D \geq 0$ on a neighborhood in E of the generic point of $\operatorname{center}_E F$,
$$\operatorname{mult}_{F'}((g|_{E'})^* D) = e \operatorname{mult}_F D.$$

(8) $\widehat{Y'}$ is the formal completion of Y' along E', $\varpi_1, \dots, \varpi_m$ are generators of the pullbacks of $\widehat{\mathcal{D}}_1, \dots, \widehat{\mathcal{D}}_m$ to $\widehat{Y'}$ near the generic point of $\operatorname{center}_{E'} F'$, and for any $i \in I$, ϕ_i is the function on $\widehat{Y'}$ near the generic point of $\operatorname{center}_{E'} F'$ such that the pullback of $\widetilde{s}_i$ to $\widehat{Y'}$ is $\phi_i \varpi_1^{\mu_{i,1}} \cdots \varpi_m^{\mu_{i,m}}$, where $\mu_i = (\mu_{i,1}, \dots, \mu_{i,m})$.

For a valuation w' on Y' with center on E', $w'(E')$ is the value of a local equation of E', and $w'(\phi_i)$ is defined by continuity. For a positive rational number c, let $\xi_{F,c}$ be the valuation on C_E given by
$$\xi_{F,c}(\psi) := \min\{jc + \nu_F(\psi_j) \mid \psi_j \neq 0\}$$
for $\psi = \sum_{j \geq 0} \psi_j$ with $\psi_j \in H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$, where $\operatorname{gr}_E \mathcal{O}_{X,x}$ is identified with $\bigoplus_{j \geq 0} H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$ by Proposition 10.2(1).

In Set-up 10.3, the isomorphism ι in (5) exists, since $\widehat{\mathcal{D}}(\lambda_0)$ and $\mathcal{O}_{\widehat{\mathfrak{Y}}}(-\mathfrak{E})$ both extend $\mathcal{L}$, hence are isomorphic by Proposition 10.2(2), and the restriction to $\mathfrak{E}$ of such an isomorphism differs from the given isomorphism $\mathcal{D}(\lambda_0) \cong \mathcal{L}$ by a constant in $H^0(\mathfrak{E}, \mathcal{O}_{\mathfrak{E}}^*) = \mathbb{C}^*$, which we remove by rescaling. The function $\xi := \xi_{F,c}$ is a valuation. For non-zero ψ, ψ', let j_0 (resp. j_0') be the smallest j with $\xi(\psi_j) = \xi(\psi)$ (resp. $\xi(\psi_j') = \xi(\psi')$). Since $\mathfrak{E}$ is integral, $\sigma\sigma' \neq 0$ and $\nu_F(\sigma\sigma') = \nu_F(\sigma) + \nu_F(\sigma')$ for non-zero sections σ, σ' (see Definition 9.1), so in the component $\sum_{j+j'=j_0+j_0'} \psi_j \psi_{j'}'$ of $\psi\psi'$ of degree $j_0 + j_0'$, we have $\xi(\psi_{j_0}\psi_{j_0'}') = \xi(\psi) + \xi(\psi')$ while $\xi(\psi_j \psi_{j'}') > \xi(\psi) + \xi(\psi')$ for the other non-zero terms. Thus $\xi(\psi\psi') = \xi(\psi) + \xi(\psi')$, and $\xi(\psi + \psi') \geq \min\{\xi(\psi), \xi(\psi')\}$ follows from $\xi(\psi_j + \psi_j') \geq \min\{\xi(\psi_j), \xi(\psi_j')\}$ for any j, as
$$\nu_F(\sigma + \sigma') \geq \min\{\nu_F(\sigma), \nu_F(\sigma')\}$$
(see Definition 9.1).

10.2. **Gauss valuations on the cone and the upper bound.**

Lemma 10.4. *Let n, $(X \ni x, B)$, E, f, a, ρ, $\mathfrak{E}$, $\mathcal{L}$, and q be as in Set-up 10.3(1)(2), let F be a prime divisor over E, let c be a positive rational number, and let $\xi_{F,c}$ be as in Set-up 10.3. Then* $\operatorname{center}_{C_E}(\xi_{F,c}) = o$,

$$a(\xi_{F,c}, C_E, B_{C_E}) = ac + a(F, E, B_E), \quad \text{and} \quad \widehat{\operatorname{vol}}(\xi_{F,c}, C_E, B_{C_E}) = a\Psi_{E,B_E}(F, ac).$$

Proof. Let $\psi = \sum_{j\geq 0} \psi_j$ with $\psi_j \in H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$. Since $\nu_F(\psi_j) \geq 0$, we have $\xi_{F,c}(\psi_j) \geq jc$, and $\xi_{F,c}(\psi_0) = 0$ if $\psi_0 \neq 0$, because $H^0(\mathfrak{E}, \mathcal{O}_{\mathfrak{E}}) = \mathbb{C}$ by Construction-Theorem 6.2(1). Thus $\xi_{F,c} \geq 0$ on $\operatorname{gr}_E \mathcal{O}_{X,x}$, and $\xi_{F,c}(\psi) > 0$ if and only if $\psi_0 = 0$, that is, ψ belongs to the ideal of o. Hence $\operatorname{center}_{C_E}(\xi_{F,c}) = o$. For a real number z we let

$$\mathfrak{a}_z(\xi_{F,c}) := \{\psi \in \operatorname{gr}_E \mathcal{O}_{X,x} \mid \xi_{F,c}(\psi) \geq z\}. \tag{10.3}$$

Since $\xi_{F,c}(\psi) = \min_j \xi_{F,c}(\psi_j)$, we have

$$\mathfrak{a}_z(\xi_{F,c}) = \bigoplus_{j\geq 0} \big(\mathfrak{a}_z(\xi_{F,c}) \cap H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})\big), \tag{10.4}$$

and

$$H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}) \subset \mathfrak{a}_z(\xi_{F,c}) \quad \text{for any } j \geq z/c.$$

As

$$\dim_{\mathbb{C}} H^0(\mathfrak{E}, \mathcal{L}^{\otimes j}) < +\infty \quad \text{for any } j,$$

the quotient of $\operatorname{gr}_E \mathcal{O}_{X,x}$ by $\mathfrak{a}_z(\xi_{F,c})$ is supported at o. Thus the colength in $\mathcal{O}_{C_E,o}$ of the valuation ideal of $\xi_{F,c}$ at level z is

$$\dim_{\mathbb{C}} \operatorname{gr}_E \mathcal{O}_{X,x}/\mathfrak{a}_z(\xi_{F,c}) < +\infty,$$

and

$$\operatorname{vol}(\xi_{F,c}) = \lim_{z\to+\infty} \frac{n!}{z^n} \dim_{\mathbb{C}} \operatorname{gr}_E \mathcal{O}_{X,x}/\mathfrak{a}_z(\xi_{F,c}). \tag{10.5}$$

Let Q be a sufficiently divisible positive integer such that Construction-Theorem 6.2(3) holds with Q in place of q, let $\pi : C_E \to C(E, P_Q)$ be the finite morphism there, and let

$$\mathcal{V} := \bigoplus_{k\geq 0} H^0(\mathfrak{E}, \mathcal{L}^{\otimes kQ}) \subset \operatorname{gr}_E \mathcal{O}_{X,x}$$

be the coordinate ring of $C(E, P_Q)$. By Proposition 10.2(1) with Q in place of q, the element of $H^0(\mathfrak{E}, \mathcal{L}^{\otimes kQ})$ corresponding to $t \in H^0(E, \mathcal{O}_E(kP_Q))$ under the isomorphisms of Construction-Theorem 6.2(3) and Proposition 10.2(1) is $\rho^* t$. Since $\mathcal{L}^{\otimes kQ}$ is the pullback of a line bundle on E, Definition 9.1 with 1 in place of q shows that $\nu_F(\rho^* t)$ is the multiplicity along F of the pullback of $\operatorname{div}(t)$. Hence the restriction v of $\xi_{F,c}$ to $C(E, P_Q)$ is

$$v\Big(\sum_{k\geq 0} t_k\Big) = \min\{kQc + \nu_F(\rho^* t_k) \mid t_k \neq 0\}, \quad t_k \in H^0(E, \mathcal{O}_E(kP_Q)),$$

that is, $v = v_{F,Qc}$ is the Gauss valuation on $C(E, P_Q)$ with base prime divisor F and weight Qc. Here $\mathfrak{a}_z(v) := \mathfrak{a}_z(\xi_{F,c}) \cap \mathcal{V}$ is the set of elements of $\mathcal{V}$ with v-value at least z, and

$$\operatorname{vol}(v) = \lim_{z\to+\infty} \frac{n!}{z^n} \dim_{\mathbb{C}} \mathcal{V}/\mathfrak{a}_z(v).$$

By Construction-Theorem 6.2(3) with Q in place of q, P_Q is an ample Cartier divisor on E and $-(K_E + B_E) \sim_{\mathbb{R}} \frac{a}{Q} P_Q$. Lemma 5.4, with $E, B_E, P_Q, Qc, \frac{a}{Q}$ in place of S, B, P, c, r, implies

$$\begin{aligned} a(v, C(E, P_Q), (B_E)_{C(E,P_Q)}) &= ac + a(F, E, B_E), \\ \widehat{\operatorname{vol}}(v, C(E, P_Q), (B_E)_{C(E,P_Q)}) &= \frac{a}{Q}\Psi_{E,B_E}(F, ac). \end{aligned} \tag{10.6}$$

Let $h : \widetilde{E} \to E$ be a resolution such that F is a divisor on $\widetilde{E}$. By the formula for v, near the generic point of F, v is the monomial valuation on the total space of $h^*\mathcal{O}_E(-P_Q)$ with weights Qc and 1 on a fiber coordinate and on a local equation of F. Since Qc is rational, v is a positive multiple of a divisorial valuation, hence so is $\xi_{F,c}$, as π is finite. Since π is finite and

$$K_{C_E} + B_{C_E} = \pi^*\big(K_{C(E,P_Q)} + (B_E)_{C(E,P_Q)}\big),$$

we have $a(\xi_{F,c}, C_E, B_{C_E}) = a(v, C(E,P_Q), (B_E)_{C(E,P_Q)})$. By (10.6), the first equality of the lemma holds.

For $r \in \{0, 1, \dots, Q-1\}$ let

$$M_r := \bigoplus_{k\geq 0} H^0(\mathfrak{E}, \mathcal{L}^{\otimes(kQ+r)}),$$

a $\mathcal{V}$-submodule of $\operatorname{gr}_E \mathcal{O}_{X,x}$. Then

$$\operatorname{gr}_E \mathcal{O}_{X,x} = \bigoplus_{r=0}^{Q-1} M_r \quad \text{and} \quad M_0 = \mathcal{V}.$$

By Construction-Theorem 6.2(1), $\operatorname{gr}_E \mathcal{O}_{X,x}$ is a domain, and the degrees of the non-zero homogeneous elements of $\operatorname{gr}_E \mathcal{O}_{X,x}$ form a submonoid of $\mathbb{N}$ which generates $\mathbb{Z}$, hence contains any sufficiently large integer. Thus for any r there exist non-zero homogeneous elements $\psi_r^+ \in M_r$ and $\psi_r^- \in M_{Q-r}$, where $M_Q := M_0$. Then

$$\psi_r^+ \mathcal{V} \subset M_r \quad \text{and} \quad \psi_r^- M_r \subset \mathcal{V}.$$

Let z be a real number. Since $\xi_{F,c}$ is a valuation, for $\psi \in \mathcal{V}$ and $\psi' \in M_r$ we have

$$\xi_{F,c}(\psi_r^+\psi) = \xi_{F,c}(\psi_r^+) + v(\psi) \quad \text{and} \quad v(\psi_r^-\psi') = \xi_{F,c}(\psi_r^-) + \xi_{F,c}(\psi').$$

Hence multiplication by ψ_r^+ and by ψ_r^- induce injective $\mathbb{C}$-linear maps

$$\mathcal{V}/\mathfrak{a}_{z-\xi_{F,c}(\psi_r^+)}(v) \to M_r/(M_r \cap \mathfrak{a}_z(\xi_{F,c})) \quad \text{and} \quad M_r/(M_r \cap \mathfrak{a}_z(\xi_{F,c})) \to \mathcal{V}/\mathfrak{a}_{z+\xi_{F,c}(\psi_r^-)}(v),$$

so

$$\dim_{\mathbb{C}} \mathcal{V}/\mathfrak{a}_{z-\xi_{F,c}(\psi_r^+)}(v) \leq \dim_{\mathbb{C}} M_r/(M_r \cap \mathfrak{a}_z(\xi_{F,c})) \leq \dim_{\mathbb{C}} \mathcal{V}/\mathfrak{a}_{z+\xi_{F,c}(\psi_r^-)}(v). \tag{10.7}$$

Since $\mathfrak{a}_z(\xi_{F,c})$ is spanned by homogeneous elements,

$$\mathfrak{a}_z(\xi_{F,c}) = \bigoplus_{r=0}^{Q-1} (M_r \cap \mathfrak{a}_z(\xi_{F,c})),$$

so

$$\dim_{\mathbb{C}} \operatorname{gr}_E \mathcal{O}_{X,x}/\mathfrak{a}_z(\xi_{F,c}) = \sum_{r=0}^{Q-1} \dim_{\mathbb{C}} M_r/(M_r \cap \mathfrak{a}_z(\xi_{F,c})).$$

Since $\dim C(E,P_Q) = n$, for any real number y we have

$$\lim_{z\to+\infty} \frac{n!}{z^n} \dim_{\mathbb{C}} \mathcal{V}/\mathfrak{a}_{z+y}(v) = \operatorname{vol}(v).$$

Thus (10.7) implies

$$\operatorname{vol}(\xi_{F,c}) = Q \operatorname{vol}(v).$$

By the first equality and (10.6),

$$\begin{aligned}
\widehat{\operatorname{vol}}(\xi_{F,c}, C_E, B_{C_E}) &= a(v, C(E,P_Q), (B_E)_{C(E,P_Q)})^n \cdot Q \operatorname{vol}(v) \\
&= Q\widehat{\operatorname{vol}}(v, C(E,P_Q), (B_E)_{C(E,P_Q)}) \\
&= a\Psi_{E,B_E}(F, ac).
\end{aligned}$$

The lemma follows. □

Proposition 10.5. *Notation and conditions as in Set-up 10.3. Let $c > 0$ be a rational number and w' a positive multiple of a divisorial valuation on Y' such that*

(i) $\mathrm{center}_{Y'} w'$ *is the generic point of* $\mathrm{center}_{E'} F'$,
(ii) $w'(E') = ec$, *and*
(iii) $w'(\phi_i) \geq e\nu_F(s_i)$ *for any* $i \in I$ *with* $s_i \neq 0$.

Let w be the valuation on X induced by $\frac{1}{e}w'$ via $f \circ g : Y' \to X$. Then $\mathrm{center}_X w = x$,

$$a(w, X, B) = ac + \frac{1}{e}a(w', Y', E' + B_{Y'}), \qquad \mathrm{vol}(w) \leq \mathrm{vol}(\xi_{F,c}),$$

and

$$\frac{\widehat{\mathrm{vol}}(X \ni x, B)}{a} \leq \Big(\frac{ac + \frac{1}{e}a(w', Y', E' + B_{Y'})}{ac + a(F, E, B_E)}\Big)^n \Psi_{E,B_E}(F, ac).$$

Proof. For $i \in I$, $s_i \in H^0(\mathfrak{E}, \mathcal{D}(\mu_i))$, and $q\nu_F(s_i) \geq k_i$ if $s_i \neq 0$. Since $Y' \to \rho^{-1}(U)$ is étale and $\mathfrak{E}$ is Cartier, E' is Cartier. Let $\eta' := \mathrm{center}_{Y'} w'$, and let x_0 be a local equation of E' at η'. As in Set-up 10.3, w' is extended by continuity to functions on $\widehat{Y'}$ near η', and for such functions ζ_1, ζ_2 we have $w'(\zeta_1 + \zeta_2) \geq \min\{w'(\zeta_1), w'(\zeta_2)\}$ and $w'(\zeta_1\zeta_2) \geq w'(\zeta_1) + w'(\zeta_2)$, and $w' \geq 0$ on such functions regular at η'. The morphism $Y' \to \rho^{-1}(U) \subset \mathfrak{Y}$ induces a morphism $\widehat{Y'} \to \widehat{\mathfrak{Y}}$, by which we pull back sections on $\widehat{\mathfrak{Y}}$. As in Construction-Theorem 6.2, X is affine and $\mathfrak{a}_j \subset H^0(X, \mathcal{O}_X)$.
Step 1. *The center and the log discrepancy of w.* For $j \geq 0$ and $\chi \in \mathfrak{a}_j$,

$$\rho^* f^* \chi \in H^0(\mathfrak{Y}, \mathcal{O}_{\mathfrak{Y}}(-j\mathfrak{E})),$$

so $(f \circ g)^*\chi$ is divisible by x_0^j near η'. Since $w'(x_0) = w'(E') = ec$,

$$w(\chi) \geq jc \quad \text{for any } j \geq 0 \text{ and any } \chi \in \mathfrak{a}_j. \tag{10.8}$$

Since $(f \circ g)(\eta') \in f(E) = \{x\}$, $w \geq 0$ on $\mathcal{O}_{X,x}$. Since $E = f^{-1}(x)$ as sets, $\mathfrak{a}_1$ is the ideal of x in $H^0(X, \mathcal{O}_X)$, so (10.8) with $j = 1$ implies that $w > 0$ on the maximal ideal of $\mathcal{O}_{X,x}$. Thus $\mathrm{center}_X w = x$. Since $f \circ g$ is generically finite, w is a positive multiple of a divisorial valuation.

Let $V \subset E$ be the closed subset on which E is not Cartier. Since $Y' \to \rho^{-1}(U)$ is étale and ρ is an isomorphism over $Y \backslash V$, g is étale over $U \backslash V$, and $\mathrm{codim}_Y V \geq 2$. As $E' = g^*E$ and $B_{Y'} = g^*B_Y$, we have

$$K_{Y'} + E' + B_{Y'} = g^*(K_U + E + B_Y).$$

Let w_Y be the valuation on Y induced by w' via g, so that $w = \frac{1}{e}w_Y$ as valuations on X, and $w_Y(E) = w'(E') = ec$. Since g is finite, $a(w', Y', E' + B_{Y'}) = a(w_Y, Y, E + B_Y)$. Since $K_Y + E + B_Y = f^*(K_X + B) + aE$,

$$a(w_Y, X, B) = a(w_Y, Y, E + B_Y) + aw_Y(E) = a(w', Y', E' + B_{Y'}) + aec.$$

Dividing by e, we obtain the formula for $a(w, X, B)$.
Step 2. *Lifting sections with large $\xi_{F,c}$-value.* Let $z > 0$ be a real number. We show that for any integer $j \geq 0$ and any non-zero $s \in H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$ with $\xi_{F,c}(s) = jc + \nu_F(s) \geq z$, there exists $\chi \in \mathfrak{a}_j$ such that $w(\chi) \geq z$ and the class of χ in $\mathfrak{a}_j/\mathfrak{a}_{j+1}$ corresponds to s under Proposition 10.2(1).

Since $\mathcal{L}^{\otimes q} = \rho^*(\mathcal{O}_Y(-qE)|_E)$ and $-qE|_E$ is an ample Cartier divisor on E, $\mathcal{L}$ is nef. As $\mathcal{L} \cong \mathcal{D}(\lambda_0)$, we have $\mathcal{L}^{\otimes j} \cong \mathcal{D}(j\lambda_0)$ and $j\lambda_0 \in \Sigma$. Let

$$\kappa := \max\{0, \lceil q(z - jc)\rceil\}.$$

Since $q\nu_F(s)$ is a non-negative integer and $q\nu_F(s) \geq q(z - jc)$, we have $q\nu_F(s) \geq \kappa$, so s belongs to the component of $\mathcal{A}_F(\Sigma)$ of degree $(j\lambda_0, \kappa)$. Since the s_i $(i \in I)$ are homogeneous generators of $\mathcal{A}_F(\Sigma)$, there exist a finite set $\mathcal{N} \subset \mathbb{N}^I$ and $\gamma_\alpha \in \mathbb{C}$ $(\alpha \in \mathcal{N})$ such that

$$s = \sum_{\alpha \in \mathcal{N}} \gamma_\alpha s^\alpha, \quad s^\alpha := \prod_{i \in I} s_i^{\alpha_i},$$

and for any $\alpha = (\alpha_i)_{i\in I} \in \mathcal{N}$,

$$\sum_{i\in I} \alpha_i \mu_i = j\lambda_0, \quad \sum_{i\in I} \alpha_i k_i = \kappa, \quad \text{and } \alpha_i = 0 \text{ for any } i \in I \text{ with } s_i = 0.$$

Let

$$\Phi := \sum_{\alpha\in\mathcal{N}} \gamma_\alpha \prod_{i\in I} \widetilde{s}_i^{\alpha_i},$$

a section of $\widehat{\mathcal{D}}(j\lambda_0)$, which we identify with $\mathcal{O}_{\widehat{\mathfrak{Y}}}(-j\mathfrak{E})$ by $\iota^{\otimes j}$ (Set-up 10.3(5)). Since ι restricts to $\mathcal{D}(\lambda_0) \cong \mathcal{L}$ on $\mathfrak{E}$ and $\widetilde{s}_i$ restricts to s_i, the image of Φ in $H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$ is s. Near η', ι sends $\varpi_1^{\lambda_{0,1}} \cdots \varpi_m^{\lambda_{0,m}}$ to ωx_0 for a formal function ω regular at η', where $\lambda_0 = (\lambda_{0,1}, \dots, \lambda_{0,m})$, so $w'(\omega) \geq 0$. By Set-up 10.3(8), the pullback of Φ to Y' is

$$\omega^j x_0^j \sum_{\alpha\in\mathcal{N}} \gamma_\alpha \prod_{i\in I} \phi_i^{\alpha_i}$$

near η'. By (ii) and (iii),

$$w'(\Phi) \geq jec + \min_{\alpha\in\mathcal{N}} \sum_{i\in I} \alpha_i w'(\phi_i) \geq jec + e \min_{\alpha\in\mathcal{N}} \sum_{i\in I} \alpha_i \frac{k_i}{q} = e\Big(jc + \frac{\kappa}{q}\Big) \geq ez.$$

Let $N > j$ be an integer such that $Nc \geq z$. Let $\Phi_j := \Phi$. For $k = j, j+1, \dots, N-1$, given

$$\Phi_k \in H^0(\widehat{\mathfrak{Y}}, \mathcal{O}_{\widehat{\mathfrak{Y}}}(-k\mathfrak{E})),$$

by the surjectivity in Proposition 10.2(1), there exists $\chi_k \in \mathfrak{a}_k$ such that the image of $\rho^* f^* \chi_k - \Phi_k$ in $H^0(\mathfrak{E}, \mathcal{L}^{\otimes k})$ is 0, and we let

$$\Phi_{k+1} := \Phi_k - \rho^* f^* \chi_k \in H^0(\widehat{\mathfrak{Y}}, \mathcal{O}_{\widehat{\mathfrak{Y}}}(-(k+1)\mathfrak{E})).$$

Let $\chi := \sum_{k=j}^{N-1} \chi_k \in \mathfrak{a}_j$. Since $\chi_k \in \mathfrak{a}_{j+1}$ for $k > j$ and the image of Φ_j in $H^0(\mathfrak{E}, \mathcal{L}^{\otimes j})$ is s, the class of χ in $\mathfrak{a}_j/\mathfrak{a}_{j+1}$ corresponds to s. Moreover,

$$\rho^* f^* \chi = \Phi - \Phi_N \quad \text{on } \widehat{\mathfrak{Y}}.$$

Near η', the pullback of Φ_N to Y' is x_0^N times a formal function regular at η', so $w'(\Phi_N) \geq eNc \geq ez$. Thus

$$w'((f\circ g)^*\chi) \geq \min\{w'(\Phi), w'(\Phi_N)\} \geq ez,$$

that is, $w(\chi) \geq z$.

Step 3. *The colength inequality.* Since f is an isomorphism over $X\backslash\{x\}$, $\mathfrak{a}_j/\mathfrak{a}_{j+1}$ is supported at x, so

$$\mathfrak{a}_j/\mathfrak{a}_{j+1} = \mathfrak{a}_j\mathcal{O}_{X,x}/\mathfrak{a}_{j+1}\mathcal{O}_{X,x} \quad \text{for any } j \geq 0.$$

For a real number $z > 0$, let $\mathfrak{a}_z(w) \subset \mathcal{O}_{X,x}$ be the valuation ideal of w at level z, and let

$$\operatorname{gr}_E \mathfrak{a}_z(w) := \bigoplus_{j\geq 0} \big((\mathfrak{a}_z(w) \cap \mathfrak{a}_j\mathcal{O}_{X,x}) + \mathfrak{a}_{j+1}\mathcal{O}_{X,x}\big)/\mathfrak{a}_{j+1}\mathcal{O}_{X,x} \subset \operatorname{gr}_E \mathcal{O}_{X,x}.$$

By (10.4), the ideal $\mathfrak{a}_z(\xi_{F,c})$ of (10.3) is spanned by homogeneous elements. By **Step 2**,

$$\mathfrak{a}_z(\xi_{F,c}) \subset \operatorname{gr}_E \mathfrak{a}_z(w).$$

Let N be an integer such that $Nc \geq z$. By (10.8), $\mathfrak{a}_N\mathcal{O}_{X,x} \subset \mathfrak{a}_z(w)$. For $0 \leq k \leq N$ let $\overline{\mathfrak{a}}_k$ be the image of $\mathfrak{a}_k\mathcal{O}_{X,x}$ in $\mathcal{O}_{X,x}/\mathfrak{a}_z(w)$. Then $\overline{\mathfrak{a}}_0 = \mathcal{O}_{X,x}/\mathfrak{a}_z(w)$, $\overline{\mathfrak{a}}_N = 0$, and since $\mathfrak{a}_{k+1} \subset \mathfrak{a}_k$,

$$\overline{\mathfrak{a}}_k/\overline{\mathfrak{a}}_{k+1} \cong \mathfrak{a}_k\mathcal{O}_{X,x}/\big((\mathfrak{a}_z(w) \cap \mathfrak{a}_k\mathcal{O}_{X,x}) + \mathfrak{a}_{k+1}\mathcal{O}_{X,x}\big),$$

which is the degree k part of

$$\operatorname{gr}_E \mathcal{O}_{X,x}/\operatorname{gr}_E \mathfrak{a}_z(w),$$

and the degree k part is 0 for $k \geq N$, since $\mathfrak{a}_k \mathcal{O}_{X,x} \subset \mathfrak{a}_z(w)$. As the residue field of $\mathcal{O}_{X,x}$ is $\mathbb{C}$, we obtain

$$\ell(\mathcal{O}_{X,x}/\mathfrak{a}_z(w)) = \dim_{\mathbb{C}} \operatorname{gr}_E \mathcal{O}_{X,x}/\operatorname{gr}_E \mathfrak{a}_z(w) \leq \dim_{\mathbb{C}} \operatorname{gr}_E \mathcal{O}_{X,x}/\mathfrak{a}_z(\xi_{F,c}),$$

where ℓ denotes the length. Multiplying by $\frac{n!}{z^n}$, letting $z \to +\infty$, and using (10.5), we obtain

$$\operatorname{vol}(w) \leq \operatorname{vol}(\xi_{F,c}). \tag{10.9}$$

Since $\operatorname{center}_X w = x$, **Step 1**, (10.9), and Lemma 10.4 imply

$$\begin{aligned}
\widehat{\operatorname{vol}}(X \ni x, B) &\leq a(w, X, B)^n \operatorname{vol}(w) \\
&\leq \Big(\frac{a(w, X, B)}{a(\xi_{F,c}, C_E, B_{C_E})}\Big)^n \widehat{\operatorname{vol}}(\xi_{F,c}, C_E, B_{C_E}) \\
&= \Big(\frac{ac + \frac{1}{e} a(w', Y', E' + B_{Y'})}{ac + a(F, E, B_E)}\Big)^n a \Psi_{E,B_E}(F, ac).
\end{aligned}$$

Dividing by a, the proposition follows. □

11. Valuations and rounding of coefficients on index one covers

11.1. **Construction of the valuations.** Let Y be a normal variety and E a normal prime divisor on Y which is Cartier. Since E is a normal Cartier prime divisor, Y is regular at the codimension 1 points of E, so any Weil divisor D on Y with no component E is Cartier there, and restricting D at the codimension 1 points of E defines a Weil divisor $D|_E$ on E.

Theorem 11.1. *Let $n \geq 2$ be an integer, $\mathcal{U} \to T$ a flat affine morphism between varieties whose fibers U_t are normal Cohen–Macaulay varieties of dimension $n-1$, and $h : \mathcal{Z} \to \mathcal{U}$ a projective morphism such that $\mathcal{Z} \to T$ is smooth and $h_t := h|_{\mathcal{Z}_t} : \mathcal{Z}_t \to U_t$ is birational for any $t \in T$. Let J be a finite set and $\mathcal{F}_j$ $(j \in J)$ prime divisors on $\mathcal{Z}$ whose fibers $F_{j,t}$ are prime divisors. Let $\mathcal{B}_1, \dots, \mathcal{B}_m \geq 0$ be relative Mumford divisors on $\mathcal{U} \to T$, $\Lambda \subset [0,1)^m$ a finite set, and $v_0 > 0$ a real number. Then there exists a positive integer k_0 depending only on $\mathcal{U} \to T$, $\mathcal{Z}$, $\{\mathcal{F}_j\}_{j \in J}$, $\mathcal{B}_1, \dots, \mathcal{B}_m$, Λ, and v_0 satisfying the following. Let $t \in T$ be a closed point, and let $j \in J$, $b \in \Lambda$, and $V := \operatorname{center}_{U_t} F_{j,t}$. Let Y be a normal variety containing an open neighborhood of the generic point of V in U_t as a reduced Cartier divisor E, and $B_1, \dots, B_m \geq 0$ Weil divisors on Y with no component E such that $B_i|_E = \mathcal{B}_i|_{U_t}$ near the generic point of V for $i = 1, \dots, m$, and let $B := \sum_{i=1}^m b_i B_i$ and $B_{U_t} := \sum_{i=1}^m b_i \mathcal{B}_i|_{U_t}$. Assume that there exist a closed point $y \in V \cap E$ and an $\mathbb{R}$-divisor $\Delta \geq 0$ on Y such that*

(i) $(Y, E + B)$ is plt near y,
(ii) (Y, Δ) is klt near y, and
(iii) $\widehat{\operatorname{vol}}(Y \ni y, \Delta) \geq v_0$.

Let I be a finite set, and ϕ_i $(i \in I)$ functions on the formal completion of Y along E near the generic point of V whose restrictions $\overline{\phi}_i$ to E are non-zero. Let

$$M := \max\big(\{0\} \cup \{\operatorname{mult}_{F_{j,t}}(h_t^* \operatorname{div}(\overline{\phi}_i)) \mid i \in I\}\big).$$

Then $K_{U_t} + B_{U_t}$ is $\mathbb{R}$-Cartier near the generic point of V, and for any integer $k > k_0 + M$, there exists a divisorial valuation w on Y with value group $\mathbb{Z}$ such that

(1) $\operatorname{center}_Y w$ is the generic point of V,
(2) $w(E) = k$,
(3) $a(w, Y, E + B) = a(F_{j,t}, U_t, B_{U_t})$, and
(4) for any $i \in I$,

$$w(\phi_i) = \operatorname{mult}_{F_{j,t}}(h_t^* \operatorname{div}(\overline{\phi}_i)),$$

where w is extended by continuity to the completion along E of the local ring of Y at the generic point of V.

Proof. We let $F := F_{j,t}$, let κ be the residue field of $\mathcal{O}_{U_t,V}$, and let $d := \dim \mathcal{O}_{U_t,V}$. Since h_t is birational and F is a prime divisor on $\mathcal{Z}_t$, $1 \le d \le n-1$. Let $x_0 \in \mathcal{O}_{Y,V}$ be a local equation of E, so $\mathcal{O}_{Y,V}/(x_0) = \mathcal{O}_{U_t,V}$.

Step 1. In this step we write b as a convex combination of rational vectors and bound the Cartier indices of the corresponding pairs.

Let $\Pi_b \subset \mathbb{R}^m$ be the smallest affine subspace defined over $\mathbb{Q}$ containing b. Since $\Pi_b \cap \mathbb{Q}^m$ is dense in Π_b, $b \in [0,1)^m$, and $\Pi_b \subset \{b' \in \mathbb{R}^m \mid b'_i = 0\}$ for any i with $b_i = 0$, there exist a finite set $Q_b \subset \Pi_b \cap \mathbb{Q}^m \cap [0,1)^m$ and real numbers $\lambda_{b'} > 0$ ($b' \in Q_b$) such that

$$\sum_{b' \in Q_b} \lambda_{b'} = 1 \quad \text{and} \quad \sum_{b' \in Q_b} \lambda_{b'} b' = b.$$

We fix Q_b and the $\lambda_{b'}$ once for any $b \in \Lambda$, and let r_0 be a positive integer such that $r_0 b' \in \mathbb{Z}^m$ for any $b \in \Lambda$ and any $b' \in Q_b$. For $b' \in \mathbb{R}^m$ let $B(b') := \sum_{i=1}^m b'_i B_i$ and $B_{U_t}(b') := \sum_{i=1}^m b'_i \mathcal{B}_i|_{U_t}$. Thus $B = B(b)$, $B_{U_t} = B_{U_t}(b)$, and $B(b') \ge 0$ for any $b' \in Q_b$.

Let $P_1, \dots, P_{m_1}$ be the prime divisors through y which are components of $\sum_{i=1}^m B_i$, and write $B_i = \sum_{i'=1}^{m_1} c_{ii'} P_{i'}$ near y with $c_{ii'} \in \mathbb{Z}$. Then $b' \to (\sum_{i=1}^m b'_i c_{ii'})_{1 \le i' \le m_1}$ is a $\mathbb{Q}$-linear map $\mathbb{R}^m \to \mathbb{R}^{m_1}$, and the preimage of an affine subspace defined over $\mathbb{Q}$ is an affine subspace defined over $\mathbb{Q}$. By Lemma 2.9(1), the set W_b of $b' \in \mathbb{R}^m$ such that $K_Y + B(b')$ is $\mathbb{R}$-Cartier near y is either empty or an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$. Since E is Cartier and $K_Y + E + B$ is $\mathbb{R}$-Cartier near y, $b \in W_b$. Thus $\Pi_b \subset W_b$, and $Q_b \subset W_b$.

Let $b' \in Q_b$. Since $b' \in \mathbb{Q}^m \cap W_b$, $K_Y + B(b')$ is $\mathbb{Q}$-Cartier near y, so $r_0(K_Y + B(b'))$ is a $\mathbb{Q}$-Cartier Weil divisor near y. Let r_1 be the positive integer of Theorem 2.17 for the dimension n and the real number v_0. Since (Y, Δ) is klt near y and $\widehat{\mathrm{vol}}(Y \ni y, \Delta) \ge v_0$, Theorem 2.17 for the closed subvariety $\{y\}$ implies that $r_1 r_0(K_Y + B(b'))$ is Cartier near y. We let

$$r := r_0 r_1,$$

which depends only on n, v_0, and the sets Q_b ($b \in \Lambda$). Since $y \in V$, any open neighborhood of y in Y contains the generic point of V, so $\mathcal{O}_Y(r(K_Y + B(b')))$ is invertible at the generic point of V for any $b' \in Q_b$.

Possibly shrinking Y to a neighborhood of the generic point of V, we may assume that x_0 is a regular function on Y defining E, that E is an open subset of U_t, that $B_i|_E = \mathcal{B}_i|_E$ for $i = 1, \dots, m$, and that $\mathcal{O}_Y(r(K_Y + B(b')))$ is invertible for any $b' \in Q_b$.

Let $b' \in Q_b$. The variety Y and the prime divisor E are regular at any codimension 1 point of E, and $B(b')$ is Cartier there, so the Poincaré residue along E and the multiplication by x_0^{-r} define an isomorphism at the codimension 1 points of E

$$\mathcal{O}_Y(r(K_Y + B(b')))|_E \xrightarrow{x_0^{-r}} \mathcal{O}_Y(r(K_Y + E + B(b')))|_E \to \mathcal{O}_E\Big(r\Big(K_E + \sum_{i=1}^m b'_i \mathcal{B}_i|_E\Big)\Big).$$

The first sheaf is invertible and the last sheaf is reflexive on the normal variety E, so the composite isomorphism extends to an isomorphism $\mathrm{res}_{b'}$ over E. The Poincaré residue of $(dx_0/x_0) \wedge \vartheta$ is $\vartheta|_E$, so for any rational differential $(n-1)$-form ϑ on Y which is regular at the generic point of E,

$$\mathrm{res}_{b'}\left((dx_0 \wedge \vartheta)^{\otimes r}\right) = (\vartheta|_E)^{\otimes r} \tag{11.1}$$

as rational sections, since both sides are equal at the generic point of E. Since $\mathrm{res}_{b'}$ is an isomorphism and $\mathcal{O}_Y(r(K_Y + B(b')))$ is invertible, $r(K_{U_t} + B_{U_t}(b'))$ is Cartier near the generic point of V for any $b' \in Q_b$, and since $b = \sum_{b' \in Q_b} \lambda_{b'} b'$, $K_{U_t} + B_{U_t}$ is $\mathbb{R}$-Cartier near the generic point of V.

Step 2. In this step we construct w.

Since U_t is affine, after a general linear change of coordinates of a closed embedding $U_t \subset \mathbb{A}^{n-1+c}$ for some $c \in \mathbb{N}$, there exist coordinates $z_1, \dots, z_{n-1-d}, x_1, \dots, x_{d+c}$ on $\mathbb{A}^{n-1+c}$ such that the

restrictions of $z_1, \dots, z_{n-1-d}$ to V form a transcendence basis of κ over $\mathbb{C}$. Let $R_0 := \mathcal{O}_{\mathbb{A}^{n-1+c},V}$, a regular local ring of dimension $d+c$, and let $\mathfrak{b}_0 \subset R_0$ be the ideal of U_t, a prime ideal of height c with $R_0/\mathfrak{b}_0 = \mathcal{O}_{U_t,V}$.

Since the restrictions of $z_1, \dots, z_{n-1-d}$ to U_t are algebraically independent, $dz_1, \dots, dz_{n-1-d}$ are linearly independent at the generic point of U_t, so by the Jacobian criterion at the generic point of U_t, the Jacobian matrix of a set of generators of $\mathfrak{b}_0$ with respect to $x_1, \dots, x_{d+c}$ has rank c modulo $\mathfrak{b}_0$. Since $\mathbb{C}$ is infinite, general linear combinations $\overline{f}_1, \dots, \overline{f}_c$ of the generators of $\mathfrak{b}_0$ satisfy the following:

(1) $\overline{f}_1, \dots, \overline{f}_c$ is a regular sequence in R_0 (choose $\overline{f}_i$ outside the minimal primes of $(\overline{f}_1, \dots, \overline{f}_{i-1})$, which have height $i-1 < c$).
(2) Possibly reordering $x_1, \dots, x_{d+c}$, the minor

$$\theta_0 := \det(\partial \overline{f}_i / \partial x_{d+i'})_{1 \le i,i' \le c}$$

is not in $\mathfrak{b}_0$.

If $c = 0$ we let $\theta_0 := 1$. Let

$$\mathfrak{c}_0 := (\overline{f}_1, \dots, \overline{f}_c) \subset \mathfrak{b}_0.$$

The localization of R_0 at $\mathfrak{b}_0$ is a regular local ring of dimension c in which $d\overline{f}_1, \dots, d\overline{f}_c$ are linearly independent, so

$$\mathfrak{c}_0 (R_0)_{\mathfrak{b}_0} = \mathfrak{b}_0 (R_0)_{\mathfrak{b}_0}.$$

Since $R_0/\mathfrak{c}_0$ is Cohen–Macaulay, $\mathfrak{c}_0$ has no embedded primes, and the $\mathfrak{b}_0$-primary component of $\mathfrak{c}_0$ is $\mathfrak{b}_0$. Thus $(\mathfrak{c}_0 : \mathfrak{b}_0)$ contains the intersection of the other primary components of $\mathfrak{c}_0$, which is not contained in $\mathfrak{b}_0$. We fix

$$\psi_0 \in (\mathfrak{c}_0 : \mathfrak{b}_0) \backslash \mathfrak{b}_0,$$

and let $\overline{\psi}_0 := \psi_0|_{U_t} \neq 0$. Since $\theta_0|_{U_t} \neq 0$, by the Jacobian criterion the restrictions of $z_1, \dots, z_{n-1-d}, x_1, \dots, x_d$ to U_t define a morphism $\pi : U_t \to \mathbb{A}^{n-1}$ which is étale at the generic point of U_t. We let

$$e := \operatorname{mult}_F \operatorname{div}(\theta_0|_{U_t}) \ge 0.$$

Nonzero polynomials in $z_1, \dots, z_{n-1-d}$ are units in R_0, so $\mathbb{C}(z) := \mathbb{C}(z_1, \dots, z_{n-1-d}) \subset R_0$. Since κ is finite separable over $\mathbb{C}(z)$, by Hensel's lemma the completion $\widehat{R}_0$ contains a unique coefficient field containing $\mathbb{C}(z)$, which we also denote by κ. For $i = 1, \dots, d+c$ let $\widehat{x}_i := x_i - \varrho_i$, where $\varrho_i \in \kappa$ is the element with $x_i - \varrho_i$ in the maximal ideal of $\widehat{R}_0$. Then $\widehat{R}_0 = \kappa[[\widehat{x}_1, \dots, \widehat{x}_{d+c}]]$, and the continuous κ-derivation $\partial/\partial \widehat{x}_i$ of $\widehat{R}_0$ restricts to the $\mathbb{C}(z)$-derivation $\partial/\partial x_i$ of R_0.

We fix lifts to $\mathcal{O}_{Y,V}$ of the restrictions of $z_1, \dots, z_{n-1-d}, x_1, \dots, x_{d+c}$ to U_t and denote them by $\widetilde{z}_1, \dots, \widetilde{z}_{n-1-d}, \widetilde{x}_1, \dots, \widetilde{x}_{d+c}$. Since the residues of $\widetilde{z}_1, \dots, \widetilde{z}_{n-1-d}$ are algebraically independent, $z_i \to \widetilde{z}_i$ embeds $\mathbb{C}(z)$ into $\mathcal{O}_{Y,V}$, and $\widehat{\mathcal{O}}_{Y,V}$ contains a unique coefficient field containing $\mathbb{C}(z)$, which maps onto the coefficient field κ of $\widehat{\mathcal{O}}_{U_t,V}$. We identify both with κ. Let

$$R := \kappa[[x_0, \widehat{x}_1, \dots, \widehat{x}_{d+c}]] \to \widehat{\mathcal{O}}_{Y,V}$$

be the continuous κ-algebra homomorphism sending x_0 to x_0 and $\widehat{x}_i$ to $\widetilde{x}_i - \varrho_i$. It is surjective, since $\widehat{\mathcal{O}}_{Y,V}$ is complete and the images generate the maximal ideal. Let $\mathfrak{b}$ be the kernel. Then $R/x_0R = \widehat{R}_0$ and

$$R/(\mathfrak{b} + x_0 R) = \widehat{\mathcal{O}}_{U_t,V} = \widehat{R}_0/\mathfrak{b}_0\widehat{R}_0.$$

Since x_0 is a nonzerodivisor of $\mathcal{O}_{Y,V}$ and $\mathcal{O}_{Y,V}/(x_0) = \mathcal{O}_{U_t,V}$ is Cohen–Macaulay,

$$\operatorname{depth} \mathcal{O}_{Y,V} = \dim \mathcal{O}_{Y,V} = d+1,$$

so $\widehat{\mathcal{O}}_{Y,V} = R/\mathfrak{b}$ is Cohen–Macaulay, and $\mathfrak{b}$ has height c.

Let $f_1, \dots, f_c \in \mathfrak{b}$ be lifts of $\overline{f}_1, \dots, \overline{f}_c \in \mathfrak{b}_0\widehat{R}_0$, and let $\mathfrak{c} := (f_1, \dots, f_c) \subset \mathfrak{b}$. Since $\widehat{R}_0$ is flat over R_0, $\overline{f}_1, \dots, \overline{f}_c$ is a regular sequence in $\widehat{R}_0$, so $x_0, f_1, \dots, f_c$ is a regular sequence in R, and

x_0 is a nonzerodivisor on $R/\mathfrak{c}$. Applying $\mathrm{Hom}_{R/\mathfrak{c}}(R/\mathfrak{b},-)$, which sends an $R/\mathfrak{c}$-module to its submodule annihilated by $\mathfrak{b}$, to the exact sequence

$$0 \to R/\mathfrak{c} \xrightarrow{x_0} R/\mathfrak{c} \to \widehat{R}_0/\mathfrak{c}_0\widehat{R}_0 \to 0,$$

we obtain the exact sequence

$$(\mathfrak{c} : \mathfrak{b})/\mathfrak{c} \to (\mathfrak{c}_0\widehat{R}_0 : \mathfrak{b}_0\widehat{R}_0)/\mathfrak{c}_0\widehat{R}_0 \to \mathrm{Ext}^1_{R/\mathfrak{c}}(R/\mathfrak{b}, R/\mathfrak{c}),$$

where the first map is reduction modulo x_0. By [BH98, Lemma 3.1.16],

$$\mathrm{Ext}^1_{R/\mathfrak{c}}(R/\mathfrak{b}, R/\mathfrak{c}) \cong \mathrm{Ext}^{c+1}_R(R/\mathfrak{b}, R),$$

which vanishes by [BH98, Theorems 3.3.7(a) and 3.3.10(c)(ii)], since $R/\mathfrak{b}$ is a Cohen–Macaulay R-module of dimension $\dim R - c$. Since $\widehat{R}_0$ is flat over R_0,

$$(\mathfrak{c}_0\widehat{R}_0 : \mathfrak{b}_0\widehat{R}_0) = (\mathfrak{c}_0 : \mathfrak{b}_0)\widehat{R}_0 \ni \psi_0.$$

Thus there exists $\psi \in (\mathfrak{c} : \mathfrak{b})$ whose reduction modulo x_0 is in $\psi_0 + \mathfrak{c}_0\widehat{R}_0$, and $\psi\mathfrak{b} \subset \mathfrak{c}$.

Let $\mathcal{K}_0$ be the completion of $\mathbb{C}(U_t)$ with respect to the valuation $g \to \mathrm{mult}_F \mathrm{div}(g)$, and let μ_0 be the extension of $g \to \mathrm{mult}_F \mathrm{div}(g)$ to $\mathcal{K}_0$. Since $\mathrm{mult}_F \mathrm{div}(g) \geq 1$ for any non-zero g in the maximal ideal of $\mathcal{O}_{U_t,V}$, the inclusion $\mathcal{O}_{U_t,V} \subset \mathcal{K}_0$ extends to a continuous homomorphism $\widehat{\mathcal{O}}_{U_t,V} \to \mathcal{K}_0$. For $g \in R$ we denote by $g|_{U_t} \in \mathcal{K}_0$ the image of g under the composite

$$R \to \widehat{R}_0 \to \widehat{\mathcal{O}}_{U_t,V} \to \mathcal{K}_0.$$

In particular $g|_{U_t} = 0$ for $g \in \mathfrak{c}$. Let ζ be an indeterminate, and let $\mathcal{K}$ be the completion of $\mathcal{K}_0(\zeta)$ with respect to the Gauss valuation μ given by

$$\mu\Big(\sum_q g_q\zeta^q\Big) = \min_q \big(\mu_0(g_q) + qk\big) \qquad (g_q \in \mathcal{K}_0).$$

Then $\mu(\mathcal{K}^\times) = \mathbb{Z}$. For $\delta = (\delta_1, \dots, \delta_c) \in \mathcal{K}^c$ let $\mu(\delta) := \min_i \mu(\delta_i)$. For δ with $\mu(\delta) \geq 1$, let $\iota_\delta : R \to \mathcal{K}$ be the continuous κ-algebra homomorphism with

$$\iota_\delta(x_0) = \zeta, \quad \iota_\delta(\widehat{x}_i) = \widehat{x}_i|_{U_t} \ (1 \leq i \leq d), \quad \text{and} \quad \iota_\delta(\widehat{x}_{d+i}) = \widehat{x}_{d+i}|_{U_t} + \delta_i \ (1 \leq i \leq c),$$

which converges since the images of $x_0, \widehat{x}_1, \dots, \widehat{x}_{d+c}$ have value at least 1 and coefficients in κ have value 0. For any $g \in R$, $\iota_\delta(g) - g|_{U_t}$ is a convergent sum of terms each of which contains ζ or some δ_i as a factor, with all other factors of value at least 0. Thus

$$\mu(\iota_\delta(g) - g|_{U_t}) \geq \min\{k, \mu(\delta)\} \quad \text{for any } g \in R. \tag{11.2}$$

Assume that $k > 2e$. We construct δ with $\iota_\delta(f_i) = 0$ for $i = 1, \dots, c$. Let

$$\mathbf{J} := \big(\iota_0(\partial f_i/\partial \widehat{x}_{d+i'})\big)_{1 \leq i,i' \leq c}.$$

Since $\det(\partial f_i/\partial \widehat{x}_{d+i'})$ reduces to θ_0 modulo x_0, (11.2) implies

$$\mu(\det \mathbf{J} - \theta_0|_{U_t}) \geq k > e = \mu(\theta_0|_{U_t}),$$

so $\mu(\det \mathbf{J}) = e$ and the entries of $\mathbf{J}^{-1}$ have value at least $-e$. Let $\mathbb{B} := \{\delta \in \mathcal{K}^c \mid \mu(\delta) \geq k - e\}$ and

$$\mathcal{N}(\delta) := \delta - \mathbf{J}^{-1}(\iota_\delta(f_i))_{1 \leq i \leq c} \qquad (\delta \in \mathbb{B}).$$

Since $f_i|_{U_t} = 0$, (11.2) implies $\mu(\iota_0(f_i)) \geq k$, so $\mu(\mathcal{N}(0)) \geq k - e$. For $\delta, \delta' \in \mathbb{B}$, the Taylor expansion of f_i in $\widehat{x}_{d+1}, \dots, \widehat{x}_{d+c}$ shows that

$$\iota_\delta(f_i) - \iota_{\delta'}(f_i) - \mathbf{J}(\delta - \delta')$$

is a sum of terms each of which contains some $\delta_{i'} - \delta'_{i'}$ and some $\delta_{i''}$ or $\delta'_{i''}$ as factors, with all other factors of value at least 0. Thus

$$\mu(\mathcal{N}(\delta) - \mathcal{N}(\delta')) \geq \mu(\delta - \delta') + k - 2e \geq \mu(\delta - \delta') + 1.$$

Hence $\mathcal{N}(\mathbb{B}) \subset \mathbb{B}$, and since $\mathcal{K}$ is complete, the sequence $\mathcal{N}^q(0)$ $(q \in \mathbb{N})$ converges to some $\delta \in \mathbb{B}$ with $\mathcal{N}(\delta) = \delta$, that is, $\iota_\delta(f_i) = 0$ for $i = 1, \dots, c$. We let $\iota := \iota_\delta$. Then $\iota(\mathfrak{c}) = 0$ and, by (11.2),

$$\mu(\iota(g) - g|_{U_t}) \geq k - e \quad \text{for any } g \in R. \tag{11.3}$$

Assume that $k - e > \operatorname{mult}_F \operatorname{div}(\overline{\psi}_0)$. Since the reduction of ψ modulo x_0 is in $\psi_0 + \mathfrak{c}_0\widehat{R}_0$, $\psi|_{U_t} = \overline{\psi}_0$, so (11.3) implies $\iota(\psi) \neq 0$. Since $\psi\mathfrak{b} \subset \mathfrak{c}$ and $\iota(\mathfrak{c}) = 0$, $\iota(\mathfrak{b}) = 0$. Thus ι induces a homomorphism $\widehat{\mathcal{O}}_{Y,V} \to \mathcal{K}$, and by composition a homomorphism $\mathcal{O}_{Y,V} \to \mathcal{K}$, also denoted by ι. We have $\iota(\widetilde{z}_i) = z_i|_{U_t}$, $\iota(\widetilde{x}_i) = x_i|_{U_t}$ for $1 \leq i \leq d$, and $\iota(x_0) = \zeta$. Since π is étale at the generic point of U_t and ζ is transcendental over $\mathcal{K}_0$, the image of $\mathcal{O}_{Y,V}$ has transcendence degree $n = \dim Y$ over $\mathbb{C}$, so ι is injective on $\mathcal{O}_{Y,V}$ and extends to the rational functions on Y. For a non-zero rational function g on Y we let

$$w(g) := \mu(\iota(g)) \in \mathbb{Z}.$$

Then w is a valuation on Y, $w \geq 0$ on $\mathcal{O}_{Y,V}$, and $w > 0$ on the maximal ideal of $\mathcal{O}_{Y,V}$, since $\mu(\zeta) = k$ and $\mu(\widehat{x}_i|_{U_t}), \mu(\delta) \geq 1$. Thus $\operatorname{center}_Y w$ is the generic point of V, and

$$w(E) = w(x_0) = \mu(\zeta) = k.$$

Since $w \geq 0$ on $\mathcal{O}_{Y,V}$ and $w(x_0) = k \geq 1$, $w(g) \geq qk$ for any $g \in x_0^q\mathcal{O}_{Y,V}$ and any $q \in \mathbb{N}$, so w extends by continuity to the completion of $\mathcal{O}_{Y,V}$ along E, and the extension is $\mu \circ \iota$. For $i \in I$, the reduction of ϕ_i modulo x_0 is $\overline{\phi}_i$, so if $k > e + M$, then (11.3) implies

$$w(\phi_i) = \operatorname{mult}_F(h_t^* \operatorname{div}(\overline{\phi}_i)).$$

We show that w is divisorial with value group $\mathbb{Z}$. Let ϖ be a rational function on U_t with $\operatorname{mult}_F \operatorname{div}(\varpi) = 1$, and let $u_1, \dots, u_{d-1}$ be rational functions on U_t with $\operatorname{mult}_F \operatorname{div}(u_i) = 0$ whose restrictions to F, together with the restrictions of $z_1, \dots, z_{n-1-d}$, define a rational map $F \dashrightarrow \mathbb{A}^{n-2}$ which is étale at the generic point of F. Write $\varpi = \varpi_1/\varpi_2$ and $u_i = u_{i,1}/u_{i,2}$ with non-zero $\varpi_1, \varpi_2, u_{i,1}, u_{i,2} \in \mathcal{O}_{U_t,V}$, let $\widetilde{\varpi}_1, \widetilde{\varpi}_2, \widetilde{u}_{i,1}, \widetilde{u}_{i,2}$ be lifts of them to $\mathcal{O}_{Y,V}$, and let

$$\widetilde{\varpi} := \widetilde{\varpi}_1/\widetilde{\varpi}_2, \quad \widetilde{u}_i := \widetilde{u}_{i,1}/\widetilde{u}_{i,2}, \quad \widetilde{\tau} := x_0/\widetilde{\varpi}^k, \quad \text{and} \quad \tau := \zeta/\varpi^k.$$

Assume that $k - e > \operatorname{mult}_F \operatorname{div}(g) + 1$ for

$$g \in \{\varpi_1, \varpi_2\} \cup \{u_{i,1}, u_{i,2} \mid 1 \leq i \leq d-1\}.$$

By (11.3), $\iota(\widetilde{g}) = g(1 + \alpha_g)$ with $\mu(\alpha_g) \geq 1$ for any such g with lift $\widetilde{g}$. Thus in $\mathcal{K}$,

$$\iota(\widetilde{\varpi}) = \varpi(1 + \alpha_0), \quad \iota(\widetilde{u}_i) = u_i + \alpha_i', \quad \text{and} \quad \iota(\widetilde{\tau}) = \tau(1 + \alpha_0)^{-k},$$

where

$$1 + \alpha_0 := (1 + \alpha_{\varpi_1})(1 + \alpha_{\varpi_2})^{-1} \quad \text{and} \quad \alpha_i' := u_i\big((1 + \alpha_{u_{i,1}})(1 + \alpha_{u_{i,2}})^{-1} - 1\big),$$

so $\mu(\alpha_0) \geq 1$ and $\mu(\alpha_i') \geq 1$. In particular $w(\widetilde{\varpi}) = 1$, so the value group of w is $\mathbb{Z}$. The residues of $\widetilde{z}_1, \dots, \widetilde{z}_{n-1-d}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau}$ with respect to w are the residues of $z_1, \dots, z_{n-1-d}, u_1, \dots, u_{d-1}, \tau$ with respect to μ, which are algebraically independent over $\mathbb{C}$, since the residue of τ is transcendental over the residue field of $\mathcal{K}_0$. Thus the residue field of w has transcendence degree $n - 1$ over $\mathbb{C}$, and by the equality case of the Abhyankar inequality there exists a prime divisor G over Y with $w(g) = \operatorname{mult}_G \operatorname{div}(g)$ for any non-zero rational function g on Y. Since $w(\widetilde{\varpi}) = 1$, $\widetilde{\varpi}$ is a uniformizer at the generic point of G. Since the residues of $\widetilde{z}_1, \dots, \widetilde{z}_{n-1-d}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau}$ are rational functions on G which define a rational map $G \dashrightarrow \mathbb{A}^{n-1}$ étale at the generic point of G, the form

$$\Omega_G := d\widetilde{z}_1 \wedge \cdots \wedge d\widetilde{z}_{n-1-d} \wedge d\widetilde{\varpi} \wedge d\widetilde{u}_1 \wedge \cdots \wedge d\widetilde{u}_{d-1} \wedge d\widetilde{\tau}$$

generates the differential n-forms at the generic point of G.

Step 3. In this step we compute $a(w, Y, E + B)$.

Let

$$\eta := dz_1 \wedge \cdots \wedge dz_{n-1-d} \wedge dx_1 \wedge \cdots \wedge dx_d,$$

a rational differential $(n-1)$-form on U_t, and let

$$\eta_Y := d\widetilde{z}_1 \wedge \cdots \wedge d\widetilde{z}_{n-1-d} \wedge d\widetilde{x}_1 \wedge \cdots \wedge d\widetilde{x}_d,$$

a rational differential $(n-1)$-form on Y. It is regular near the generic point of E and $\eta_Y|_E = \eta$. Let

$$\Omega_F := dz_1 \wedge \cdots \wedge dz_{n-1-d} \wedge d\varpi \wedge du_1 \wedge \cdots \wedge du_{d-1}.$$

Since ϖ is a uniformizer at the generic point of F and the restrictions of $z_1, \dots, z_{n-1-d}, u_1, \dots, u_{d-1}$ to F define a rational map étale at the generic point of F, Ω_F generates the differential $(n-1)$-forms at the generic point of F, and we may write

$$\eta = \theta_\eta \Omega_F, \qquad \theta_\eta := \det\Big(\frac{\partial(x_1, \dots, x_d)}{\partial(\varpi, u_1, \dots, u_{d-1})}\Big),$$

where the partial derivatives are taken with respect to $\varpi, u_1, \dots, u_{d-1}$ with $z_1, \dots, z_{n-1-d}$ fixed. We let $\beta := \operatorname{mult}_F \operatorname{div}(\theta_\eta)$.

Let ξ_0 be the rational function on Y with $dx_0 \wedge \eta_Y = \xi_0 \Omega_G$. We claim that $w(\xi_0) = k + \beta$. Since $\mathcal{K}$ is a complete discrete valuation field of characteristic 0 with uniformizer ϖ whose residue field is separable algebraic over the field generated by the residues of $z_1, \dots, z_{n-1-d}, u_1, \dots, u_{d-1}, \tau$, there exist continuous $\mathbb{C}(z)$-derivations $\partial_0 := \varpi\partial_\varpi$, $\partial_i := \partial_{u_i}$ $(1 \le i \le d-1)$, and $\partial_d := \tau\partial_\tau$ of $\mathcal{K}$ which do not decrease μ and satisfy $\partial_0(\varpi) = \varpi$, $\partial_i(u_i) = 1$ $(1 \le i \le d-1)$, $\partial_d(\tau) = \tau$, and ∂_i vanishes on the other elements of $\varpi, u_1, \dots, u_{d-1}, \tau$. They act on a coefficient field coefficientwise, and ∂_d vanishes on $\mathcal{K}_0$. Via ι they restrict to $\mathbb{C}(z)$-derivations $\mathbb{C}(Y) \to \mathcal{K}$, which are determined by their values on the separating transcendence basis $\widetilde{\varpi}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau}$ of $\mathbb{C}(Y)$ over $\mathbb{C}(z)$. For rational functions $g_0, \dots, g_d$ on Y let

$$\mathbf{D}(g_0, \dots, g_d) := \det(\partial_i(g_{i'}))_{0 \le i, i' \le d}.$$

By the chain rule,

$$\iota(\xi_0) = \pm \frac{\mathbf{D}(x_0, \widetilde{x}_1, \dots, \widetilde{x}_d)}{\mathbf{D}(\widetilde{\varpi}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau})}.$$

Since the ∂_i do not decrease μ and $\mu(\alpha_0), \mu(\alpha_i') \ge 1$, the matrix $(\partial_i(\iota(\widetilde{g}_{i'})))_{0 \le i, i' \le d}$, where $\widetilde{g}_0, \dots, \widetilde{g}_d$ denote $\widetilde{\varpi}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau}$, is the product of the identity matrix plus a matrix with entries of value at least 1 and the diagonal matrix with entries $\iota(\widetilde{\varpi}), 1, \dots, 1, \iota(\widetilde{\tau})$. For example,

$$\partial_0(\iota(\widetilde{\varpi})) = \iota(\widetilde{\varpi})\Big(1 + \frac{\partial_0(\alpha_0)}{1 + \alpha_0}\Big) \quad \text{and} \quad \partial_i(\iota(\widetilde{\varpi})) = \iota(\widetilde{\varpi})\frac{\partial_i(\alpha_0)}{1 + \alpha_0} \quad (i \ne 0).$$

Thus

$$\mu(\mathbf{D}(\widetilde{\varpi}, \widetilde{u}_1, \dots, \widetilde{u}_{d-1}, \widetilde{\tau})) = w(\widetilde{\varpi}) + w(\widetilde{\tau}) = 1.$$

On the other hand, $\iota(x_0) = \zeta = \varpi^k \tau$, and ∂_d vanishes on $\iota(\widetilde{x}_1), \dots, \iota(\widetilde{x}_d) \in \mathcal{K}_0$, so

$$\mathbf{D}(x_0, \widetilde{x}_1, \dots, \widetilde{x}_d) = \pm \varpi^k \tau \cdot \varpi \theta_\eta,$$

whose value is $k + 1 + \beta$. Thus $w(\xi_0) = k + 1 + \beta - 1 = k + \beta$, and the claim holds.

Let $b' \in Q_b$. Let $\sigma_{b'}$ be a generator of

$$\mathcal{O}_{U_t}\big(r\big(K_{U_t} + B_{U_t}(b')\big)\big)$$

at the generic point of V (see **Step 1**), and let $\chi_{b'}$ be the rational function on U_t with $\eta^{\otimes r} = \chi_{b'}\sigma_{b'}$. Then $\chi_{b'} \in \mathcal{O}_{U_t, V} \backslash \{0\}$, since η is regular near the generic point of V and $rB_{U_t}(b') \ge 0$. Let $\widetilde{\sigma}_{b'}$ be a generator of $\mathcal{O}_Y(r(K_Y + B(b')))$ at the generic point of V with $\operatorname{res}_{b'}(\widetilde{\sigma}_{b'}) = \sigma_{b'}$, and let $\widetilde{\chi}_{b'}$ be the rational function on Y with

$$(dx_0 \wedge \eta_Y)^{\otimes r} = \widetilde{\chi}_{b'}\widetilde{\sigma}_{b'}.$$

By (11.1), $\widetilde{\chi}_{b'}|_E = \chi_{b'}$. Assume that $k - e > \operatorname{mult}_F \operatorname{div}(\chi_{b'})$. Then (11.3) implies

$$w(\widetilde{\chi}_{b'}) = \operatorname{mult}_F \operatorname{div}(\chi_{b'}).$$

For a rational differential n-form $\omega = \xi\Omega_G$ on Y with $\omega^{\otimes r} = g\widetilde{\sigma}_{b'}$, we have $a(G, Y, B(b')) = 1 + w(\xi) - w(g)/r$. For a rational differential $(n-1)$-form $\omega' = \xi'\Omega_F$ on U_t with $\omega'^{\otimes r} = g'\sigma_{b'}$, we have

$$a\big(F, U_t, B_{U_t}(b')\big) = 1 + \operatorname{mult}_F \operatorname{div}(\xi') - \frac{\operatorname{mult}_F \operatorname{div}(g')}{r}.$$

Applying the first formula to $\omega = dx_0 \wedge \eta_Y = \xi_0\Omega_G$ with $g = \widetilde{\chi}_{b'}$, the second to $\omega' = \eta = \theta_\eta\Omega_F$ with $g' = \chi_{b'}$, and using $a(w, Y, E + B(b')) = a(G, Y, B(b')) - w(x_0)$ and $w(x_0) = k$, we obtain

$$a(w, Y, E+B(b')) = 1+k+\beta-\frac{\operatorname{mult}_F \operatorname{div}(\chi_{b'})}{r}-k = 1+\beta-\frac{\operatorname{mult}_F \operatorname{div}(\chi_{b'})}{r} = a\big(F, U_t, B_{U_t}(b')\big).$$

Near the generic point of V,

$$K_Y + E + B = \sum_{b'\in Q_b} \lambda_{b'}(K_Y + E + B(b')) \quad \text{and} \quad K_{U_t} + B_{U_t} = \sum_{b'\in Q_b} \lambda_{b'}\big(K_{U_t} + B_{U_t}(b')\big)$$

are convex combinations of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors. Since log discrepancies are affine in such combinations,

$$a(w, Y, E + B) = a(F, U_t, B_{U_t}).$$

In **Steps 2** and 3 we used $k > e + M$ and $k > k_1$, where

$$\begin{aligned} k_1 := \max\big(\{1, 2e, e + \operatorname{mult}_F \operatorname{div}(\overline{\psi}_0)\} \cup \{e + \operatorname{mult}_F \operatorname{div}(\chi_{b'}) \mid b' \in Q_b\} \\ \cup \{e + 1 + \operatorname{mult}_F \operatorname{div}(g) \mid g \in \{\varpi_1, \varpi_2\} \cup \{u_{i,1}, u_{i,2} \mid 1 \le i \le d-1\}\}\big). \end{aligned} \tag{11.4}$$

Since $e \ge 0$, $k > k_1 + M$ implies all of these. Thus **Steps 1–3** prove the conclusion of the theorem for t, j, and b and any integer $k > k_1 + M$.

Step 4. In this step we bound k_1 uniformly. By **Step 1**, r and the sets Q_b ($b \in \Lambda$) depend only on n, v_0, and Λ. The integer k_1 depends only on the orders along $F_{j,t}$ of finitely many functions on U_t and $\mathcal{Z}_t$: θ_0, $\overline{\psi}_0$, $\chi_{b'}$ ($b' \in Q_b$), ϖ_1, ϖ_2, $u_{i,1}$, $u_{i,2}$ ($1 \le i \le d-1$), and **Steps 2** and 3 only use the properties of the functions above and of $z_1, \dots, z_{n-1-d}, x_1, \dots, x_{d+c}, \overline{f}_1, \dots, \overline{f}_c, \sigma_{b'}$ listed there, none of which involves Y.

After a stratification of T into finitely many locally closed subsets and finite surjective base changes of the strata, the choices of **Steps 1–3** are made once on $\mathcal{U}$ and $\mathcal{Z}$ over each stratum T', along the relative center $h(\mathcal{F}_j)$ and for any $j \in J$ and $b \in \Lambda$. The ideals $\mathfrak{b}_{\mathcal{U}}$, $\mathfrak{c}_{\mathcal{U}}$, $(\mathfrak{c}_{\mathcal{U}} : \mathfrak{b}_{\mathcal{U}})$ on $\mathbb{A}^{n-1+c} \times T'$ corresponding to $\mathfrak{b}_0$, $\mathfrak{c}_0$, $(\mathfrak{c}_0 : \mathfrak{b}_0)$ and the sheaves

$$\mathcal{O}_{\mathcal{U}}\Big(r\Big(K_{\mathcal{U}/T} + \sum_{i=1}^{m} b_i'\mathcal{B}_i\Big)\Big)$$

are computed from finite presentations by finitely many Hom, tensor and image operations. By generic flatness for the kernels and cokernels of the maps in the finite presentations, after shrinking T', their formation and the conditions of **Step 2** (algebraic independence, regular sequence, $\theta_0 \ne 0$, $\overline{\psi}_0 \ne 0$) commute with restriction to the fibers over T'. Since U_t is normal, the restrictions are the corresponding ideals on $\mathbb{A}^{n-1+c}$ and reflexive sheaves on U_t. The set of $t \in T'$ for which the generic point of V belongs to the invertibility locus of the sheaves

$$\mathcal{O}_{U_t}\big(r\big(K_{U_t} + B_{U_t}(b')\big)\big) \quad (b' \in Q_b)$$

is constructible, and for the other t there exists no Y as in the theorem by **Step 1**.

Let g be one of the finitely many functions above on $\mathcal{Z}$ over T', and let $\varpi_{\mathcal{F}}$ be a local equation of $\mathcal{F}_j$ at the generic point of $\mathcal{F}_j$. On an open subset of $\mathcal{Z}$ containing the generic point of $\mathcal{F}_j$ we may write

$$g = \varpi_{\mathcal{F}}^{\operatorname{mult}_{\mathcal{F}_j} \operatorname{div}(g)} \cdot g'$$

with a unit g'. After shrinking T', the open subset contains the generic point of $F_{j,t}$ for any $t \in T'$, $\varpi_{\mathcal{F}}|_{\mathcal{Z}_t}$ is a local equation of the prime divisor $F_{j,t}$, and $g'|_{\mathcal{Z}_t}$ is a unit at the generic point of $F_{j,t}$, so

$$\operatorname{mult}_{F_{j,t}} \operatorname{div}(g|_{\mathcal{Z}_t}) = \operatorname{mult}_{\mathcal{F}_j} \operatorname{div}(g) \quad \text{for any } t \in T'.$$

Repeating the argument on the complement of T', k_1 is constant on each of finitely many strata, and we let k_0 be the maximum of the constants over all strata, $j \in J$, and $b \in \Lambda$. Then k_0 depends only on $\mathcal{U} \to T$, $\mathcal{Z}$, $\{\mathcal{F}_j\}_{j\in J}$, $\mathcal{B}_1, \dots, \mathcal{B}_m$, Λ, and v_0, and the theorem follows. □

11.2. **Volume lower bounds and rounding of coefficients.** In this subsection we prove that, for a Kollár component E of $(X \ni x, B)$ with $\delta(E, B_E) \geq \kappa$, a positive lower bound of $\widehat{\operatorname{vol}}(X \ni x, B)/a(E, X, B)$ implies positive lower bounds, depending also on n, κ, and $t \in (0, 1]$, of $\widehat{\operatorname{vol}}(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'}))$ and $\widehat{\operatorname{vol}}(Y' \ni y', B_{Y'} + (1-t)E')$ on the index 1 cover of $(Y \ni y, E)$ at any closed point $y \in E$ (Lemma 11.2), and that for a sequence of germs $(Y' \ni y', E' + B_{Y'})$ with $B_{Y'} \in \Gamma$ for a DCC set Γ and with $\widehat{\operatorname{vol}}(Y' \ni y', B_{Y'} + \frac{1}{2}E')$ and $\widehat{\operatorname{vol}}(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'}))$ bounded from below, after passing to a subsequence the coefficients of $B_{Y'}$ can be rounded up to a finite set, changing the log discrepancies by a factor tending to 1 (Lemma 11.3).

Lemma 11.2. *Let $n \geq 2$ be an integer, $\kappa > 0$ a real number, $(X \ni x, B)$ a klt germ of dimension n, and E a Kollár component of $(X \ni x, B)$ with plt blow-up $f : Y \to X$. Let $y \in E$ be a closed point, $g : (Y' \ni y') \to (Y \ni y)$ an index 1 cover of $(Y \ni y, E)$, $E' := g^*E$, and $B_{Y'} := g^*B_Y$. Assume that $\delta(E, B_E) \geq \kappa$. Then, possibly shrinking Y near y and replacing Y' with the preimage, the following hold.*

(1) We have

$$\widehat{\operatorname{vol}}(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'})) \geq \Big(\frac{n-1}{n}\Big)^{n-1} \kappa^{n-1} \frac{\widehat{\operatorname{vol}}(X \ni x, B)}{a(E, X, B)}.$$

(2) For any real number $t \in (0, 1]$,

$$\widehat{\operatorname{vol}}(Y' \ni y', B_{Y'} + (1-t)E') \geq nt\kappa^{n-1} \frac{\widehat{\operatorname{vol}}(X \ni x, B)}{a(E, X, B)}.$$

Proof. Let $r := \operatorname{ind}_y(Y, E)$ and $L := -(K_E + B_E)$. Since f is a plt blow-up, E is a $\mathbb{Q}$-Cartier prime divisor on Y, $(Y, E + B_Y)$ is plt, and $\operatorname{Diff}_E(B_Y) = B_E$. By Definition-Lemma 2.16, possibly shrinking Y near y and replacing Y' with the preimage, E' is a normal Cartier prime divisor, $(Y', E' + B_{Y'})$ is plt, and

$$\widehat{\operatorname{vol}}(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'})) = r\widehat{\operatorname{vol}}(E \ni y, B_E). \tag{11.5}$$

(1) Since E is a Kollár component of $(X \ni x, B)$, (E, B_E) is a klt log Fano pair of dimension $n - 1 \geq 1$. Since $\delta(E, B_E) \geq \kappa$, Lemma 2.13 with $d = n - 1$ implies

$$\kappa^{n-1} L^{n-1} \leq \Big(\frac{n}{n-1}\Big)^{n-1} \widehat{\operatorname{vol}}(E \ni y, B_E).$$

By Lemma 2.8(2) and (3),

$$\widehat{\operatorname{vol}}(X \ni x, B) \leq \widehat{\operatorname{vol}}(E, X, B) = a(E, X, B)L^{n-1}.$$

Since $r \geq 1$, by (11.5) we have

$$\begin{aligned}\widehat{\operatorname{vol}}(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'})) &\geq \widehat{\operatorname{vol}}(E \ni y, B_E) \geq \Big(\frac{n-1}{n}\Big)^{n-1} \kappa^{n-1} L^{n-1} \\ &\geq \Big(\frac{n-1}{n}\Big)^{n-1} \kappa^{n-1} \frac{\widehat{\operatorname{vol}}(X \ni x, B)}{a(E, X, B)}.\end{aligned}$$

(2) Fix a real number $t \in (0, 1]$ and let $D := B_{Y'} + (1-t)E'$. Write $B_{Y'} = \sum_{j=1}^m b_j P_j$, where $P_1, \dots, P_m$ are the distinct irreducible components of $B_{Y'}$ and $b_j > 0$ for any j. Possibly

shrinking Y near y, we may assume that $y' \in P_j$ for any j. Since E' is a normal Cartier prime divisor,

$$\mathrm{Diff}_{E'}(B_{Y'}) = B_{Y'}|_{E'} = \sum_{j=1}^{m} b_j P_j|_{E'}.$$

Since E' is Cartier and $(Y', E' + B_{Y'})$ is plt, $(Y' \ni y', D)$ is a klt germ.

We prove

$$\widehat{\mathrm{vol}}(Y' \ni y', D) \geq \frac{n^n t}{(n-1)^{n-1}} \widehat{\mathrm{vol}}(E' \ni y', \mathrm{Diff}_{E'}(B_{Y'})). \tag{11.6}$$

When D is a $\mathbb{Q}$-divisor, (11.6) is [LZ21, Theorem 4.6]: the coefficient of E' in D is $1-t$ and $(D-(1-t)E')|_{E'} = \mathrm{Diff}_{E'}(B_{Y'})$. In general, let W' be the set of $b' \in \mathbb{R}^m$ such that $K_{Y'} + E' + \sum_{j=1}^m b'_j P_j$ is $\mathbb{R}$-Cartier near y'. Since E' is Cartier, $W' \times \mathbb{R}$ is the set W of Lemma 2.9 for the prime divisors $P_1, \dots, P_m, E'$. By Lemma 2.9(1), W' is an affine subspace of $\mathbb{R}^m$ defined over $\mathbb{Q}$, and $b := (b_1, \dots, b_m) \in W'$. Thus there exist $b^{(k)} \in W' \cap \mathbb{Q}^m$ for $k \in \mathbb{N}^+$ such that $\lim_{k\to+\infty} b^{(k)} = b$. Let $t_k \in (0,1]$ be rational numbers such that $\lim_{k\to+\infty} t_k = t$, and let

$$D_k := \sum_{j=1}^{m} b_j^{(k)} P_j + (1-t_k)E'.$$

Since $b_j > 0$ for any j, $D_k \geq 0$ for any $k \gg 0$. Since $K_{Y'} + D_k$ is $\mathbb{R}$-Cartier near y' and D_k is a $\mathbb{Q}$-divisor, $K_{Y'} + D_k$ is $\mathbb{Q}$-Cartier near y'. Let Q run over the prime divisors in $\bigcup_j \mathrm{Supp}(P_j|_{E'})$. Then

$$\sum_j b_j^{(k)} P_j|_{E'} = \sum_Q \Big(\sum_j b_j^{(k)} \,\mathrm{mult}_Q(P_j|_{E'}) \Big) Q,$$

and its coefficients tend to those of $\mathrm{Diff}_{E'}(B_{Y'})$ as $k \to +\infty$. Moreover,

$$K_{E'} + \sum_{j=1}^{m} b_j^{(k)} P_j|_{E'} = \Big(K_{Y'} + E' + \sum_{j=1}^{m} b_j^{(k)} P_j \Big)\Big|_{E'}$$

is $\mathbb{R}$-Cartier near y', and $(E' \ni y', \mathrm{Diff}_{E'}(B_{Y'}))$ is klt by [Kol23, Theorem 11.17.1]. Fix a real number $\epsilon \in (0,1)$. By Lemma 2.9(2) for $P_1, \dots, P_m, E'$ on Y' and for the prime divisors Q on E', both at the closed point y', for any $k \gg 0$, $(Y' \ni y', D_k)$ is a klt germ,

$$\widehat{\mathrm{vol}}(Y' \ni y', D_k) \leq (1+\epsilon)^n \widehat{\mathrm{vol}}(Y' \ni y', D),$$

and

$$\widehat{\mathrm{vol}}\Big(E' \ni y', \sum_{j=1}^{m} b_j^{(k)} P_j|_{E'}\Big) \geq (1-\epsilon)^{n-1} \widehat{\mathrm{vol}}(E' \ni y', \mathrm{Diff}_{E'}(B_{Y'})).$$

Since $(D_k - (1-t_k)E')|_{E'} = \sum_{j=1}^m b_j^{(k)} P_j|_{E'}$, by [LZ21, Theorem 4.6],

$$\widehat{\mathrm{vol}}(Y' \ni y', D_k) \geq \frac{n^n t_k}{(n-1)^{n-1}} \widehat{\mathrm{vol}}\Big(E' \ni y', \sum_{j=1}^{m} b_j^{(k)} P_j|_{E'}\Big).$$

Thus

$$(1+\epsilon)^n \widehat{\mathrm{vol}}(Y' \ni y', D) \geq \frac{n^n t_k}{(n-1)^{n-1}} (1-\epsilon)^{n-1} \widehat{\mathrm{vol}}(E' \ni y', \mathrm{Diff}_{E'}(B_{Y'}))$$

for any $k \gg 0$. Letting $k \to +\infty$ and then $\epsilon \to 0$, we obtain (11.6).

By (11.6) and (1),

$$\widehat{\mathrm{vol}}(Y' \ni y', D) \geq \frac{n^n t}{(n-1)^{n-1}} \Big(\frac{n-1}{n}\Big)^{n-1} \kappa^{n-1} \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{a(E, X, B)} = nt\kappa^{n-1} \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{a(E, X, B)}.$$

The lemma follows. □

Lemma 11.3. *Let $n \geq 2$ be an integer, $\Gamma \subset [0,1]$ a DCC set, and ϵ_1, ϵ_2 positive real numbers. For any positive integer i, let $(X_i \ni x_i, S_i + B_i)$ be a germ of dimension n, where S_i is a normal Cartier prime divisor through x_i and $B_i \geq 0$ is an $\mathbb{R}$-divisor with no component S_i, and let $B_{S_i} := \mathrm{Diff}_{S_i}(B_i)$. Assume that for any i, $B_i \in \Gamma$, $(X_i, S_i + B_i)$ is plt near x_i, and*

$$\widehat{\mathrm{vol}}\Big(X_i \ni x_i, B_i + \frac{1}{2}S_i\Big) \geq \epsilon_1, \quad \widehat{\mathrm{vol}}(S_i \ni x_i, B_{S_i}) \geq \epsilon_2.$$

Then, possibly passing to a subsequence and shrinking X_i near x_i, there exist a finite set $\Gamma_0 \subset [0,1]$, $\mathbb{R}$-divisors B_i^+ on X_i, and real numbers $\sigma_i \geq 0$ with $\lim_{i\to+\infty} \sigma_i = 0$, such that for any i:

(1) $B_i^+ \geq B_i$, $\mathrm{Supp}\, B_i^+ = \mathrm{Supp}\, B_i$, and $B_i^+ \in \Gamma_0$.
(2) $K_{X_i} + S_i + B_i^+$ is $\mathbb{R}$-Cartier and $(X_i, S_i + B_i^+)$ is plt.
(3) For any divisorial valuation w on X_i with $\mathrm{center}_{X_i} w \in S_i$,

$$(1 - \sigma_i)a(w, X_i, S_i + B_i) \leq a(w, X_i, S_i + B_i^+).$$

Proof. Possibly shrinking X_i near x_i, we may assume that $(X_i, S_i + B_i)$ is plt and that any irreducible component of B_i contains x_i. Write $B_i = \sum_{j=1}^{m_i} b_{i,j}B_{i,j}$, where $B_{i,j}$ are the distinct irreducible components of B_i and $b_{i,j} \in \Gamma\backslash\{0\}$. Since S_i is a normal Cartier prime divisor,

$$B_{S_i} = B_i|_{S_i} = \sum_{j=1}^{m_i} b_{i,j}B_{i,j}|_{S_i}.$$

Since S_i is Cartier, $B_{i,j} \not\subset S_i$, and $x_i \in B_{i,j} \cap S_i$, $B_{i,j}|_{S_i}$ is a non-zero Weil divisor on S_i with $B_{i,j}|_{S_i} \geq 0$ for any j.

Since S_i is Cartier and $(X_i, S_i + B_i)$ is plt, $(X_i \ni x_i, B_i + \frac{1}{2}S_i)$ is a klt germ. By [Kol23, Theorem 11.17.1], (S_i, B_{S_i}) is klt.

Let Γ' be the closure of $\Gamma \cup \{\frac{1}{2}\}$ in $[0,1]$, which is a DCC set. We let $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma\backslash\{0\} \neq \emptyset$, and $\gamma := 1$ otherwise. Thus $b_{i,j} \geq \gamma > 0$ for any i, j. Since Γ' is DCC, there exists a real number $\tau > 0$ such that

$$\Gamma' \cap \Big(\frac{1}{2}, \frac{1}{2} + \tau\Big) = \emptyset.$$

By Theorem 2.18, there exists a positive integer M depending only on n and ϵ_1 such that $\mathrm{mld}(X_i \ni x_i, B_i + \frac{1}{2}S_i) \leq M$ for any i. The germs $(X_i \ni x_i, B_i + \frac{1}{2}S_i)$ are klt of dimension n, their coefficients belong to the closed DCC set Γ', all their irreducible components contain x_i, and $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i + \frac{1}{2}S_i) \geq \epsilon_1$. Let $c_1 > 0$ be the real number given by Theorem 2.19 for $d = n$, ϵ_1, M, $\mathcal{I} = \Gamma'$ and the sequence $(X_i \ni x_i, B_i + \frac{1}{2}S_i)$, and fix a real number t with $0 < t < \min\{c_1, \tau\}$. By Theorem 2.19, possibly passing to a subsequence, there exist a finite set $\mathcal{I}_t \subset \Gamma'$ and a map $p_t : \Gamma' \to \mathcal{I}_t$ such that $\gamma' \leq p_t(\gamma') \leq \gamma' + t$ for any $\gamma' \in \Gamma'$, $K_{X_i} + p_t(B_i + \frac{1}{2}S_i)$ is $\mathbb{R}$-Cartier for any i, and

$$\lim_{i\to+\infty} \Big\| p_t\Big(B_i + \frac{1}{2}S_i\Big) - \Big(B_i + \frac{1}{2}S_i\Big) \Big\| = 0.$$

Since $p_t(\frac{1}{2}) \in \Gamma' \cap [\frac{1}{2}, \frac{1}{2} + t]$ and $t < \tau$, $p_t(\frac{1}{2}) = \frac{1}{2}$. Let

$$\Gamma_0 := \mathcal{I}_t, \quad B_i^+ := \sum_{j=1}^{m_i} p_t(b_{i,j})B_{i,j}, \quad D_i := B_i^+ - B_i, \quad \text{and} \quad \mu_i := \|D_i\|.$$

Then $\Gamma_0 \subset [0,1]$ is finite, $B_i^+ \in \Gamma_0$, and $B_i^+ \geq B_i$ with $\mathrm{Supp}\, B_i^+ = \mathrm{Supp}\, B_i$ since $p_t(b_{i,j}) \geq b_{i,j} > 0$. This proves (1). Since $p_t(B_i + \frac{1}{2}S_i) = B_i^+ + \frac{1}{2}S_i$, we have $\lim_{i\to+\infty} \mu_i = 0$, and since S_i is Cartier, $K_{X_i} + S_i + B_i^+$ is $\mathbb{R}$-Cartier. Thus $D_i = (K_{X_i} + S_i + B_i^+) - (K_{X_i} + S_i + B_i)$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor, and $D_i \geq 0$ by (1). Moreover,

$$0 \leq p_t(b_{i,j}) - b_{i,j} \leq \mu_i \leq \frac{\mu_i}{\gamma} b_{i,j}$$

for any j, so

$$0 \le D_i|_{S_i} = \sum_{j=1}^{m_i}(p_t(b_{i,j}) - b_{i,j})B_{i,j}|_{S_i} \le \frac{\mu_i}{\gamma}B_{S_i}. \tag{11.7}$$

Let $\epsilon_3 := \min\{\epsilon_2, \gamma\}$. The germ $(S_i \ni x_i, B_{S_i})$ is klt of dimension $n-1 \ge 1$ and

$$\widehat{\mathrm{vol}}(S_i \ni x_i, B_{S_i}) \ge \epsilon_2 \ge \epsilon_3,$$

and $\mathrm{mult}_Q B_{S_i} \ge \gamma \ge \epsilon_3$ for any prime divisor $Q \subset \mathrm{Supp}\, B_{S_i}$ since $B_{S_i} = \sum_j b_{i,j}B_{i,j}|_{S_i}$, $b_{i,j} \ge \gamma$, and $B_{i,j}|_{S_i} \ge 0$ is a Weil divisor. Let $c_2 > 0$ be the real number given by Theorem 2.20 for $d = n-1$ and ϵ_3. By Theorem 2.20, for any i there exists a $\mathbb{Q}$-divisor $B'_{S_i} \ge (1+c_2)B_{S_i}$ on S_i such that $(S_i \ni x_i, B'_{S_i})$ is a klt germ. Possibly shrinking X_i near x_i, we may assume that (S_i, B'_{S_i}) is klt. Let $G_i := B'_{S_i} - B_{S_i}$. Then $G_i \ge c_2 B_{S_i}$, and $G_i = (K_{S_i} + B'_{S_i}) - (K_{S_i} + B_{S_i})$ is $\mathbb{R}$-Cartier.

Let

$$\sigma_i := \frac{2\mu_i}{\gamma c_2}.$$

Then $\lim_{i\to+\infty}\sigma_i = 0$, and possibly passing to a subsequence, we may assume that $\sigma_i < 1$ for any i. If $\mu_i = 0$, then $B_i^+ = B_i$, and (2) and (3) hold. Assume that $\mu_i > 0$, and let $\Delta_i := B_i + \sigma_i^{-1}D_i$. Then $\Delta_i \ge 0$ has no component S_i, $K_{X_i} + S_i + \Delta_i$ is $\mathbb{R}$-Cartier, and

$$\mathrm{Diff}_{S_i}(\Delta_i) = \Delta_i|_{S_i} = B_{S_i} + \sigma_i^{-1}D_i|_{S_i}.$$

By (11.7),

$$\sigma_i^{-1}D_i|_{S_i} \le \frac{\gamma c_2}{2\mu_i}\cdot\frac{\mu_i}{\gamma}B_{S_i} = \frac{c_2}{2}B_{S_i} \le \frac{1}{2}G_i.$$

Then for any prime divisor F over S_i,

$$a\Big(F, S_i, B_{S_i} + \frac{1}{2}G_i\Big) = \frac{1}{2}a(F, S_i, B_{S_i}) + \frac{1}{2}a(F, S_i, B'_{S_i}) > 0.$$

We have $\frac{1}{2}G_i - \sigma_i^{-1}D_i|_{S_i} \ge 0$, and this $\mathbb{R}$-divisor is $\mathbb{R}$-Cartier as the difference of $K_{S_i} + B_{S_i} + \frac{1}{2}G_i$ and $K_{S_i} + \mathrm{Diff}_{S_i}(\Delta_i) = (K_{X_i} + S_i + \Delta_i)|_{S_i}$. Thus for any prime divisor F over S_i,

$$a(F, S_i, \mathrm{Diff}_{S_i}(\Delta_i)) \ge a\Big(F, S_i, B_{S_i} + \frac{1}{2}G_i\Big) > 0,$$

so $(S_i, \mathrm{Diff}_{S_i}(\Delta_i))$ is klt. By [Kol23, Theorem 11.17.1], possibly shrinking X_i near x_i, $(X_i, S_i + \Delta_i)$ is plt.

Since

$$S_i + B_i^+ = (1-\sigma_i)(S_i + B_i) + \sigma_i(S_i + \Delta_i)$$

and $(X_i, S_i + B_i)$ and $(X_i, S_i + \Delta_i)$ are plt, $(X_i, S_i + B_i^+)$ is plt. This proves (2).

Let w be a divisorial valuation on X_i with $\mathrm{center}_{X_i} w \in S_i$. Then w is λ times the divisorial valuation of a prime divisor F over X_i for some real number $\lambda > 0$. If $F = S_i$, then both log discrepancies in (3) are 0. Otherwise, since $\mathrm{center}_{X_i} F \in S_i$, F is exceptional over X_i, and since $(X_i, S_i + \Delta_i)$ is plt,

$$0 < a(w, X_i, S_i + \Delta_i) = a(w, X_i, S_i + B_i) - \sigma_i^{-1}w(D_i).$$

Thus $w(D_i) < \sigma_i a(w, X_i, S_i + B_i)$. Since $a(w, X_i, S_i + B_i^+) = a(w, X_i, S_i + B_i) - w(D_i)$, we obtain (3). The lemma follows. □

12. Bounding $\widehat{\mathrm{vol}}(X \ni x, B)/a(E, X, B)$ by $\Phi(E, B_E)$

Theorem 12.1. *Let n, ϵ, κ, and C be as in Set-up 4.8(1), and $\Gamma \subset [0,1]$ a DCC set. Then for any real number $\tau > 0$, there exists a real number $a_0 > 0$ depending only on n, Γ, ϵ, κ, C, and τ satisfying the following. Let $(X \ni x, B)$ and E satisfy Set-up 4.8(2)(3). Assume that $B \in \Gamma$ and $a(E, X, B) \leq a_0$. Then*

$$\frac{\widehat{\mathrm{vol}}(X \ni x, B)}{a(E, X, B)} < \Phi(E, B_E) + \tau.$$

Proof. Suppose that the theorem fails for τ. Then for any positive integer i, there exist $(X_i \ni x_i, B_i)$ and E_i satisfying Set-up 4.8(2)(3) in place of $(X \ni x, B)$ and E such that $B_i \in \Gamma$,

$$a_i := a(E_i, X_i, B_i) \leq \frac{1}{i} \quad \text{and} \quad \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} - \Phi(E_i, B_{E_i}) \geq \tau.$$

For any i, we let $f_i : Y_i \to X_i$ be the plt blow-up of E_i and $L_i := -(K_{E_i} + B_{E_i})$. We let θ be as in Set-up 4.8(1). By Set-up 4.8(2)(3),

$$\frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} \geq \frac{\epsilon \mathrm{mld}(X_i \ni x_i, B_i)}{a_i} \geq \frac{\epsilon}{C}. \tag{12.1}$$

Step 1. In this step we fix finitely many families which contain the pairs (E_i, B_{E_i}), the index 1 covering stacks of E_i, and generators of the filtered algebras of the lc places F_j below.

By Lemma 7.2, possibly passing to a subsequence, there exist $\mathcal{E} \to T$ and $\mathcal{B}_1, \dots, \mathcal{B}_m$ satisfying the conditions of Set-up 5.13 with $d = n - 1$ (with $\mathcal{E}$ and E_t in place of $\mathcal{S}$ and S_t), closed points $t_i \in T$, and $b_i \in \mathcal{P}$ such that $E_i \cong E_{t_i}$ by an isomorphism mapping B_{E_i} onto $B_{t_i}(b_i)$ for any i, and

$$b_\infty := \lim_{i \to +\infty} b_i \in \mathcal{P}.$$

We identify E_i with E_{t_i}. Below, T is replaced several times by the members of a finite stratification into locally closed subvarieties and then by finite surjective covers of these. Each time, we pass to a subsequence such that all t_i belong to one member, replace T by the cover of the member, and replace t_i by a closed point over t_i.

The set $\mathcal{P}_0 := \{b_i \mid i \in \mathbb{N}^+\} \cup \{b_\infty\}$ is a compact subset of $\mathcal{P}$. By Theorem 7.1 with $\frac{\tau}{3}$ in place of ϵ, after replacing T, there exist a smooth variety $\mathcal{Z}$ with a projective morphism $\mathcal{Z} \to \mathcal{E}$ such that $\mathcal{Z} \to T$ is smooth and $\mathcal{Z}_t \to E_t$ is birational for any $t \in T$, a finite set J, and, for any $j \in J$, a prime divisor $\mathcal{F}_j$ on $\mathcal{Z}$, a $\mathbb{Q}$-linear combination $\mathcal{G}_j \geq 0$ of relative Mumford divisors on $\mathcal{E} \to T$, and a positive rational number u_j, satisfying conditions (1)–(3) of Theorem 7.1. For any i and $j \in J$, we denote the prime divisor F_{j,t_i} over E_i by F_j, and we let

$$\beta_{i,j} := a(F_j, E_i, B_{E_i}).$$

By Theorem 7.1(3), for any i,

$$\min\left(\{L_i^{n-1}\} \cup \{\Psi_{E_i, B_{E_i}}(F_j, u_j) \mid j \in J\}\right) < \Phi(E_i, B_{E_i}) + \frac{\tau}{3}. \tag{12.2}$$

By Proposition 4.9(3), $\mathrm{ind}_y(Y_i, E_i) \leq \theta^{-1}$ for any closed point $y \in E_i$. In particular, $\theta^{-1} \geq 1$. We let

$$q := \mathrm{lcm}(1, 2, \dots, \lfloor \theta^{-1} \rfloor).$$

Then $\mathrm{ind}_y(Y_i, E_i)$ divides q for any closed point $y \in E_i$, and qE_i is Cartier near E_i.

Let $\mathcal{V} \subset \mathcal{E}$ be the union of the set of points at which $\mathcal{E} \to T$ is not smooth and $\bigcup_{k=1}^m \mathrm{Supp}\, \mathcal{B}_k$. Since $\mathcal{E} \to T$ is flat, for any $t \in T$,

$$\mathcal{V} \cap E_t = \mathrm{Sing}(E_t) \cup \bigcup_{k=1}^m \mathrm{Supp}(\mathcal{B}_k|_{E_t}) \neq E_t.$$

By Theorem 8.3 for $\mathcal{E} \to T$, $\mathcal{V}$, and q, after replacing T, there exist line bundles $\mathcal{H}'_1, \dots, \mathcal{H}'_{N_0}$ on $\mathcal{E}$ and a finite set $\mathfrak{P}$ of pairs satisfying the conclusions of Theorem 8.3 with N_0 in place of m.

For any i, let $\rho_i : \mathfrak{Y}_i \to Y_i$ be the index 1 covering stack of E_i, $\mathfrak{E}_i := \rho_i^* E_i$, and $\mathcal{L}_i := \mathcal{O}_{\mathfrak{E}_i}(-\mathfrak{E}_i)$.

For a closed point $y \in E_i$, we let $r := \mathrm{ind}_y(Y_i, E_i)$, $\pi_y : (\widetilde{Y} \ni \widetilde{y}) \to (Y_i \ni y)$ an index 1 cover of $(Y_i \ni y, E_i)$, and $\widetilde{E} := \pi_y^* E_i$. Near y, $\mathfrak{Y}_i$ is the quotient stack of $\widetilde{Y}$ by μ_r. Thus near the closed point over y, $\mathfrak{E}_i$ is the quotient stack $[\widetilde{E}/\mu_r]$ of the normal variety $\widetilde{E}$ (Definition-Lemma 2.16(1)), and the coarse map ρ_i is $[\widetilde{E}/\mu_r] \to \widetilde{E}/\mu_r$, which is separated since μ_r is finite. Hence $\mathfrak{E}_i$ is a normal tame Deligne–Mumford stack, and $\mathfrak{E}_i$ is separated since E_i is separated and separatedness of ρ_i is local on E_i. Moreover, the stabilizer of any geometric point of $[\widetilde{E}/\mu_r]$ is a subgroup of μ_r acting on the fiber of $\mathcal{L}_i$ through the restriction of the character by which μ_r acts on the fiber of $\mathcal{L}_i$ at $\widetilde{y}$, so (a) and (b) below imply the conditions on stabilizers at all geometric points. The stack $\mathfrak{E}_i$ with coarse map $\rho_i : \mathfrak{E}_i \to E_i$ and the line bundle $\mathcal{L}_i$ satisfy the conditions of Theorem 8.3(1) on $\mathfrak{S}$, ρ, and $\mathcal{D}$ with t_i in place of t, since:

(a) The stabilizer of $\mathfrak{E}_i$ at the closed point over y is μ_r, and $r \le \theta^{-1}$ divides q.

(b) μ_r acts faithfully on the fiber of $\mathcal{L}_i$ at the closed point over y, since otherwise, for some integer $1 \le r' < r$, μ_r acts trivially on the fiber of $\mathcal{O}_{\widetilde{Y}}(-r'\widetilde{E})$ at $\widetilde{y}$, so applying the Reynolds operator of μ_r to a local generator of $\mathcal{O}_{\widetilde{Y}}(-r'\widetilde{E})$ at $\widetilde{y}$ yields a μ_r-invariant local generator at $\widetilde{y}$, and $r'E_i$ is Cartier at y, which contradicts $r = \mathrm{ind}_y(Y_i, E_i)$.

(c) ρ_i is an isomorphism over $E_i \backslash \mathcal{V}$: for $y \in E_i \backslash \mathcal{V}$, E_i is smooth at y and $y \notin \mathrm{Supp}\, B_{E_i}$, so near $\widetilde{y}$ the ramification divisor of $\pi_y|_{\widetilde{E}}$ is $-\mathrm{Diff}_{\widetilde{E}}(\pi_y^* B_{Y_i}) \le 0$ by Definition-Lemma 2.16(2) and the Hurwitz formula, hence $\pi_y|_{\widetilde{E}}$ is étale near $\widetilde{y}$ by the purity of the branch locus [Gro03, Exposé X, Théorème 3.1], and $r = 1$ since $\widetilde{y}$ is the only point over y, so E_i is Cartier near y (Definition 10.1).

Hence, by Theorem 8.3 and since $\mathfrak{P}$ is finite, possibly passing to a subsequence, there exist $(\mathfrak{S}', \mathcal{D}') \in \mathfrak{P}$ with coarse map $\rho' : \mathfrak{S}' \to \mathcal{E}$ and, for any i, a vector $p_i \in \mathbb{Z}^{N_0}$ and an isomorphism $\mathfrak{E}_i \cong \mathfrak{S}'_{t_i}$ over E_i under which

$$\mathcal{L}_i \cong \big(\mathcal{D}' \otimes \rho'^* \mathcal{H}'(p_i)\big)|_{\mathfrak{S}'_{t_i}}.$$

Let $\lambda' \in \{0, \dots, q-1\}^{N_0}$ be a vector with $\mathcal{D}'^{\otimes q} \cong \rho'^* \mathcal{H}'(\lambda')$ given by Theorem 8.3, and $\mathcal{M}_0 := \mathcal{H}'(\lambda')$.

By Theorem 9.2 for $\mathcal{E} \to T$, $\rho' : \mathfrak{S}' \to \mathcal{E}$ in place of $\rho : \mathfrak{S} \to \mathcal{E}$, q, $\mathcal{D}_0 := \mathcal{D}'$, $\mathcal{M}_0$, $\mathcal{Z}$, J, $\mathcal{F}_j$, and $\mathcal{G}_j$, after replacing T, there exist line bundles $\mathcal{H}_1, \dots, \mathcal{H}_{N_0}$ on $\mathcal{E}$, a rational polyhedral cone $\Sigma \subset \mathbb{R}^{1+N_0}$, finite sets I_j ($j \in J$), and a real number $\Theta > 0$ satisfying conditions (1)–(3) of Theorem 9.2 with N_0 in place of m, where $\mathcal{D}_k := \rho'^* \mathcal{H}_k$ for $1 \le k \le N_0$. We regard I_j as a finite set of sections: any $s \in I_j$ is a section of $\mathcal{D}(\lambda_s)$ on $\mathfrak{S}'$ with $\lambda_s \in \Sigma \cap \mathbb{Z}^{1+N_0}$. Since the restrictions $\mathcal{H}_k|_{E_{t_i}}$ form a basis of $\mathrm{Pic}(E_{t_i})$, there exist integers $\lambda_{i,1}, \dots, \lambda_{i,N_0}$ such that

$$\mathcal{H}'(p_i)|_{E_{t_i}} \cong \bigotimes_{k=1}^{N_0} \mathcal{H}_k^{\otimes \lambda_{i,k}}|_{E_{t_i}}.$$

Let $\lambda_i := (1, \lambda_{i,1}, \dots, \lambda_{i,N_0})$. Then

$$\mathcal{L}_i \cong \mathcal{D}(\lambda_i)|_{\mathfrak{S}'_{t_i}}. \tag{12.3}$$

For any i and $j \in J$, let $I_{i,j}$ be the set of $s \in I_j$ such that $s|_{\mathfrak{S}'_{t_i}} \ne 0$. By Theorem 9.2(1), $\mathcal{D}(\lambda)|_{\mathfrak{E}_i}$ is nef if and only if $\lambda \in \Sigma \cap \mathbb{Z}^{1+N_0}$. By Theorem 9.2(2), the restrictions of the $s \in I_{i,j}$ generate $\mathcal{A}_{F_j}(\Sigma \cap \mathbb{Z}^{1+N_0})$ on $\mathfrak{E}_i$, since removing zero elements does not change the generated algebra. By Theorem 9.2(3),

$$\nu_{F_j}(s|_{\mathfrak{E}_i}) \le \Theta \quad \text{for any } s \in I_{i,j}. \tag{12.4}$$

By Proposition 10.2(2) and (12.3), there exist extensions $\widehat{\mathcal{D}}_k$ of $\mathcal{D}_k|_{\mathfrak{E}_i}$ ($0 \le k \le N_0$) to the formal completion of $\mathfrak{Y}_i$ along $\mathfrak{E}_i$ with an isomorphism $\widehat{\mathcal{D}}(\lambda_i) \cong \mathcal{O}(-\mathfrak{E}_i)$ restricting to (12.3), and, for any $s \in I_{i,j}$, a section $\widehat{s}$ of $\widehat{\mathcal{D}}(\lambda_s)$ restricting to $s|_{\mathfrak{E}_i}$. Thus, for any i and $j \in J$, conditions (1)–(5) of Set-up 10.3 hold for $(X_i \ni x_i, B_i)$, E_i, f_i, q, with $1 + N_0$ in place of m, $(\mathcal{D}_k|_{\mathfrak{E}_i})_{0 \le k \le N_0}$ in place

of $(\mathcal{D}_k)_{1\le k\le m}$, λ_i in place of λ_0, $\Sigma\cap\mathbb{Z}^{1+N_0}$ in place of Σ, F_j in place of F, and $s\in I_{i,j}$ in place of $i\in I$.

Step 2. In this step we construct cyclic charts of Y_i near the generic point of $\operatorname{center}_{E_i}F_j$ whose central divisors are open subsets of the fibers of finitely many families, and prime divisors over the fibers of the families dominating F_j.

We fix a finite cover of $\mathcal{E}$ by affine open subsets on which $\mathcal{M}_0,\mathcal{H}_1,\dots,\mathcal{H}_{N_0}$ are trivial, and fix trivializations of $\mathcal{M}_0,\mathcal{H}_1,\dots,\mathcal{H}_{N_0}$ on any member $\mathcal{W}$ of the cover. Via $\mathcal{D}_0^{\otimes q}\cong\rho'^*\mathcal{M}_0$, $\mathcal{D}_0^{\otimes q}$ is trivialized over $\rho'^{-1}(\mathcal{W})$. We let $\mathcal{U}_{\mathcal{W}}$ be the closed substack of the total space of $\mathcal{D}_0$ over $\rho'^{-1}(\mathcal{W})$ consisting of the ℓ with $\ell^{\otimes q}=1$. Then $\mathcal{U}_{\mathcal{W}}\to\rho'^{-1}(\mathcal{W})$ is a μ_q-torsor. Thus $\mathcal{U}_{\mathcal{W}}$ is étale over $\mathfrak{S}'$, and $\mathcal{U}_{\mathcal{W}}\to T$ is flat with normal fibers. Since $\mathcal{D}_0$ is faithful on stabilizers and, étale locally on $\mathcal{W}$, $\rho'^{-1}(\mathcal{W})$ is the quotient stack of a scheme finite over the étale neighborhood by a stabilizer group of $\mathfrak{S}'$ [Ols16, Theorem 11.3.1], étale locally on $\mathcal{W}$ the stack $\mathcal{U}_{\mathcal{W}}$ is the quotient of a scheme finite over the étale neighborhood by a free action of a finite group. Thus $\mathcal{U}_{\mathcal{W}}$ is finite over $\mathcal{W}$, hence an affine scheme.

By [Gro66, Théorème 9.7.7(iv)], for a morphism $\mathcal{X}\to T$ of finite type, the set of $t\in T$ such that $\mathcal{X}_t$ is geometrically integral is locally constructible. Taking a finite cover of T over which the irreducible components of the generic fibers of the $\mathcal{U}_{\mathcal{W}}$ are geometrically integral and using Noetherian induction, after replacing T, any irreducible component $\mathcal{U}$ of any $\mathcal{U}_{\mathcal{W}}$ has geometrically integral fibers U_t, and $(\mathcal{U}_{\mathcal{W}})_t$ is the disjoint union of the fibers of the irreducible components of $\mathcal{U}_{\mathcal{W}}$. For such $\mathcal{U}$, the fiber U_{t_i} is Cohen–Macaulay for any i: U_{t_i} is étale over $\mathfrak{S}'_{t_i}\cong\mathfrak{E}_i$, which near the closed point over $y\in E_i$ is the quotient stack of $\widetilde{E}$ by μ_r, and $(\widetilde{E},\operatorname{Diff}_{\widetilde{E}}(\pi_y^*B_{Y_i}))$ is klt by Definition-Lemma 2.16(2), since (E_i,B_{E_i}) is klt. On the other hand, the set of points of $\mathcal{U}$ at which the fiber is Cohen–Macaulay is open by [Gro66, Théorème 12.1.1(vi)] with $\mathcal{O}_{\mathcal{U}}$ in place of $\mathcal{F}$, so by Chevalley's theorem the set of $t\in T$ such that U_t is Cohen–Macaulay is constructible. Thus, after replacing T, U_t is a normal Cohen–Macaulay variety of dimension $n-1$ for any $t\in T$. We let $\mathfrak{C}$ be the set of irreducible components $\mathcal{U}\to T$ of the $\mathcal{U}_{\mathcal{W}}$. Then $\mathfrak{C}$ is finite. Since the irreducible components of any $\mathcal{U}_{\mathcal{W}}$ are pairwise disjoint, they are open and closed in $\mathcal{U}_{\mathcal{W}}$, so any $\mathcal{U}\in\mathfrak{C}$ is flat and affine over T.

Let $\mathcal{Q}_1,\dots,\mathcal{Q}_{N_2}$ be the irreducible components of the preimages of $\bigcup_{k=1}^m\operatorname{Supp}\mathcal{B}_k$ in the members of $\mathfrak{C}$. By [Gro66, Théorème 9.7.7(iv)] and Noetherian induction, after replacing T, the following holds for any $t\in T$ and any $\mathcal{U}\in\mathfrak{C}$: the fibers $Q_{\alpha,t}$ of the $\mathcal{Q}_\alpha\subset\mathcal{U}$ are pairwise distinct prime divisors on U_t, and any prime divisor on U_t contained in the preimage of $\bigcup_k\operatorname{Supp}(\mathcal{B}_k|_{E_t})$ is one of the $Q_{\alpha,t}$. By generic flatness and Noetherian induction, after replacing T once more, $\mathcal{Q}_\alpha\to T$ is flat for any α. Since the fibers U_t are normal, the $\mathcal{Q}_\alpha$ are relative Mumford divisors on $\mathcal{U}\to T$.

Let $\mathcal{U}\in\mathfrak{C}$ with $\mathcal{U}\subset\mathcal{U}_{\mathcal{W}}$, and $\mathcal{Z}_{\mathcal{U}}$ the normalization of the irreducible component of $\mathcal{Z}\times_{\mathcal{E}}\mathcal{U}$ dominating $\mathcal{U}$. Since $\mathcal{U}\to\mathcal{W}$ is finite of degree at most q, so is

$$\mathcal{Z}_{\mathcal{U}}\to\mathcal{Z}\times_{\mathcal{E}}\mathcal{W}.$$

Thus for any $j\in J$, at most q prime divisors on $\mathcal{Z}_{\mathcal{U}}$ map onto $\mathcal{F}_j\cap(\mathcal{Z}\times_{\mathcal{E}}\mathcal{W})$, and the ramification index e along any of them satisfies $1\le e\le q$.

Possibly replacing T with the members of a stratification of T into irreducible locally closed subsets, we may assume that T is irreducible. Over the geometric generic point of T, we take a resolution of the fiber of $\mathcal{Z}_{\mathcal{U}}$ which is an isomorphism over the non-singular locus, together with the strict transforms of the prime divisors dominating the $\mathcal{F}_j$ and their multiplicities in the pullbacks of the $\mathcal{F}_j$. After a finite surjective cover of T, the resolution and the strict transforms are defined over the function field of T, and they spread out over a dense open subset $T^\circ\subset T$ such that, over any closed point $t\in T^\circ$, the fiber of the spread-out resolution is a resolution which is an isomorphism over the non-singular locus, and the fibers of the spread-out divisors are the strict transforms of the prime divisors dominating the $F_{j,t}$, with their multiplicities in the pullbacks of the $F_{j,t}$. Repeating the argument over $T\backslash T^\circ$ (Noetherian induction), after replacing

T, there exist a smooth variety $\mathcal{Z}'$ with a projective morphism $h' : \mathcal{Z}' \to \mathcal{U}$ factoring through $\mathcal{Z}_{\mathcal{U}}$ such that $\mathcal{Z}' \to T$ is smooth and $h'_t := h'|_{Z'_t} : Z'_t \to U_t$ is birational for any $t \in T$, a finite set J' with a map $J' \to J$, and, for any $j' \in J'$, a prime divisor $\mathcal{F}'_{j'}$ on $\mathcal{Z}'$ whose fibers $F'_{j',t}$ are prime divisors and an integer $1 \le e_{j'} \le q$, satisfying the following for any $t \in T$ and $j \in J$:

(i) For any $j' \in J'$ over j, the morphism $Z'_t \to \mathcal{Z}_t$ induced by $\mathcal{Z}' \to \mathcal{Z}$ maps $F'_{j',t}$ dominantly to $F_{j,t}$, and the ramification index along $F'_{j',t}$ is $e_{j'}$.
(ii) Any prime divisor on Z'_t dominating $F_{j,t}$ is $F'_{j',t}$ for some $j' \in J'$ over j.

We fix such $\mathcal{Z}'$, J', $\mathcal{F}'_{j'}$, and $e_{j'}$ for any $\mathcal{U} \in \mathfrak{C}$, and suppress the dependence on $\mathcal{U}$ from the notation.

Now fix i and $j \in J$. We construct $\mathcal{U} \in \mathfrak{C}$, Y', g, E', $B_{Y'}$, y', y, V, $j' \in J'$, and $e_{i,j}$, and suppress the dependence on (i,j) of all but $e_{i,j}$ from the notation. Let ζ be the generic point of $\mathrm{center}_{E_i} F_j$ and $\mathcal{W}$ a member of the cover such that $\zeta \in \mathcal{W}$. By (12.3) and the trivializations of $\mathcal{H}_1, \dots, \mathcal{H}_{N_0}$, and $\mathcal{M}_0$ on $\mathcal{W}$, the line bundle

$$\mathcal{L}_i^{\otimes q} = \rho_i^*(\mathcal{O}_{Y_i}(-qE_i)|_{E_i})$$

is trivialized over $\rho_i^{-1}(E_i \cap \mathcal{W})$. Let ψ_0 be the nowhere vanishing section of $\mathcal{O}_{Y_i}(-qE_i)|_{E_i \cap \mathcal{W}}$ corresponding to 1. Let $Y_i^\circ \subset Y_i$ be an affine open subset such that $\zeta \in Y_i^\circ$, $Y_i^\circ \cap E_i \subset \mathcal{W}$, and $\mathcal{O}_{Y_i}(-qE_i)|_{Y_i^\circ}$ has a generator ψ. Then $\psi|_{E_i} = v\psi_0$ for a nowhere vanishing function v on $Y_i^\circ \cap E_i$. Since Y_i° is affine, $v = \widetilde{v}|_{E_i}$ for a regular function $\widetilde{v}$ on Y_i°. Possibly replacing Y_i° with the open subset on which $\widetilde{v} \neq 0$ and ψ with $\psi/\widetilde{v}$, we may assume that $\psi|_{E_i} = \psi_0$.

Let Y'' be the closed substack of the total space of $\mathcal{O}_{\mathfrak{Y}_i}(-\mathfrak{E}_i)$ over $\rho_i^{-1}(Y_i^\circ)$ consisting of the ℓ with $\ell^{\otimes q} = \rho_i^*\psi$, where

$$\mathcal{O}_{\mathfrak{Y}_i}(-q\mathfrak{E}_i) = \rho_i^*\mathcal{O}_{Y_i}(-qE_i).$$

Since $\rho_i^*\psi$ generates $\mathcal{O}_{\mathfrak{Y}_i}(-q\mathfrak{E}_i)$, Y'' is a μ_q-torsor over $\rho_i^{-1}(Y_i^\circ)$, so it is étale over $\rho_i^{-1}(Y_i^\circ)$ and normal. Since ρ_i is an isomorphism away from $\mathfrak{E}_i$ and $\mathcal{L}_i$ is faithful on stabilizers by **Step 1**, $\mathcal{O}_{\mathfrak{Y}_i}(-\mathfrak{E}_i)$ is faithful on stabilizers, so as for $\mathcal{U}_{\mathcal{W}}$, Y'' is a scheme finite over Y_i°. Since $\psi|_{E_i} = \psi_0$, (12.3) and the trivializations of $\mathcal{H}_1, \dots, \mathcal{H}_{N_0}$ induce an isomorphism from the preimage of $\mathfrak{E}_i$ in Y'' onto the preimage of $Y_i^\circ \cap E_i$ in $(\mathcal{U}_{\mathcal{W}})_{t_i}$, under which the tautological section of the pullback of $\mathcal{L}_i$ corresponds to the tautological section of the pullback of $\mathcal{D}_0$.

Since the morphism

$$(\mathcal{U}_{\mathcal{W}})_{t_i} \to E_{t_i} \cap \mathcal{W}$$

is finite and surjective on any irreducible component, there exist some $\mathcal{U} \in \mathfrak{C}$ with $\mathcal{U} \subset \mathcal{U}_{\mathcal{W}}$ and a prime divisor on Z'_{t_i} dominating F_j, which by (ii) is F'_{j',t_i} for some $j' \in J'$ over j, and $\mathrm{center}_{U_{t_i}}(F'_{j',t_i})$ dominates $\mathrm{center}_{E_i} F_j \ni \zeta$. We let

$$V := \mathrm{center}_{U_{t_i}}(F'_{j',t_i}), \qquad e_{i,j} := e_{j'},$$

y' a closed point of V over Y_i°, regarded as a closed point of Y'', Y' the connected component of Y'' containing y', $g : Y' \to Y_i^\circ$ the induced finite morphism, $E' := g^*E_i$, $B_{Y'} := g^*B_{Y_i}$, and $y := g(y')$. Then g factors through the étale morphism $Y' \to \rho_i^{-1}(Y_i^\circ)$, and E' is a reduced Cartier divisor on Y' which contains an open neighborhood of the generic point of V in U_{t_i} as an open subset. Moreover, the fiber product $Y' \times_{\mathfrak{Y}_i} \widetilde{Y}$ is a μ_r-torsor over Y' and étale over $\widetilde{Y}$, where $\pi_y : \widetilde{Y} \to Y_i$ is as in **Step 1**. Thus $(Y' \ni y', E' + B_{Y'})$ and $(\widetilde{Y} \ni \widetilde{y}, \widetilde{E} + \pi_y^* B_{Y_i})$ have a common étale neighborhood, so by Definition-Lemma 2.16(1)(2), the following hold:

(A) E' is a normal Cartier divisor near y', $(Y', E' + B_{Y'})$ is plt near y', and $B_{Y'} \in \Gamma$, since g is étale in codimension 1.
(B) $K_{E'} + \mathrm{Diff}_{E'}(B_{Y'}) = (g|_{E'})^*(K_{E_i} + B_{E_i})$ near y'. Thus by the Hurwitz formula, $\mathrm{Diff}_{E'}(B_{Y'}) \le (g|_{E'})^* B_{E_i}$, and by (i),

$$a(F'_{j',t_i}, E', \mathrm{Diff}_{E'}(B_{Y'})) = e_{i,j}\beta_{i,j}. \tag{12.5}$$

By (i), F'_{j',t_i} is as in Set-up 10.3 with $e = e_{i,j}$.

For any $s \in I_{i,j}$, write $\lambda_s = (\lambda_{s,0}, \dots, \lambda_{s,N_0})$. Over $\rho'^{-1}(\mathcal{W})$, the pullback of $\mathcal{D}_0$ to $\mathcal{U}_{\mathcal{W}}$ has the tautological nowhere vanishing section, and $\mathcal{H}_1, \dots, \mathcal{H}_{N_0}$ are trivialized. In the trivializations above, the pullback of s to $\mathcal{U}_{\mathcal{W}}$ is a regular function $\phi_{\mathcal{W},s}$, and we let $\overline{\phi}_s := \phi_{\mathcal{W},s}|_{U_{t_i}}$. Since the q-th power of the tautological section is 1 in the trivialization of $\mathcal{M}_0$, $\overline{\phi}_s^q$ is the pullback of the function on $E_{t_i} \cap \mathcal{W}$ which represents $s^{\otimes q}|_{\mathfrak{S}'_{t_i}}$, regarded as a section of

$$\mathcal{M}_0^{\otimes \lambda_{s,0}} \otimes \bigotimes_{k=1}^{N_0} \mathcal{H}_k^{\otimes q\lambda_{s,k}},$$

in the trivializations. Hence, by Definition 9.1, (i), and (12.4), $\overline{\phi}_s \neq 0$ and

$$\operatorname{mult}_{F'_{j',t_i}} ((h'_{t_i})^* \operatorname{div}(\overline{\phi}_s)) = e_{i,j} \nu_{F_j}(s|_{\mathfrak{E}_i}) \leq q\Theta. \tag{12.6}$$

We choose trivializations of $\widehat{\mathcal{D}}_1, \dots, \widehat{\mathcal{D}}_{N_0}$ on the formal completion $\widehat{Y'}$ of Y' along E' near the generic point of V which restrict to the trivializations of $\mathcal{H}_1, \dots, \mathcal{H}_{N_0}$, and the trivialization of

$$\widehat{\mathcal{D}}_0 \cong \widehat{\mathcal{D}}(\lambda_i) \otimes \bigotimes_{k=1}^{N_0} \widehat{\mathcal{D}}_k^{\otimes(-\lambda_{i,k})} \cong \mathcal{O}(-\mathfrak{E}_i) \otimes \bigotimes_{k=1}^{N_0} \widehat{\mathcal{D}}_k^{\otimes(-\lambda_{i,k})}$$

given by the tautological section of $g^*\mathcal{O}_{\mathfrak{Y}_i}(-\mathfrak{E}_i)$ and the trivializations of $\widehat{\mathcal{D}}_1, \dots, \widehat{\mathcal{D}}_{N_0}$. For $s \in I_{i,j}$, let ϕ_s be as in Set-up 10.3(8), with the generators of the pullbacks of $\widehat{\mathcal{D}}_0, \dots, \widehat{\mathcal{D}}_{N_0}$ given by the trivializations above in place of $\varpi_1, \dots, \varpi_m$. Since the isomorphism from the preimage of $\mathfrak{E}_i$ in Y'' onto an open subset of $(\mathcal{U}_{\mathcal{W}})_{t_i}$ above matches the tautological sections, $\phi_s|_{E'} = \overline{\phi}_s$. Thus Set-up 10.3 holds with Y_i° in place of U, F'_{j',t_i} in place of F', $e = e_{i,j}$, and $s \in I_{i,j}$ in place of $i \in I$.

Step 3. In this step we bound local volumes on Y' from below and round the coefficients of $B_{Y'}$ to a finite set.

By Lemma 11.2(2) with $\frac{1}{2}$ in place of t and Lemma 11.2(1), both applied to π_y, and by (12.1),

$$\widehat{\operatorname{vol}}\Big(\widetilde{Y} \ni \widetilde{y}, \pi_y^* B_{Y_i} + \frac{1}{2}\widetilde{E}\Big) \geq v_0 := \frac{n}{2}\kappa^{n-1}\frac{\epsilon}{C}$$

and

$$\widehat{\operatorname{vol}}\big(\widetilde{E} \ni \widetilde{y}, \operatorname{Diff}_{\widetilde{E}}(\pi_y^* B_{Y_i})\big) \geq v_1 := \Big(\frac{n-1}{n}\Big)^{n-1}\kappa^{n-1}\frac{\epsilon}{C}.$$

Since $(Y' \ni y', E' + B_{Y'})$ and $(\widetilde{Y} \ni \widetilde{y}, \widetilde{E} + \pi_y^* B_{Y_i})$ have a common étale neighborhood, by Lemma 2.10(2),

$$\widehat{\operatorname{vol}}\Big(Y' \ni y', B_{Y'} + \frac{1}{2}E'\Big) \geq v_0 \quad \text{and} \quad \widehat{\operatorname{vol}}\big(E' \ni y', \operatorname{Diff}_{E'}(B_{Y'})\big) \geq v_1. \tag{12.7}$$

We let $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma\backslash\{0\} \neq \emptyset$, and $\gamma := 1$ otherwise. Let $Y'_0 \subset Y'$ be an open neighborhood of y' such that $E'_0 := E' \cap Y'_0$ is a normal prime Cartier divisor contained in U_{t_i}, any irreducible component of $B_{Y'}|_{Y'_0}$ contains y', and $Q_{\alpha,t_i} \cap E'_0 = \emptyset$ for any $\alpha \notin \Omega$, where Ω is the set of α with $\mathcal{Q}_\alpha \subset \mathcal{U}$ and $y' \in Q_{\alpha,t_i}$. Since Y'_0 contains the generic point of V, $a(w, Y', E' + B_{Y'}) = a(w, Y'_0, E'_0 + B_{Y'}|_{Y'_0})$ for any divisorial valuation w on Y' whose center is the generic point of V. Write

$$B_{Y'}|_{Y'_0} = \sum_{k=1}^{N_1} b'_k D_k,$$

where the D_k are distinct prime divisors and $b'_k \in \Gamma\backslash\{0\}$, so $b'_k \geq \gamma$. Since E'_0 is a normal Cartier divisor, $\operatorname{Diff}_{E'_0}(B_{Y'}|_{Y'_0}) = \sum_k b'_k D_k|_{E'_0}$, where $D_k|_{E'_0}$ is the restriction defined at the beginning of

Section 11, and $D_k|_{E'_0} \neq 0$ since $y' \in D_k \cap E'_0$. By (B),

$$\mathrm{Supp}(D_k|_{E'_0}) \subset (g|_{E'})^{-1}(\mathrm{Supp}\, B_{E_i}) \subset (g|_{E'})^{-1}\Big(\bigcup_{k'=1}^{m} \mathrm{Supp}(\mathcal{B}_{k'}|_{E_{t_i}})\Big),$$

so by **Step 2** there exist $c_{k,\alpha} \in \mathbb{N}$ such that

$$D_k|_{E'_0} = \sum_{\alpha\in\Omega} c_{k,\alpha} Q_{\alpha,t_i}|_{E'_0} \quad \text{for any } k.$$

Since $(E'_0, \mathrm{Diff}_{E'_0}(B_{Y'}|_{Y'_0}))$ is klt by (A), for any $\alpha \in \Omega$,

$$\gamma \sum_{k=1}^{N_1} c_{k,\alpha} \leq \sum_{k=1}^{N_1} b'_k c_{k,\alpha} = \mathrm{mult}_{Q_{\alpha,t_i}} \mathrm{Diff}_{E'_0}(B_{Y'}|_{Y'_0}) < 1.$$

Thus $\sum_k c_{k,\alpha} < \gamma^{-1}$, and since $D_k|_{E'_0} \neq 0$ for any k,

$$N_1 \leq \sum_{k=1}^{N_1} \sum_{\alpha\in\Omega} c_{k,\alpha} \leq \gamma^{-1} N_2.$$

For $1 \leq k \leq N_1$, let

$$\mathcal{T}_k := \sum_{\alpha\in\Omega} c_{k,\alpha} \mathcal{Q}_\alpha.$$

Then $\mathcal{T}_1, \dots, \mathcal{T}_{N_1}$ are relative Mumford divisors on $\mathcal{U} \to T$ with $D_k|_{E'_0} = \mathcal{T}_k|_{U_{t_i}}$ on E'_0. Hence $(\mathcal{T}_1, \dots, \mathcal{T}_{N_1})$ belongs to the finite set of tuples

$$\Big(\sum_{\alpha} c_{k,\alpha} \mathcal{Q}_\alpha\Big)_{1\leq k\leq N_1}$$

with $N_1 \leq \gamma^{-1} N_2$, $c_{k,\alpha} \in \mathbb{N}$, and $\sum_k c_{k,\alpha} < \gamma^{-1}$ for any α, where the $\mathcal{Q}_\alpha$ are contained in one member of $\mathfrak{C}$.

For any $j \in J$, by (A) and (12.7), the germs $(Y'_0 \ni y', E'_0 + B_{Y'}|_{Y'_0})$, indexed by i, satisfy the conditions of Lemma 11.3 with v_0 and v_1 in place of ϵ_1 and ϵ_2, and we let $\Gamma_{0,j}$ be the finite set Γ_0 of Lemma 11.3 for j. Since J is finite, possibly passing to subsequences and shrinking Y'_0 near y', for $\Gamma_0 := \bigcup_{j\in J} \Gamma_{0,j}$, there exist, for any i and j, an $\mathbb{R}$-divisor $B^+_{Y'} = \sum_k b^+_k D_k$ on Y'_0 and a real number $\sigma_{i,j} \geq 0$ with $\lim_{i\to+\infty} \sigma_{i,j} = 0$ satisfying the following:

(1) $b^+_k \in \Gamma_0$ and $b^+_k \geq b'_k$ for any k.
(2) $K_{Y'_0} + E'_0 + B^+_{Y'}$ is $\mathbb{R}$-Cartier and $(Y'_0, E'_0 + B^+_{Y'})$ is plt.
(3) For any divisorial valuation w on Y'_0 with $\mathrm{center}_{Y'_0} w \in E'_0$,

$$(1 - \sigma_{i,j}) a(w, Y'_0, E'_0 + B_{Y'}|_{Y'_0}) \leq a(w, Y'_0, E'_0 + B^+_{Y'}). \tag{12.8}$$

Since any D_k contains $y' \in E'_0$ and $(Y'_0, E'_0 + B^+_{Y'})$ is plt, $b^+_k < 1$ for any k. Thus

$$b^+ := (b^+_1, \dots, b^+_{N_1}) \in \Lambda_{N_1} := (\Gamma_0 \cap [0,1))^{N_1},$$

which is a finite subset of $[0,1)^{N_1}$.

Step 4. In this step we construct divisorial valuations w' on Y' with $w'(E') = k_{i,j}$ and bounded log discrepancy, and we conclude.

For any $\mathcal{U} \in \mathfrak{C}$ and any tuple $(\mathcal{T}_1, \dots, \mathcal{T}_{N_1})$ in **Step 3**, let $k_0(\mathcal{U}, \mathcal{T}_1, \dots, \mathcal{T}_{N_1})$ be the positive integer given by Theorem 11.1 for $\mathcal{U} \to T$, $h' : \mathcal{Z}' \to \mathcal{U}$, $\{\mathcal{F}'_{j'}\}_{j'\in J'}$, $\mathcal{T}_1, \dots, \mathcal{T}_{N_1}$ in place of $\mathcal{B}_1, \dots, \mathcal{B}_m$, Λ_{N_1} in place of Λ, and v_0. Let k_0 be the maximum of the finitely many integers $k_0(\mathcal{U}, \mathcal{T}_1, \dots, \mathcal{T}_{N_1})$. Then k_0 depends only on the members of $\mathfrak{C}$ with their $\mathcal{Z}'$ and $\mathcal{F}'_{j'}$, the finitely many tuples, Γ_0, and v_0, and in particular not on i.

Fix i and $j \in J$, and let

$$k_{i,j} := \left\lceil \frac{e_{i,j}u_j}{a_i} \right\rceil, \quad c := \frac{k_{i,j}}{e_{i,j}}, \quad \text{and} \quad u_{i,j} := a_i c.$$

Then

$$u_j \le u_{i,j} < u_j + \frac{a_i}{e_{i,j}} \le u_j + a_i. \tag{12.9}$$

Since $\lim_{i\to+\infty} a_i = 0$, $u_j > 0$, $e_{i,j} \ge 1$, and $\lim_{i\to+\infty} \sigma_{i,j} = 0$, for any $i \gg 0$ and $j \in J$, we have $k_{i,j} > k_0 + q\Theta$ and $\sigma_{i,j} < 1$. Let $i \gg 0$. By **Step 2**, **Step 3**, (12.7), and (12.6), the conditions of Theorem 11.1 hold with the following in place of the data there:

(1) t_i in place of t.
(2) j' in place of j.
(3) b^+ in place of b.
(4) Y'_0, E'_0, $D_1, \dots, D_{N_1}$, and y' in place of Y, E, $B_1, \dots, B_m$, and y.
(5) $\Delta := B_{Y'}|_{Y'_0} + \frac{1}{2}E'_0$, which is klt near y' since E'_0 is Cartier, $(Y'_0, E'_0 + B_{Y'}|_{Y'_0})$ is plt near y' by (A), and $b'_k \le b^+_k < 1$ for any k.
(6) $s \in I_{i,j}$ in place of $i \in I$.
(7)
$$M = \max\left(\{0\} \cup \{e_{i,j}\nu_{F_j}(s|_{\mathfrak{E}_i}) \mid s \in I_{i,j}\}\right) \le q\Theta < k_{i,j} - k_0.$$

Thus there exists a divisorial valuation w' on Y' with value group $\mathbb{Z}$ such that $\operatorname{center}_{Y'} w'$ is the generic point of V, $w'(E') = k_{i,j} = e_{i,j}c$,

$$a(w', Y'_0, E'_0 + B^+_{Y'}) = a\Big(F'_{j',t_i}, U_{t_i}, \sum_{k=1}^{N_1} b^+_k \mathcal{T}_k|_{U_{t_i}}\Big),$$

and $w'(\phi_s) = e_{i,j}\nu_{F_j}(s|_{\mathfrak{E}_i})$ for any $s \in I_{i,j}$. Near the generic point of V, we have

$$\sum_k b^+_k \mathcal{T}_k|_{U_{t_i}} = \operatorname{Diff}_{E'_0}(B^+_{Y'}) \ge \operatorname{Diff}_{E'_0}(B_{Y'}|_{Y'_0}),$$

since $B^+_{Y'} \ge B_{Y'}|_{Y'_0}$. Hence, by (12.5) and (12.8),

$$\frac{a(w', Y', E' + B_{Y'})}{e_{i,j}} \le \frac{a(w', Y'_0, E'_0 + B^+_{Y'})}{e_{i,j}(1-\sigma_{i,j})} \le \frac{a(F'_{j',t_i}, E', \operatorname{Diff}_{E'}(B_{Y'}))}{e_{i,j}(1-\sigma_{i,j})} = \frac{\beta_{i,j}}{1-\sigma_{i,j}}.$$

By Proposition 10.5 and since

$$u_{i,j} + \frac{\beta_{i,j}}{1-\sigma_{i,j}} \le \frac{u_{i,j} + \beta_{i,j}}{1-\sigma_{i,j}},$$

$$\frac{\widehat{\operatorname{vol}}(X_i \ni x_i, B_i)}{a_i} \le \Big(\frac{u_{i,j} + \frac{\beta_{i,j}}{1-\sigma_{i,j}}}{u_{i,j} + \beta_{i,j}}\Big)^n \Psi_{E_i, B_{E_i}}(F_j, u_{i,j}) \le (1-\sigma_{i,j})^{-n} \Psi_{E_i, B_{E_i}}(F_j, u_{i,j}). \tag{12.10}$$

We bound the dependence of $\Psi_{E_i,B_{E_i}}(F_j, u)$ on u. For any $t \in T$ and $j \in J$, $a(F_{j,t}, E_t, B_t(b))$ is an affine function of b on the affine subspace of the $b \in \mathbb{R}^m$ such that $K_{E_t} + B_t(b)$ is $\mathbb{R}$-Cartier, and by Theorem 7.1(2) the restriction of the function $a(F_{j,t}, E_t, B_t(\cdot))$ to $\mathcal{P}$ does not depend on t. Since $\mathcal{P}_0$ is compact and J is finite, there exists a real number $\beta_0 > 0$ such that

$$\beta_{i,j} \le \beta_0 \quad \text{for any } i \text{ and } j \in J. \tag{12.11}$$

For any closed point $y \in E_i$, by Lemma 2.13 with $n-1$ in place of d and since $\delta(E_i, B_{E_i}) \ge \kappa$,

$$\kappa^{n-1} L_i^{n-1} \le \Big(\frac{n}{n-1}\Big)^{n-1} \widehat{\operatorname{vol}}(E_i \ni y, B_{E_i}) \le n^{n-1},$$

where $\widehat{\mathrm{vol}}(E_i \ni y, B_{E_i}) \le (n-1)^{n-1}$ follows from Lemma 2.8(1) when $n \ge 3$, and from $\widehat{\mathrm{vol}}(E_i \ni y, B_{E_i}) = 1 - \mathrm{mult}_y B_{E_i} \le 1$ when $n = 2$, in which case E_i is a smooth curve and the valuations on E_i centered at y are the positive multiples of $\mathrm{mult}_y \mathrm{div}(\cdot)$. Thus

$$L_i^{n-1} \le n^{n-1}\kappa^{1-n}. \tag{12.12}$$

By Definition 5.3,

$$\Psi_{E_i,B_{E_i}}(F_j, u) = (\beta_{i,j} + u)^n \chi(u),$$

where $\chi(u)$ is the second factor in Definition 5.3 with E_i, B_{E_i}, L_i, and F_j in place of S, B, L, and F, and h as there, and $0 \le \chi(u) \le L_i^{n-1}u^{-n}$. Differentiating the integral in $\chi(u)$ under the integral sign, since $0 \le \mathrm{vol}(h^*L_i - zF_j) \le L_i^{n-1}$ for any real number $z \ge 0$, and $n\int_0^{+\infty}(z+u)^{-n-1}dz = u^{-n}$,

$$|\chi'(u)| \le \frac{2nL_i^{n-1}}{u^{n+1}}.$$

By (12.11) and (12.12), for any $u \in [u_j, u_j + 1]$,

$$\Big|\frac{d}{du}\Psi_{E_i,B_{E_i}}(F_j, u)\Big| \le \eta_j := n^n\kappa^{1-n}\Big(\frac{(\beta_0 + u_j + 1)^{n-1}}{u_j^n} + \frac{2(\beta_0 + u_j + 1)^n}{u_j^{n+1}}\Big)$$

and

$$\Psi_{E_i,B_{E_i}}(F_j, u_j) \le \omega_j := n^{n-1}\kappa^{1-n}(\beta_0 + u_j)^n u_j^{-n}.$$

Since $a_i \le 1$, by (12.9) and (12.10),

$$\frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} \le \Psi_{E_i,B_{E_i}}(F_j, u_j) + \big((1-\sigma_{i,j})^{-n} - 1\big)\omega_j + (1-\sigma_{i,j})^{-n}\eta_j a_i.$$

Since J is finite, $\lim_{i\to+\infty} a_i = 0$, and $\lim_{i\to+\infty}\sigma_{i,j} = 0$ for any j, for any $i \gg 0$,

$$\frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} < \Psi_{E_i,B_{E_i}}(F_j, u_j) + \frac{\tau}{3} \quad \text{for any } j \in J.$$

By Theorem 6.1(1), $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)/a_i \le L_i^{n-1}$. Thus by (12.2), for any $i \gg 0$,

$$\frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a_i} < \Phi(E_i, B_{E_i}) + \frac{2\tau}{3},$$

which contradicts the choice of $(X_i \ni x_i, B_i)$ and E_i. □

13. Proof of the main theorem

Proof of Theorem 1.4. Assume that $\mathcal{R}_{n,\Gamma}$ does not satisfy the ACC. Then there exist klt germs $(X_i \ni x_i, B_i)$ of dimension n with $B_i \in \Gamma$, $i \in \mathbb{N}^+$, such that the sequence

$$R_i := \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{\mathrm{mld}(X_i \ni x_i, B_i)}$$

is strictly increasing. By Theorem 1.2, $R_i \le n^{n-1}$ for any i. Thus the limit $R_\infty := \lim_{i\to+\infty} R_i$ exists and

$$0 < R_1 \le R_i < R_\infty \le n^{n-1} \quad \text{for any } i.$$

Possibly shrinking X_i to a neighborhood of x_i, we may assume that any irreducible component of B_i contains x_i. We write

$$B_i = \sum_j b_{i,j}B_{i,j},$$

where $B_{i,j}$ are the distinct irreducible components of B_i. Then $b_{i,j} \in \Gamma\backslash\{0\}$ and $x_i \in B_{i,j}$ for any i, j.

Let $\mathcal{I}$ be the closure of $\Gamma \cup \{0\}$ in $[0,1]$, which is a DCC set. We let $\gamma := \min(\Gamma\backslash\{0\})$ if $\Gamma\backslash\{0\} \neq \emptyset$, and $\gamma := 1$ otherwise. Then $b_{i,j} \geq \gamma$ for any i, j, that is,

$$B_i \geq \gamma \operatorname{Supp} B_i \quad \text{for any } i.$$

Step 1. We show that

$$\lim_{i\to+\infty} \widehat{\operatorname{vol}}(X_i \ni x_i, B_i) = 0.$$

Assume not. Possibly passing to a subsequence, there exists a real number $\lambda > 0$ such that $\widehat{\operatorname{vol}}(X_i \ni x_i, B_i) \geq \lambda$ for any i.

By Theorem 2.20, with n in place of d and $\min\{\lambda, \gamma\}$ in place of ϵ, there exist a real number $c > 0$ and, for any i, a $\mathbb{Q}$-divisor B_i^+ such that $(X_i \ni x_i, B_i^+)$ is a klt germ and

$$B_i^+ - B_i \geq cB_i.$$

By Theorem 2.18, with n in place of d and λ in place of ϵ, there exists a positive integer M such that $\operatorname{mld}(X_i \ni x_i, B_i) \leq M$ for any i. Let $t_0 > 0$ be the real number given by Theorem 2.19 for M, $\mathcal{I}$, and the sequence $(X_i \ni x_i, B_i)$, with n in place of d and λ in place of ϵ, and let $t := t_0/2$. By Theorem 2.19, possibly passing to a subsequence, there exist a finite set $\mathcal{I}_t \subset \mathcal{I}$ and a map $p := p_t : \mathcal{I} \to \mathcal{I}_t$ satisfying Theorem 2.19(1)–(3), where $p(B_i) := \sum_j p(b_{i,j})B_{i,j}$.

Let

$$s_i := \frac{\|p(B_i) - B_i\|}{c\gamma}.$$

Since $b \leq p(b) \leq b + t$ for any $b \in \mathcal{I}$ by Theorem 2.19(1), $B_i \geq \gamma \operatorname{Supp} B_i$, and $B_i^+ - B_i \geq cB_i$, we have

$$0 \leq p(B_i) - B_i \leq \|p(B_i) - B_i\| \operatorname{Supp} B_i \leq \frac{\|p(B_i) - B_i\|}{\gamma} B_i \leq s_i(B_i^+ - B_i),$$

and $\lim_{i\to+\infty} s_i = 0$ by Theorem 2.19(3). Possibly passing to a subsequence, we may assume that $s_i \leq 1 - 2^{-1/n}$ for any i, so that $0 \leq s_i < 1$ and $(1 - s_i)^n \geq \frac{1}{2}$.

Possibly shrinking X_i to a neighborhood of x_i, we may assume that (X_i, B_i^+) is klt. Since $K_{X_i} + B_i$, $K_{X_i} + B_i^+$, and $K_{X_i} + p(B_i)$ are $\mathbb{R}$-Cartier, the $\mathbb{R}$-divisors $p(B_i) - B_i \geq 0$ and $s_i(B_i^+ - B_i) - (p(B_i) - B_i) \geq 0$ are $\mathbb{R}$-Cartier, so for any valuation v on X_i,

$$0 \leq v(p(B_i) - B_i) \leq s_i v(B_i^+ - B_i).$$

Since (X_i, B_i^+) is klt, by (2.1), $a(v, X_i, B_i^+) = a(v, X_i, B_i) - v(B_i^+ - B_i) \geq 0$, so

$$v(p(B_i) - B_i) \leq s_i a(v, X_i, B_i).$$

Since $a(v, X_i, p(B_i)) = a(v, X_i, B_i) - v(p(B_i) - B_i)$, we obtain

$$(1 - s_i)a(v, X_i, B_i) \leq a(v, X_i, p(B_i)) \leq a(v, X_i, B_i). \tag{13.1}$$

Since $s_i < 1$ and (X_i, B_i) is klt, the left inequality of (13.1) for the valuations of prime divisors over X_i implies that $(X_i \ni x_i, p(B_i))$ is a klt germ. Taking the infimum of (13.1) over valuations centered at x_i and over prime divisors with center x_i, we obtain

$$\widehat{\operatorname{vol}}(X_i \ni x_i, p(B_i)) \geq (1 - s_i)^n \widehat{\operatorname{vol}}(X_i \ni x_i, B_i) \quad \text{and} \quad \operatorname{mld}(X_i \ni x_i, p(B_i)) \leq \operatorname{mld}(X_i \ni x_i, B_i).$$

Let

$$R_i' := \frac{\widehat{\operatorname{vol}}(X_i \ni x_i, p(B_i))}{\operatorname{mld}(X_i \ni x_i, p(B_i))}.$$

Since $0 \leq B_i \leq p(B_i)$ and $K_{X_i} + B_i$ is $\mathbb{R}$-Cartier, $R_i' \leq R_i$ by Proposition 3.1. Thus

$$\frac{1}{2}R_1 \leq (1 - s_i)^n R_i \leq R_i' \leq R_i < R_\infty \quad \text{for any } i.$$

Since $\lim_{i\to+\infty} s_i = 0$, $\lim_{i\to+\infty} R_i' = R_\infty$. Since $p(B_i) \in \mathcal{I}_t$,

$$R_i' \in \mathcal{R}_{n,\mathcal{I}_t} \cap \left[\frac{1}{2}R_1, +\infty\right),$$

which is a finite set by Theorem 1.3, as $\mathcal{I}_t \subset [0,1]$ is finite. Thus $R_i' = R_\infty$ for any $i \gg 0$, which contradicts $R_i' < R_\infty$. This proves **Step 1**.

Step 2. Let $\epsilon := R_1$. Then $\widehat{\mathrm{vol}}(X_i \ni x_i, B_i) \geq \epsilon \mathrm{mld}(X_i \ni x_i, B_i)$ for any i. By Theorem 4.7, there exist real numbers $C \geq 1$ and $\kappa > 0$ and, for any i, a Kollár component E_i of $(X_i \ni x_i, B_i)$ such that

$$a(E_i, X_i, B_i) \leq C\mathrm{mld}(X_i \ni x_i, B_i) \quad \text{and} \quad \delta(E_i, B_{E_i}) \geq \kappa.$$

Since

$$\mathrm{mld}(X_i \ni x_i, B_i) = \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{R_i} \leq \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{\epsilon},$$

Step 1 implies that

$$\lim_{i \to +\infty} a(E_i, X_i, B_i) = 0.$$

Step 3. For any i, $(X_i \ni x_i, B_i)$ and E_i satisfy the conditions of Set-up 4.8 with n, ϵ, κ, C. Let θ be as in Set-up 4.8(1). By **Step 2**, possibly passing to a subsequence, we may assume that $a(E_i, X_i, B_i) \leq \theta/2$ for any i. By Proposition 4.9(5),

$$\mathrm{mld}(X_i \ni x_i, B_i) = a(E_i, X_i, B_i), \quad \text{so} \quad R_i = \frac{\widehat{\mathrm{vol}}(X_i \ni x_i, B_i)}{a(E_i, X_i, B_i)}.$$

Let $\Phi_i := \Phi(E_i, B_{E_i})$. By Theorem 6.1(1), $\Phi_i \leq R_i$ for any i.

Let $\tau > 0$ be a real number. Since $\lim_{i \to +\infty} a(E_i, X_i, B_i) = 0$, Theorem 12.1 and $\Phi_i \leq R_i$ imply that

$$0 \leq R_i - \Phi_i < \tau \quad \text{for any } i \gg 0.$$

Since $\tau > 0$ is arbitrary, $\lim_{i \to +\infty} \Phi_i = R_\infty$. By Theorem 6.1(2), the set $\{\Phi_i\}_{i \geq 1}$ satisfies the ACC, so possibly passing to a subsequence, we may assume that $(\Phi_i)_{i \geq 1}$ is decreasing. Then $\Phi_i \geq R_\infty$ for any i, whereas $\Phi_i \leq R_i < R_\infty$, a contradiction. The theorem follows. □

References

[ACV03] D. Abramovich, A. Corti, and A. Vistoli, *Twisted bundles and admissible covers*, Comm. Algebra **31** (2003), no. 8, 3547–3618.

[AOV08] D. Abramovich, M. Olsson, and A. Vistoli, *Tame stacks in positive characteristic*, Ann. Inst. Fourier (Grenoble) **58** (2008), no. 4, 1057–1091.

[Alp13] J. Alper, *Good moduli spaces for Artin stacks*, Ann. Inst. Fourier (Grenoble) **63** (2013), no. 6, 2349–2402.

[Bir21] C. Birkar, *Singularities of linear systems and boundedness of Fano varieties*, Ann. of Math. (2) **193** (2021), no. 2, 347–405.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[Blu18] H. Blum, *Existence of valuations with smallest normalized volume*, Compos. Math. **154** (2018), no. 4, 820–849.

[BJ20] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. **365** (2020), 107062, 57 pp.

[BL21] H. Blum and Y. Liu, *The normalized volume of a singularity is lower semicontinuous*, J. Eur. Math. Soc. (JEMS) **23** (2021), no. 4, 1225–1256.

[BC11] S. Boucksom and H. Chen, *Okounkov bodies of filtered linear series*, Compos. Math. **147** (2011), no. 4, 1205–1229.

[BH98] W. Bruns and J. Herzog, *Cohen–Macaulay rings*, revised ed., Cambridge Stud. Adv. Math. **39**, Cambridge University Press, Cambridge (1998).

[Cad07] C. Cadman, *Using stacks to impose tangency conditions on curves*, Amer. J. Math. **129** (2007), no. 2, 405–427.

[CLZ25] S. R. Choi, Z. Li, and C. Zhou, *Variation of cones of divisors in a family of varieties—Fano type case*, arXiv:2504.04109.

[Cut14] S. D. Cutkosky, *Asymptotic multiplicities of graded families of ideals and linear series*, Adv. Math. **264** (2014), 55–113.

[Ful93] W. Fulton, *Introduction to toric varieties*, Annals of Mathematics Studies **131**, Princeton University Press, Princeton, NJ (1993).

[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. **28** (1966), 5–255.
[Gro03] A. Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Documents Mathématiques (Paris) **3**, Société Mathématique de France, Paris (2003).
[HMX14] C. D. Hacon, J. McKernan, and C. Xu, *ACC for log canonical thresholds*, Ann. of Math. (2) **180** (2014), no. 2, 523–571.
[Han26] J. Han, *A sharp inequality between local volumes and minimal log discrepancies*, arXiv:2608.16726.
[HLQ24] J. Han, J. Liu, and L. Qi, *ACC for local volumes*, arXiv:2408.15090.
[HLS19] J. Han, J. Liu, and V. V. Shokurov, *ACC for minimal log discrepancies of exceptional singularities*, arXiv:1903.04338.
[HLS26] J. Han, J. Liu, and V. V. Shokurov, *ACC for minimal log discrepancies of exceptional singularities*, Peking Math. J. **9** (2026), no. 2, 225–257.
[HLQ23] J. Han, Y. Liu, and L. Qi, *ACC for local volumes and boundedness of singularities*, J. Algebraic Geom. **32** (2023), no. 3, 519–583.
[HK00] Y. Hu and S. Keel, *Mori dream spaces and GIT*, Michigan Math. J. **48** (2000), no. 1, 331–348.
[JM12] M. Jonsson and M. Mustaţă, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.
[KKMS73] G. Kempf, F. Knudsen, D. Mumford, and B. Saint-Donat, *Toroidal embeddings. I*, Lecture Notes in Math. **339**, Springer-Verlag, Berlin (1973).
[Kim26] D. Kim, *On the normalized local volume of a non-closed point*, arXiv:2604.27484.
[Kle05] S. L. Kleiman, *The Picard scheme*, Fundamental algebraic geometry, Math. Surveys Monogr. **123**, Amer. Math. Soc., Providence, RI, 2005, pp. 235–321.
[Kol96] J. Kollár, *Rational curves on algebraic varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **32**, Springer-Verlag, Berlin (1996).
[Kol13] J. Kollár, *Singularities of the minimal model program*, with a collaboration of S. Kovács, Cambridge Tracts in Mathematics **200**, Cambridge University Press, Cambridge (2013).
[Kol23] J. Kollár, *Families of varieties of general type*, Cambridge Tracts in Math. **231**, Cambridge Univ. Press, Cambridge (2023).
[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).
[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **48**, Springer-Verlag, Berlin (2004).
[LM09] R. Lazarsfeld and M. Mustaţă, *Convex bodies associated to linear series*, Ann. Sci. Éc. Norm. Supér. (4) **42** (2009), no. 5, 783–835.
[Li18] C. Li, *Minimizing normalized volumes of valuations*, Math. Z. **289** (2018), no. 1–2, 491–513.
[LLX20] C. Li, Y. Liu, and C. Xu, *A guided tour to normalized volume*, Geometric analysis, Progr. Math. **333**, Birkhäuser/Springer, Cham, 2020, pp. 167–219.
[LX18] C. Li and C. Xu, *Stability of valuations: higher rational rank*, Peking Math. J. **1** (2018), no. 1, 1–79.
[LX20] C. Li and C. Xu, *Stability of valuations and Kollár components*, J. Eur. Math. Soc. (JEMS) **22** (2020), no. 8, 2573–2627.
[LiuJ26] J. Liu, *Boundedness of K-semistable log cone singularities with coefficients bounded away from zero*, preprint (2026), https://jihaoliu.org/papers/boundedness-of-k-semistable-log-cone-singularities-with-coefficients-bounded-away-from-zero.html.
[Liu26] Y. Liu, *Optimal bounds for local volumes of threefold singularities*, J. Lond. Math. Soc. (2) **114** (2026), no. 4, Paper No. e70738.
[LMS26] Y. Liu, J. Moraga, and H. Süß, *On the boundedness of singularities via normalized volume*, J. Inst. Math. Jussieu **25** (2026), no. 6, 3033–3081.
[LW24] Y. Liu and Y. Wu, *K-semistability of log Fano cone singularities*, arXiv:2408.05189.
[LX19] Y. Liu and C. Xu, *K-stability of cubic threefolds*, Duke Math. J. **168** (2019), no. 11, 2029–2073.
[LZ26] Y. Liu and C. Zhou, *Nonproportional wall crossing for K-stability*, Proc. Lond. Math. Soc. (3) **132** (2026), no. 4, Paper No. e70157.
[LZ21] Y. Liu and Z. Zhuang, *Birational superrigidity and K-stability of singular Fano complete intersections*, Int. Math. Res. Not. IMRN **2021**, no. 1, 382–401.
[Mat89] H. Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1989).
[Ols16] M. Olsson, *Algebraic spaces and stacks*, Amer. Math. Soc. Colloq. Publ. **62**, Amer. Math. Soc., Providence, RI (2016).
[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.
[Xu20] C. Xu, *A minimizing valuation is quasi-monomial*, Ann. of Math. (2) **191** (2020), no. 3, 1003–1030.

[XZ21] C. Xu and Z. Zhuang, *Uniqueness of the minimizer of the normalized volume function*, Camb. J. Math. **9** (2021), no. 1, 149–176.
[XZ24] C. Xu and Z. Zhuang, *Boundedness of log Fano cone singularities and discreteness of local volumes*, arXiv:2404.17134.
[XZ25] C. Xu and Z. Zhuang, *Stable degenerations of singularities*, J. Amer. Math. Soc. **38** (2025), no. 3, 585–626.
[ZS60] O. Zariski and P. Samuel, *Commutative algebra. Vol. II*, The University Series in Higher Mathematics, D. Van Nostrand Co., Princeton, NJ (1960).
[Zhu24a] Z. Zhuang, *On boundedness of singularities and minimal log discrepancies of Kollár components*, J. Algebraic Geom. **33** (2024), no. 3, 521–565.
[Zhu24b] Z. Zhuang, *On boundedness of singularities and minimal log discrepancies of Kollár components, II*, Geom. Topol. **28** (2024), no. 8, 3909–3933.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China
Email address: liujihao@math.pku.edu.cn